

Markov chains revisited

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Preface — please read!

What this book is intended to be. Although I’ve disguised it very well, this book is actually written for those interested in the applications of Markov chains (discrete-time and continuous-time processes that satisfy the Markov property and take values in countable sets). I have two main goals in writing it:

1. To equip graduate students and researchers with the tools and intuition necessary to overcome the theoretical headaches encountered in the modelling of real-world phenomena using Markov chains with infinite state spaces. My aim here is not so much to introduce novel results as to gather scattered ones and make the existing theory more accessible.

“Although only a special case of the strong Markov property is involved in the culminating Theorem 8, a “prodigious” amount of preparation, it seems, enters into the proof of this intuitively obvious result. One wonders if the present theory of stochastic process is not still too difficult for applications.”

Kai Lai Chung in (Chung, 1967).

2. To survey and consolidate the disparate numerical techniques available for the practical use of these type of models (this material is currently absent from the manuscript).

I also have several other aims:

1. Help abridge the gap between undergraduate-level texts, such as Norris’s fine book (Norris, 1997), and more advanced treatments of the type in (Chung, 1967; Rogers and Williams, 2000a). To this end, I introduce early on the path space and path law of a chain and give the strong Markov property in its most applicable (albeit ugly) form which involves these concepts.
2. Give a comprehensive account of the long-term behaviour of chains and the associated Foster-Lyapunov criteria similar to that given in (Meyn and Tweedie, 2009) for general discrete-time Markov processes (but in the much simplified and more accessible setting of countable state spaces). In particular, an account free of the irreducibility assumptions pervasive in the introductory Markov chain literature. These assumptions are generally not necessary and, unfortunately, can hinder the practical use of the results that feature them. Not just because they limit the applicability of the stated result to the irreducible case, but also because, frustratingly, arguing in practice that irreducible chains are indeed irreducible sometimes proves very challenging.
3. Highlight the edges of the theory where unresolved issues and open questions creep.

“The mistakes and unresolved difficulties of the past in mathematics have always been the opportunities of its future; and should analysis ever appear to be without flaw or blemish, its perfection might only be that of death. ”

Eric Temple Bell in (Bell, 1945).

What this book currently is. A rigorous and largely self-contained account of (a) the bread-and-butter concepts and techniques in Markov chain theory and (b) the long-term behaviour of chains. As much as possible, the treatment is probabilistic instead of analytical (I stay away from semigroup theory). Personally, I tend to find that the intuition lies with the former and not the latter. As stated above, this manuscript is geared towards those interested in the use of Markov chains as models of real-life phenomena. For this reason, I focus on the type of chains most commonly encountered in practice (time-homogeneous, minimal, and right-continuous in

the discrete topology) and choose a starting point very familiar to this audience: the (Kendall-) Gillespie Algorithm commonly used to simulate these chains. In order to keep the prerequisite knowledge and technical complications to a minimum, I take a ‘bare-bones’ approach that keeps the focus on chains (instead of more general processes) to an almost pathological degree: I use the ‘jump and hold’ structure of chains extensively; almost no martingale theory; minimal coupling; no stochastic calculus; and, even though regeneration and renewal (of course!) feature in the book, they do so exclusively in the context of chains. I have also taken some extra steps to avoid imposing certain assumptions encountered in other texts that sometimes prove to be stumbling blocks in practice (e.g., irreducibility of the state space, boundedness of test functions and of stopping times).

“The traveller often has the choice between climbing a peak or using a cable car.”

William Feller in (Feller, 1971).

To facilitate the cable car experience for those who do not wish to drag themselves through the proofs, I try to give the main results and ideas at the beginning of each section and leave the more technical arguments until the end (some sections are more amenable to this than others). This said, to those with the luxury of time, I heartily recommend climbing.

This is not a finished product! Read at your own peril. This book is a work in progress to which I plan to add material over time. To facilitate its use in the interim, I will post updated drafts on arXiv. If you find any mistakes or typos in the latest draft posted, or have some general thoughts on the book, please let me know about them. I can be reached at

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Prerequisites. The only prerequisites are a competence with the measure theoretic foundations of modern probability and a familiarity with conditional expectation, its properties, and discrete-time martingales; all to a level no further than that of (Williams, 1991, Chapters 1–10). See the “Some important notation and preliminaries” section for details.

How to use this book. This paragraph is for the non-expert reader, the expert reader of course will be able to judge this matter for him-or-herself! The manuscript, as it stands, consists of two parallel developments of the theory, the first for discrete-time chains and the second for continuous-time ones. Each is intended to be read linearly, although sections marked with an ‘*’ are of little importance to later sections. Readers not interested in their contents can skip these sections without jeopardising their future comprehension. Readers whose interests lie entirely within the continuous-time case may be tempted to skip Sections 1–25 dealing with the discrete-time one. I, however, would advise against this. Not only are the proofs for the continuous-time case often reliant on, or analogous to, those for the discrete-time one, but discrete-time chains are technically simpler and the underlying ideas are exactly the same (ignore, however, any mention of periodicity in the discrete-time treatment). To minimise the amount of repetition, I often give the intuitive discussion only for the discrete-time case (for instance, compare Sections 2 and 12 with Section 30).

Referencing within the book. Within individual sections, objects (equations, theorems, examples, etc.) are labelled sequentially with a single number. Outside a section, the section’s number is prefixed to the labels of that section’s objects. For instance, Equation (7) in Section 1 would be referenced as “(7)” throughout Section 1 but as “(1.7)” throughout Sections 2, 3,...

On the book’s format. Readers acquainted with David Williams’s books might find the formatting of this manuscript suspiciously familiar and they would be right to do so. You know what they say about imitation and flattery.

Bland assurances and exercises. I, too, will occasionally adhere to the following convention (a convention I used to find despicable as a student, it appears that I may have lived long enough to become the villain):

“Convention dictates that Itô’s formula should be proved for $n = 1$, the general case being left as an exercise, amid bland assurances that only the notation is any more difficult.”

Chris Rogers and David Williams in (Rogers and Williams, 2000b).

The first few applications in the book of a given result tend to be quite detailed while, as is perhaps inevitable, later applications are more cavalier or shamelessly left as an exercise for the reader. If you find yourself hopelessly lost in one of these bland assurances or offhand applications, you can always email me at the address given above and I’ll try my best to get back to you. Who knows, you may have found a mistake (in which case I would be in your debt!). Indeed, in my experience, such applications and assurances tend to be the weakest points of books and articles.

“‘Obvious’ is the most dangerous word in mathematics.”

Eric Temple Bell in (Bell, 1952).

I claim no originality of the results presented in this book. Even though I may have ironed out certain corners of the theory while writing this book, by and large everything I have (so far!) covered in the manuscript is well-known within the community.

“Nothing of me is original. I am the combined effort of everybody I’ve ever known.”

Chuck Palahniuk in Invisible Monsters.

Markov chain theory is a classical and very mature subject; the literature on it is vast. Many of the results I give in this manuscript can be found verbatim, or close to, elsewhere. Whenever the history of a given result (concept, approach, etc.) seems clear to me, I append to the end of the section containing the result (concept, approach, etc.) a ‘Notes and references’ blurb commenting on this history. However, regrettably, these are mostly lacking, at least for the time being.

“We hope that we always have told the truth, but realize that it is seldom the whole truth. It is not our intention to give anyone less than his full measure of credit and we apologize in advance to anyone who may feel slighted.”

Robert M. Blumenthal and Ronald K. Gettoor in (Blumenthal and Gettoor, 2007).

Acknowledgements. There are many I should thank here, and I will in due course.

Juan Kuntz
London, July 2022

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0. Some important notation and preliminaries. The symbol

‘ $:=$ ’

is used to define whatever precedes as whatever follows it, and vice versa for ‘ $=$ ’.

Numbers.

$\mathbb{N}$ denotes the set of natural numbers (including zero),

$\mathbb{Z}$ that of integers,

$\mathbb{Z}_+$ that of positive integers,

$\mathbb{R}$ that of real numbers,

$[0, \infty)$ that of non-negative real numbers,

$(0, \infty)$ that of positive real numbers,

and $\mathbb{Q}$ that of rational numbers.

Extended numbers.

$\mathbb{N}_E := \mathbb{N} \cup \{\infty\}$ denotes the set of extended natural numbers,

$\mathbb{Z}_E := \mathbb{Z} \cup \{-\infty, \infty\}$ that of extended integers,

$\mathbb{R}_E := \mathbb{R} \cup \{-\infty, \infty\}$ that of extended real numbers,

$[0, \infty] := [0, \infty) \cup \{\infty\}$ that of non-negative numbers,

$(0, \infty] := (0, \infty) \cup \{\infty\}$ that of positive real numbers,

Whenever I say that a number (function, random variable, etc.) is ‘non-negative’, I mean that it takes values in, $[0, \infty]$ instead of $[0, \infty)$, while ‘non-negative and finite’ means that it takes values in, $[0, \infty)$. Similarly with ‘positive’, ‘non-positive’, and ‘negative’.

We order the extended numbers as usual: for all $a, b \in \mathbb{R}$, $a \geq b$ if and only if $a - b \in [0, \infty)$ and

$$-\infty \leq c \leq \infty \quad \forall c \in \mathbb{R}_E.$$

The rules we use to manipulate them are also the usual:

infinity times -1 is minus infinity: $\infty \cdot (-1) = -\infty$,

infinity times a positive number $a \in (0, \infty]$ is itself: $\infty \cdot a = \infty$,

any real number $a \in \mathbb{R}$ divided by infinity is zero: $a/\infty = 0$,

infinity plus a number $a \neq -\infty$ is itself: $a + \infty = \infty$,

and infinity times zero is zero: $\infty \cdot 0 = 0$.

Inf/sup/min/max/floor/ceiling. The infimum (resp. supremum) of the empty set $\emptyset$ is plus infinity (resp. minus infinity):

$$\inf \emptyset = \infty, \quad \sup \emptyset = -\infty.$$

For any numbers $a, b \in \mathbb{R}_E$,

$$a \wedge b := \min\{a, b\} \quad a \vee b := \max\{a, b\},$$

$\lfloor a \rfloor$ denotes the largest number in $\mathbb{Z}_E$ no greater than a , and $\lceil a \rceil$ the smallest no smaller than a .

Sigma-algebras.

2^Ω denotes power set $\{A : A \subseteq \Omega\}$ of any given set Ω .

$\mathcal{B}(\mathbb{R})$ denotes the Borel sigma-algebra on $\mathbb{R}$ (topologised with the euclidean topology): the smallest sigma-algebra on $\mathbb{R}$ that contains all of its open subsets (i.e., the sigma-algebra *generated* by the collection of open subsets).

$\mathcal{B}(\Omega) := \{A \cap \Omega : A \in \mathcal{B}(\mathbb{R})\}$ denotes the Borel sigma-algebra on Ω , for any Borel subset Ω in $\mathcal{B}(\mathbb{R})$ of $\mathbb{R}$.

$\mathcal{B}(\mathbb{R}_E)$ denotes the sigma-algebra on $\mathbb{R}_E$ generated by $\mathcal{B}(\mathbb{R}) \cup \{\{-\infty\}, \{\infty\}\}$.

Functions. Throughout the following, let f be a function mapping from Ω to $\mathcal{X}$ and let $\mathcal{F}$ and $\mathcal{G}$ be sigma-algebras on Ω and $\mathcal{X}$, respectively.

$f = g$ means that $f(\omega) = g(\omega)$ for all ω in Ω and similarly for $f \geq g$ and $f \leq g$.

For any subset B of the co-domain $\mathcal{X}$, $\{f \in B\}$ denotes the pre-image of B :

$$\{f \in B\} := \{\omega \in \Omega : f(\omega) \in B\}.$$

f is $\mathcal{F}/\mathcal{G}$ -measurable if it is a measurable mapping from $(\Omega, \mathcal{F})$ to $(\mathcal{X}, \mathcal{G})$:

$$\{f \in B\} \in \mathcal{F} \quad \forall B \in \mathcal{G}.$$

If B is a singleton $\{x\}$, then we write $\{f = x\}$ instead of $\{f \in B\}$. Similarly for $\{f \geq x\}$, $\{f \leq x\}$, and $\mathbb{R}_E$ -valued functions f .

For any a subset A of Ω , 1_A denotes the indicator function of A :

$$1_A(\omega) := \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{otherwise} \end{cases}$$

For a single point ω in Ω , we use the shorthand $1_\omega := 1_{\{\omega\}}$.

Sometimes, we will encounter $\mathbb{R}_E$ -valued functions f that are only defined on a subset A of Ω . In these cases, I will use $1_A f$ to denote the function on Ω defined as

$$(1) \quad (1_A f)(x) := \begin{cases} f(x) & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

(2) LEMMA. Suppose that A belongs to $\mathcal{F}$ and that f maps from A to $\mathbb{R}_E$. Then $1_A f$ in (1) is $\mathcal{F}/\mathcal{B}(\mathbb{R}_E)$ -measurable if

$$\{\omega \in A : f(\omega) \in B\} \in \mathcal{F}, \quad \forall B \in \mathcal{B}(\mathbb{R}_E).$$

Proof. Fix any B in $\mathcal{B}(\mathbb{R}_E)$. If 0 does not belong to B , then

$$\{1_A f \in B\} = \{\omega \in A : f(\omega) \in B\} \in \mathcal{F}.$$

Otherwise,

$$\{1_A f \in B\} = (\Omega \setminus A) \cup \{\omega \in A : f(\omega) \in B\},$$

and it follows that $\{1_A f \in B\}$ belongs to $\mathcal{F}$ as sigma-algebras are closed under unions and intersections. $\square$

Well-defined integrals and sums. For any unsigned measure ρ on $(\Omega, \mathcal{F})$ and $\mathbb{R}_E$ valued function f on Ω , I say that the integral

$$\rho(f) := \int f(x) \rho(dx)$$

is *well-defined* if

$$\rho(f \wedge 0) = \int (f(x) \wedge 0) \rho(dx) > -\infty \quad \text{or} \quad \rho(f \vee 0) = \int (f(x) \vee 0) \rho(dx) < \infty.$$

The terminology also applies to sums: setting ρ to be the counting measure on a countable set $\mathcal{S}$, we have that

$$\sum_{\omega \in \Omega} f(\omega)$$

is well-defined if

$$\sum_{\omega \in \Omega} f(\omega) \wedge 0 > -\infty \quad \text{or} \quad \sum_{\omega \in \Omega} f(\omega) \vee 0 < \infty.$$

Countable sets and vector notation.

I always use the power set $2^{\mathcal{S}}$ of a countable set $\mathcal{S}$ as a sigma-algebra on $\mathcal{S}$: whenever I say a measure or a distribution on $\mathcal{S}$, we really mean a measure or distribution on $(\mathcal{S}, 2^{\mathcal{S}})$.

Given any measure ρ on $\mathcal{S}$, I commit the slight abuse of notation of using ρ to denote both the measure and its density with respect to the counting measure on $(\mathcal{S}, 2^{\mathcal{S}})$. That is, I write $\rho(x) := \rho(\{x\})$ for each x in $\mathcal{S}$, so that

$$\rho(A) = \sum_{x \in A} \rho(x)$$

for each subset A of $\mathcal{S}$, and

$$\rho(f) = \sum_{x \in \mathcal{S}} f(x) \rho(x),$$

for every $\mathbb{R}_E$ -valued function f on $\mathcal{S}$ such that the above is well-defined.

I treat $\mathbb{R}_E$ -valued functions f on $\mathcal{S}$ as ‘column vectors’ $(f(x))_{x \in \mathcal{S}}$, and measures ρ on $\mathcal{S}$ as ‘row vectors’, of (extended) real numbers indexed by the elements of $\mathcal{S}$. In particular, given a matrix $A := (a(x, y))_{x, y \in \mathcal{S}}$ of (extended) real numbers indexed by $\mathcal{S}$, I use Af to denote the $\mathbb{R}_E$ -valued function defined by

$$Af(x) := \sum_{y \in \mathcal{S}} a(x, y) f(y) \quad \forall x \in \mathcal{S},$$

and ρA to denote the measure defined by

$$\rho A(x) := \sum_{y \in \mathcal{S}} \rho(y) a(y, x) \quad \forall x \in \mathcal{S},$$

assuming, of course, that the above sums are all well-defined.

Preliminaries. Throughout the text, I assume that the reader is familiar with the following:

Sigma-algebras, generating sets, measurable functions, and the operations that preserve measurability (e.g., the composition of two measurable functions are measurable).

The Lebesgue integral, properties thereof (linearity, etc.), and the standard machine.

Fatou's lemma and the monotone, dominated, and bounded convergence theorems.

Product measures, Tonelli's theorem, and Fubini's theorem.

Probability triplets and the definition of random variables as measurable functions.

The modes of convergence of random variables (almost surely, in probability, etc.).

The definition of conditional expectation as a random variables and its properties (see below in particular).

The definition of a discrete-time martingale.

For those who are not, I recommend ([Williams, 1991](#); [Tao, 2011](#)).

Properties of conditional expectation. I will make repeated use of the following:

(3) **THEOREM.** *Suppose that X is non-negative ($X(\omega) \geq 0$ for all $\omega \in \Omega$) or integrable ($\mathbb{E}[|X|] < \infty$) $\mathbb{R}_E$ -valued random variable on some probability triplet $(\Omega, \mathcal{F}, \mathbb{P})$.*

(i) *If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X|\mathcal{G}] = X$, a.s.*

(ii) *(Linearity) If X_1 and X_2 are integrable random variables and $a_1, a_2 \in \mathbb{R}$,*

$$\mathbb{E}[a_1X_1 + a_2X_2|\mathcal{G}] = a_1\mathbb{E}[X_1|\mathcal{G}] + a_2\mathbb{E}[X_2|\mathcal{G}], \quad a.s.$$

If $X_1, X_2, \dots$ are non-negative random variables and $a_1, a_2, \dots$ non-negative constants, then

$$\mathbb{E}\left[\sum_{n=1}^{\infty} a_n X_n | \mathcal{G}\right] = \sum_{n=1}^{\infty} a_n \mathbb{E}[X_n | \mathcal{G}], \quad a.s.$$

(iii) *(Positivity) If X is non-negative, then $\mathbb{E}[X|\mathcal{G}]$ is non-negative, a.s.*

(iv) *(Tower property) If $\mathcal{H}$ is a sigma-algebra contained in $\mathcal{G}$, then*

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}] | \mathcal{H}] = \mathbb{E}[X|\mathcal{H}], \quad a.s.$$

(v) *(Take out what is known) Suppose that Z is $\mathcal{G}$ -measurable. If Z is bounded, or if both Z and X are non-negative, then*

$$\mathbb{E}[ZX|\mathcal{G}] = Z\mathbb{E}[X|\mathcal{G}], \quad a.s.$$

Proof. The above properties are proven in ([Williams, 1991](#), Section 9.7) for the $\mathbb{R}$ -valued integrable case. However, the proofs given therein hold almost verbatim our slightly more general case. $\square$

Part I

Theory

“We have not succeeded in answering all our problems. The answers we have found only serve to raise a whole set of new questions. In some ways we feel we are as confused as ever, but we believe we are confused on a higher level and about more important things.”

Posted outside the mathematics reading room in Tromsø University, according to Bernt Øksendal (Øksendal, 2003).

Introduction: Markov processes and their long-term behaviour

This book revolves around Markov processes: collections of random variables $(X_t)_{t \in \mathbb{T}}$ (referred to as *processes*) that are indexed by an ordered set $\mathbb{T}$ of time points, take values in some *state space*, and satisfy the Markov property. Informally, the Markov property says that we are equally well-equipped to predict what will happen in the future if we only know what is going on right now (that is, the current *state* of the process) or if we know what is going on right now *and* what happened in the past. In our case, the indexing set will either be the natural numbers, in which case we describe the process as *discrete-time*, or the non-negative real numbers, in which case we classify it as *continuous-time*. If the state space is a countable set, we refer to the process as a *chain* (beware, it is equally as common for authors to call a Markov process a ‘chain’ if it is a discrete-time process regardless of whether of the state space is countable or not). A Markov process is said to be *time-homogeneous* if the probability that it transitions from any given subset of the state space to any other over a given interval of time depends only on the interval’s length and not on its endpoints. Using the words ‘a Markov process’ when one means ‘a time-homogeneous Markov process’ is nearly a tradition in the field as John F. C. Kingman amusingly pointed out in a lovely talk he gave at Imperial College in 2013. We will adhere to this ‘tradition’ (throughout the book, we only treat the time-homogeneous case).

One of the most well-understood and extensively studied aspects of Markov processes is their *long-term* (or *asymptotic* or *long-run*) behaviour. Without getting into technicalities (we will have plenty of these later!), the possible long-term behaviours can be roughly grouped into four types (see also Fig. 1):

1. The sample paths of the process diverge to infinity in a finite amount of time. Because discrete-time processes can only change state once per step and single step transitions of infinite size are usually not allowed, only continuous-time processes exhibit this behaviour. In this case, the process is said to be *explosive* and the time at which this event occurs is called the *explosion time*.
2. The process diverges to infinity in an infinite amount of time. In this case, the process is said to be *transient*. Processes that are not explosive nor transient are said to be *recurrent*.
3. The paths do not tend to infinity and keep revisiting certain small subsets of the state space (for euclidean-valued processes ‘small’ typically means compact in the usual topology, while for chains it usually means finite). However, the excursions from these states get larger and larger over time and the frequency of the visits decays to zero. In this case, the mass of the distribution describing process’s location spreads out over the entire state space and the probability that the process lies inside any given small set tends to zero as time progresses. In this case, the process is said to be *null recurrent*.

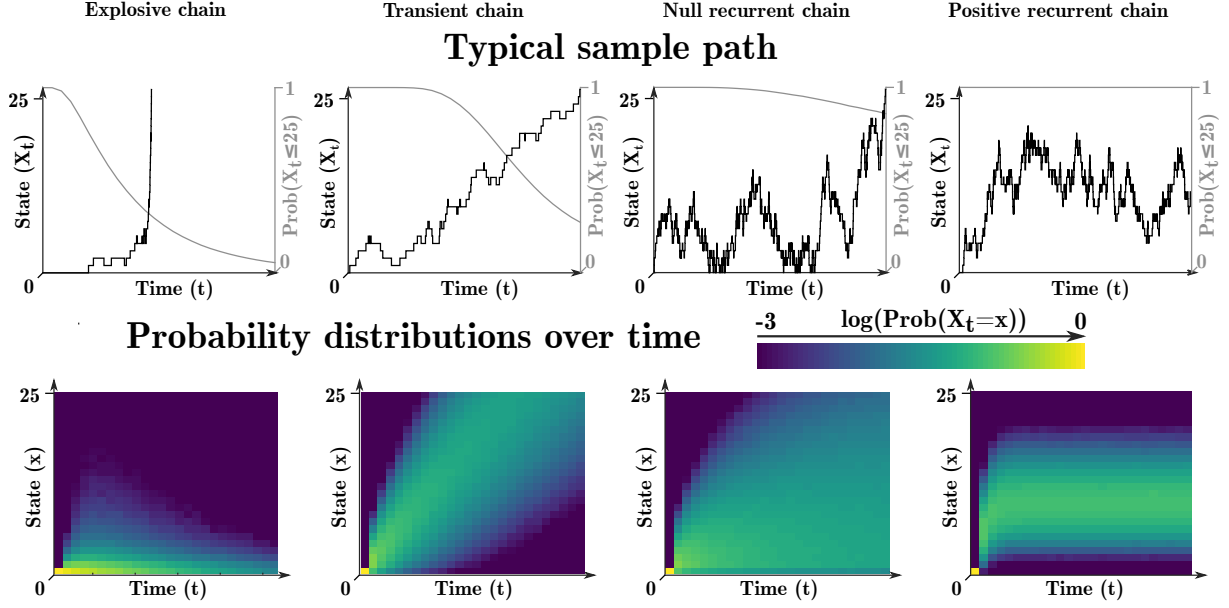


Figure 1: The long-term behaviours of continuous-time Markov chains. (Top, black) Typical sample paths $t \mapsto X_t(\omega)$ of an explosive chain, a transient chain, a null recurrent chain, and a positive recurrent chain. For all four chains, the state space is $\mathbb{N}$ but the plot only displays states 0–25. (Top, shaded) The probability $\text{Prob}(X_t \leq 25)$ of the chains being within the plotted states at time t . (Bottom) Corresponding probability distributions (in log-scale) as a function of time; chains started in state 0.

In the explosive case, the chain lingers awhile in certain states ($\sim 0-5$). However, once it ventures out of these states, its path shoots off towards infinity. Thus, the probability mass quickly escapes towards infinity with any mass remaining in the plotted region being concentrated in states 0–5. In the transient case, the paths steadily march towards infinity. Correspondingly, the probability mass moves away from the lower states and heads towards infinity. In the null recurrent case, the paths keep returning certain states (e.g. 0), however these visits become ever rarer as time progresses. Consequently, the probability mass does not distinctly move away from these states but rather spreads out over an ever growing region in such a way that the probability of being in any given state tends to zero. In the positive recurrent case, the chain keeps visiting certain states (e.g. 10) and the frequency of these visits does not decay to zero. After a bit of time, the probability distribution stabilises around a *stationary distribution*.

4. None of the above. In this case, for each path, there are small subsets of the state space for which the frequency of the path’s visits does not degenerate to zero. Consequently, the mass of the distribution of the process’s location does not vanish over time and instead concentrates around these small subsets. The process is then said to be *positive recurrent*.

The above can be viewed as progressive levels of stability of Markov processes. In this book, we often focus on the fourth type of behaviour (which we refer to as *stable*) since it is the one typically exhibited by Markovian models of real-life phenomena. Or at least, it is often hoped to be exhibited by these models! I find the argument ‘the physical phenomenon exhibits no such unstable behaviour, and thus this model of it cannot either’ sometimes given to be flawed. Models are abstractions of real life and can be far from faithful representations of it. An unstable model of stable phenomena is not something impossible; it is merely an indication of poor modelling choices. Indeed, modelling is a difficult task and mistakes happen. Moreover, many fitting algorithms involve automated sweeps of parameter sets, and these algorithms can easily step through sets that lead to poor models. For these reasons, I believe care must be taken to rule out unstable behaviours even in models of stable phenomena. This can be done in practice using the Foster-Lyapunov criteria discussed throughout the book.

If the process is positive recurrent, then it has at least one *stationary* (or *invariant* or *steady-state*) distribution. That is, a probability distribution π on the state space such that sampling the process’s starting location from π ensures that the process is stationary (i.e. that

its finite-dimensional distributions are unchanged by shifts in time).

In the stable case, stationary distributions play an important role for all other starting distributions too. Various ergodic theorems, or generalised laws of large numbers, tell us that the *time* (or *empirical*) averages of the process converge to averages with respect to stationary distributions. In particular, in the continuous-time case, we have that for each sample path $t \mapsto X_t(\omega)$, there exists a stationary distribution π such that the fraction of time the path spends in any given region A of the state space converges to π 's mass in A :

$$(4) \quad \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T 1_A(X_t(\omega)) dt = \pi(A).$$

If $\mathbb{T} = \mathbb{N}$, then replace $\int_0^T 1_A(X_t(\omega)) dt$ with $\sum_{t=0}^{T-1} 1_A(X_t(\omega))$. The stationary distribution in question generally depends on the sample path, but not on A . Furthermore, if the process is *aperiodic*¹ (meaning that the distribution of the chain's location does not oscillate in time), then the distribution of the chain's location at time t converges to some stationary distribution as t approaches infinity. Which stationary distribution in particular depends on the starting distribution. If there is only one stationary distribution, then the probability that the chain lies in any particular region of the state space converges to the total fraction of time that any given path spends in the region. That is, after enough time has passed

$$(5) \quad \text{time averages} \approx \text{space averages}.$$

For this reason, aperiodic positive recurrent Markov processes with a unique stationary distribution are said to be *ergodic*.

The stationary distributions are intimately tied to the *closed irreducible sets* contained in the state space. These are sets that the process, once inside, cannot escape from (hence, 'closed') and that do not contain any proper subsets also fulfilling this property (hence 'irreducible', these sets cannot be 'reduced' further). All states not contained within a closed irreducible set are transient in the sense that the process will eventually leave each such state and never return. This implies that the paths either diverge to infinity or get absorbed into one of the closed irreducible sets. If the process is positive recurrent, then the former is not possible and each sample path $t \mapsto X_t(\omega)$ will eventually enter a closed irreducible set $\mathcal{I}$. Because the path proceeds to spend the remainder of time inside $\mathcal{I}$, it must be the case that the stationary distribution π featuring in (4) has support contained in $\mathcal{I}$. The irreducibility of these sets further implies that there is no more than one such distribution and that it has support on the entire set. It is known as the *ergodic distribution* associated with the closed irreducible set.

As mentioned before, if the process is positive recurrent and aperiodic, then the distribution of its location converges to a stationary distribution. This limiting distribution is just the weighted combination of the ergodic distributions where the weight given to each one of these distributions is the probability that the process ever enters the irreducible set associated with that ergodic distribution. Consequently, if the starting distribution is confined to a closed irreducible set, we recover (5), hence justifying the name 'ergodic distribution'.

Even though we have so far discussed Markov processes in general, this book is not intended to be a survey of general Markov process theory. For this, there is an abundance of far better resources: (Ethier and Kurtz, 1986; Kallenberg, 2001; Meyn and Tweedie, 1992, 1993a,b; Rogers and Williams, 2000a,b) are all great. Instead, the main aim of the following Sections 1–50 is to formalise the above discussion for the cases of discrete-time and continuous-time chains. By focusing on countable state spaces, we avoid a host of technical questions that arise with uncountable ones and often prove to only be a distraction from important ideas guiding the theory. In fact, virtually every aspect of the behaviour of Markov processes is also encountered in

¹Note that continuous-time Markov chains are always aperiodic, and hence why there often is no mention of periodicity in texts discussing them, see Section 38 for details.

the study of chains. For this reason, possessing a mastery of theory pertinent to the chains proves to be a great boon even when dealing with more complicated cases (I am a firm believer of that paradigm pervasive throughout science and mathematics that goes ‘a thorough understanding of simple situations is vital to tackle more complicated ones’).

“Seulement, alors qu’autrefois il suffisait de deux heures d’exposé pour traiter l’intégrale d’Itô, et qu’ensuite les belles applications commençaient, il faut à présent un cours de six mois sur les définitions. Que peut on y faire? Les mathématiques et les mathématiciens ont pris cette tournure. Il est temps de commencer.”

Paul-André Meyer ([Meyer, 1976](#)) on the dilemma of modern math education.

Discrete-time chains I: the basics

The least technically involved type of Markov processes are discrete-time Markov processes on countable state spaces, or *discrete-time Markov chains*, or *discrete-time chains* for short. Our motivation behind their study is threefold:

1. Discrete-time chains are of interest in their own right and serve as models for a host of real life phenomena.
2. Many proofs in the continuous-time case require discrete-time chains.
3. By avoiding the (mostly technical) complications arising from an uncountable state space or time axis, discrete-time chains provide us with the ideal probabilistic playground to (a) develop our understanding of the Markov property and its consequences; (b) acquire an intuition for the veracity of statements regarding more general Markov processes; and (c) get exposure to the types of proofs and arguments pervasive in Markov chain theory. Moreover, as we will see when studying the long-term behaviour of continuous-time chains (Sections 38–50), many proofs in the continuous-time case are analogous to their discrete-time counterparts (to the point of being a verbatim copy except for some symbol and reference-call changes).

The aim of this chapter is to introduce discrete-time chains, their associated statistics (time-varying law, path law, stopping distributions, occupation measures, etc.), and the bread-and-butter tools (Chapman-Kolmogorov equation, martingale property, stopping times, Dynkin’s formula, strong Markov property, etc.) that form the foundation upon which the rest of discrete-time chain theory rests.

The treatment here is geared towards those interested in proving results on Markov chains instead of those just interested in using them to study particular models of interest and, for this reason, is a bit technical at points. In particular, the measure-theoretic machinery required to introduce, and to comfortably use, the strong Markov property in its full splendour is heavier than that needed throughout the rest of the chapter. Past a certain point, working on Markov chain theory without the strong Markov property is like trying to wade your way through neck deep water instead of learning how to swim: sometimes do-able but needlessly tiring and frustrating. My advice is to bite the bullet and work your way through the details, even if only once in your life.

Overview of the chapter. We begin by defining the chains through an algorithm used to simulate them in practice (Section 1). Next, we formalise the idea of the chain’s history-to-date with the filtration it generates and prove our first version of the Markov property stating that, conditioned on the present, the chain’s past and future are independent (Section 2). We then pause for a moment to introduce a very simple, but very instructive, example (Section 3). We resume our treatment of the theory in Section 4 where we introduce the time-varying law (describing the chain’s location as a function of time) and use the Markov property to derive a recursion expressing the time-varying law in terms of the *one-step matrix* (describing the chain’s next position conditioned on its present position) and the *initial distribution* (describing the chain’s starting location).

In Section 5, we take a very important step: we stop viewing the chain as a sequence of random variables each taking values in the state space and instead regard it as a single random variable taking values in the space of possible sequences of states (the *path space*). We then take a moment to give an equivalent description of the chains in terms of martingales (Section 6) which, as we will see in Section 9, proves key when studying the statistics of the chain at random points in time (e.g. the first time that the chain enters some set of interest) instead of deterministic one (e.g. time-step number 10). Capitalising on our new path space and martingale perspectives,

we show in Section 7 that we are not doing anything wrong by using the particular definition of the chain given in Section 1 instead other definitions found elsewhere: all these definitions are equivalent in the sense that they lead to the same distribution, or *path law*, on the path space.

Next up are stopping times (Section 8), which are the points in time that events of a certain type occur. In particular, events whose occurrence we can deduce the instant they happen by continuously monitoring the chain (e.g. the first time the chain visits a given set, but not the last time it does). These events are non-predictive: their occurrence does not tell us anything about the future of the chain that cannot be gleaned from the chain's position at the moment of occurrence. For this reason, stopping times combine seamlessly with the Markov property and we are able to generalise many of the results featuring deterministic times so that they also apply to stopping times. In particular, we prove Dynkin's formula (Section 9), a generalisation of the recurrence satisfied by the time-varying (Section 4), and the strong Markov property (Section 12), a generalisation of the Markov property (Section 2). Lastly, given any stopping time ς , we introduce (Section 10) its *stopping distribution* (describing the joint statistics of ς and of the chain's location at ς) and its *occupation measure* (describing the states visited by the chain prior to ς and the times of visit) and we characterise (Section 11) them in terms of sets linear equations for the special, but important, case of an *exit time* (i.e. the point in time when the chain first ventures outside of a given set of interest).

1. The chain's definition. A discrete-time Markov chain (or *discrete-time chain* for short) is a sequence $X = (X_n)_{n \in \mathbb{N}}$ of random variables that take values in a countable set $\mathcal{S}$ (the *state space*) and satisfy the Markov property. Out of convenience, we will assume throughout the book that the chain has been constructed in a particular way (in Section 7, we will show that we lose nothing by picking this particular construction). The aim of this section is to describe the construction.

One-step matrices. For the definition, we need a *one-step matrix* $P = (p(x, y))_{x, y \in \mathcal{S}}$ describing where the chain will go next given its current location. In particular, $p(x, y)$ denotes the probability that the chain next visits y if it currently lies at x . For this reason, the one-step matrix must satisfy:

$$(1) \quad p(x, y) \geq 0, \quad \forall x, y \in \mathcal{S}, \quad \sum_{y \in \mathcal{S}} p(x, y) = 1, \quad \forall x \in \mathcal{S}.$$

The technical set-up. Next, we require a measurable space $(\Omega, \mathcal{F})$, a collection $(\mathbb{P}_x)_{x \in \mathcal{S}}$ of probability measures on $(\Omega, \mathcal{F})$, an $\mathcal{F}/2^{\mathcal{S}}$ -measurable function X_0 from Ω to $\mathcal{S}$, and a sequence of $\mathcal{F}/\mathcal{B}((0, 1))$ -measurable functions $U_1, U_2, \dots$ from Ω to $(0, 1)$ such that, under $\mathbb{P}_x$,

1. $\{X_0 = x\}$ occurs almost surely: $\mathbb{P}_x(\{X_0 = x\}) = 1$;
2. for each $n > 0$, the random variable U_n is uniformly distributed on $(0, 1)$: $\mathbb{P}_x(\{U_n \leq u\}) = u$, for all u in $(0, 1)$;
3. and the random variables $X_0, U_1, U_2, \dots$ are independent:

$$\mathbb{P}_x(\{X_0 = y, U_1 \leq u_1, \dots, U_n \leq u_n\}) = \mathbb{P}_x(\{X_0 = y\}) \mathbb{P}_x(\{U_1 \leq u_1\}) \dots \mathbb{P}_x(\{U_n \leq u_n\})$$

for all y in $\mathcal{S}$, $u_1, \dots, u_n$ in $(0, 1)$, and $n > 0$;

for each x in $\mathcal{S}$. For those of you inclined to such things, there's a detailed construction of this space, the measures, and the random variables at the end of the section.

Below, we will set the chain's starting location to be X_0 . For this reason, $\mathbb{P}_x$ will describe the statistics of the chain were it to start from x . To instead describe its statistics were its

starting location be sampled from a given probability distribution $\gamma = (\gamma(x))_{x \in \mathcal{S}}$ on $\mathcal{S}$ (the *initial distribution*), we define the probability measure $\mathbb{P}_\gamma$ on $(\Omega, \mathcal{F})$ via

$$(2) \quad \mathbb{P}_\gamma(A) := \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x(A) \quad \forall A \in \mathcal{F}.$$

Under this measure, the starting location of the chain has law γ :

$$\mathbb{P}_\gamma(\{X_0 = y\}) = \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x(\{X_0 = y\}) = \sum_{x \in \mathcal{S}} \gamma(x) 1_y(x) = \gamma(y) \quad \forall y \in \mathcal{S}.$$

However, the random variables $U_1, U_2, \dots$ are still uniformly distributed:

$$\mathbb{P}_\gamma(\{U_n \leq u\}) = \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x(\{U_n \leq u\}) = u \left(\sum_{x \in \mathcal{S}} \gamma(x) \right) = u \quad \forall u \in (0, 1), \quad n > 0;$$

and $X_0, U_1, U_2, \dots$ are still independent:

$$\begin{aligned} \mathbb{P}_\gamma(\{X_0 = y, U_1 \leq u_1, \dots, U_n \leq u_n\}) &= \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x(\{X_0 = y, U_1 \leq u_1, \dots, U_n \leq u_n\}) \\ &= \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x(\{X_0 = y\}) \mathbb{P}_x(\{U_1 \leq u_1\}) \dots \mathbb{P}_x(\{U_n \leq u_n\}) \\ &= \left(\sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x(\{X_0 = y\}) \right) u_1 \dots u_n \\ &= \mathbb{P}_\gamma(\{X_0 = y\}) \mathbb{P}_\gamma(\{U_1 \leq u_1\}) \dots \mathbb{P}_\gamma(\{U_n \leq u_n\}), \end{aligned}$$

for all y in $\mathcal{S}$, $u_1, \dots, u_n$ in $(0, 1)$, and $n > 0$. Throughout the book, we use $\mathbb{E}_x$ (resp. $\mathbb{E}_\gamma$) to denote expectation with respect to $\mathbb{P}_x$ (resp. $\mathbb{P}_\gamma$).

An algorithmic definition. Armed with the one-step matrix and the random variables $X_0, U_1, U_2, \dots$, we define our chain $X = (X_n)_{n \in \mathbb{N}}$ recursively by running Algorithm 1 below. It goes as follows: sample a state x from the initial distribution γ and start the chain at x (i.e., set $X_0 := x$). Next, sample y from the probability distribution $p(x, \cdot)$ in (1) and update the chain's state to y . Repeat these steps starting from y instead of x . All random variables sampled must be independent of each other.

Algorithm 1 An algorithm to construct discrete-time chains on $\mathcal{S} = \{x_0, x_1, \dots\}$

```

1: for  $n = 1, 2, \dots$  do
2:   sample  $U_n \sim \text{uni}_{(0,1)}$  independently of  $(X_0, U_1, \dots, U_{n-1})$ 
3:    $i := 0$ 
4:   while  $U_n > \sum_{j=0}^i p(X_{n-1}, x_j)$  do
5:      $i := i + 1$ 
6:   end while
7:    $X_n := x_i$ 
8: end for

```

The underlying space. In the above, we glossed over the details of the space $(\Omega, \mathcal{F})$, the measures $(\mathbb{P}_x)_{x \in \mathcal{S}}$, and the random variables $X_0, U_1, U_2, \dots$. We take a moment here to argue their existence. To do so, we require the following.

(3) THEOREM. For each natural number n , let ρ_n be a probability measure on a measurable space $(\Omega_n, \mathcal{G}_n)$. Define the sequence space

$$\Omega = \prod_{n=0}^{\infty} \Omega_n,$$

the coordinate functions

$$W_n : \Omega \rightarrow \Omega_n \quad W_n(\omega) := \omega_n \quad \forall \omega \in \Omega, \quad n \geq 0,$$

and let $\mathcal{F}$ be the sigma-algebra on Ω generated by $\prod_{n=0}^{\infty} \mathcal{G}_n$. There exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ such that

$$\mathbb{P} \left(\left(\prod_{n=0}^k A_n \right) \times \left(\prod_{n=k+1}^{\infty} \Omega_n \right) \right) = \prod_{n=0}^k \rho_n(A_n),$$

for all $k \geq 0$ and A_0 in $\mathcal{G}_0, \dots, A_k$ in $\mathcal{G}_k$. In particular, under $\mathbb{P}$, $W_0, W_1, \dots$ is a sequence of independent random variables such that W_n has law ρ_n for each $n \geq 0$.

Proof. This is a consequence of Carathéodory's extension theorem, see (Williams, 1991, Appendix A9). The proof given therein is phrased for the case that $\Omega_n := \mathbb{R}$ and $\mathcal{F}_n := \mathcal{B}(\mathbb{R})$ for all $n \geq 0$, but it holds identically in our slightly more general setting. $\square$

To construct the random variables in Algorithm 1, let $(\Omega, \mathcal{F})$ be as in the theorem's premise after defining

$$\Omega_0 := \mathcal{S}, \quad \mathcal{G}_0 := 2^{\mathcal{S}}, \quad \Omega_n := (0, 1), \quad \mathcal{G}_n := \mathcal{B}((0, 1)), \quad \forall n > 0,$$

and set X_0 to be W_0 and U_n to be W_n for each $n > 0$. Given any state x , set ρ_0 to be the point mass 1_x at x and ρ_n to be the uniform measure on $(0, 1)$ for each $n > 0$. The theorem shows that a measure $\mathbb{P}_x$ on $(\Omega, \mathcal{F})$ satisfying the desired properties (1–3 listed after (1)) exists.

A word of warning. None of the probability spaces in $\{(\Omega, \mathcal{F}, \mathbb{P}_x)\}_{x \in \mathcal{S}}$ will be complete. To see why, pick any (possible non-measurable) subset A of $(0, 1)$. If $(\Omega, \mathcal{F}, \mathbb{P}_x)$ were to be complete for a given x , then, for any other $y \neq x$, the set $B := \{y\} \times A \times \prod_{n=2}^{\infty} (0, 1)$ would belong to $\mathcal{F}$ because B is a subset of $\{y\} \times \prod_{n=1}^{\infty} (0, 1)$ and the latter has $\mathbb{P}_x$ measure of zero:

$$\mathbb{P}_x \left(\{y\} \times \prod_{n=1}^{\infty} (0, 1) \right) = 1_x(y) \prod_{n=1}^{\infty} \text{uni}_{(0,1)}((0, 1)) = 1_x(y) = 0.$$

Define a real-valued function ρ on $2^{(0,1)}$ via

$$\rho(A) := \mathbb{P}_y \left(\{y\} \times A \times \prod_{n=2}^{\infty} (0, 1) \right), \quad \forall A \subseteq (0, 1)$$

It is straightforward to check that ρ is a probability measure on $((0, 1), 2^{(0,1)})$ and that it coincides with the uniform distribution on $\mathcal{B}((0, 1))$. However, this is impossible as it would contradict the existence of non-Lebesgue-measurable sets, e.g., see (Tao, 2011).

The moral here is that, if we wish to work with collections of probability measures (one per possible starting state of the chain) on a *single* measurable space, then we must resign ourselves to working with incomplete measures. In particular, this means that we must take some care when applying bounded and dominated convergence to sequences of random variables that only converge $\mathbb{P}_\gamma$ -almost surely (instead of pointwise). The way to mitigate this issue is to note that,

for any $\mathcal{F}/\mathcal{B}(\mathbb{R})$ -measurable real-valued functions $V_1, V_2, \dots$ on a (possibly incomplete) space $(\Omega, \mathcal{F}, \mathbb{P})$, the set

$$(4) \quad A := \{\omega \in \Omega : (V_n(\omega))_{n \in \mathbb{Z}_+} \text{ converges}\} = \bigcap_{k=1}^{\infty} \bigcup_{l=1}^{\infty} \bigcap_{n=1}^l \bigcap_{m=1}^l \left\{ \omega \in \Omega : |V_n(\omega) - V_m(\omega)| \leq \frac{1}{k} \right\}$$

is measurable ($A \in \mathcal{F}$). Thus, even though the almost sure limit of $(V_n)_{n \in \mathbb{Z}_+}$ may not be measurable (or even defined everywhere), the sequence $(1_A V_n)_{n \in \mathbb{Z}_+}$ does converge everywhere and its limit is measurable (the pointwise limit of measurable functions is measurable). We then apply bounded or dominated convergence to $(1_A V_n)_{n \in \mathbb{Z}_+}$ instead of $(V_n)_{n \in \mathbb{Z}_+}$ and exploit that $\mathbb{P}_\gamma(A) = 1$ to compute the expectation of the limit:

$$\lim_{n \rightarrow \infty} \mathbb{E}_\gamma[V_n] = \lim_{n \rightarrow \infty} \mathbb{E}_\gamma[1_A V_n] = \mathbb{E}_\gamma \left[\lim_{n \rightarrow \infty} 1_A V_n \right],$$

under the usual integrability conditions for dominated convergence. In fact, many authors tacitly write ‘ $\lim_{n \rightarrow \infty} V_n$ ’ when they really mean ‘ $\lim_{n \rightarrow \infty} 1_A V_n$ ’.

2. The Markov property. The Markov property states that conditioned on the present, the chain’s past and future are independent. The ‘present’ is described by the position X_n of the chain X at the current time-step n . For the time being, we limit the chain’s ‘future’ to be its position X_{n+1} at the next time step $n+1$ (we will overcome this restriction in Section 12). To formalise the notion of its ‘past’, we employ the filtration generated by the chain.

The filtration generated by the chain. Throughout the book, we use $(\mathcal{F}_n)_{n \in \mathbb{N}}$ to denote the *filtration* generated by X :

(1) **DEFINITION** (Filtration generated by the chain). *The filtration $(\mathcal{F}_n)_{n \in \mathbb{N}}$ generated by X is the increasing sequence of sigma-algebras defined by*

$$\mathcal{F}_n := \text{the sigma-algebra generated by } (X_0, X_1, \dots, X_n) \quad \forall n \geq 0,$$

meaning that $\mathcal{F}_n$ is the smallest sigma-algebra such that $X_0, \dots, X_n$ are $\mathcal{F}_n$ -measurable random variables.

An event A belongs to $\mathcal{F}_n$ if and only if observing the chain up until time n is sufficient to deduce whether A has occurred. Formally, this is the case if and only if we are able to express the event’s indicator function in terms of $X_0, X_1, \dots, X_n$:

$$1_A(\omega) = g(X_0(\omega), X_1(\omega), \dots, X_n(\omega)) \quad \forall \omega \in \Omega,$$

for some $g : \mathcal{S}^n \rightarrow \{0, 1\}$. Similarly, a random variable Y is $\mathcal{F}_n$ -measurable if and only if it can be expressed in terms of $X_0, \dots, X_n$ (i.e. if we can compute $Y(\omega)$ from $X_0(\omega), \dots, X_n(\omega)$, for each ω in Ω). For these reasons, $\mathcal{F}_n$ represents all information that can be gleaned from observing the chain up until time n and the filtration is viewed as the history of the chain.

The Markov property. The process X defined via Algorithm 1 is referred to as a Markov chain because it satisfies the Markov property:

(2) **THEOREM** (The Markov property). *For any initial distribution γ ,*

$$(3) \quad \mathbb{P}_\gamma(\{X_{n+1} = y\} | \mathcal{F}_n) = \mathbb{P}_\gamma(\{X_{n+1} = y\} | X_n) = p(X_n, y) \quad \forall y \in \mathcal{S}, \quad n \geq 0, \quad \mathbb{P}_\gamma\text{-a.s.},$$

where $(\mathcal{F}_n)_{n \in \mathbb{N}}$ denotes the filtration generated by the chain (Definition 1).

Proof. Enumerate the state space $\{x_0, x_1, \dots\}$ as in Algorithm 1. We will show that

$$(4) \quad \mathbb{P}_\gamma(\{X_{n+1} = x_i\} | \mathcal{F}_n) = p(X_n, x_i) \quad \forall i \geq 0, \quad \mathbb{P}_\gamma\text{-almost surely.}$$

The other inequality in (3) then follows by taking expectations conditional on X_n and using the tower property of conditional expectation (Theorem 0.3i, iv). Because the collection of events

$$\{\{X_0 = z_0, \dots, X_{n-1} = z_n\} : z_0, \dots, z_n \in \mathcal{S}\}$$

generates $\mathcal{F}_n$, to prove (4) it suffices to show that

$$\mathbb{P}_\gamma(\{X_0 = z_0, \dots, X_n = z_n, X_{n+1} = x_i\}) = \mathbb{E}_\gamma[1_{\{X_0=z_0, \dots, X_n=z_n\}} p(X_n, x_i)] \quad \forall z_0, \dots, z_n \in \mathcal{S}, \quad i \geq 0.$$

The chain's definition in Algorithm 1 implies that

$$\{X_n = z_n, X_{n+1} = x_i\} = \left\{ X_n = z_n, \sum_{j=0}^{i-1} p(z_n, x_j) \leq U_{n+1} < \sum_{j=0}^i p(z_n, x_j) \right\} \quad \forall z_n \in \mathcal{S}, \quad i \geq 0.$$

Because $X_1, \dots, X_n$ are functions of $(X_0, U_1, \dots, U_n)$ and U_{n+1} is independent of $(X_0, U_1, \dots, U_n)$, X_n and U_{n+1} are independent. For this reason, (4) follows from the above:

$$\begin{aligned} \mathbb{P}_\gamma(\{X_0 = z_0, \dots, X_n = z_n, X_{n+1} = x_i\}) \\ &= \mathbb{P}_\gamma(\{X_0 = z_0, \dots, X_n = z_n\}) \mathbb{P}_\gamma\left(\left\{\sum_{j=0}^{i-1} p(z_n, x_j) \leq U_{n+1} < \sum_{j=0}^i p(z_n, x_j)\right\}\right) \\ &= \mathbb{P}_\gamma(\{X_0 = z_0, \dots, X_n = z_n\}) p(z_n, x_i) \\ &= \mathbb{E}_\gamma[1_{\{X_0=z_0, \dots, X_n=z_n\}} p(X_n, x_i)] \quad \forall z_0, \dots, z_n \in \mathcal{S}, \quad i \geq 0. \end{aligned}$$

□

In this time-homogeneous setting², Markov property really means two things: conditioned on the present, (a) the chain's future and past are independent and (b) the chain starts 'afresh' from its current location.

To see (a), note that $\mathcal{F}_n$ is generated by $\mathcal{F}_{n-1}$ and X_n . For this reason, the first equality in (3) tells us that the chain's future (for now, modelled by X_{n+1}) conditioned on its past ($\mathcal{F}_{n-1}$) and present (X_n) equals that conditioned on only its present. In other words, if we are aware of the chain's present, knowledge of its past does not improve our ability to predict its future. Because

$$\begin{aligned} \mathbb{P}_\gamma(\text{future} | \text{past and present}) &= \mathbb{P}_\gamma(\text{future} | \text{present}) \\ &\Leftrightarrow \\ \mathbb{P}_\gamma(\text{past and future} | \text{present}) &= \mathbb{P}_\gamma(\text{past} | \text{present}) \mathbb{P}_\gamma(\text{future} | \text{present}), \end{aligned}$$

it follows that the chain's past and future are independent (a). For a formal proof, use

(5) PROPOSITION (Characterising conditional independence, [Kallenberg, 2001](#), Proposition 6.6).
For any probability triplet $(\Omega, \mathcal{F}, \mathbb{P})$ and sigma-algebras $\mathcal{G}, \mathcal{H}, \mathcal{I} \subseteq \mathcal{F}$,

$$\begin{aligned} \mathbb{P}(I | \mathcal{GH}) &= \mathbb{P}(I | \mathcal{H}) \quad \mathbb{P}\text{-almost surely, } \forall I \in \mathcal{I} \\ &\Leftrightarrow \\ \mathbb{P}(G \cap I | \mathcal{H}) &= \mathbb{P}(G | \mathcal{H}) \mathbb{P}(I | \mathcal{H}) \quad \mathbb{P}\text{-almost surely, } \forall G \in \mathcal{G}, \quad I \in \mathcal{I}, \end{aligned}$$

where $\mathcal{GH}$ denotes the sigma-algebra generated by $\mathcal{G}$ and $\mathcal{H}$.

²(b) falters in the time-inhomogeneous case.

to show that (3) holds for all x in $\mathcal{S}$ if and only if $\mathcal{F}_{n-1}$ and X_{n+1} are conditionally independent:

$$(6) \quad \mathbb{P}_\gamma(A \cap \{X_{n+1} = y\} | X_n) = \mathbb{P}_\gamma(A | X_n) \mathbb{P}_\gamma(\{X_{n+1} = y\} | X_n) \quad \mathbb{P}_\gamma\text{-almost surely,}$$

for any given A in $\mathcal{F}_{n-1}$ and y in $\mathcal{S}$.

To see (b), notice that setting $n := 0$ and $\gamma := 1_x$ in (3) and taking expectations with respect to $\mathbb{P}_x$, we find that $p(x, y) = \mathbb{P}_x(\{X_1 = y\})$. Plugging this back into (3), we find that

$$(7) \quad \mathbb{P}_\gamma(\{X_{n+1} = y\} | X_n) = \mathbb{P}_{X_n}(\{X_1 = y\}) \quad \forall y \in \mathcal{S}, \quad \mathbb{P}_\gamma\text{-almost surely}$$

That is, the statistics of the chain's future (X_{n+1}) conditioned on X_n are identical to those of the chain started at X_n . Of course, X_{n+1} is only a small sliver of the chain's future. As we will show in Section 12, (a) and (b) also hold for the entirety of the chain's future. For the time being however, X_{n+1} is all we need.

Before proceeding, it is worth pointing out that the second equality in (3) justifies the name 'one-step matrix' afforded to P : $p(x, y)$ is the probability that the chain next transitions to y if it currently lies at x , regardless of when this transition occurs and of the chain's history up until this point.

We finish the section with one final re-writing of (3) that will prove of great use later on: For any bounded real-valued function f on $\mathcal{S}$ and natural number n ,

$$(8) \quad \mathbb{E}_\gamma[f(X_{n+1}) | \mathcal{F}_n] = \mathbb{E}_\gamma[f(X_{n+1}) | X_n] = Pf(X_n) \quad \mathbb{P}_\gamma\text{-almost surely,}$$

where

$$Pf(x) := \sum_{y \in \mathcal{S}} p(x, y) f(y) \quad \forall x \in \mathcal{S}.$$

(9) EXERCISE. Show that (3) holds if and only if (8) holds for all bounded real-valued functions f on $\mathcal{S}$. Hint: express f as $f \vee 0 - f \wedge 0$ and use Theorem 0.3ii.

3. Gambler's ruin. At various points in the book, we illustrate aspects of the theory using the famous gambler's ruin problem: A gambler, say her name is Alice, plays a game with multiple rounds. In each round, a coin is tossed. If it lands on heads, Alice wins a pound. Otherwise, she loses one. The coin may be biased so that the probability that it lands on heads is $a \in (0, 1)$. Alice has bad credit; if she goes broke, no one will lend her money and she will no longer be allowed to play. Let X_n denote Alice's wealth right after the n th round of betting and X_0 denote her initial wealth. Clearly, $(X_n)_{n \in \mathbb{N}}$ is a Markov chain with one-step matrix

$$p(0, y) = 1_0(y) \quad \forall y \geq 0, \quad p(x, y) = (1 - a)1_{x-1}(y) + a1_{x+1}(y), \quad \forall x > 0, \quad y \geq 0.$$

4. The time-varying law and its difference equation. For any state x in $\mathcal{S}$, let

$$(1) \quad p_n(x) := \mathbb{P}_\gamma(\{X_n = x\})$$

denote the probability that the chain is in x at time n if its initial position was sampled from γ . Taking expectations of (2.3), we find that the *time-varying law* $p = (p_n)_{n \in \mathbb{N}} = ((p_n(x))_{x \in \mathcal{S}})_{n \in \mathbb{N}}$ of X is the only solution to the difference equation

$$(2) \quad p_{n+1}(x) = \sum_{x' \in \mathcal{S}} p_n(x') p(x', x) \quad \forall x \in \mathcal{S}, \quad n \in \mathbb{N}, \quad p_0(x) = \gamma(x) \quad \forall x \in \mathcal{S},$$

or, in matrix notation,

$$p_{n+1} = p_n P \quad \forall n \in \mathbb{N}, \quad p_0 = \gamma.$$

This is the first of the *analytical characterisations* that we will encounter in this book. We have taken an object (p) defined probabilistically (1) in terms of the chain (X) and the underlying

probability measure ($\mathbb{P}_\gamma$) and derived an equivalent description of the object involving analytical equations and inequalities but no probability. This non-probabilistic description is what we call the analytical characterisation of p . As we will see throughout the book, these types of characterisations are a recurring theme in the Markov chain theory. For ease of reference, I re-state the above as:

(3) **THEOREM** (Analytical characterisation of the time-varying law). *The time-varying law $p = (p_n)_{n \in \mathbb{N}}$ of X defined by (1) is the unique solution of (2).*

Iterating (2.3) and using the tower and take-out-what-is-known properties of conditional expectation (Theorem 0.3iv, v), we obtain an expression for the joint distribution of $X_0, X_1, \dots, X_n$:

$$(4) \quad \mathbb{P}_\gamma(\{X_0 = x_0, X_1 = x_1, \dots, X_n = x_n\}) = \gamma(x_0)p(x_0, x_1) \dots p(x_{n-1}, x_n),$$

for all natural numbers n and states $x_0, x_1, \dots, x_n$. Marginalising the above, we obtain expressions for the joint law of any finite subset of $(X_n)_{n \in \mathbb{N}}$ (the collection of these distributions is often referred to as the *finite-dimensional distributions*). In particular, we find

$$p_n(x) = \sum_{x' \in \mathcal{S}} \gamma(x') p_n(x', x) \quad \forall x \in \mathcal{S},$$

or $p_n = \gamma P_n$ in matrix notation, for the distribution $p_n = (p_n(x))_{x \in \mathcal{S}}$ of the chain at time n in terms the initial distribution γ and of the n -step matrix $P_n = (p_n(x, y))_{x, y \in \mathcal{S}}$ defined by:

$$(5) \quad p_n(x, y) := \sum_{x_1 \in \mathcal{S}} \dots \sum_{x_{n-1} \in \mathcal{S}} p(x, x_1) \dots p(x_{n-1}, y) \quad \forall x, y \in \mathcal{S} \quad n > 0.$$

Out of notational convenience, we use P_0 to denote the identity matrix $(1_x(y))_{x, y \in \mathcal{S}}$ on $\mathcal{S}$. The nomenclature here is motivated by the equation

$$(6) \quad \mathbb{P}_\gamma(\{X_{n+m} = y\} | X_n) = p_m(X_n, y) \quad \forall y \in \mathcal{S}, \quad n, m \geq 0, \quad \mathbb{P}_\gamma\text{-almost surely.}$$

obtained by replacing in (4) ‘ n ’ with ‘ $n + m$ ’ and ‘ x_{n+m} ’ with ‘ y ’, multiplying both sides by $1_{\{X_n = x_n\}}$, summing over all $x_0, \dots, x_{n-1}, x_{n+1}, \dots, x_{n+m-1}$ in $\mathcal{S}$, and taking expectations. In short, $p_m(x, y)$ is the probability that the chain will be at y in m steps if it currently lies in x .

(7) **EXERCISE.** *Prove (6).*

A good point to end this section is the celebrated *Chapman-Kolmogorov equation*: for any natural numbers n and m ,

$$(8) \quad p_{n+m}(x, y) = \sum_{z \in \mathcal{S}} p_n(x, z) p_m(z, y) \quad \forall x, y \in \mathcal{S},$$

or $P_{n+m} = P_n P_m$ in matrix notation. It follows directly from the definition of the n -step matrix in (5).

5. The path space and path law. Up until now, we have viewed the chain as a sequence of random variables each taking values in the state space $\mathcal{S}$. Sometimes, it is very useful to instead think of it as a single random variable taking values in the space of all possible sequences of states (the *path space*):

$$\mathcal{P} := \mathcal{S}^{\mathbb{N}} = \mathcal{S} \times \mathcal{S} \times \dots,$$

formally, the space of all functions from $\mathbb{N}$ to $\mathcal{S}$. To do so, we need to assign a sigma-algebra to $\mathcal{S}^{\mathbb{N}}$. We pick the *cylinder sigma-algebra* $\mathcal{E}$ generated by the collection of subsets of $\mathcal{P}$ of the form

$$(1) \quad \{x_0\} \times \{x_1\} \times \dots \times \{x_n\} \times \mathcal{S} \times \mathcal{S} \times \dots,$$

where n is any natural number and $x_0, x_1, \dots, x_n$ are any states in $\mathcal{S}$. Let $Y := (Y_n)_{n \in \mathbb{N}}$ be a sequence of random variables Y_n taking values in $\mathcal{S}$ and defined on some underlying measurable space $(\Omega, \mathcal{F})$. A simple exercise shows that the collection Y (viewed as a function from Ω to $\mathcal{P}$) is $\mathcal{F}/\mathcal{E}$ -measurable (i.e. a random variable taking values in $\mathcal{P}$) if and only if each Y_n is $\mathcal{F}/2^{\mathcal{S}}$ -measurable where $2^{\mathcal{S}}$ denotes the power set of $\mathcal{S}$. In particular, the chain $X := (X_n)_{n \in \mathbb{N}}$ defined in Section 1 is $\mathcal{F}/\mathcal{E}$ -measurable. For this reason,

$$(2) \quad \mathbb{L}_\gamma(A) := \mathbb{P}_\gamma(\{X \in A\}) \quad \forall A \in \mathcal{E},$$

is a well-defined probability measure on $(\mathcal{P}, \mathcal{E})$ known as the *path law* of X (some authors simply say the *law* of X), where

$$\{X \in A\} := \{\omega \in \Omega : X(\omega) \in A\}$$

denotes the preimage of A under X . Just as with $\mathbb{P}_x$ and $\mathbb{E}_x$, we write throughout the book $\mathbb{L}_x$ as a shorthand for $\mathbb{L}_\gamma$ with $\gamma = 1_x$.

(3) EXERCISE. *Convince yourself that $\mathbb{L}_\gamma$ is a probability measure on $(\mathcal{P}, \mathcal{E})$.*

For example, (4.4) shows that

$$(4) \quad \mathbb{L}_\gamma(\{x_0\} \times \{x_1\} \times \dots \times \{x_n\} \times \mathcal{S} \times \mathcal{S} \times \dots) = \gamma(x_0)p(x_0, x_1) \dots p(x_{n-1}, x_n)$$

for all natural numbers n and states $x_0, x_1, \dots, x_n$. Remarkably, the above is all we need to fully specify $\mathbb{L}_\gamma$:

(5) THEOREM. *The path law $\mathbb{L}_\gamma$ defined in (2) is the only measure on $(\mathcal{P}, \mathcal{E})$ satisfying (4) for all natural numbers n and states $x_0, x_1, \dots, x_n$.*

To argue the above, we need the following consequence of Dynkin's π - λ lemma:

(6) LEMMA (We lose nothing by working with π -systems that generate a sigma-algebra instead of the entire sigma-algebra, Williams, 1991, Lemma 1.6). *Suppose that $\mathcal{H}$ is a π -system on a set Ω , that is, $\mathcal{H}$ is set of subsets of Ω that is closed under finite intersections:*

$$A, B \in \mathcal{H} \Rightarrow A \cap B \in \mathcal{H}.$$

If $\mathcal{H}$ generates a sigma-algebra $\mathcal{G}$ and μ_1, μ_2 are two measures on $(\Omega, \mathcal{G})$ satisfying $\mu_1(\Omega) = \mu_2(\Omega) < \infty$ and $\mu_1(A) = \mu_2(A)$ for all A in $\mathcal{H}$, then $\mu_1(A) = \mu_2(A)$ for all A in $\mathcal{G}$.

(7) EXERCISE. *Using Lemma 6 prove Theorem 5. Hint: show that*

$$\{\{x_0\} \times \{x_1\} \times \dots \times \{x_n\} \times \mathcal{S} \times \mathcal{S} \times \dots : x_0, x_1, \dots, x_n \in \mathcal{S}, n \in \mathbb{N}\} \cup \emptyset$$

is a π -system.

6. The martingale characterisation. Martingales play an important role in most areas of modern probability and Markov theory is no exception. Indeed, if we look at them from the right angle, it is not too difficult to see that Markov chains can be equivalently defined in terms of martingales. Here is that angle:

(1) THEOREM (Martingale characterisation). *Suppose that $X = (X_n)_{n \in \mathbb{N}}$ is a sequence of random-variables on a probability triplet $(\Omega, \mathcal{F}, \mathbb{P})$ that take values in $\mathcal{S}$ and let $(\mathcal{F}_n)_{n \in \mathbb{N}}$ denote the filtration generated by the sequence (Definition 2.1). The sequence satisfies the Markov property (2.3) (with $\mathbb{P}$ replacing $\mathbb{P}_\gamma$) if and only if*

$$(2) \quad M_n := g(n)f(X_n) - g(0)f(X_0) - \sum_{m=0}^{n-1} (g(m+1)Pf(X_m) - g(m)f(X_m)) \quad \forall n \geq 0,$$

defines an $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -adapted $\mathbb{P}$ -martingale, for every bounded real-valued function f on $\mathcal{S}$ and real-valued function g on $\mathbb{N}$. In particular, if X is the chain introduced in Section 1, then (2) defines an $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -adapted $\mathbb{P}_\gamma$ -martingale for every bounded f, g and initial distribution γ .

Proof. Clearly, M_n is $\mathcal{F}_n$ -adapted, and there are no integrability issues since g , f , and, consequently, Pf are bounded functions. Suppose that $(X_n)_{n \in \mathbb{N}}$ satisfies the Markov property. To prove that (2) defines a martingale we need to show that

$$\mathbb{E}[M_n | \mathcal{F}_{n-1}] = M_{n-1}, \quad \forall n \geq 0, \quad \mathbb{P}\text{-almost surely.}$$

Applying (2.8), we obtain

$$\mathbb{E}[g(n)f(X_n) | \mathcal{F}_{n-1}] = g(n)\mathbb{E}[f(X_n) | \mathcal{F}_{n-1}] = g(n)Pf(X_{n-1}), \quad \mathbb{P}_\gamma\text{-almost surely;}$$

and so,

$$\mathbb{E}[M_n | \mathcal{F}_{n-1}] = \mathbb{E}_\gamma[g(n)f(X_n) | \mathcal{F}_{n-1}] - g(n)Pf(X_{n-1}) + M_{n-1}^f = M_{n-1}^f, \quad \mathbb{P}_\gamma\text{-almost surely.}$$

Conversely, suppose that (2) holds for every bounded f, g . Pick any y in $\mathcal{S}$ and n in $\mathbb{N}$ and set $f := 1_y$ and $g := 1$ in (2) so that

$$M_{n+1} - M_n = 1_y(X_{n+1}) - p(X_n, y).$$

Conditioning on $\mathcal{F}_n$ and taking expectations, we find that

$$\mathbb{P}(\{X_{n+1} = y\} | \mathcal{F}_n) = p(X_n, y) \quad \mathbb{P}\text{-almost surely.}$$

The Markov property (2.3) follows by conditioning on X_n , taking expectations, and applying the tower property of conditional expectation (in particular, Theorem 0.3i, iv). $\square$

7. Other definitions of the chain*. Armed with the path space, path law, and martingale characterisation, we can now answer a question we have so far avoided: our definition of the chain via Algorithm 1 seems rather specific; does this matter?

In short, no, it does not. In long, depending on what text you pick up, you will find a discrete-time chain $(X_n)_{n \in \mathbb{N}}$ on a countable state space $\mathcal{S}$ with one-step matrix P and initial distribution γ defined as a sequence of random variables taking values in $\mathcal{S}$, defined on a probability triplet $(\Omega, \mathcal{F}, \mathbb{P}_\gamma)$, and satisfying $\mathbb{P}_\gamma(\{X_0 = \cdot\}) = \gamma(\cdot)$ and

- (a) the Markov property (2.3), where $(\mathcal{F}_n)_{n \in \mathbb{N}}$ denotes the filtration generated by the sequence (Definition 2.1),
- (b) or (4.4) for every n in $\mathbb{N}$ and all states $x_0, \dots, x_n$ in $\mathcal{S}$,
- (c) or the martingale property: (6.2) defines an $\mathcal{F}_n$ -adapted $\mathbb{P}_\gamma$ -martingale for every bounded real-valued function f on $\mathcal{S}$ and real-valued function g on $\mathbb{N}$.

The above three statements are equivalent:

(1) **THEOREM** (Equivalent definitions of discrete-time chains). *If $X := (X_n)_{n \in \mathbb{N}}$ is a sequence of $\mathcal{S}$ -valued random-variables on a probability triplet $(\Omega, \mathcal{F}, \mathbb{P}_\gamma)$ with $\mathbb{P}_\gamma(\{X_0 = x\}) = \gamma(x)$ for all x in $\mathcal{S}$, then statements (a), (b), or (c) are equivalent.*

Proof. Previously, we already showed that (a) $\Rightarrow$ (b) (Section 4) and (a) $\Leftrightarrow$ (c) (Theorem 6.1) and, so, it suffices to show here that (b) $\Rightarrow$ (a). If we can argue that

$$(2) \quad \mathbb{E}[f(X_{n+1}) | \mathcal{F}_n] = Pf(X_n) \quad \mathbb{P}_\gamma\text{-almost surely}$$

for any bounded real-valued function f on $\mathcal{S}$, then the rest of the (a) follows by conditioning on X_n , taking expectations, and applying the tower property (Theorem 0.3i, iv). Picking any $x_0, \dots, x_n$ in $\mathcal{S}$ and applying (4.4) we obtain

$$\begin{aligned} \mathbb{E} [f(X_{n+1}) 1_{\{X_0=x_0\}} \dots 1_{\{X_n=x_n\}}] &= \sum_{x \in \mathcal{S}} f(x) \mathbb{E} [1_{\{X_0=x_0\}} \dots 1_{\{X_n=x_n\}} 1_{\{X_{n+1}=x\}}] \\ &= \sum_{x \in \mathcal{S}} f(x) \gamma(x_0) p(x_0, x_1) \dots p(x_n, x) \\ &= \gamma(x_0) p(x_0, x_1) \dots p(x_{n-1}, x_n) P f(x_n) \\ &= \mathbb{E} [P f(X_n) 1_{\{X_0=x_0\}} \dots 1_{\{X_n=x_n\}}], \end{aligned}$$

and (2) follows because the collection of events $\{\{X_0 = x_0, \dots, X_n = x_n\} : x_0, \dots, x_n \in \mathcal{S}\}$ generates $\mathcal{F}_n$. $\square$

Because the path law is fully specified by (5.4) (Theorem 5.5), Theorem 1 shows that the path law (5.2) of a discrete-time chain with one-step matrix P and initial distribution γ is the same regardless of whether we use (a), (b), or (c) above in its definition. For this reason, the particular construction of the chain does not matter as long as we are only interested in questions that can be answered by observing the entire path of the chain. I focus on the construction given in Algorithm 1 because I find it straightforward to work with and because the algorithm itself is an easy-to-implement recipe for simulating chains in practice.

A final observation: the time-varying law does not characterise the path law. In contrast with (a–c) above, the time-varying law $(p_n)_{n \in \mathbb{N}}$ of the chain in (4.1) is *not* enough to uniquely define a distribution on the path space. For instance, using Theorem 1.3 we can build a sequence of independent random variables $(W_n)_{n \in \mathbb{N}}$ such that the law of W_n is p_n for each n in $\mathbb{N}$ —something absurd for all but the most trivial of chains (think of (2.3)).

8. Stopping times. Given the filtration $(\mathcal{F}_n)_{n \in \mathbb{N}}$ generated by the chain (Definition 2.1), a random variable is said to be an $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping time if it is the time at which some event occurs with the event being such that we are able to deduce whether it has occurred by any given point in time if we have observed the chain's path up until (and including) said point. For example, the first (or second, or k th) time that the chain visits a given set of interest is a stopping time, but the last time it visits the set is not. Formally:

(1) **DEFINITION** (Discrete-time stopping times). *Given the filtration $(\mathcal{F}_n)_{n \in \mathbb{N}}$ generated by the chain, a random variable $\varsigma : \Omega \rightarrow \mathbb{N}_E$ is said to be an $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping time if and only if the event $\{\varsigma \leq n\}$ belongs to $\mathcal{F}_n$, for each n in $\mathbb{N}$. With a stopping time ς , we associate the sigma-algebra $\mathcal{F}_\varsigma$ defined by*

$$\forall A \in \mathcal{F}, \quad A \in \mathcal{F}_\varsigma \Leftrightarrow A \cap \{\varsigma \leq n\} \in \mathcal{F}_n \quad \forall n \geq 0.$$

(2) **EXERCISE.** *Convince yourself that $\mathcal{F}_\varsigma$ is a sigma-algebra.*

The name ‘stopping time’ afforded to these random variables stems from the occurrence such an event being used in practice to signal that one should stop what they are doing and take a certain action. For instance, our gambler Alice (Section 3) would probably be wise in walking away from the game the moment her winnings reached an amount she finds satisfactory. In this vein, the chain is often said to *stop* at ς .

The *pre- ς sigma-algebra* $\mathcal{F}_\varsigma$ formalises the chain's history up until (and including) ς : it is the collection of events whose occurrence we are able to deduce by tracking the chain's position up until the stopping time ς . For instance, if ς is the second time that the chain visits a given subset A , then the event that the chain visits A once (or at least once, or twice, or at least

twice) belongs to $\mathcal{F}_\varsigma$ but the event that the chain visits A three (or four, or at least four, or ...) does not belong to $\mathcal{F}_\varsigma$. From this vantage point, the following useful lemma is almost trivial:

(3) LEMMA. *If ς and ϑ are two $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping times, the*

- (i) *The events $\{\varsigma \leq \vartheta\}$, $\{\varsigma < \vartheta\}$, and $\{\varsigma = \vartheta\}$ belong to $\mathcal{F}_\vartheta$.*
- (ii) *Given any event A in $\mathcal{F}_\varsigma$, the event $A \cap \{\varsigma \leq \vartheta\}$ belongs to $\mathcal{F}_\vartheta$.*
- (iii) *In particular, if $\varsigma \leq \vartheta$, then $\mathcal{F}_\varsigma \subseteq \mathcal{F}_\vartheta$.*

Part (i) states that we are able to decide whether the event associated with ς has occurred by (or before, or at) ϑ if we have observed the chain up until ϑ . Part (ii) states that anything we are able to deduce from observing the chain up until ς , we are able to deduce from observing the chain up until ϑ as long as ς is no greater than ϑ . Part (iii) then follows directly from (ii): if ϑ is always greater than ς , then we are able to deduce by time ϑ anything we can deduce by time ς .

Proof. (i) Fix any natural number n and note that

$$\{\varsigma < \vartheta\} \cap \{\vartheta \leq n\} = \bigcap_{m=0}^{n-1} \{\varsigma \leq m\} \cap \{m < \vartheta\} \cap \{\vartheta \leq n\} = \bigcap_{m=0}^{n-1} \{\varsigma \leq m\} \cap (\Omega \setminus \{\vartheta \leq m\}) \cap \{\vartheta \leq n\}.$$

Because sigma-algebras are closed under finite intersections and complements, the above belongs to $\mathcal{F}_n$ and it follows that $\{\varsigma < \vartheta\}$ belongs to $\mathcal{F}_\vartheta$. An analogous argument shows that $\{\vartheta < \varsigma\}$ also belongs to $\mathcal{F}_\vartheta$. Given that sigma-algebras are closed under complements and intersections, it follows that $\{\vartheta \leq \varsigma\} = \Omega \setminus \{\vartheta < \varsigma\}$ and $\{\varsigma = \vartheta\} = \{\vartheta \leq \varsigma\} \cap (\Omega \setminus \{\vartheta < \varsigma\})$ also belong to $\mathcal{F}_\vartheta$.

(ii) For any natural number n ,

$$A \cap \{\varsigma \leq \vartheta\} \cap \{\vartheta \leq n\} = (A \cap \{\varsigma \leq n\}) \cap (\{\varsigma \leq \vartheta\} \cap \{\vartheta \leq n\}).$$

Because (i) shows that $\{\varsigma \leq \vartheta\}$ belongs to $\mathcal{F}_\vartheta$, and because sigma-algebras are closed under intersections, the above implies that $A \cap \{\varsigma \leq \vartheta\} \cap \{\vartheta \leq n\}$ belongs to $\mathcal{F}_n$, and the result follows.

(iii) Follows directly from (ii) as, in this case, $\{\varsigma \leq \vartheta\} = \Omega$. □

(4) EXERCISE. *Show that the event $\{\varsigma < \infty, X_\varsigma = x\}$ belongs to $\mathcal{F}_\varsigma$, for any x in $\mathcal{S}$. Hint: notice that Lemma 3i implies that $\{\varsigma = n\}$ belongs to $\mathcal{F}_n$ for all $n \geq 0$.*

An important example: hitting times. The *hitting time* (or *first passage time*) σ_A of a subset A of the state space is the first time that the chain X visits the subset (or infinity if it never does). Formally,

$$(5) \quad \sigma_A(\omega) := \inf\{n \in \mathbb{N} : X_n(\omega) \in A\} \quad \forall \omega \in \Omega,$$

where we are adhering to our convention that $\inf \emptyset = \infty$. We use the shorthand $\sigma_x := \sigma_{\{x\}}$ for any x in $\mathcal{S}$.

Because we are able to deduce whether the chain has visited a set by observing the chain up until the (and including) the moment it does, hitting times are $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping times:

(6) PROPOSITION. *Let A be any subset of $\mathcal{S}$ and $(\mathcal{F}_n)_{n \in \mathbb{N}}$ denote the filtration generated by the chain (Definition 2.1). The hitting time σ_A of A defined in (5) is an $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping time.*

Proof. This follows immediately from the hitting time's definition as it implies that

$$\{\sigma_A \leq 0\} = \{X_0 \in A\} \in \mathcal{F}_0, \quad \{\sigma_A \leq n\} = \bigcup_{m=0}^n \{X_m \in A\} \in \mathcal{F}_n \quad \forall n > 0.$$

□

9. Dynkin's formula. As we will see in both this section and Section 12, the non-predictive nature of stopping times allows us to generalise the results we have gathered this far so that they apply to stopping times instead of just deterministic times. Here, we make use of the martingale characterisation (Theorem 6.1) to prove *Dynkin's formula*: a generalisation of the recurrence (4.2) satisfied by the time-varying law. In particular, summing over n in (4.2), we obtain

$$\mathbb{E}_\gamma [1_{\{X_n=x\}}] = p_n(x) = \gamma(x) + \sum_{m=0}^{n-1} (p_{m+1}P(x) - p_m(x)) = \gamma(x) + \mathbb{E}_\gamma \left[\sum_{m=0}^{n-1} (p(X_m, x) - 1_{\{X_m=x\}}) \right].$$

If f is bounded real-valued function $\mathcal{S}$, then multiplying both sides by $f(x)$ and summing over x in $\mathcal{S}$ we obtain the following the integral version of (4.2):

$$\mathbb{E}_\gamma [f(X_n)] = \gamma(f) + \mathbb{E}_\gamma \left[\sum_{m=0}^{n-1} (Pf(X_m) - f(X_m)) \right].$$

Dynkin's formula tells us that the same equation holds if we replace n with a stopping time.

(1) **THEOREM (Dynkin's Formula).** *Let f be a bounded real-valued function on $\mathcal{S}$, g be a real-valued function on $\mathbb{N}$, and $(\mathcal{F}_n)_{n \in \mathbb{N}}$ denote the filtration generated by the chain (Definition 2.1). If ς is an $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping time and $\varsigma_n := \varsigma \wedge n$ for any given $n \geq 0$,*

$$\mathbb{E}_\gamma [g(\varsigma_n)f(X_{\varsigma_n})] = g(0)\mathbb{E}_\gamma [f(X_0)] + \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma_n-1} g(m+1)Pf(X_m) - g(m)f(X_m) \right].$$

The proof of Dynkin's formula consists of combining the martingale characterisation of the chain (Theorem 6.1) with Doob's optional stopping theorem:

(2) **THEOREM (Doob's optional stopping).** *Let $(\Omega, \mathcal{G}, \mathbb{P})$ be a probability triplet, $(\mathcal{G}_n)_{n \in \mathbb{N}}$ be a filtration contained in $\mathcal{G}$, and $(M_n)_{n \in \mathbb{N}}$ be a $(\mathcal{G}_n)_{n \in \mathbb{N}}$ -adapted $\mathbb{P}$ -martingale. For any bounded $(\mathcal{G}_n)_{n \in \mathbb{N}}$ -stopping time ς , $\mathbb{E}[M_\varsigma] = \mathbb{E}[M_0]$.*

Proof. Note that for any natural number n , $\varsigma \wedge n \neq \varsigma \wedge (n-1)$ only if $\varsigma > n-1$ in which case $\varsigma \geq n$. For this reason,

$$\begin{aligned} \mathbb{E}[M_{\varsigma \wedge n} - M_{\varsigma \wedge (n-1)}] &= \mathbb{E}[1_{\{\varsigma > n-1\}}(M_n - M_{n-1})] = \mathbb{E}[\mathbb{E}[1_{\{\varsigma > n-1\}}(M_n - M_{n-1})|\mathcal{G}_n]] \\ &= \mathbb{E}[1_{\{\varsigma > n-1\}}(\mathbb{E}[M_n|\mathcal{G}_n] - M_{n-1})] = 0. \end{aligned}$$

where the final equality follows from the martingale property of $(M_n)_{n \in \mathbb{N}}$ and the others from the tower rule and take-out-what-is-known properties of conditional expectation (Theorem 0.3i, iv, v). Thus,

$$\mathbb{E}[M_{\varsigma \wedge n}] = \mathbb{E}[M_{\varsigma \wedge (n-1)}] = \mathbb{E}[M_{\varsigma \wedge (n-2)}] = \cdots = \mathbb{E}[M_0] \quad \forall n \geq 0.$$

Because ς is bounded, there exists an n such that $\varsigma \wedge n = \varsigma$ and the result follows. $\square$

Proof of Theorem 1. Because f and $g \circ \varsigma_n$ are bounded, all random variables in the equation are well-defined and Theorem 6.1 shows that $M = (M_n)_{n \in \mathbb{N}}$ in (6.2) is an $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -adapted $\mathbb{P}_\gamma$ -martingale. Because ς_n is a bounded $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping time, Doob's optional stopping (Theorem 2) then tells us that $\mathbb{E}_\gamma [M_{\varsigma_n}] = \mathbb{E}_\gamma [M_0] = 0$. $\square$

10. Stopping distributions and occupation measures. With a stopping time ς we associate a *stopping distribution* μ and *occupation measure* ν defined by

$$(1) \quad \mu(n, x) := \mathbb{P}_\gamma(\{\varsigma = n, X_\varsigma = x\}), \quad \forall n \geq 0, \quad x \in \mathcal{S},$$

$$(2) \quad \nu(n, x) := \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma-1} 1_n(m) 1_x(X_m) \right] = \mathbb{E}_\gamma \left[1_x(X_n) \sum_{m=0}^{\varsigma-1} 1_n(m) \right]$$

$$= \mathbb{P}_\gamma(\{\varsigma > n, X_n = x\}), \quad \forall n \geq 0, \quad x \in \mathcal{S}.$$

In other words, $\mu(n, x)$ is the probability that the chain stops at time n while in state x and $\nu(n, x)$ is the probability that the chain is in state x at time n and that it has not yet stopped.

The mass of the stopping distribution is simply the probability that the stopping time is finite:

$$(3) \quad \mu(\mathbb{N}, \mathcal{S}) = \sum_{n=0}^{\infty} \sum_{x \in \mathcal{S}} \mathbb{P}_\gamma(\{\varsigma = n, X_\varsigma = x\}) = \mathbb{P}_\gamma(\{\varsigma < \infty\});$$

while that of the occupation measure is the mean stopping time:

$$(4) \quad \nu(\mathbb{N}, \mathcal{S}) = \sum_{n=0}^{\infty} \sum_{x \in \mathcal{S}} \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma-1} 1_n(m) 1_x(X_m) \right] = \mathbb{E}_\gamma \left[\left(\sum_{m=0}^{\varsigma-1} \sum_{n=0}^{\infty} 1_n(m) \right) \left(\sum_{x \in \mathcal{S}} 1_x(X_m) \right) \right]$$

$$= \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma-1} 1 \right] = \mathbb{E}_\gamma[\varsigma].$$

The stopping distribution and the occupation measure are tied together by a set of linear equations:

(5) LEMMA. *The pair (μ, ν) satisfies*

$$(6) \quad \begin{aligned} \mu(0, x) + \nu(0, x) &= \gamma(x), & \forall x \in \mathcal{S}, \\ \mu(n, x) + \nu(n, x) &= \sum_{y \in \mathcal{S}} \nu(n-1, y) p(y, x), & \forall n > 0, \quad x \in \mathcal{S}. \end{aligned}$$

Proof. The first set of equations follow directly from the definitions of μ and ν . The second set requires a bit more work. Pick any positive integer n and state x in $\mathcal{S}$. Setting $g := 1_n$ and $f := 1_x$ in Dynkin's formula (Theorem 9.1) yields

$$(7) \quad \mathbb{E}_\gamma[1_n(\varsigma_k) 1_x(X_{\varsigma_k})] + \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma_k-1} 1_n(m) 1_x(X_m) \right] = \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma_k-1} 1_n(m+1) p(X_m, x) \right] \quad \forall k > 0,$$

where ς_k denotes the minimum $\varsigma \wedge k$ of ς and k . For each ω in Ω , the sequence $(\varsigma_k(\omega))_{k \in \mathbb{Z}_+}$ is increasing and has limit $\varsigma(\omega)$. Thus, monotone convergence implies that

$$(8) \quad \lim_{k \rightarrow \infty} \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma_k-1} 1_n(m) 1_x(X_m) \right] = \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma-1} 1_n(m) 1_x(X_m) \right],$$

$$(9) \quad \lim_{k \rightarrow \infty} \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma_k-1} 1_n(m+1) P(X_m, x) \right] = \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma-1} 1_n(m+1) P(X_m, x) \right].$$

Similarly, bounded convergence yields

$$(10) \quad \lim_{n \rightarrow \infty} \mathbb{E}_\gamma[1_n(\varsigma_k) 1_x(X_{\varsigma_k})] = \mathbb{E}_\gamma[1_n(\varsigma) 1_x(X_\varsigma)] = \mathbb{P}_\gamma(\{\varsigma = n, X_\varsigma = x\}).$$

Putting (7)–(10) together we have that

$$\mathbb{P}_\gamma(\{\varsigma = n, X_\varsigma = x\}) + \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma-1} 1_n(m) 1_x(X_m) \right] = \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma-1} 1_n(m+1) P(X_m, x) \right].$$

Using Tonelli's Theorem, we obtain the second equation in (6). $\square$

The marginals. The *time marginal* μ_T of the stopping distribution

$$(11) \quad \mu_T(n) := \sum_{x \in \mathcal{S}} \mu(n, x) = \mathbb{P}_\gamma \left(\bigcup_{x \in \mathcal{S}} \{\zeta = n, X_\zeta = x\} \right) = \mathbb{P}_\gamma(\{\zeta = n\}),$$

is the distribution of the exit time itself. Technically, it is the distribution of ζ restricted to $\mathbb{N}$ because ζ may take the value ∞ (indicating that the chain never stops). However, we recover the full distribution of ζ from the above using $\mathbb{P}_\gamma(\{\zeta = \infty\}) = 1 - \mathbb{P}_\gamma(\{\zeta < \infty\}) = 1 - \mu_T(\mathbb{N})$. Similarly, it is straightforward to check that the time marginal $\nu_T(n) := \nu(n, \mathcal{S})$ is equal to one minus the cumulative distribution function of ζ .

The *space marginals* μ_S and ν_S of the stopping distribution and occupation measure,

$$(12) \quad \mu_S(x) := \sum_{n=0}^{\infty} \mu(n, x) = \mathbb{P}_\gamma \left(\bigcup_{n=0}^{\infty} \{\zeta = n, X_\zeta = x\} \right) = \mathbb{P}_\gamma(\{\zeta < \infty, X_\zeta = x\}),$$

$$(13) \quad \nu_S(x) := \sum_{n=0}^{\infty} \nu(n, x) = \mathbb{E}_\gamma \left[\sum_{m=0}^{\zeta-1} \left(\sum_{n=0}^{\infty} 1_n(m) \right) 1_x(X_m) \right] = \mathbb{E}_\gamma \left[\sum_{m=0}^{\zeta-1} 1_x(X_m) \right],$$

tell us where the chain stops and where it spends time before stopping, respectively. Explicitly, $\mu_S(x)$ is the probability that the chain stops in x , while $\nu_S(x)$ denotes the expected number of visits the chain makes to x before stopping. Summing over n in (6), we find that the space marginals of the exit distribution and occupation measure also satisfy a set of linear equations:

(14) COROLLARY. *The pair (μ_S, ν_S) satisfies*

$$\mu_S(x) + \nu_S(x) = \gamma(x) + \sum_{y \in \mathcal{S}} \nu_S(y) p(y, x) \quad \forall x \in \mathcal{S}.$$

11. Exit times*. In applications, we are often interested in how long the chain takes to exit a given subset of the state space and what part of the domain's boundary does the chain cross to exit. To study this problem, we single out a subset $\mathcal{D}$ of the state space $\mathcal{S}$ and refer to it as the *domain*. The *exit time* σ from $\mathcal{D}$ is the first instant that chain first lies outside of $\mathcal{D}$:

$$(1) \quad \sigma(\omega) := \inf \{n \geq 0 : X_n(\omega) \notin \mathcal{D}\} \quad \forall \omega \in \Omega.$$

Equivalently, the exit time is the hitting time $\sigma_{\mathcal{D}^c}$ (8.5) of the domain's complement $\mathcal{D}^c$. Proposition 8.6 shows that σ is an $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping time (8.1), where $(\mathcal{F}_n)_{n \in \mathbb{N}}$ denotes the filtration generated by the chain (Definition 2.1).

The exit distribution and occupation measure. Let μ and ν be as in (10.1)–(10.2) with σ replacing ζ . In this case, the $\mu(n, x)$ is the probability that the chain first exits the domain at time n by moving to state x , and we refer to μ as the *exit distribution*. Similarly, $\nu(n, x)$ denotes the probability that the chain is in state x at time n and that it has not yet exited the domain. Because the chain lies outside the domain at the time of exit, the support of the exit distribution μ is contained outside of the domain. Similarly, because before exiting the chain lies inside the domain, the support of the occupation measure ν is contained outside of the domain. In other words,

$$(2) \quad \mu(\mathbb{N}, \mathcal{D}) = 0, \quad \nu(\mathbb{N}, \mathcal{D}^c) = 0.$$

Combining the above with Lemma 10.5, we obtain an equivalent description of the exit distribution and occupation measure in terms of a linear recursion:

(3) **THEOREM** (Analytical characterisation of μ and ν). *Let σ denote the exit time (1) from the domain $\mathcal{D}$. Its exit distribution μ (10.1) and occupation measure ν (10.2) are given by*

$$(4) \quad \begin{aligned} \nu(n, x) &= 1_{\mathcal{D}}(x) \hat{p}_n(x), & \forall n > 0, & \quad \nu(0, x) = 1_{\mathcal{D}}(x) \gamma(x), & \forall x \in \mathcal{S}, \\ \mu(n, x) &= 1_{\mathcal{D}^c}(x) (\hat{p}_n(x) - \hat{p}_{n-1}(x)), & \forall n > 0, & \quad \mu(0, x) = 1_{\mathcal{D}^c}(x) \gamma(x), & \forall x \in \mathcal{S}, \end{aligned}$$

where $\hat{p}_n = (\hat{p}_n(x))_{x \in \mathcal{S}}$ is defined by the difference equation

$$(5) \quad \hat{p}_{n+1}(x) = 1_{\mathcal{D}^c}(x) \hat{p}_n(x) + \sum_{x' \in \mathcal{D}} \hat{p}_n(x') p(x', x) \quad \forall x \in \mathcal{S}, \quad n \geq 0, \quad \hat{p}_0(x) = \gamma(x) \quad \forall x \in \mathcal{S}.$$

The ideas behind Theorem 3 are simple: consider a second chain $\hat{X}$ which is identical to X except that every state outside of the domain $\mathcal{D}$ has been turned into an absorbing state (i.e. a state the chain cannot leave). In other words, $\hat{X}$ has one-step matrix $\hat{P} := (\hat{p}(x, y))_{x, y \in \mathcal{S}}$ with

$$(6) \quad \hat{p}(x, y) := \begin{cases} p(x, y) & \text{if } x \in \mathcal{D} \\ 1_x(y) & \text{if } x \notin \mathcal{D} \end{cases} \quad \forall x, y \in \mathcal{S}.$$

Theorem 4.3 shows that the time-varying law of $\hat{X}$ is $\hat{p} = (\hat{p}_n)_{n \in \mathbb{N}}$ in (5). Suppose that we built $\hat{X}$ by running Algorithm 1 with $\hat{P}$ replacing P but keeping the same $X_0, U_1, U_2, \dots$. Because X and $\hat{X}$ are updated using the same rules for as long as they remain inside the domain $\mathcal{D}$, the two chains are identical up until (and including) the moment that they simultaneously exit $\mathcal{D}$. Thus, X and $\hat{X}$ leave $\mathcal{D}$ at the same time and via the same state. For this reason, the probability $\mu(\{0, 1, \dots, n\}, x)$ that X has exited the domain by time n via state x is also the probability that $\hat{X}$ exited by n via x . Because $\hat{X}$ gets stuck in the first state it visits outside of $\mathcal{D}$, it follows that $\mu(\{0, 1, \dots, n\}, x)$ is the probability $\hat{p}_n(x)$ that $\hat{X}$ is in state x at time n . Using $\mu(n, x) = \mu(\{0, 1, \dots, n\}, x) - \mu(\{0, 1, \dots, n-1\}, x)$, we obtain the second equation in (4). The equation for the occupation measure follows from similar reasons. The key here is that $\hat{X}$ is in a state belonging to the domain only if it has not yet exited. Thus, the probability $\nu(n, x)$ that $\hat{X}$ is in a state x belonging to $\mathcal{D}$ at time n and that it has not yet exited $\mathcal{D}$ is simply the probability $\hat{p}_n(x)$ that it is in state x at time n .

The space marginals. Just as with the joint exit distribution μ and occupation measure ν of an exit time, their space marginals μ_S and ν_S are fully specified by a set of linear equations:

(7) **THEOREM** (Analytical characterisation of μ_S and ν_S). *Let σ denote the exit time (1) from the domain $\mathcal{D}$. The space marginal μ_S in (10.12) of its exit distribution is given by*

$$(8) \quad \mu_S(x) = \begin{cases} \gamma(x) + \sum_{z \in \mathcal{D}} \nu_S(z) p(z, x) & \forall x \notin \mathcal{D} \\ 0 & \forall x \in \mathcal{D} \end{cases}.$$

The space marginal ν_S in (10.13) of its occupation measure is the minimal non-negative solution to the equations

$$(9) \quad \nu_S(x) = \begin{cases} \gamma(x) + \sum_{z \in \mathcal{D}} \nu_S(z) p(z, x) & \forall x \in \mathcal{D} \\ 0 & \forall x \notin \mathcal{D} \end{cases}.$$

By minimal I mean that, if $\rho = (\rho(x))_{x \in \mathcal{S}}$ is any non-negative solution of (9), then $\rho(x) \geq \nu_S(x)$ for all x in $\mathcal{D}$.

Proof. Given that Corollary 10.14 and (2) imply that μ_S and ν_S satisfy (8,9), all that remains to be shown is the minimality of ν_S . Let $\rho = (\rho(x))_{x \in \mathcal{S}}$ be any other non-negative solution of

(9). If x lies outside of the domain, $\nu_S(x) = \rho(x) = 0$ and the result is trivial. Otherwise,

$$\begin{aligned}
 \rho(x) &= \gamma(x) + \sum_{z_0 \in \mathcal{D}} \rho(z_0)p(z_0, x) = \gamma(x) + \sum_{z_0 \in \mathcal{D}} \gamma(z_0)p(z_0, x) \\
 &+ \sum_{z_0 \in \mathcal{D}} \sum_{z_1 \in \mathcal{D}} \rho(z_0)p(z_0, z_1)p(z_1, x) \\
 &= \cdots = \gamma(x) + \sum_{l=1}^m \sum_{z_0 \in \mathcal{D}} \cdots \sum_{z_{l-1} \in \mathcal{D}} \gamma(z_0)p(z_0, z_1) \cdots p(z_{l-1}, x) \\
 &+ \sum_{z_0 \in \mathcal{D}} \cdots \sum_{z_m \in \mathcal{D}} \rho(z_0)p(z_0, z_1)p(z_1, z_2) \cdots p(z_m, x) \\
 (10) \quad &\geq \gamma(x) + \sum_{l=1}^m \sum_{z_0 \in \mathcal{D}} \cdots \sum_{z_{l-1} \in \mathcal{D}} \gamma(z_0)p(z_0, z_1) \cdots p(z_{l-1}, x) \quad \forall x \in \mathcal{D}.
 \end{aligned}$$

However, it follows from (4.4) and the definition of the exit time σ (1) that

$$\begin{aligned}
 \sum_{z_0 \in \mathcal{D}} \cdots \sum_{z_{l-1} \in \mathcal{D}} \gamma(z_0)p(z_0, z_1) \cdots p(z_{l-1}, x) &= \mathbb{P}_\gamma(\{X_0 \in \mathcal{D}, \dots, X_{l-1} \in \mathcal{D}, X_l = x\}) \\
 &= \mathbb{P}_\gamma(\{X_l = x, \sigma > l\}) \quad \forall x \in \mathcal{D}.
 \end{aligned}$$

Using the above, we rewrite (10) as

$$\rho(x) \geq \sum_{l=0}^m \mathbb{P}_\gamma(\{X_l = x, \sigma > l\}) \quad \forall l > 0, \quad x \in \mathcal{D}.$$

Taking the limit $m \rightarrow \infty$ and comparing the above with the definitions of the occupation measure ν and its marginal ν_S in (10.2) and (10.13), we have the desired $\rho(x) \geq \nu_S(x)$. $\square$

You may be wondering whether we can shed the minimality requirement from the characterisation (7). In other words, can the equations (9) have other non-negative solutions aside from ν_S ? They can:

(11) EXAMPLE. Consider the exit from $\{1, 2\}$ of a chain with state space $\{0, 1, 2\}$, initial distribution 1_1 , and one-step matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

The chain starts at 1, jumps to 0 in the first step, and then remains at 0 forever. Thus, we have that the space marginal of the occupation measure is $\nu_S = 1_1$. Simple algebraic manipulations tell us that $(\rho(x))_{x \in \mathcal{S}}$ solves (9) if and only if

$$(12) \quad \rho = 1_1 + \alpha 1_2 \quad \text{for some } \alpha \in [0, \infty].$$

In other words, (9) has infinitely many solutions including ν_S . However, out of all these solutions, ν_S is the minimal one:

$$\nu_S(x) = 1_1(x) \leq 1_1(x) + \alpha 1_2(x) = \rho(x), \quad \forall x \in \{0, 1, 2\}.$$

The issue in the above example is that we have chosen to include in our state space the closed set $\{2\}$ which is not accessible from the support of the initial distribution 1_1 . For this reason, the state 2 is irrelevant to any question regarding the exit of the chain from $\{1, 2\}$. By

ruling out such an extraneous closed set, it is straightforward to show that there are no other non-negative solutions of (9) also satisfying the condition

$$\gamma(\mathcal{S} \setminus \mathcal{D}) + \sum_{z \in \mathcal{D}} \sum_{x \notin \mathcal{D}} \rho(z) p(z, x) = \mathbb{P}_x(\{\sigma < \infty\})$$

obtained by summing (8) over x , see (Kuntz, 2017, Corollary 2.11). However, the question remains...

An open question. When does (9) have multiple non-negative solutions?

Ruin probabilities. We close this chapter by revisiting our gambler Alice (Section 3) and applying Theorem 7 to work out her ruin probabilities. In particular, suppose that Alice aims to leave the game with K pounds in her pocket and will carry on playing until she either accumulates this amount of money or goes broke trying (at which point, she is excluded from the game). Suppose that X_0 belongs $\{1, 2, \dots, K-1\}$ and consider the problem of computing the probability that Alice is successful in her venture. The time of exit σ (1) from the domain $\{1, 2, \dots, K-1\}$ is the betting round that Alice goes broke or accumulates K pounds, whichever comes first. For this reason, $\mu_S(K)$ is the probability that Alice is successful, while $\mu_S(0)$ is the probability she is not. The equations (9) satisfied by ν_S read

$$(13) \quad \begin{cases} \gamma(1) = \nu_S(1) - (1-a)\nu_S(2) \\ \gamma(x) = \nu_S(x) - a\nu_S(x-1) - (1-a)\nu_S(x+1), & \forall x = 2, \dots, K-2, \\ \gamma(K-1) = \nu_S(K-1) - a\nu_S(K-2) \end{cases}$$

while those (8) satisfied by μ_S read

$$(14) \quad \mu_S(0) = (1-a)\nu_S(1), \quad \mu_S(K) = a\nu_S(K-1).$$

If the coin is unbiased ($a = 1/2$), then multiplying (13) by x and summing over $x \in \{1, \dots, K-1\}$ yields

$$(15) \quad \mathbb{E}_\gamma[X_0] = \sum_{x=1}^{K-1} x\gamma(x) = \frac{K\nu_S(K-1)}{2}.$$

The above and (14) imply that the probability $\mu_S(K)$ that Alice is successful in her strategy is $\mathbb{E}_\gamma[X_0]/K$ (i.e. her *ruin probability* is $1 - \mathbb{E}_\gamma[X_0]/K$).

If the coin is biased ($a \neq 1/2$), we may re-arrange the middle equations in (13) into

$$\nu_S(x) - \nu_S(x-1) = \alpha(\nu_S(x+1) - \nu_S(x)) + \frac{\gamma(x)}{a}, \quad \forall x = 2, \dots, K-2,$$

where $\alpha := (1-a)/a$. Iterating the above, we have that

$$(16) \quad \nu_S(2) - \nu_S(1) = \alpha^{K-3}(\nu_S(K-1) - \nu_S(K-2)) + \frac{1}{a} \sum_{y=2}^{K-2} \alpha^{y-2} \gamma(y).$$

The remaining equations in (13) and (14) imply that

$$\nu_S(2) - \nu_S(1) = \frac{1}{1-a}(\alpha^{-1}\mu_S(0) - \gamma(1)), \quad \nu_S(K-1) - \nu_S(K-2) = \frac{1}{a}(\gamma(K-1) - \alpha\mu_S(K)).$$

Combining the above with (16) yields

$$(17) \quad \mu_S(0) + \alpha^K \mu_S(K) = \sum_{y=1}^{K-1} \alpha^y \gamma(y) = \mathbb{E}_\gamma[\alpha^{X_0}].$$

Adding up the equations (13), we find that

$$1 = \mathbb{P}_\gamma(\{X_0 \in \{1, \dots, K-1\}\}) = \sum_{x=1}^{K-1} \gamma(x) = (1-a)\nu_S(1) + a\nu_S(K-1) = \mu_S(0) + \mu_S(K),$$

where the last equation follows from (14). In other words, Alice will eventually either amass K pounds or go broke (with probability one). Combining this fact with (17), we find the probability that Alice is successful is

$$(18) \quad \mu_S(K) = \frac{1 - \mathbb{E}_\gamma[\alpha^{X_0}]}{1 - \alpha^K}.$$

In other words, in the biased case, her ruin probability is $1 - \frac{1 - \mathbb{E}_\gamma[\alpha^{X_0}]}{1 - \alpha^K}$.

Alice develops a full-blown gambling addiction and abandons her strategy: the only thing stopping her now from betting is bankruptcy. Taking the limit $K \rightarrow \infty$ in (15) and (18), we recover the classical result that the probability that Alice will now go broke is one unless the coin is biased in her favour ($a > 1/2$), in which case it is $\mathbb{E}_\gamma[\alpha^{X_0}]$.

12. The strong Markov property. To finish the chapter, we prove the *strong Markov property*, a generalisation the Markov property (2.3) that applies to $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping times ς (instead of just deterministic times n). However, if we try to replace n with ς in (2.3) we immediately run into difficulties: X_ς may only be partially defined as ς may be infinite, so what does it mean to condition on X_ς ?

To circumvent this issue, we re-write (2.3) in an uglier, but easier-to-manipulate form: recall (2.7), i.e.

$$\mathbb{P}_\gamma(\{X_{n+1} = y\} | \mathcal{F}_n) = \mathbb{P}_{X_n}(\{X_1 = y\}) \quad \forall y \in \mathcal{S}, \quad \mathbb{P}_\gamma\text{-almost surely.}$$

For a (say, bounded) real-valued function f on $\mathcal{S}$, multiply both sides by $f(y)$ and sum over y in $\mathcal{S}$ to find that

$$\mathbb{E}_\gamma[f(X_{n+1}) | \mathcal{F}_n] = \mathbb{E}_{X_n}[f(X_1)], \quad \mathbb{P}_\gamma\text{-almost surely.}$$

Pick any x in $\mathcal{S}$ and (for the time being, bounded) $\mathcal{F}_n/\mathcal{B}(\mathbb{R})$ -measurable random variable Z , multiply both sides by $Z1_{\{n < \infty, X_n = x\}}$ (note that $\{n < \infty\} = \Omega$), apply the take-out-what-is-known property of conditional expectation (Theorem 0.3v), take expectations, and apply the tower property (Theorem 0.3iv) to obtain

$$(1) \quad \mathbb{E}_\gamma[Z1_{\{n < \infty, X_n = x\}}f(X_{n+1})] = \mathbb{E}_\gamma[Z1_{\{n < \infty, X_n = x\}}] \mathbb{E}_x[f(X_1)] \quad \forall x \in \mathcal{S}.$$

Setting $Z := 1$ and $f := 1_y$ and re-arranging we recover (2.3), proving the equivalence of (1) and (2.3). However, replacing n with ς in (1) does not raise any red flags (our convention (0.1) regarding partially-defined functions is important here). We obtain

$$(2) \quad \mathbb{E}_\gamma[Z1_{\{\varsigma < \infty, X_\varsigma = x\}}f(X_{\varsigma+1})] = \mathbb{E}_\gamma[Z1_{\{\varsigma < \infty, X_\varsigma = x\}}] \mathbb{E}_x[f(X_1)] \quad \forall x \in \mathcal{S}$$

for $\mathcal{F}_\varsigma/\mathcal{B}(\mathbb{R})$ -measurable Z (c.f. Definition 8.1) and every bit of the equation makes sense.

All of this re-writing aside, (2) is still just a fancy way of saying that (a) the chain's future conditioned on its past and present equals that conditioned on only the present (i.e. the past and future are conditionally independent given the present, see Proposition 2.5) and (b) the chain is 'born anew' from its current location. The only difference is that we have moved the dividing line between past and the future from n to ς .

In (2), we are still using the location $X_{\varsigma+1}$ of the chain one-step after the present to represent the chain's future, while in reality $X_{\varsigma+1}$ captures only a sliver of the future. We also take the opportunity here to rectify this sorry state of affairs: we show that (2) holds for every appropriately-measurable functional F of the chain's present and future positions $(X_\varsigma, X_{\varsigma+1}, \dots)$ instead of just $f(X_{\varsigma+1})$. To this end, we need the ς -shifted chain:

Shifting the chain left by ς . To describe the chain's future from ς onwards, we use the *shifted chain* X^ς defined as the function mapping from $\{\varsigma < \infty\}$ to the path space $\mathcal{P}$ (c.f. Section 5) with

$$(3) \quad X^\varsigma(\omega) := (X_{\varsigma(\omega)}(\omega), X_{\varsigma(\omega)+1}(\omega), X_{\varsigma(\omega)+2}(\omega), \dots) \quad \forall \omega \in \{\varsigma < \infty\}.$$

This $\mathcal{P}$ -valued function X^ς describes the chain *shifted leftwards* in time by ς :

$$X_n^\varsigma(\omega) = X_{\varsigma(\omega)+n}(\omega) \quad \forall \omega \in \{\varsigma < \infty\}, \quad n \geq 0.$$

The strong Markov property. We are now in great shape to state and prove the strong Markov property. Note that whenever we write $1_{\{\varsigma < \infty\}}F(X^\varsigma)$ in what follows, we are using the notation for partially-defined functions introduced in (0.1).

(4) **THEOREM** (The strong Markov property). *Let $(\mathcal{F}_n)_{n \in \mathbb{N}}$ be the filtration generated by the chain (Definition 2.1), $\mathcal{E}$ be the cylinder sigma-algebra on the path space $\mathcal{P}$ (Section 5), ς be an $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping time (Definition 8.1), and $\mathcal{F}_\varsigma$ be the pre- ς sigma-algebra (Definition 8.1). Suppose that Z is an $\mathcal{F}_\varsigma/\mathcal{B}(\mathbb{R}_E)$ -measurable random variable, F is a $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable function, and that Z and F are both non-negative, or both bounded. If x is any state in $\mathcal{S}$, then $Z1_{\{\varsigma < \infty, X_\varsigma = x\}}F(X^\varsigma)$ is $\mathcal{F}/\mathcal{B}(\mathbb{R}_E)$ -measurable, where X^ς denotes the ς -shifted chain (3). Moreover,*

$$(5) \quad \mathbb{E}_\gamma [Z1_{\{\varsigma < \infty, X_\varsigma = x\}}F(X^\varsigma)] = \mathbb{E}_\gamma [Z1_{\{\varsigma < \infty, X_\varsigma = x\}}] \mathbb{E}_x [F(X)] \quad \forall x \in \mathcal{S}.$$

We do this proof in two steps, beginning by proving the measurability of $Z1_{\{\varsigma < \infty, X_\varsigma = x\}}F(X^\varsigma)$ then moving on to (5).

Step 1: $Z1_{\{\varsigma < \infty, X_\varsigma = x\}}F(X^\varsigma)$ is $\mathcal{F}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Because of Exercise 8.4, and because the product of measurable functions is measurable, all we need to argue here is that $1_{\{\varsigma < \infty\}}F(X^\varsigma)$ is $\mathcal{F}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Due to Lemma 0.2, this consists of proving that

$$\{\varsigma < \infty, F(X^\varsigma) \in A\} \in \mathcal{F}, \quad \forall A \in \mathcal{B}(\mathbb{R}_E).$$

However,

$$\{\varsigma < \infty, F(X^\varsigma) \in A\} \in \mathcal{F} = \bigcup_{n=0}^{\infty} \{\varsigma = n, \Theta_n(X) \in F^{-1}(A)\},$$

where, for each $n \geq 0$, $\Theta_n : \mathcal{P} \rightarrow \mathcal{P}$ denotes n -shift operator defined by

$$\Theta_n((x_m)_{m \in \mathbb{N}}) := (x_n, x_{n+1}, x_{n+2}, \dots) \quad \forall (x_m)_{m \in \mathbb{N}} \in \mathcal{P}.$$

Given that $\{\varsigma = n\}$ belongs to $\mathcal{F}_n \subseteq \mathcal{F}$ (Lemma 8.3*i*) and F is $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable by assumption, it suffices to show that Θ_n is $\mathcal{E}/\mathcal{E}$ measurable. Because the cylinder sets (5.1) generate $\mathcal{E}$, this follows easily. $\square$

Step 2: (5) holds. Given Step 1 and our assumption that Z and F are either both non-negative or both bounded, every term in (5) is well-defined. Because $\mathbb{E}_x [F(X)] = \mathbb{L}_x(F)$, where $\mathbb{L}_x$ denotes the path law (5.2) starting from x , and because of the definition of the Lebesgue integral, it suffices to show that

$$(6) \quad \mathbb{P}_\gamma (A \cap \{\varsigma < \infty, X_\varsigma = x, X^\varsigma \in B\}) = \mathbb{P}_\gamma (A \cap \{\varsigma < \infty, X_\varsigma = x\}) \mathbb{L}_x(B)$$

for all A in $\mathcal{F}_\varsigma$, B in $\mathcal{E}$, and x in $\mathcal{S}$ —if you are not following this, go read about the *standard machine* in (Williams, 1991, Section 5.12).

Fix any A in $\mathcal{F}_\zeta$ and x in $\mathcal{S}$. If $c := \mathbb{P}_\gamma(A \cap \{\zeta < \infty, X_\zeta = x\}) = 0$, then both sides of (6) are zero and we are done. Suppose that $c \neq 0$ and consider the map $\tilde{\mathbb{L}} : \mathcal{E} \rightarrow \mathbb{R}_E$ defined by

$$\tilde{\mathbb{L}}(B) := c^{-1} \mathbb{P}_\gamma(\{A \cap \{\zeta < \infty, X_\zeta = x, X^\zeta \in B\}) \quad \forall B \in \mathcal{E}.$$

Clearly, $\tilde{\mathbb{L}}$ is non-negative and $\tilde{\mathbb{L}}(\mathcal{P}) = c/c = 1$. Furthermore, Tonelli's Theorem implies that $\tilde{\mathbb{L}}$ is countably additive. In other words, $\tilde{\mathbb{L}}$ is a probability measure. Suppose that we are able to show that

$$(7) \quad \tilde{\mathbb{L}}(\{x_0\} \times \{x_1\} \times \cdots \times \{x_m\} \times \mathcal{S} \times \mathcal{S} \times \cdots) = \mathbb{L}_x(\{x_0\} \times \{x_1\} \times \cdots \times \{x_m\} \times \mathcal{S} \times \mathcal{S} \times \cdots)$$

for all natural numbers m and states $x_0, x_1, \dots, x_n$ in $\mathcal{S}$. Theorem 5.5 then shows that $\tilde{\mathbb{L}}$ is $\mathbb{L}_x$ and (6) follows.

To prove (7), re-write this equation as

$$\begin{aligned} & \mathbb{P}_\gamma(A \cap \{\zeta < \infty, X_\zeta = x, X_\zeta = x_0, X_{\zeta+1} = x_1, \dots, X_{\zeta+m} = x_m\}) \\ &= \mathbb{P}_\gamma(A \cap \{\zeta < \infty, X_\zeta = x\}) \mathbb{P}_x(\{X_0 = x_0, X_1 = x_1, \dots, X_m = x_m\}). \end{aligned}$$

If $x \neq x_0$ then both sides of the above are zero and we are done. Suppose that $x = x_0$ and fix any $l > 0$. Using the definitions in Algorithm 1, we can express X_{n+l} as some measurable function of X_n and $U_{n+1}, \dots, U_{n+l}$:

$$X_{n+l} = f_l(X_n, U_{n+1}, \dots, U_{n+l}) \quad \forall j > 0, \quad n \geq 0$$

and this function depends on l but not on n . Because $(X_1, \dots, X_n)$ is a function of X_0 and $(U_1, \dots, U_n)$ and these are independent of $(U_{n+1}, \dots, U_{n+j})$, the random variable

$$f_l(x, U_{n+1}, \dots, U_{n+l})$$

is independent of $(X_0, \dots, X_n)$ and, consequently, of $\mathcal{F}_n$. Because $A \cap \{\zeta = n, X_n = x\}$ belongs to $\mathcal{F}_n$ (Lemma 8.3(i)) and $(U_n)_{n \in \mathbb{Z}_+}$ is i.i.d., it follows that

$$\begin{aligned} & \mathbb{P}_\gamma(A \cap \{\zeta = n, X_n = x, X_{n+1} = x_1, \dots, X_{n+m} = x_m\}) \\ &= \mathbb{P}_\gamma(A \cap \{\zeta = n, X_n = x, f_1(x, U_{n+1}) = x_1, \dots, f_m(x, U_{n+1}, \dots, U_{n+m}) = x_m\}) \\ &= \mathbb{P}_\gamma(A \cap \{\zeta = n, X_n = x\}) \mathbb{P}_\gamma(\{f_1(x, U_{n+1}) = x_1, \dots, f_m(x, U_{n+1}, \dots, U_{n+m}) = x_m\}) \\ &= \mathbb{P}_\gamma(A \cap \{\zeta = n, X_n = x\}) \mathbb{P}_\gamma(\{f_1(x, U_1) = x_1, \dots, f_m(x, U_1, \dots, U_l) = x_m\}) \\ &= \mathbb{P}_\gamma(A \cap \{\zeta = n, X_n = x\}) \mathbb{P}_x(\{f_1(X_0, U_1) = x_1, \dots, f_m(X_0, U_1, \dots, U_m) = x_m\}) \\ &= \mathbb{P}_\gamma(A \cap \{\zeta = n, X_n = x\}) \mathbb{P}_x(\{X_1 = x_1, \dots, X_m = x_m\}), \end{aligned}$$

Summing both sides over n in $\mathbb{N}$ then yields (7), so completing the proof. $\square$

Measurable functions on $\mathcal{P}$. We finish this section with the following technical lemma that will cover our measurability-checking necessities for functionals on $\mathcal{P}$ throughout the remainder of our treatment of discrete-time chains.

(8) LEMMA. *The following are $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable functions:*

(i) *For any real-valued function f on $\mathcal{S}$,*

$$F_-(x) := \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} f(x_m), \quad F_+(x) := \limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} f(x_m), \quad \forall x \in \mathcal{P}.$$

(ii) *For any subset A of $\mathcal{S}$ and positive integer k ,*

$$F(x) := \inf \left\{ n > 0 : \sum_{m=1}^n 1_A(x_m) = k \right\} \quad \forall x \in \mathcal{P}.$$

(iii) For any non-negative function f on $\mathcal{S}$,

$$G(x) := \sum_{n=0}^{F((x_m)_{m \in \mathbb{N}})-1} f(x_n) \quad \forall x \in \mathcal{P},$$

where F is as in (ii).

Proof. (i) The definition of the cylinder sigma-algebra $\mathcal{E}$ in Section 5 implies that $(x_m)_{m \in \mathbb{N}} \mapsto f(x_n)$ is $\mathcal{E}/\mathcal{B}(\mathbb{R})$ -measurable for each $n \in \mathbb{N}$. Because finite sums and products of measurable functions are measurable, we have that $(x_m)_{m \in \mathbb{N}} \mapsto \frac{1}{n} \sum_{m=0}^{n-1} f(x_m)$ is $\mathcal{E}/\mathcal{B}(\mathbb{R})$ -measurable for each $n \in \mathbb{Z}_+$. Because the limit infimum and supremum of measurable functions are measurable, the result follows.

(ii) By definition, $F(x)$ denotes the time that the path x enters the set A for the k th time and so

$$\begin{aligned} F &= \infty \cdot 1_{\{g_\infty(x) < k\}} + \sum_{n=1}^{\infty} n 1_{\{g_{n-1}(x) < k\}} 1_{\{g_n(x) = k\}} \\ &= \infty \cdot \lim_{N \rightarrow \infty} 1_{\{g_N(x) < k\}} + \lim_{N \rightarrow \infty} \sum_{n=1}^N n 1_{\{g_{n-1}(x) < k\}} 1_{\{g_n(x) = k\}}, \end{aligned}$$

where $g_n(x) := \sum_{m=1}^n 1_A(x_m)$ denotes the number of times the path $x = (x_m)_{m \in \mathbb{N}}$ has entered A by time n (with $g_0 := 0$). Because any finite sum of measurable functions is measurable, g_n is $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable, for each $n \in \mathbb{Z}_+$. The result then follows from the above by exploiting once again that the sum, product, limit superior, and limit inferior of measurable functions are all measurable.

(iii) Because any finite sum of measurable functions is measurable, $g_n((x_m)_{m \in \mathbb{N}}) := \sum_{m=0}^n f(x_m)$ is $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable for any $n \in \mathbb{N}$. Furthermore, as the limit of measurable functions is measurable, $g_\infty := \lim_{n \rightarrow \infty} g_n$ is also $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Pick any $A \in \mathcal{B}(\mathbb{R}_E)$ and note that

$$\left\{ (n, (x_m)_{m \in \mathbb{N}}) \in \mathbb{N}_E \times \mathcal{P} : \sum_{m=0}^n f(x_m) \in A \right\} = (\{\infty\} \times \{g_\infty \in A\}) \cup \left(\bigcup_{n=0}^{\infty} \{n\} \times \{g_n \in A\} \right).$$

Because the right-hand side belongs to the product sigma-algebra $2^{\mathbb{N}_E} \times \mathcal{E}$, it follows that

$$H(n, (x_m)_{m \in \mathbb{N}}) := \sum_{m=0}^n f(x_m)$$

defines an $2^{\mathbb{N}_E} \times \mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable function, for each $n \in \mathbb{N}$. Because $G((x_m)_{m \in \mathbb{N}}) = H(F((x_m)_{m \in \mathbb{N}}) - 1, (x_n)_{n \in \mathbb{N}})$, with F as in (ii), the result then follows because the composition of measurable functions is measurable. $\square$

Discrete-time chains II: the long-term behaviour

Armed with the tools introduced throughout the previous chapter (Sections 1–12), we are ready to start chipping away at the long-term behaviour of discrete-time chains. The aim of this chapter (Sections 13–25) is to formalise the discussion given in the introduction to Part I for the case of discrete-time chains.

Overview of the chapter. We begin in Section 13 by introducing entrance times and using them to (a) describe whether the chain is able to travel between two given states; and (b) define the *closed communicating classes*: subsets of the state space that the chain, once inside, is unable to leave but is free to explore within. We then describe (Section 14) the states that the chain will keep revisiting (the *recurrent* states) and those that it not (the *transient* states). Upon each visit to a recurrent state, the chain starts ‘afresh’ in the sense that the segments of the chain’s path delimited by the visits form an i.i.d. sequence. We prove this important *regenerative property* of discrete-time chains in Section 15. Combining this property with the law of large numbers, we find (Section 16) that the fraction of time-steps each path spends in any given state settles down as the number of time-steps approaches infinity. Of course, for transient states the fraction converges to zero as the visits to these states eventually stop. For recurrent states, the limiting value may be zero—in which case the frequency of the visits degenerates to zero and the state is said to be *null recurrent*—or non-zero—in which case the frequency remains bounded away from zero and the state is said to be *positive recurrent*. As we will see, the average time it takes the chain to return to a recurrent state differentiates these two cases: it is infinite for null recurrent states and finite for positive recurrent ones.

In Section 17, we introduce the *stationary distributions* of the chain: the starting distributions that turn the chain into a stationary process (i.e. one whose finite-dimensional statistics do not change with time). We also show that there exists a stationary distribution with support on a given state if and only if that state is positive recurrent and that positive recurrence is a *class property*: a state within a closed communicating class is positive recurrent if and only if every state within the class is positive recurrent. Moreover, as we will see, there is only one stationary distribution with support contained inside any given positive recurrent class (known as the *ergodic distribution* associated with that class) and the set of stationary distributions is simply the convex hull of these ergodic distributions. In Section 18, we revisit the limiting value of the fraction of time-steps that the chain spends in any given positive recurrent state x and express it in terms of the ergodic distribution π associated with the class $\mathcal{C}$ that x belongs to (it is $\pi(x)$ if the chain’s path ever enters $\mathcal{C}$ and zero otherwise). Using these facts, we then see why chains whose paths all enter the positive recurrent closed communicating classes (i.e. *positive Tweedie recurrent* chains) are typically viewed as the *stable* chains.

We then turn our attention to the long-term behaviour of the chain’s statistics and focus on the question ‘does the time-varying law converge and to what?’. Here, we require the notion of a *periodic chain* (Section 19): one for which the possibility of return to at least one positive recurrent state is a periodic phenomenon (e.g. the chain may return to the state after an even number of steps but not an odd number). We then settle the question in Section 20: if the chain is periodic, the time-varying law does not converge but oscillates instead. Otherwise, the time-varying law converges to a convex combination of the ergodic distributions, where the weight awarded to each distribution is the probability that the chain ever enters the closed communicating class associated with it. This marks the end of our treatment on the basics of the long-term behaviour of discrete-time chains. In my experience, the material presented throughout Sections 1–20 forms a good base camp of sorts from which one can approach other more advanced or specialised summits of Markov chain theory.

At this point there are many directions we could take: further investigate the limiting behaviour of the pathwise averages and derive pertinent central limit theorems and laws of the

iterated logarithm, study the relationship between the rate of convergence the time-varying law and the tails of the return times, delve into the notion of quasistationarity describing the medium-term behaviour of the chain, figure out how to compute in practice time-varying laws, occupation measures, and stopping/exit/stationary distributions introduced throughout the sections, start working on the theory for the continuous-time case or the general state space case, etc. The remainder of this chapter is dedicated to two such subjects:

1. (Geometric recurrence and convergence: Section 21) We study the question of when does the time-varying law converge geometrically fast and prove Kendall's theorem: a key result in this area showing that the answer lies in the tails of the return time distributions.
2. (Foster-Lyapunov criteria: Sections 23–25). We derive a series of analytical inequalities involving the one-step matrix that yield information on the entrance and return times of a chain and, consequently, on its recurrence properties and stationary distributions. Before we start with these however, we introduce in Section 22 certain *geometric trials arguments* that will be important for the proofs of Sections 23–25.

13. Entrance times, accessibility, and closed communicating classes. Before discussing which bits of the state space the chain might visit in the long-run, we need to discuss which bits it will ever visit in the first place. Here, we require *entrance times*:

(1) DEFINITION (Entrance times). *Given any subset A of $\mathcal{S}$ and positive integer k , the k th entrance time ϕ_A^k is the time-step when the chain enters the set A for the k th time or infinity if it never does:*

$$\phi_A^0(\omega) := 0 \quad \forall \omega \in \Omega, \quad \phi_A^k(\omega) := \inf \left\{ n > 0 : \sum_{m=1}^n 1_A(X_m(\omega)) = k \right\} \quad \forall \omega \in \Omega, \quad k > 0,$$

where we are adhering to our convention that $\inf \emptyset = \infty$. For the first entrance time, we write ϕ_A instead of ϕ_A^1 . Additionally, we use the shorthands $\phi_x := \phi_{\{x\}}$ (resp. $\phi_x^k := \phi_{\{x\}}^k$) to denote the first (resp. k th) entrance time of a state x .

You may be more familiar with the recursive definition of entrance times:

$$(2) \quad \phi_A^0(\omega) := 0 \quad \forall \omega \in \Omega, \quad \phi_A^k(\omega) := \inf \{ n > \phi_A^{k-1}(\omega) : X_n(\omega) \in A \} \quad \forall \omega \in \Omega, \quad k > 0.$$

However, it is not difficult to check that the two definitions are equivalent:

(3) EXERCISE. *Use induction to show that (1) and (2) are equivalent. Hint: the chain has visited A exactly k times by ϕ_A^k and so $\sum_{m=1}^{\phi_A^k} 1_A(X_m) = k$ whenever ϕ_A^k is finite.*

The difference between entrance times and hitting times (Section 8) is that—somewhat inconsistently with our nomenclature for exit times (Section 11)—we do not consider starting inside A as ‘entering’ A (hence, $\phi_A > 0$ by definition). If the chain starts in x , then ϕ_x is the first time that the chain *returns* to x . For this reason, ϕ_x is often referred to as the *first return time*. Similarly, ϕ_x^k is known as the k th *return time*. Of course, entrance times are stopping times because we are able to deduce whether the chain has entered a given set by continuously monitoring it:

(4) EXERCISE. *Using similar arguments to those given in the proof of Proposition 8.6, show that entrance times are $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping times, where $(\mathcal{F}_n)_{n \in \mathbb{N}}$ denotes the filtration generated by the chain (Definition 2.1).*

Accessibility, communicating classes, and closed sets. Equipped with entrance times, we now are able to consider the question of whether the chain may reach any region of the state space from any other or if it is limited in some way. In particular, if the chain is able to travel from one subset of the state A to another B , we say that B is *accessible* from A . Formally:

(5) DEFINITION (Accessibility). *A state y is accessible from a state x (or $x \rightarrow y$ for short) if and only if $\mathbb{P}_x(\{\phi_y < \infty\}) > 0$. Similarly, we say that a subset B of the state space is accessible from another subset A (or $A \rightarrow B$) if for every x in A there exists a y in B such that $x \rightarrow y$ (equivalently, if $\mathbb{P}_\gamma(\{\phi_B < \infty\}) > 0$ for all initial distributions γ with support in A).*

It is straightforward to characterise accessibility in terms of the one-step matrix:

(6) LEMMA. *$x \rightarrow y$ if and only if $p(x, y) > 0$ or there exists some states $x_1, \dots, x_l$ such that*

$$p(x, x_1)p(x_1, x_2) \dots p(x_l, y) > 0.$$

Proof. By the definition of entrance times,

$$\{\phi_y < \infty\} = \{X_1 \in A\} \cup \{X_1 \notin A, X_2 \in A\} \cup \{X_1 \notin A, X_2 \notin A, X_3 \in A\} \cup \dots$$

and the lemma follows by taking expectations and applying (4.4). $\square$

It follows from the lemma that $\rightarrow$ is a transitive relation on both $\mathcal{S}$ and its power set. Two sets are said to *communicate* if each is accessible from the other. A set to be a *communicating class* if all its subsets communicate:

(7) DEFINITION (Communicating class). *A subset $\mathcal{C}$ of $\mathcal{S}$ is a communicating class if and only if $x \rightarrow y$ for each pair x, y in $\mathcal{C}$ (equivalently $A \rightarrow B$ for all subsets A, B of $\mathcal{C}$).*

A subset of the state space is said to be *closed* if no state outside the set is accessible from a state inside the set.

(8) DEFINITION (Closed set). *A subset $\mathcal{C}$ of $\mathcal{S}$ is closed if and only if for each x in $\mathcal{C}$ and y in $\mathcal{S}$, $x \rightarrow y$ implies that y belongs to $\mathcal{C}$.*

Closed sets are those from which the chain may never escape.

(9) PROPOSITION. *If the chains starts in a closed set $\mathcal{C}$ (i.e. the initial distribution γ satisfies $\gamma(\mathcal{C}) = 1$), then the chain remains in $\mathcal{C}$:*

$$\mathbb{P}_\gamma(\{X_n \in \mathcal{C} \ \forall n \in \mathbb{N}\}) = 1.$$

Proof. Because Lemma 6 implies that $\sum_{y \in \mathcal{C}} p(x, y) = \sum_{y \in \mathcal{S}} p(x, y) = 1$ for all x in $\mathcal{C}$, (4.4) implies that

$$\begin{aligned} & \mathbb{P}_\gamma(\{X_0 \in \mathcal{C}, X_1 \in \mathcal{C}, \dots, X_{n-2} \in \mathcal{C}, X_{n-1} \in \mathcal{C}, X_n \in \mathcal{C}\}) \\ &= \sum_{x_0 \in \mathcal{C}} \sum_{x_1 \in \mathcal{C}} \dots \sum_{x_{n-2} \in \mathcal{C}} \sum_{x_{n-1} \in \mathcal{C}} \sum_{x_n \in \mathcal{C}} \mathbb{P}_\gamma(\{X_0 = x_0, X_1 = x_1, \dots, X_{n-2} = x_{n-2}, X_{n-1} = x_{n-1}, X_n = x_n\}) \\ &= \sum_{x_0 \in \mathcal{C}} \sum_{x_1 \in \mathcal{C}} \dots \sum_{x_{n-2} \in \mathcal{C}} \sum_{x_{n-1} \in \mathcal{C}} \sum_{x_n \in \mathcal{C}} \gamma(x_0) p(x_0, x_1) \dots p(x_{n-2}, x_{n-1}) p(x_{n-1}, x_n) \\ &= \sum_{x_0 \in \mathcal{C}} \sum_{x_1 \in \mathcal{C}} \dots \sum_{x_{n-2} \in \mathcal{C}} \sum_{x_{n-1} \in \mathcal{C}} \gamma(x_0) p(x_0, x_1) \dots p(x_{n-2}, x_{n-1}) \left(\sum_{x_n \in \mathcal{C}} p(x_{n-1}, x_n) \right) \\ &= \sum_{x_0 \in \mathcal{C}} \sum_{x_1 \in \mathcal{C}} \dots \sum_{x_{n-2} \in \mathcal{C}} \sum_{x_{n-1} \in \mathcal{C}} \gamma(x_0) p(x_0, x_1) \dots p(x_{n-2}, x_{n-1}) = \dots = \sum_{x_0 \in \mathcal{C}} \gamma(x_0) = 1 \quad \forall n \in \mathbb{N}. \end{aligned}$$

Taking the limit $n \rightarrow \infty$ in the above and applying downwards monotone convergence then completes the proof. $\square$

Closed communicating classes in particular are very important for Markov chain theory: they are the irreducible sets mentioned in the introduction to Part I. For this reason, the chain and its one-step matrix are said to be *irreducible* if the whole state space $\mathcal{S}$ is a single closed communicating class (they are said to be φ -*irreducible* if there is a single closed communicating class and it is accessible from all states in the state space).

Two useful consequences of the strong Markov property. We finish the section by proving two propositions that will be of use later. The first concerns the *hitting probabilities* of the chain meaning the chance $\mathbb{P}_x(\{\phi_A < \infty\})$ that the chain ever enters a set A for the various possible starting states x .

(10) PROPOSITION (The hitting probabilities satisfy a set of linear equations). *For all states x in $\mathcal{S}$ and subsets A of $\mathcal{S}$,*

$$\mathbb{P}_x(\{\phi_A < \infty\}) = \sum_{y \in A} p(x, y) + \sum_{z \notin A} p(x, z) \mathbb{P}_z(\{\phi_A < \infty\}).$$

The second proposition formalises the intuitive idea that the chain may visit at most one of two disjoint closed sets (for it enters one of these sets, it will never leave the set, and, consequently, never visit the other one):

(11) PROPOSITION (The chain visits at most one of two disjoint closed sets). *If $\mathcal{C}_1$ and $\mathcal{C}_2$ are two disjoint closed sets, then*

$$\mathbb{P}_\gamma(\{\phi_{\mathcal{C}_1} < \infty, \phi_{\mathcal{C}_2} < \infty\}) = 0.$$

The proofs of both propositions require the following consequence of the strong Markov property.

(12) LEMMA. *For any subset A of the state space and $(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping time ς ,*

$$\begin{aligned} \mathbb{P}_\gamma(\{\varsigma < \phi_A < \infty\}) &= \sum_{x \in \mathcal{S}} \mathbb{P}_\gamma(\{\varsigma < \phi_A, X_\varsigma = x\}) \mathbb{P}_x(\{\phi_A < \infty\}), \\ \mathbb{P}_\gamma(\{\varsigma < \phi_A = \infty\}) &= \sum_{x \in \mathcal{S}} \mathbb{P}_\gamma(\{\varsigma < \phi_A, X_\varsigma = x\}) \mathbb{P}_x(\{\phi_A = \infty\}). \end{aligned}$$

Proof. Note that X_n does not belong to A for $n = 1, 2, \dots, \varsigma$ if $\varsigma < \phi_A$, and so

$$\begin{aligned} \phi_A &= \inf \left\{ n > 0 : \sum_{m=1}^n 1_A(X_m) = 1 \right\} = \inf \left\{ n > \varsigma : \sum_{m=\varsigma+1}^n 1_A(X_m) = 1 \right\} \\ (13) \quad &= \varsigma + \inf \left\{ n > 0 : \sum_{m=1}^n 1_A(X_m^\varsigma) = 1 \right\} \quad \text{on} \quad \{\varsigma < \phi_A\}, \end{aligned}$$

where X^ς denotes the ς -shifted chain in (12.3).

Lemma 12.8ii shows that the function $F((x_m)_{m \in \mathbb{N}}) := \inf \{n > 0 : \sum_{m=1}^n 1_A(x_m) = 1\}$ is $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable, where $\mathcal{E}$ denotes the cylinder sigma-algebra on the path space (Section 5). Given that we are able to deduce whether the chain has entered A before time ς from observing the chain up until ς , the event $\{\varsigma < \phi_A\}$ belongs to the pre- ς sigma-algebra $\mathcal{F}_\varsigma$ introduced in Definition 8.1 (for a formal argument, use Lemma 8.3i and Exercise 4). For these reasons, applying the strong Markov property (Theorem 12.4) yields the first equality given in the premise:

$$\begin{aligned} \mathbb{P}_\gamma(\{\varsigma < \phi_A < \infty\}) &= \mathbb{E}_\gamma[1_{\{\varsigma < \phi_A\}} 1_{\{\phi_A - \varsigma < \infty\}}] = \sum_{x \in \mathcal{S}} \mathbb{E}_\gamma[1_{\{\varsigma < \phi_A\}} 1_{\{\varsigma < \infty, X_\varsigma = x\}} 1_{\mathbb{N}}(F(X^\varsigma))] \\ &= \sum_{x \in \mathcal{S}} \mathbb{E}_\gamma[1_{\{\varsigma < \phi_A\}} 1_{\{\varsigma < \infty, X_\varsigma = x\}}] \mathbb{E}_x[1_{\mathbb{N}}(F(X))] \\ &= \sum_{x \in \mathcal{S}} \mathbb{P}_\gamma(\{\varsigma < \phi_A, X_\varsigma = x\}) \mathbb{P}_x(\{\phi_A < \infty\}) \end{aligned}$$

where the second and fourth equality follow from (13) and the fact that $\{\varsigma < \phi_A\} \subseteq \{\varsigma < \infty\}$ (as ς is strictly less than ϕ_A only if the ς is finite). The second equality in the premise follows by replacing ' $< \infty$ ' and ' $1_{\mathbb{N}}$ ' with ' $= \infty$ ' and ' 1_{∞} ' in the above. $\square$

Proof of Proposition 10. Given that

$$\mathbb{P}_x(\{\phi_A < \infty\}) = \mathbb{P}_x(\{\phi_A = 1\}) + \mathbb{P}_x(\{1 < \phi_A < \infty\}) = \sum_{y \in A} p(x, y) + \mathbb{P}_x(\{1 < \phi_A < \infty\}),$$

the proposition follows directly by setting $\varsigma := 1$ and $\gamma := 1_x$ in Lemma 12 and noting that, due to ϕ_A 's definition, $X_1 \in A$ on $\{\phi_A = 1\}$ and $X_1 \notin A$ on $\{\phi_A > 1\}$. $\square$

Proof of Proposition 11. Because the sets are disjoint, the chain cannot enter both at the same time:

$$\mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}_1} = \phi_{\mathcal{C}_2} < \infty\}) = \mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}_1} = \phi_{\mathcal{C}_2} < \infty, X_{\phi_{\mathcal{C}_1}} \in \mathcal{C}_2\}) = 0.$$

Thus,

$$\mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}_1} < \infty, \phi_{\mathcal{C}_2} < \infty\}) = \mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}_1} < \phi_{\mathcal{C}_2}, \phi_{\mathcal{C}_2} < \infty\}) + \mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}_2} < \phi_{\mathcal{C}_1}, \phi_{\mathcal{C}_1} < \infty\}).$$

However, setting $A := \mathcal{C}_2$ and $\varsigma := \phi_{\mathcal{C}_1}$ in Lemma 12, we find that

$$(14) \quad \mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}_1} < \phi_{\mathcal{C}_2}, \phi_{\mathcal{C}_2} < \infty\}) = \sum_{x \in \mathcal{C}_1} \mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}_1} < \phi_{\mathcal{C}_2}, X_{\phi_{\mathcal{C}_1}} = x\}) \mathbb{P}_x(\{\phi_{\mathcal{C}_2} < \infty\})$$

Because the sets are closed and disjoint,

$$\mathbb{P}_x(\{\phi_{\mathcal{C}_2} < \infty\}) = \sum_{y \in \mathcal{C}_2} \mathbb{P}_x(\{\phi_{\mathcal{C}_2} = \phi_y, \phi_y < \infty\}) \leq \sum_{y \in \mathcal{C}_2} \mathbb{P}_x(\{\phi_y < \infty\}) = 0 \quad \forall x \in \mathcal{C}_1,$$

and it follows that the right-hand side of (14) is zero. Swapping $\mathcal{C}_1$ and $\mathcal{C}_2$ throughout the above completes the proof. $\square$

14. Recurrence and transience. Our study of the long-term behaviour starts in earnest with the notions of *recurrence* and *transience*:

(1) **DEFINITION** (Recurrent and transient states). *A state x is said to be recurrent if the chain has probability one of returning to it (i.e. $\mathbb{P}_x(\{\phi_x < \infty\}) = 1$) and transient otherwise.*

Recurrent states are those the chain will keep revisiting (whenever an initial visit occurs) while transient ones are those the chain will cease visiting at some point:

(2) **THEOREM** (Characterising recurrent and transient states). *For any initial distribution γ and x in $\mathcal{S}$,*

$$(3) \quad \mathbb{P}_{\gamma}(\{\phi_x^k < \infty\}) = \mathbb{P}_{\gamma}(\{\phi_x < \infty\}) \mathbb{P}_x(\{\phi_x < \infty\})^{k-1} \quad \forall k > 0.$$

Moreover, if x is transient, then, with probability one, the chain visits x finitely many times: $\mathbb{P}_{\gamma}(\{R < \infty\}) = 1$ where

$$R := \sum_{n=1}^{\infty} 1_x(X_n)$$

denotes the total number of returns to x . If x instead is recurrent, then the probability $\mathbb{P}_{\gamma}(\{R = \infty\})$ that the chain visits x infinitely many times equals the probability $\mathbb{P}_{\gamma}(\{\phi_x < \infty\})$ that visits x at least once.

Proof. Because the number of entrances to x by time ϕ_x^k is exactly k ($\sum_{m=1}^{\phi_x^k} 1_x(X_m) = k$) and ϕ_x^{k+1} is greater than ϕ_x^k whenever the latter is finite (Exercise 13.3), we have that

$$\begin{aligned} \phi_x^{k+1} &= \inf \left\{ n > 0 : \sum_{m=1}^n 1_x(X_m) = k+1 \right\} = \inf \left\{ n > \phi_x^k : \sum_{m=\phi_x^k+1}^n 1_x(X_m) = 1 \right\} \\ &= \phi_x^k + \inf \left\{ n > 0 : \sum_{m=1}^n 1_x(X_m^{\phi_x^k}) = 1 \right\} \quad \forall k > 0, \end{aligned}$$

where $X^{\phi_x^k}$ denotes the ϕ_x^k -shifted chain (12.3). Given the above, an application of the strong Markov property similar to that in the proof of Lemma 13.12 yields

$$\mathbb{P}_\gamma(\{\phi_x^{k+1} < \infty\}) = \mathbb{P}_\gamma(\{\phi_x^k < \infty\})\mathbb{P}_x(\{\phi_x < \infty\}) \quad \forall k > 0.$$

Iterating the above equation backwards, we obtain (3).

Next, the total number of returns the chain makes to the state is equal to the number of return times that are finite:

$$(4) \quad R = \sum_{k=1}^{\infty} 1_{\{\phi_x^k < \infty\}},$$

Because ϕ_x^k is finite only if $\phi_x^1, \dots, \phi_x^{k-1}$ are finite too, monotone convergence implies that

$$\begin{aligned} (5) \quad \mathbb{P}_\gamma(\{R = \infty\}) &= \mathbb{P}_\gamma\left(\left\{\sum_{k=1}^{\infty} 1_{\{\phi_x^k < \infty\}} = \infty\right\}\right) = \mathbb{P}_\gamma(\{\phi_x^k < \infty \quad \forall k \in \mathbb{Z}_+\}) \\ &= \lim_{k \rightarrow \infty} \mathbb{P}_\gamma(\{\phi_x^k < \infty\}) = \mathbb{P}_\gamma(\{\phi_x < \infty\}) \lim_{k \rightarrow \infty} \mathbb{P}_x(\{\phi_x < \infty\})^k \\ &= \begin{cases} \mathbb{P}_\gamma(\{\phi_x < \infty\}) & \text{if } x \text{ is recurrent} \\ 0 & \text{if } x \text{ is transient} \end{cases}. \end{aligned}$$

□

The concepts of transience and recurrence are examples of *class properties* in the sense that they hold for any one state in a communicating class if and only if they hold for every state in the communicating class:

(6) **THEOREM** (Transience and recurrence are class properties). *If x is recurrent and $x \rightarrow y$, then y is also recurrent and*

$$\mathbb{P}_x(\{\phi_y < \infty\}) = \mathbb{P}_y(\{\phi_x < \infty\}) = 1.$$

For this reason, (a) if y is transient and $x \rightarrow y$, then x is also transient and (b) a state in a communicating class is recurrent (resp. transient) if and only if all states in the class are recurrent (resp. transient). Moreover, a state is recurrent only if it belongs to a closed communicating class, and so the set $\mathcal{R}$ of all recurrent states equals the union of one or more closed communicating classes.

Hence, we say that a communicating class is *recurrent* (resp. *transient*) if any one (and therefore all) of its states is recurrent (resp. transient). For chains, we use the following slightly more elaborate classification:

(7) **DEFINITION** (Recurrent and transient chains). *If the set $\mathcal{R}$ of all recurrent states is empty, then the chain is said to be transient. Otherwise, the chain is said to be Tweedie recurrent if, regardless of the initial distribution, the chain has probability one of entering $\mathcal{R}$:*

$$\mathbb{P}_\gamma(\{\phi_{\mathcal{R}} < \infty\}) = 1 \quad \text{for all initial distributions } \gamma,$$

where $\phi_{\mathcal{R}}$ denotes the first entrance time to $\mathcal{R}$ (Definition 13.1). If, additionally, there exists only one communicating class, then the chain is said to be Harris recurrent. If the class is the entire state space, then the chain is simply called recurrent.

We finish the section with the proof of Theorem 6 and a useful corollary thereof:

(8) COROLLARY. *If the chain ever enters a recurrent closed communicating class $\mathcal{C}$, it will eventually visit every state in $\mathcal{C}$:*

$$1_{\{\phi_{\mathcal{C}} < \infty\}} = 1_{\{\phi_x < \infty\}} \quad \forall x \in \mathcal{C}, \quad \mathbb{P}_{\gamma}\text{-almost surely.}$$

Proof. Because the chain must enter $\mathcal{C}$ to enter a state x in $\mathcal{C}$ ($\phi_{\mathcal{C}} \leq \phi_x$ by definition), we need only show that

$$\mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}} < \phi_x = \infty\}) = 0.$$

Setting $\varsigma := \phi_{\mathcal{C}}$ and $A := \{x\}$ in Lemma 13.12, we find that

$$\mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}} < \phi_x = \infty\}) = \sum_{z \in \mathcal{C}} \mathbb{P}_{\gamma}(\{\phi_{\mathcal{C}} < \phi_x, X_{\phi_{\mathcal{C}}} = z\}) \mathbb{P}_z(\{\phi_x = \infty\}) = 0$$

because $\mathbb{P}_z(\{\phi_x = \infty\}) = 1 - \mathbb{P}_z(\{\phi_x < \infty\}) = 0$ for all z, x in $\mathcal{C}$ (Theorem 6), □

A proof of Theorem 6. Here, we need the following corollary of Theorem 2:

(9) COROLLARY. *For any initial distribution γ ,*

$$\sum_{n=1}^{\infty} p_n(x) = \frac{\mathbb{P}_{\gamma}(\{\phi_x < \infty\})}{1 - \mathbb{P}_x(\{\phi_x < \infty\})},$$

where p_n denotes the time-varying law (4.2). Thus, a state x is recurrent if and only if

$$\sum_{n=1}^{\infty} p_n(x, x) = \infty \quad \forall x \in \mathcal{S},$$

where $P_n = (p_n(x, y))_{x, y \in \mathcal{S}}$ denotes the n -step matrix (4.5).

Proof. Tonelli's theorem and (3–4) imply that

$$\sum_{n=1}^{\infty} p_n(x) = \mathbb{E}_{\gamma} \left[\sum_{k=1}^{\infty} 1_x(X_k) \right] = \sum_{k=1}^{\infty} \mathbb{P}_{\gamma}(\{\phi_x^k < \infty\}) = \mathbb{P}_{\gamma}(\{\phi_x < \infty\}) \sum_{k=0}^{\infty} \mathbb{P}_x(\{\phi_x < \infty\})^k,$$

and the corollary follows from the geometric progression. □

Proof of Theorem 6. We begin by showing that if $x \rightarrow y$ and x is recurrent, then $\mathbb{P}_y(\{\phi_x < \infty\}) = 1$. The idea is that, because the chain will visit x infinitely many times (Theorem 2) and because it has positive probability of travelling from x to y , it will travel to y at some point. For the infinite returns to continue the chain must then travel back from y to x . In particular, letting $x_1, \dots, x_l$ be as in Lemma 13.6 and applying Proposition 13.10, we have that

$$\begin{aligned} \mathbb{P}_x(\{\phi_x < \infty\}) &= p(x, x) + \sum_{z \neq x} p(x, z) \mathbb{P}_z(\{\phi_x < \infty\}) \leq 1 - p(x, x_1) + p(x, x_1) \mathbb{P}_{x_1}(\{\phi_x < \infty\}) \\ &= 1 - p(x, x_1)(1 - \mathbb{P}_{x_1}(\{\phi_x < \infty\})). \end{aligned}$$

Iterating the above, we find that

$$1 = \mathbb{P}_x(\{\phi_x < \infty\}) \leq 1 - p(x, x_1)p(x_1, x_2) \dots p(x_l, y)(1 - \mathbb{P}_y(\{\phi_x < \infty\})),$$

and it follows that x is accessible from y (moreover, that $\mathbb{P}_y(\{\phi_x < \infty\}) = 1$).

Next, we argue that, if x is recurrent, then y must also be recurrent. As we have just shown, x is accessible from y . Hence, we can pick $y_1, \dots, y_k$ such that $p(y, y_1)p(y_1, y_2) \dots p(y_k, x) > 0$ (Lemma 13.6). Using the definition of the n -step matrix in (4.5) we obtain

$$p_{n+k+l+2}(y, y) \geq p(y, y_1)p(y_1, y_2) \dots p(y_k, x)p_n(x, x)p(x, x_1)p(x_1, x_2) \dots p(x_l, y) \quad \forall n \geq 0.$$

Summing over n and applying Corollary 9, we find that

$$\begin{aligned} \sum_{n=1}^{\infty} p_n(y, y) &\geq \sum_{n=1}^{\infty} p_{n+k+l+2}(y, y) \\ &\geq p(y, y_1)p(y_1, y_2) \dots p(y_k, x) \left(\sum_{n=1}^{\infty} p_n(x, x) \right) p(x, x_1)p(x_1, x_2) \dots p(x_l, y) = \infty, \end{aligned}$$

proving the recurrence of y . Given that we already know that $y \rightarrow x$, we can reverse the roles of x and y and the argument above shows that $\mathbb{P}_x(\{\phi_y < \infty\}) = 1$.

Lastly, that a recurrent state must belong to a closed communicating class follows by noting that x does not belong to a closed communicating class only if there exists a state y accessible from x that does not communicate with x . $\square$

Notes and references. While the notion of Harris recurrence is commonplace in the literature, that of Tweedie recurrence is not. Texts that touch upon the latter leave it unnamed. There is one noticeable exception: these types of chains were called *ultimately recurrent* in (Tweedie, 1975b). Presently (15/10/2019), a Google search for

“ultimately recurrent” and “markov”

seems to imply that only two other articles have ever used this term. In an effort to uniformise the terminology with “Harris recurrent chains” (and with “positive Harris recurrent chains” later on)—and in recognition of R. L. Tweedie’s exemplary work on the stability of Markov chains (work from which I learned much of what I know on this subject)—I instead call these types of chains *Tweedie recurrent*.

15. The regenerative property. In the recurrent case, the strong Markov property implies much more than never ending returns: upon each return, the chain *regenerates* in the sense that it forgets its entire past.

The idea is that the only aspect of the chain’s past and present that its future depends on is its current location. By setting the present to be the k th return time ϕ_x^k to a given state x , we ensure that the chain’s location $X_{\phi_x^k}$ at this time is independent of its past: by definition $X_{\phi_x^k}$ is x regardless of what occurred before ϕ_x^k (of course, if ϕ_x^k is infinite, it is ill-suited to represent the ‘present’, something we avoid by requiring x to be recurrent, see Theorem 14.2). For these reasons, the segment the chain’s path demarcated by two consecutive returns is independent of those demarcated by previous consecutive returns. Furthermore, given the starting-afresh-from- $X_{\phi_x^k}$ -when-conditioned-on- $X_{\phi_x^k}$ aspect of the strong Markov property, the fact that $X_{\phi_x^k}$ always equals x further implies that these path segments all have the same distribution (that of the 0-to- $(\phi_x - 1)$ path segment under $\mathbb{P}_x$). In summary, the sequence of these path segments is i.i.d., a very useful fact that opens the door to studying the long-term of chains using tools developed for i.i.d. sequences. For instance, Kolmogorov’s classical strong law of large numbers will have a starring role in the next section.

To formalise the above discussion, we define the random variable

$$(1) \quad I_k^f := 1_{\{\phi_x^{k-1} < \infty\}} \sum_{m=\phi_x^{k-1}}^{\phi_x^k-1} f(X_m),$$

where $f : \mathcal{S} \rightarrow \mathbb{R}_E$ denotes any non-negative function, k any positive integer, and we are using our convention for partially-defined random variables in (0.1).

(2) THEOREM (The regenerative property). *Let $f : \mathcal{S} \rightarrow \mathbb{R}_E$ be any non-negative function on $\mathcal{S}$. For all $k > 0$, the random variable I_k^f in (1) is $\mathcal{F}_{\phi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable, where $\mathcal{F}_{\phi_x^k}$ denotes the pre- ϕ_x^k sigma algebra (Definition 8.1). If the state x is recurrent, the sequence $(I_k^f)_{k \in \mathbb{Z}_+}$ is i.i.d. under $\mathbb{P}_x$.*

We do the proof in two steps, beginning by arguing the measurability of I_k^f . With this out of the way, we then focus on showing that $(I_k^f)_{k \in \mathbb{Z}_+}$ is i.i.d.

Step 1: I_k^f is $\mathcal{F}_{\phi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Note that

$$I_k^f = \lim_{N \rightarrow \infty} \sum_{n=1}^N \sum_{m=0}^{n-1} \sum_{l=m}^{n-1} 1_{\{\phi_x^k=n\}} 1_{\{\phi_x^{k-1}=m\}} f(X_l).$$

Because limits, finite products, and finite sums of measurable functions are measurable functions, it suffices to show that each term in the sum is $\mathcal{F}_{\phi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Because $\phi_{k-1} \leq \phi_k$ by definition, Lemma 8.3i,iii shows that $1_{\{\phi_x^{k-1}=m\}}$ is $\mathcal{F}_{\phi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Similarly, Lemma 8.3i,ii implies that $1_{\{\phi_x^k=n\}} f(X_l)$ is $\mathcal{F}_{\phi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable for all $l < n$ and the result follows. $\square$

Step 2: $(I_k^f)_{k \in \mathbb{Z}_+}$ is i.i.d. To simplify the notation, we write I_k for I_k^f throughout this proof. Suppose that x is recurrent, choose any $a_1, \dots, a_k$ in $\mathbb{R}_E$, and let

$$A_k := \{I_1 \geq a_1, \dots, I_{k-1} \geq a_{k-1}, I_k \geq a_k\}.$$

Because the return times to x are all finite with $\mathbb{P}_x$ -probability one (14.3),

$$\begin{aligned} \mathbb{P}_x(A_k) &= \mathbb{P}_x(\{I_1 \geq a_1, \dots, I_{k-1} \geq a_{k-1}, \phi_x^{k-1} < \infty, X_{\phi_x^{k-1}} = x, I_k \geq a_k\}) \\ &= \mathbb{P}_x(\{I_1 \geq a_1, \dots, I_{k-1} \geq a_{k-1}, \phi_x^{k-1} < \infty, X_{\phi_x^{k-1}} = x, G(X_{\phi_x^{k-1}}) \geq a_k\}), \end{aligned}$$

where $X_{\phi_x^{k-1}}$ denotes the ϕ_x^{k-1} -shifted chain (Section 12) and $G((x_n)_{n \in \mathbb{N}}) := \sum_{m=0}^{\inf\{n>0: x_n=x\}-1} f(x_m)$.

Because we are able to deduce the values of $I_1, \dots, I_{k-1}$ by observing the chain up until the $(k-1)$ th return, the event A_{k-1} belongs to the pre- ϕ_x^{k-1} sigma-algebra $\mathcal{F}_{\phi_x^{k-1}}$ (8.1). Formally, Lemma 8.3iii implies that

$$\mathcal{F}_{\phi_x^1} \subseteq \mathcal{F}_{\phi_x^{k-1}} \subseteq \dots \subseteq \mathcal{F}_{\phi_x^{k-1}}$$

as $\phi_x^1 \leq \phi_x^2 \leq \dots \leq \phi_x^{k-1}$ (Exercise 13.3) and it follows from Step 1 that A_{k-1} belongs to $\mathcal{F}_{\phi_x^{k-1}}$.

For these reasons (and after checking the measurability requirement on G using Lemma 12.8iii), we may apply the strong Markov property (Theorem 12.4) to obtain

$$\mathbb{P}_x(A_k) = \mathbb{P}_x(A_{k-1} \cap \{\phi_x^{k-1} < \infty, X_{\phi_x^{k-1}} = x\}) \mathbb{P}_x(\{G(X_n)_{n \in \mathbb{N}} \geq a_k\}) = \mathbb{P}_x(A_{k-1}) \mathbb{P}_x(\{I_1 \geq a_k\}).$$

Iterating the above backwards, we obtain

$$(3) \quad \mathbb{P}_x(A_k) = \mathbb{P}_x(\{I_1 \geq a_1\}) \mathbb{P}_x(\{I_1 \geq a_2\}) \dots \mathbb{P}_x(\{I_1 \geq a_k\}).$$

Setting $a_1 = \dots = a_{k-1} = 0$, we find that

$$\mathbb{P}_x(\{I_k \geq a_k\}) = \mathbb{P}_x(\{I_1 \geq a_k\}).$$

Because k and $a_1, \dots, a_k$ were arbitrary, and $\{[a, \infty] : a \in \mathbb{R}_E\}$ is a π -system that generates $\mathcal{B}(\mathbb{R}_E)$, the above and Lemma 5.6 show that the sequence $(I_k)_{k \in \mathbb{Z}_+}$ is identically distributed. For this reason, we can rewrite (3) as

$$\mathbb{P}_x(A_k) = \mathbb{P}_x(\{I_1 \geq a_1\}) \mathbb{P}_x(\{I_2 \geq a_2\}) \dots \mathbb{P}_x(\{I_k \geq a_k\}).$$

The desired independence also follows from Lemma 5.6 as $\{[a_1, \infty] \times \dots \times [a_k, \infty] : a_1, \dots, a_k \in \mathbb{R}_E\}$ is a π -system that generates $\mathcal{B}(\mathbb{R}_E^k)$. $\square$

16. The empirical distribution and positive recurrent states. Armed with the regenerative property, we can start attacking the chain's long-term behaviour more directly. Here, we use Kolmogorov's celebrated strong law of large numbers (Theorem 4) to study the limiting behaviour of the *empirical distribution* $\epsilon_N := (\epsilon_N(x))_{x \in \mathcal{S}}$ whose x -entry $\epsilon_N(x)$ denotes the fraction of the first N time-steps that the chain X spends in state x :

$$(1) \quad \epsilon_N(x) := \frac{1}{N} \sum_{n=0}^{N-1} 1_x(X_n) \quad \forall x \in \mathcal{S}.$$

From Theorem 14.2, we already know that the chain eventually stops visiting any given transient state x , and it follows that

$$\lim_{N \rightarrow \infty} \epsilon_N(x) = 0 \quad \mathbb{P}_\gamma\text{-almost surely.}$$

In the case of recurrent states x , the visits do not cease and, to understand what happens with $\epsilon_N(x)$, we must study frequency of these visits. The trick here is to notice that X does not visit state x in between two consecutive return times, and so

$$\sum_{n=\phi_x^k}^{\phi_x^{k+1}-1} 1_x(X_n) = 1 \quad \forall k > 0.$$

Breaking down the path followed by X into segments demarcated by consecutive return times and picking the return time ϕ_x^K closest to N , we find that

$$\epsilon_N(x) \approx \frac{1}{\phi_x^K} \sum_{n=0}^{\phi_x^K-1} 1_x(X_n) = \frac{1}{\phi_x^K} \sum_{k=0}^{K-1} \left(\sum_{n=\phi_x^k}^{\phi_x^{k+1}-1} 1_x(X_n) \right) = \frac{K}{\phi_x^K} \quad \mathbb{P}_x\text{-almost surely,}$$

for N large enough that $|\phi_x^K - N|/N$ is small. The regenerative property tells us that the sequence $(\phi_x^{k+1} - \phi_x^k)_{k \in \mathbb{N}}$ of elapsed times between consecutive visits is i.i.d. whenever the chain starts in x . For this reason, the law of large numbers implies that

$$\frac{K}{\phi_x^K} = \left(\frac{\phi_x^K}{K} \right)^{-1} = \left(\frac{1}{K} \sum_{k=0}^{K-1} (\phi_x^{k+1} - \phi_x^k) \right)^{-1} \approx (\mathbb{E}_x[\phi_x])^{-1} \quad \mathbb{P}_x\text{-almost surely,}$$

for large enough K . If the chain does not start in x , then noting that $\epsilon_N(x)$ is non-zero only if the chain ever visits x and applying the Markov property yields the following characterisation of $\epsilon_N(x)$'s limiting behaviour (for a formal proof, see the end of the section):

(2) THEOREM. *For all states x in $\mathcal{S}$,*

$$\epsilon_\infty(x) := \lim_{N \rightarrow \infty} \epsilon_N(x) = \frac{1_{\{\phi_x < \infty\}}}{\mathbb{E}_x[\phi_x]} \quad \mathbb{P}_\gamma\text{-almost surely.}$$

The theorem shows that the fraction $\epsilon_\infty(x)$ of all time that the chain spends in a state x is negligible unless returns to x happen quickly enough that the mean return time is finite. Or, in other words, unless x is *positive recurrent*:

(3) DEFINITION (Positive and null recurrent states). *A state x is said to be positive recurrent if its mean return time is finite ($\mathbb{E}_x[\phi_x] < \infty$). If x is recurrent but its mean return time is infinite ($\mathbb{E}_x[\phi_x] = \infty$), we say that it is null recurrent.*

Kolmogorov's law of large numbers and a proof of Theorem 2. As outlined above, the proof of Theorem 2 entails combining the regenerative property, the strong Markov property, and the law of large numbers (LLN):

(4) **THEOREM** (Kolmogorov's strong law of large numbers). *If $(W_n)_{n \in \mathbb{Z}_+}$ is a sequence of i.i.d. non-negative $\mathbb{R}_E$ -valued random variables on a probability triplet $(\Omega, \mathcal{F}, \mathbb{P})$, then their sample average converges to $\mathbb{E}[W_1]$ almost surely:*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N W_n = \mathbb{E}[W_1] \quad \mathbb{P}\text{-almost surely.}$$

Proof. The proof for the case of integrable finite-valued W_n s can be found in many books, e.g. (Williams, 1991), and we skip it. For the general case fix M in $\mathbb{N}$, apply Kolmogorov's LLN to the sequence $(W_n \wedge M)_{n \in \mathbb{Z}_+}$ to get

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N W_n \wedge M = \mathbb{E}[W_1 \wedge M] \quad \mathbb{P}\text{-almost surely.}$$

Taking the limit $M \rightarrow \infty$ and applying monotone convergence then completes the proof. $\square$

We do the proof of Theorem 2 in two steps. We begin by using the regenerative property and the LLN to argue the limit in the case that the chain starts from the state x :

Proof of Theorem 2: Step 1. For the time being, suppose that the chain starts at some state x . Note that

$$(5) \quad \epsilon_N(x) = \frac{1}{N} \sum_{n=0}^{N-1} 1_x(X_n) = \frac{1_x(X_0)}{N} + \frac{R_N}{N} \quad \forall N > 0,$$

where $R_N := \sum_{n=1}^{N-1} 1_x(X_n)$ is the total number of returns by time $N-1$ (with $R_1 := 0$). If the state is transient, then Theorem 14.2 shows that $R := \sum_{n=1}^{\infty} 1_x(X_n)$ is finite, $\mathbb{P}_x$ -almost surely, and it follows that

$$\lim_{N \rightarrow \infty} \epsilon_N(x) = \lim_{N \rightarrow \infty} \frac{R_N}{N} \leq \lim_{N \rightarrow \infty} \frac{R}{N} = 0 = \frac{1}{\mathbb{E}_x[\phi_x]} \quad \mathbb{P}_x\text{-almost surely,}$$

as $\mathbb{E}_x[\phi_x] = \infty$.

Suppose instead that x is recurrent. The chain's regenerative property (Theorem 15.2 with $f := 1$) and Kolmogorov's LLN (Theorem 4 with $W_n := 1_{\{\phi_x^n < \infty\}}(\phi_x^{n+1} - \phi_x^n)$):

$$(6) \quad \lim_{K \rightarrow \infty} \frac{\phi_x^K}{K} = \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=0}^{K-1} (\phi_x^{k+1} - \phi_x^k) = \mathbb{E}_x[\phi_x] \quad \mathbb{P}_x\text{-almost surely}$$

$$(7) \quad \Rightarrow \lim_{K \rightarrow \infty} \frac{\phi_x^{K+1} - \phi_x^K}{K} = 0 \quad \mathbb{P}_x\text{-almost surely.}$$

Because R_N is the number of visit made by time $N-1$, the R_N th return must occur by time $N-1$ and the next return must occur after this time:

$$\phi_x^{R_N} \leq N-1 < \phi_x^{R_N+1} \quad \forall N > 0, \quad \mathbb{P}_x\text{-almost surely.}$$

It follows that,

$$(8) \quad \frac{\phi_x^{R_N}}{R_N} \leq \frac{N}{R_N} \leq \frac{\phi_x^{R_N+1}}{R_N} \quad \forall N > 0, \quad \mathbb{P}_x\text{-almost surely,}$$

where we are using the convention that the two left-most terms are zero if $R_N = 0$.

Because R_N increases at most in steps of one ($R_{N+1} - R_N \in \{0, 1\}$), R_N may only approach infinity by stepping through all positive integers. Because x is recurrent, Theorem 14.2 implies that $R_N \rightarrow \infty$ as $N \rightarrow \infty$, $\mathbb{P}_x$ -almost surely, and it follows that

$$\mathbb{P}_x(\{\cup_{N=1}^{\infty} \{R_N\} = \mathbb{N}\}) = 1.$$

For this reason, (6–7) implies that

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{\phi_x^{R_N}}{R_N} &= \lim_{K \rightarrow \infty} \frac{\phi_x^K}{K} = \mathbb{E}_x[\phi_x], \\ \lim_{N \rightarrow \infty} \frac{\phi_x^{R_N+1}}{R_N} &= \lim_{K \rightarrow \infty} \frac{\phi_x^{K+1}}{K} = \lim_{K \rightarrow \infty} \frac{\phi_x^{K+1} - \phi_x^K}{K} + \lim_{N \rightarrow \infty} \frac{\phi_x^K}{K} = \mathbb{E}_x[\phi_x], \quad \mathbb{P}_x\text{-almost surely;} \end{aligned}$$

and it follows from (8) that $\epsilon_N(x) \rightarrow 1/\mathbb{E}_x[\phi_x]$ as $N \rightarrow \infty$ with $\mathbb{P}_x$ -probability one. $\square$

To finish the proof, we now transfer the result from the starting-location-is- x case to the arbitrary-initial-distribution- γ case by noting that $\epsilon_\infty(x)$ is non-zero only if ϕ_x is finite, conditioning on ϕ_x , and applying the strong Markov property:

Proof of Theorem 2: Step 2. We have now left to show that, for any initial distribution γ ,

$$(9) \quad \mathbb{P}_\gamma \left(\left\{ \liminf_{N \rightarrow \infty} \epsilon_N(x) = \frac{1_{\{\phi_x < \infty\}}}{\mathbb{E}_x[\phi_x]} \right\} \right) = 1, \quad \mathbb{P}_\gamma \left(\left\{ \limsup_{N \rightarrow \infty} \epsilon_N(x) = \frac{1_{\{\phi_x < \infty\}}}{\mathbb{E}_x[\phi_x]} \right\} \right) = 1.$$

To do so, note that, by the definition of the entrance time ϕ_x ,

$$\epsilon_N(x) = \frac{1}{N} \sum_{n=0}^{N-1} 1_x(X_n) = \frac{1_x(X_0)}{N} + \frac{1}{N} \sum_{n=1}^{N-1} 1_x(X_n) = \frac{1_x(X_0)}{N} + \frac{1_{\{\phi_x \leq N\}}}{N} \sum_{n=\phi_x}^{N-1} 1_x(X_n).$$

Given that the first term tends to zero as N grows unbounded and that

$$\limsup_{N \rightarrow \infty} \frac{1_{\{\phi_x \leq N\}}}{N} \sum_{n=\phi_x}^{\phi_x + N - 1} 1_x(X_n) \leq \limsup_{N \rightarrow \infty} \frac{1_{\{\phi_x \leq N\}} \phi_x}{N} = \limsup_{N \rightarrow \infty} \frac{1_{\{\phi_x < \infty\}} \phi_x}{N} = 0,$$

we have that

$$(10) \quad \liminf_{N \rightarrow \infty} \epsilon_N(x) = \liminf_{N \rightarrow \infty} \frac{1_{\{\phi_x < \infty\}}}{N} \sum_{n=\phi_x}^{N-1} 1_x(X_n) = \liminf_{N \rightarrow \infty} \frac{1_{\{\phi_x < \infty\}}}{N} \sum_{n=\phi_x}^{\phi_x + N - 1} 1_x(X_n),$$

$$(11) \quad \limsup_{N \rightarrow \infty} \epsilon_N(x) = \limsup_{N \rightarrow \infty} \frac{1_{\{\phi_x < \infty\}}}{N} \sum_{n=\phi_x}^{N-1} 1_x(X_n) = \limsup_{N \rightarrow \infty} \frac{1_{\{\phi_x < \infty\}}}{N} \sum_{n=\phi_x}^{\phi_x + N - 1} 1_x(X_n).$$

Given that

$$F_-(X) = \liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} 1_x(X_n), \quad F_+(X) = \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} 1_x(X_n),$$

where F_- , F_+ are as in Lemma 12.8i with $f := 1_x$, we may rewrite (10–11) as

$$(12) \quad \liminf_{N \rightarrow \infty} \epsilon_N(x) = 1_{\{\phi_x < \infty\}} F_-(X^{\phi_x}), \quad \limsup_{N \rightarrow \infty} \epsilon_N(x) = 1_{\{\phi_x < \infty\}} F_+(X^{\phi_x}),$$

where X^{ϕ_x} denotes the ϕ_x -shifted chain in (12.3). Thus,

$$\mathbb{P}_\gamma \left(\left\{ \liminf_{N \rightarrow \infty} \epsilon_N(x) = \frac{1_{\{\phi_x < \infty\}}}{\mathbb{E}_x[\phi_x]} \right\} \right) = \mathbb{P}_\gamma(\{\phi_x = \infty\}) + \mathbb{P}_\gamma \left(\left\{ \phi_x < \infty, F_-(X^{\phi_x}) = \frac{1}{\mathbb{E}_x[\phi_x]} \right\} \right).$$

Because we already showed that $F_-(X) = 1/\mathbb{E}_x[\phi_x]$ with $\mathbb{P}_x$ -probability one, applying the strong Markov property (Theorem 12.4) yields

$$\begin{aligned} \mathbb{P}_\gamma \left(\left\{ \phi_x < \infty, F_-(X^{\phi_x}) = \frac{1}{\mathbb{E}_x[\phi_x]} \right\} \right) &= \mathbb{P}_\gamma \left(\left\{ \phi_x < \infty, X_{\phi_x} = x, F_-(X^{\phi_x}) = \frac{1}{\mathbb{E}_x[\phi_x]} \right\} \right) \\ &= \mathbb{P}_\gamma(\{\phi_x < \infty, X_{\phi_x} = x\}) \mathbb{P}_x \left(\left\{ F_-(X) = \frac{1}{\mathbb{E}_x[\phi_x]} \right\} \right) \\ &= \mathbb{P}_\gamma(\{\phi_x < \infty, X_{\phi_x} = x\}) = \mathbb{P}_\gamma(\{\phi_x < \infty\}), \end{aligned}$$

and the leftmost equation in (9) follows. For the rightmost one, replace ‘lim inf’ and ‘ F_- ’ with ‘lim sup’ and ‘ F_+ ’ in all equations following (12). $\square$

17. Stationary distributions, ergodic distributions, and a Doeblin-like decomposition. A probability distribution π on $\mathcal{S}$ is said to be a stationary distribution of the chain X if sampling the initial condition from π makes X a stationary process. That is, one whose path law is invariant to time shifts:

$$(1) \quad \mathbb{P}_\pi(\{X^n \in A\}) = \mathbb{L}_\pi(A) \quad \forall A \in \mathcal{E}, \quad n \geq 0,$$

where $\mathbb{L}_\pi$ denotes the path law with initial distribution π , $\mathcal{E}$ denotes the sigma-algebra generated by the cylinder sets of the path space $\mathcal{P}$ (Section 5.5 for the definitions of $\mathbb{L}_\pi, \mathcal{P}, \mathcal{E}$), and X^n the n -shifted chain (12.3). Setting A in the above to be the slice $\{(x_m)_{m \in \mathbb{N}} \in \mathcal{P} : x_0 = x\}$ of $\mathcal{P}$, we find that

$$(2) \quad \mathbb{P}_\pi(\{X_n = x\}) = \pi(x) \quad \forall x \in \mathcal{S}, \quad n \geq 0.$$

In other words, if the chain starts with distribution π , then it remains with distribution π for all time. Applying (4.2) to the above we find that

$$(3) \quad \pi P(x) = \pi(x) \quad \forall x \in \mathcal{S},$$

or $\pi P = \pi$ for short. Conversely, marginalising over (4.4) and applying (3) yields (1) for all sets A of the form in (5.1), and it follows from Theorem 5.5 that

(4) **THEOREM** (Analytical characterisation of the stationary distributions). *A probability distribution π on $\mathcal{S}$ is a stationary distribution of X if and only if it satisfies (3).*

A stationary distribution has support on a state if and only if the state is positive recurrent. If the chain is a stationary process, then the average fraction of time-steps it spends in any given state remains constant throughout time. In particular, sampling the starting location from a stationary distribution π , taking expectations of the empirical distribution ϵ_N (Section 16), and applying (2), we obtain that

$$(5) \quad \mathbb{E}_\pi[\epsilon_N(x)] = \mathbb{E}_\pi \left[\frac{1}{N} \sum_{n=0}^{N-1} 1_x(X_n) \right] = \frac{1}{N} \sum_{n=0}^{N-1} \mathbb{P}_\pi(\{X_n = x\}) = \pi(x) \quad \forall x \in \mathcal{S}, \quad N > 0.$$

Using Theorem 16.2, taking the limit $N \rightarrow \infty$, and applying bounded convergence, we find that

$$(6) \quad \pi(x) = \lim_{N \rightarrow \infty} \mathbb{E}_\pi[\epsilon_N(x)] = \frac{\mathbb{P}_\pi(\{\phi_x < \infty\})}{\mathbb{E}_x[\phi_x]} \quad \forall x \in \mathcal{S}.$$

Thus, there exists a stationary distribution π with support on x (i.e., with $\pi(x) > 0$) only if x is positive recurrent (Definition 16.3). The intuition here is that, if the initial position is sampled from π , then the states x satisfying $\pi(x) > 0$ are those that the chain will visit often enough that the fraction of time-steps $\epsilon_N(x)$ it spends in the state does not decay to zero. Because, regardless of the starting location, visits to any non-positive-recurrent state x eventually become so rare that $\epsilon_N(x)$ collapses to zero (Theorem 16.2), it follows that x must be positive recurrent for $\pi(x)$ to be non-zero.

The converse is also true and the trick in arguing it involves the stopping distributions and occupation measures of the Section 10. In particular, let μ_S and ν_S be the space marginals (10.12–10.13) of the stopping distribution and occupation measure associated with the entrance time ϕ_x of any given recurrent state x and set the chain's starting location to be x . At the moment of entry, the chain is at x . Because x is recurrent, it follows that μ_S is the point mass 1_x at x . Because we fixed the initial distribution γ to also be 1_x , Corollary 10.14 then shows that $\nu_S = \nu_S P$. Recall that the mass of ν_S is the mean return time, c.f. (10.4). Hence, if x is positive recurrent, then we obtain a probability distribution π satisfying that $\pi = \pi P$ by normalising ν_S (i.e. $\pi := \nu_S / \nu_S(\mathcal{S})$). Moreover,

$$\pi(x) = \frac{\nu_S(x)}{\nu_S(\mathcal{S})} = \frac{\mathbb{E}_x \left[\sum_{n=0}^{\phi_x-1} 1_x(X_n) \right]}{\mathbb{E}_x [\phi_x]} = \frac{\mathbb{E}_x [1_x(X_0)]}{\mathbb{E}_x [\phi_x]} = \frac{\mathbb{P}_x (\{X_0 = x\})}{\mathbb{E}_x [\phi_x]} = \frac{1}{\mathbb{E}_x [\phi_x]} > 0.$$

In summary:

(7) THEOREM. *A state x is positive recurrent if and only if there exists a stationary distribution π such that $\pi(x) > 0$. Moreover, if x is positive recurrent, then one such stationary distribution is given by*

$$\pi(y) = \frac{1}{\mathbb{E}_x [\phi_x]} \mathbb{E}_x \left[\sum_{n=0}^{\phi_x-1} 1_y(X_n) \right] \quad \forall y \in \mathcal{S}.$$

A handy consequence of this theorem is that positive and null recurrence are class properties:

(8) COROLLARY. *If x is positive recurrent and $x \rightarrow y$, then y is also positive recurrent. Moreover, a state in a communicating class is positive (resp. null) recurrent if and only if all states in the class are positive (resp. null) recurrent.*

For this reason, we say that a communicating class is *positive recurrent* (resp. *null recurrent*) if each one (or, equivalently, all) of its states is positive recurrent (resp. null recurrent).

Proof. Let $x_1, \dots, x_l$ be as in (13.6) and suppose that x is positive recurrent. Let π be the stationary distribution in Theorem 7 satisfying $\pi(x) > 0$. Using (3) repeatedly we find that

$$\pi(y) = \sum_{z \in \mathcal{S}} \pi(z) p(z, y) \geq \pi(x_l) p(x_l, y) \geq \dots \geq \pi(x) p(x, x_1) p(x_1, x_2) \dots p(x_l, y) > 0.$$

Applying Theorem 7 then shows that y is positive recurrent. That positive recurrence is a class property follows immediately. For null recurrence, the result then follows because recurrence is a class property (Theorem 14.6) and because recurrent states are either positive recurrent or null recurrent. $\square$

Ergodic distributions; Doeblin-like decomposition; set of stationary distributions. We wrap up our treatment of the stationary distributions by characterising the set thereof. Here, we require the following *Doeblin-like* decomposition of the state space:

$$(9) \quad \mathcal{S} = \left(\bigcup_{i \in \mathcal{I}} \mathcal{C}_i \right) \cup \mathcal{T} = \mathcal{R}_+ \cup \mathcal{T},$$

where $\{\mathcal{C}_i : i \in \mathcal{I}\}$ denotes the (necessarily countable) set of positive recurrent closed communicating classes, which we index with some set $\mathcal{I}$, $\mathcal{R}_+ := \cup_{i \in \mathcal{I}} \mathcal{C}_i$ the set of all positive recurrent states, and $\mathcal{T}$ that of all other states. As we have already seen in Theorem 7, no stationary distribution has support in $\mathcal{T}$ and there at least one stationary distribution per $\mathcal{C}_i$. Much more can be said:

(10) **THEOREM** (Characterising the set of stationary distributions). *Let $\{\mathcal{C}_i : i \in \mathcal{I}\}$ be the collection of positive recurrent closed communicating classes.*

- (i) *For each $\mathcal{C}_i$, there exists a single stationary distribution π_i with support contained in $\mathcal{C}_i$ (i.e., with $\pi_i(\mathcal{C}_i) = 1$); π_i is known as the ergodic distribution associated with $\mathcal{C}_i$. It has support on all of $\mathcal{C}_i$ ($\pi_i(x) > 0$ for all states x in $\mathcal{C}_i$) and can be expressed as*

$$(11) \quad \pi_i(y) = \frac{1_{\mathcal{C}_i}(y)}{\mathbb{E}_y[\phi_y]} = \frac{1}{\mathbb{E}_x[\phi_x]} \mathbb{E}_x \left[\sum_{n=0}^{\phi_x-1} 1_y(X_n) \right] \quad \forall y \in \mathcal{S},$$

where x is any state in $\mathcal{C}_i$.

- (ii) *A probability distribution π is a stationary distribution of the chain if and only if it is a convex combination of the ergodic distributions:*

$$\pi = \sum_{i \in \mathcal{I}} \theta_i \pi_i$$

for some collection $(\theta_i)_{i \in \mathcal{I}}$ of non-negative constants satisfying $\sum_{i \in \mathcal{I}} \theta_i = 1$. For any stationary distribution π , the weight θ_i featuring in the above is the mass that π awards to $\mathcal{C}_i$ or, equivalently, the probability that the chain ever enters $\mathcal{C}_i$ if its starting location was sampled from π :

$$\theta_i = \pi(\mathcal{C}_i) = \mathbb{P}_\pi(\{\phi_{\mathcal{C}_i} < \infty\}) \quad \forall i \in \mathcal{I}.$$

Proof. (i) For any state x in a recurrent closed communicating class $\mathcal{C}_i$, Proposition 13.9 and Corollary 14.8 imply that

$$\mathbb{P}_x(\{\phi_y < \infty\}) = 1_{\mathcal{C}_i}(y) \quad \forall y \in \mathcal{S}.$$

Combining the above into (6) we find that there can only exist one stationary distribution π_i with support contained in $\mathcal{C}_i$ (i.e., $\pi_i(x) = 0$ for all $x \notin \mathcal{C}_i$), as for any such π_i ,

$$(12) \quad \pi_i(x) = \frac{\mathbb{P}_{\pi_i}(\{\phi_x < \infty\})}{\mathbb{E}_x[\phi_x]} = \frac{\sum_{x' \in \mathcal{C}_i} \pi_i(x') \mathbb{P}_{x'}(\{\phi_x < \infty\})}{\mathbb{E}_x[\phi_x]} = \frac{1_{\mathcal{C}_i}(x)}{\mathbb{E}_x[\phi_x]} \quad \forall x \in \mathcal{S}.$$

On the other hand, Theorem 7 shows that the right-hand side of (11) defines such a π_i as

$$\begin{aligned} \mathbb{E}_x[\phi_x] \pi_i(y) &= \mathbb{E}_x \left[\sum_{n=0}^{\phi_x-1} 1_y(X_n) \right] = \mathbb{E}_x \left[\sum_{n=1}^{\phi_x-1} 1_y(X_n) \right] \leq \mathbb{E}_x \left[\sum_{n=1}^{\infty} 1_y(X_n) \right] = \mathbb{E}_x \left[\sum_{k=1}^{\infty} 1_{\{\phi_y^k < \infty\}} \right] \\ &= \sum_{k=1}^{\infty} \mathbb{P}_x(\{\phi_y^k < \infty\}) = \mathbb{P}_x(\{\phi_y < \infty\}) \sum_{k=1}^{\infty} \mathbb{P}_y(\{\phi_y < \infty\})^{k-1} = 0 \quad \forall y \notin \mathcal{C}_i, \end{aligned}$$

where x is any state in $\mathcal{C}_i$ and we have made use of the closedness of $\mathcal{C}_i$ and (14.3–14.4).

(ii) Given that $\mathcal{T}$ does not contain any positive recurrent states, (6) shows that $\pi(x) = 0$ for all x in $\mathcal{T}$. Because of this, we may rewrite any stationary distribution as

$$(13) \quad \pi(x) = \sum_{i \in \mathcal{I}} \pi(\mathcal{C}_i) \left(\frac{1}{\pi(\mathcal{C}_i)} 1_{\mathcal{C}_i}(x) \pi(x) \right) \quad \forall x \in \mathcal{S}.$$

Because $\mathcal{C}_i$ is a closed communicating class, we have that

$$(1_{\mathcal{C}_i}\pi)P(x) = \pi P(x) = \pi(x) = 1_{\mathcal{C}_i}(x)\pi(x) \quad \forall x \in \mathcal{C}_i.$$

Thus, assuming that $\pi(\mathcal{C}_i) > 0$, Theorem 4 shows that $1_{\mathcal{C}_i}\pi/\pi(\mathcal{C}_i)$ is a stationary distribution with support contained in $\mathcal{C}_i$. Part (i) then shows that $1_{\mathcal{C}_i}\pi/\pi(\mathcal{C}_i)$ must be equal to π_i and we can rewrite (13) as

$$\pi(x) = \sum_{i \in \mathcal{I}} \theta_i \pi_i(x) \quad \forall x \in \mathcal{S},$$

where $\theta_i := \pi(\mathcal{C}_i)$. Summing both sides over x in $\mathcal{S}$ and applying Tonelli's theorem then shows that $\sum_{i \in \mathcal{I}} \theta_i = 1$ (i.e. π is a convex combination of the ergodic distributions).

To show that $\theta_i = \mathbb{P}_\pi(\{\phi_{\mathcal{C}_i} < \infty\})$ and complete the proof, note that $\mathbb{P}_\pi(\{\phi_{\mathcal{C}_i} < \infty\}) = \mathbb{P}_\pi(\{\phi_x < \infty\})$ if x belongs to $\mathcal{C}_i$ (Corollary 14.8) and compare the above with (6) and (12). $\square$

Notes and references. The term “Doeblin decomposition” traditionally refers to (9) with $\{\mathcal{C}_i : i \in \mathcal{I}\}$ including all recurrent closed communicating classes instead of only the positive recurrent ones. In this case, $\mathcal{T}$ is the set of all transient states and, consequently, is known as the *transient set*. The motivation behind my unorthodox choice is that, for reasons that we have touched upon in the last two sections and that will become completely clear in Sections 18 and 20, only positive recurrent states typically matter in the long-run.

18. Limits of the empirical distribution and positive recurrent chains. Almost without realising it, we have derived over the last few sections a complete description of the asymptotic behaviour of the empirical distribution ϵ_N (Section 16) tracking the fraction of time that the chain spends in each state:

(1) **THEOREM** (The pointwise limits). *Let ϵ_N denote the empirical distribution (16.1), $\{\mathcal{C}_i : i \in \mathcal{I}\}$ the collection of positive recurrent closed communicating classes $\mathcal{C}_i$ (Section 17), and π_i the ergodic distribution of $\mathcal{C}_i$ (Theorem 17.10) for each i in $\mathcal{I}$. For any initial distribution γ , we have that*

$$(2) \quad \epsilon_\infty := \lim_{N \rightarrow \infty} \epsilon_N = \sum_{i \in \mathcal{I}} 1_{\{\phi_{\mathcal{C}_i} < \infty\}} \pi_i \quad \mathbb{P}_\gamma\text{-almost surely},$$

where the convergence is pointwise and $\phi_{\mathcal{C}_i}$ denotes the time of first entrance to $\mathcal{C}_i$ (Definition 13.1).

Proof. Because, with $\mathbb{P}_\gamma$ -probability one, the chain will enter a state x (i.e. $\phi_x < \infty$) in a recurrent closed communicating class $\mathcal{C}_i$ if and only if enters the class (i.e. $\phi_{\mathcal{C}_i} < \infty$) at all (Corollary 14.8), this follows directly from Theorems 16.2 and 17.10(i). $\square$

You may be wondering under what circumstances the convergence in (2) can be strengthened. The answer turns out to be surprisingly simple:

(3) **COROLLARY** (The convergence in total variation). *The chain enters the set $\mathcal{R}_+$ of positive recurrent states with probability one (i.e. $\mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}) = 1$) if and only if the limit (2) holds in total variation with $\mathbb{P}_\gamma$ -probability one:*

$$\lim_{n \rightarrow \infty} \|\epsilon_N - \epsilon_\infty\| = 0 \quad \mathbb{P}_\gamma\text{-almost surely},$$

where $\|\rho\|$ denotes the total variation norm of any signed measure $\|\rho\|$ on $\mathcal{S}$:

$$(4) \quad \|\rho\| := \sup_{A \subseteq \mathcal{S}} |\rho(A)|.$$

The trick in proving the above is the following variation of Scheffé's lemma:

(5) LEMMA (Scheffé's lemma). *Suppose that $\rho_1, \rho_2, \dots$ is a sequence of probability distributions on $\mathcal{S}$ that converge pointwise to a limit ρ . The limit ρ is a probability distribution if and only if the sequence converges in total variation.*

Proof. If the sequence converges in total variation, then ρ is a probability distribution (as $1 = \rho_n(\mathcal{S}) \rightarrow \rho(\mathcal{S})$). To prove the converse, we will show later that

$$(6) \quad \|\rho_n - \rho\| = \sum_{x \in \mathcal{S}} 1_{U_n}(x)(\rho(x) - \rho_n(x)) = \frac{1}{2} \sum_{x \in \mathcal{S}} |\rho_n(x) - \rho(x)|,$$

because ρ_n and ρ are probability distributions, where $U_n := \{x \in \mathcal{S} : \rho_n(x) < \rho(x)\}$ is the set of states for which ρ_n underestimates ρ . Because $0 \leq 1_{U_n}(x)(\rho(x) - \rho_n(x)) \leq \rho(x)$ for all x in $\mathcal{S}$ and n in $\mathbb{N}$, and because the pointwise convergence implies that

$$\lim_{n \rightarrow \infty} 1_{U_n}(x)(\rho(x) - \rho_n(x)) = 0 \quad \forall x \in \mathcal{S},$$

taking limit $n \rightarrow \infty$ in both sides of the first equation in (6) and applying dominated convergence then proves the convergence in total variation.

To finish this proof, we need to argue (6). Note that

$$\rho_n(A) - \rho(A) = (\rho_n(A \cap U_n) - \rho(A \cap U_n)) + (\rho_n(A \cap U_n^c) - \rho(A \cap U_n^c)) \quad \forall A \subseteq \mathcal{S},$$

where U_n^c denotes the complement of U_n . Because the bracketed terms have opposite signs,

$$(7) \quad \begin{aligned} \rho(U_n) - \rho_n(U_n) &\leq \sup_{A \subseteq \mathcal{S}} |\rho_n(A) - \rho(A)| \\ &\leq \sup_{A \subseteq \mathcal{S}} \max\{\rho(A \cap U_n) - \rho_n(A \cap U_n), \rho_n(A \cap U_n^c) - \rho(A \cap U_n^c)\} \\ &\leq \max\{\rho(U_n) - \rho_n(U_n), \rho_n(U_n^c) - \rho(U_n^c)\}. \end{aligned}$$

However,

$$\rho(U_n) - \rho_n(U_n) = 1 - \rho(U_n^c) - (1 - \rho_n(U_n^c)) = \rho_n(U_n^c) - \rho(U_n^c)$$

and the first equation in (6) follows from (7). The second equation then also follows, as

$$2\|\rho_n - \rho\| = \rho(U_n) - \rho_n(U_n) + 1 - \rho(U_n^c) - (1 - \rho_n(U_n^c)) = \sum_{x \in \mathcal{S}} |\rho_n(x) - \rho(x)|.$$

□

Proof of Corollary 3. The probability of both $\{\phi_{C_i} < \infty\}$ and $\{\phi_{C_j} < \infty\}$ occurring if $i \neq j$ is zero because the classes are closed sets (Proposition 13.11). Given that every positive recurrent state x belongs to a positive recurrent closed communicating class (Theorem 14.6 and Corollary 17.8), it follows that

$$\sum_{i \in \mathcal{I}} 1_{\{\phi_{C_i} < \infty\}} \pi_i(\mathcal{S}) = \sum_{i \in \mathcal{I}} 1_{\{\phi_{C_i} < \infty\}} = 1_{\{\phi_{\cup_{i \in \mathcal{I}} C_i} < \infty\}} = 1_{\{\phi_{\mathcal{R}_+} < \infty\}} \quad \mathbb{P}_\gamma\text{-almost surely.}$$

For this reason, that the limit in (2) is a probability distribution $\mathbb{P}_\gamma$ -almost surely if and only if $\mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}) = 1$ and the result follows from Theorem 1 and Lemma 5. □

Positive recurrent chains. Corollary 3 motivates the following definitions:

(8) DEFINITION (Positive recurrent chains). *A chain is positive Tweedie recurrent if*

$$\mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}) = 1 \text{ for all initial distributions } \gamma,$$

where $\mathcal{R}_+$ denotes the set of positive recurrent states. If, additionally, there is only one closed communicating class, then the chain is said to be positive Harris recurrent. If this class is the entire state space, then the chain is simply said to be positive recurrent.

The fraction of all time that the chain spends in any given finite set F is given by

$$\epsilon_\infty(F) = \sum_{x \in F} \epsilon_\infty(x) = \lim_{N \rightarrow \infty} \sum_{x \in F} \epsilon_N(x) = \lim_{N \rightarrow \infty} \sum_{x \in F} \frac{1}{N} \sum_{n=0}^{N-1} 1_x(X_n) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} 1_F(X_n),$$

$\mathbb{P}_\gamma$ -almost surely. If the chain is not positive Tweedie recurrent, then Theorem 1 tells us that there exists at least one initial distribution γ such that

$$\mathbb{P}_\gamma(\{\epsilon_\infty(F) = 0\}) = \mathbb{P}_\gamma\left(\left\{\sum_{i \in \mathcal{I}} 1_{\{\phi_{c_i} < \infty\}} \pi_i(F) = 0\right\}\right) \geq \mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} = \infty\}) > 0.$$

In other words, with this initial distribution and non-zero probability, *the chain will spend far more (indeed, infinitely more) time-steps outside F than inside*. Because this non-zero probability is bounded below by a positive constant independent of the finite set F , we have that *a non-negligible fraction of X 's paths will spend infinitely more time-steps outside any given finite set F than inside the set*:

(9) EXERCISE. *To formally argue the above sentence introduce a sequence $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ of finite subsets (or truncations) of the state space $\mathcal{S}$ that approach $\mathcal{S}$ (i.e., with $\cup_{r=1}^\infty \mathcal{S}_r = \mathcal{S}$). Applying downwards monotone convergence, show that*

$$\mathbb{P}_\gamma(A) = \lim_{r \rightarrow \infty} \mathbb{P}_\gamma(\{\epsilon_\infty(\mathcal{S}_r) = 0\}) \geq \mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} = \infty\}),$$

where $A := \bigcap_{r=1}^\infty \{\epsilon_\infty(\mathcal{S}_r) = 0\}$. *Because any finite set F is included in a large enough truncation $\mathcal{S}_r$, conclude that ω belongs to A if and only if*

$$\omega \in \{\epsilon_\infty(F) = 0\} \text{ for all finite subsets } F \text{ of } \mathcal{S}.$$

Conversely, if the chain is positive Tweedie recurrent, then, for any given initial distribution γ and ε in $(0, 1]$, we are always able to find a finite set F large enough that the chain spends at least $(1 - \varepsilon) \times 100\%$ of all time-steps inside F :

$$(10) \quad \epsilon_\infty(F) \geq 1 - \varepsilon \quad \mathbb{P}_\gamma\text{-almost surely.}$$

For this reason, I believe that positive Tweedie recurrence precisely describes what most practitioners think of when they hear the words ‘a stable chain’. Moreover, the empirical distribution converges to a probability distribution if and only if the chain is positive Tweedie recurrent (Corollary 3). As we will see later in Section 20, the same holds for the time-varying law if the chain is aperiodic (if it is periodic, the time-varying law will not converge to anything but instead oscillate).

(11) EXERCISE. *Using Theorem 1 give a formal proof that the chain is positive Tweedie recurrent if and only if, for any given initial distribution γ and ε in $(0, 1]$, there exists a finite set F such that (10) holds, $\mathbb{P}_\gamma$ -almost surely.*

Notes and references. Just as with Harris recurrence and Tweedie recurrence (Section 14), positive Harris recurrence is a frequently encountered name in the literature while positive Tweedie recurrence is not. In the past, the latter has occasionally been referred to as non-dissipativity (Kendall, 1951b; Foster, 1952; Mauldon, 1957).

19. Periodicity and lack thereof. So far, we have managed to characterise the long-term behaviour of the *time averages* taken over individual paths and described by the empirical distribution of Section 16. We now set our sights on the long-term behaviour of the *space* (or *ensemble* or *population*) averages taken across the collection of the chain's paths at given points in time and described by the time-varying law of Section 4. To proceed, we need to introduce a notion we have managed to avoid up until now: *periodicity*.

(1) **DEFINITION** (Periodic and aperiodic states). *The greatest common divisor (gcd) of a sequence $a_1, a_2, a_3, \dots$ of integers is defined as the limit of the sequence $b_1 := \gcd(a_1)$, $b_2 := \gcd(a_1, a_2)$, $b_3 := \gcd(a_1, a_2, a_3)$, $\dots$.*

The period of a state x is the greatest common divisor of the sequence of time-steps after which the chain may return to x :

$$(2) \quad d(x) := \text{greatest common divisor of } \{n > 0 : p_n(x, x) > 0\}$$

with the convention that $d(x) := 1$ if the above set is empty, where $P_n = (p_n(x, y))_{x, y \in \mathcal{S}}$ denotes the n -step matrix (4.5). A state is periodic if its period is greater than one. Otherwise, it is said to be aperiodic.

Periodic states x are those that the chain can only return after a regular number of steps: $d(x)$ steps, or $2d(x)$ steps, etc. Aperiodic states on the other hand, are characterised as follows:

(3) **PROPOSITION** (Characterising aperiodic states). *A state x is aperiodic if and only if $p_n(x, x) > 0$ for all $n > 0$ or there exists an $n(x)$ such that $p_n(x, x) > 0$ for all n greater than $n(x)$.*

To prove the proposition, we require the following number-theoretic fact:

(4) **THEOREM** (Billingsley, 1995, A.21). *Suppose that S is a set of positive integers closed under addition and of period one. Then S contains all integers past some number n .*

Proof of Proposition 3. The reverse direction follows immediately from the definition of the state's period. If the set in the right-hand side of (2) is empty, then the forward direction also follows immediately from this definition. Otherwise, note that if k, l are such that $p_k(x, x) > 0$ and $p_l(x, x) > 0$, then the Chapman-Kolmogorov equation in (4.8) implies that

$$p_{k+l}(x, x) \geq p_k(x, x)p_l(x, x) > 0.$$

In other words, $\{n > 0 : p_n(x, x) > 0\}$ is closed under addition and the result follows from Theorem 4 above. $\square$

Perhaps unsurprisingly, the period remains constant across communicating classes:

(5) **THEOREM** (The period is a class property). *If x and y belong to the same communicating class, then they have the same period.*

Proof. Because x and y communicate, there exists positive integers k and l such that $p_k(x, y) > 0$ and $p_l(y, x) > 0$ (this follows from the Chapman-Kolmogorov equation in (4.8) and Lemma 13.6). Thus, for any n such that $p_n(y, y) > 0$, the Chapman-Kolmogorov equation also implies that

$$p_{k+n+l}(x, x) \geq p_k(x, y)p_n(y, y)p_l(y, x) > 0.$$

Similarly, we have that $p_{k+l}(x, x) \geq p_k(x, y)p_l(y, x) > 0$. Thus, we have that $d(x)$ divides both $k + n + l$ and $k + l$, and so $d(x)$ must divide n . We have shown that $d(x)$ divides every integer in $\{n > 0 : p_n(y, y) > 0\}$ and so $d(x) \leq d(y)$. Reversing x and y and repeating the same argument we find that $d(y) \leq d(x)$ and we obtain that both states have the same period ($d(x) = d(y)$). $\square$

The theorem motivates us to introduce the following definition:

(6) DEFINITION (Periodic and aperiodic classes). *The period of a communicating class is the period of any one (or, equivalently, all) of its states. The class is said to be aperiodic if its period is one and periodic otherwise.*

Periodic and aperiodic chains. In Section 16, we saw that transient and null recurrent states are immaterial to the chain's long-term behaviour because visits to any such state either cease at some point or become so rare that the fraction of time the chain spends in the state approaches zero as time progresses. For this reason, we only ask for the positive recurrent states to be aperiodic in order for a chain to be deemed aperiodic:

(7) DEFINITION (Periodic and aperiodic chains). *A chain is said to be aperiodic if all of its positive recurrent states are aperiodic. Otherwise, it is said to be periodic.*

Aside from esoteric periodic chains possessing infinitely many positive recurrent classes with distinct periods, it follows from Theorem 5 that the lowest common multiple of all of the positive recurrent states' periods is finite:

$$(8) \quad d := \text{lowest common multiple of } \{d(x) : x \notin \mathcal{T}\} < \infty,$$

where $\mathcal{T}$ denotes the set of all states that are not positive recurrent. It is then a simple matter to verify that the chain obtained by sampling X every d steps starting at step $m < d$,

$$(9) \quad X_n^{m,d} := X_{m+nd} \quad \forall n \geq 0,$$

is an aperiodic chain.

(10) EXERCISE. *Given any d and m in $\mathbb{N}$, marginalise over (4.4) to obtain*

$$\mathbb{P}_\gamma \left(\left\{ X_0^{m,d} = x_0, X_1^{m,d} = x_1, \dots, X_n^{m,d} = x_n \right\} \right) = \gamma P_m(x_0) p_d(x_0, x_1) \dots p_d(x_{n-1}, x_n)$$

for all $x_0, \dots, x_n$ in $\mathcal{S}$, n in $\mathbb{N}$. Apply Theorems 5.5 and 7.1 to show that $(X_n^{m,d})_{n \in \mathbb{N}}$ is a Markov chain with one-step matrix P_d and initial distribution γP_m . Convince yourself that if d is a multiple of the period $d(x)$ (for the original chain X) of a state x , then x is an aperiodic state for $(X_n^{m,d})_{n \in \mathbb{N}}$. Conclude that if d is as in (8), then $(X_n^{m,d})_{n \in \mathbb{N}}$ is aperiodic.

For this reason, the long-term behaviour of a periodic chain can be pieced together from the behaviour of each of its periodic components $X_n^{m,d} := (X_n^{m,d})_{n \in \mathbb{N}}$ and we (mostly) focus on the aperiodic case for the remainder of our treatment of discrete-time chains.

Before proceeding, you should take a moment to convince yourself that sampling the chain every $d(x)$ steps does not alter the recurrence properties of the state x :

(11) EXERCISE. *For any given state x , let $(X_n^{m,d})_{n \in \mathbb{N}}$ be as in (9) with d being the state's period $d(x)$. Let $\phi_x^{m,d}$ denote the first entrance time to x of $(X_n^{m,d})_{n \in \mathbb{N}}$:*

$$(12) \quad \phi_x^{m,d} := \inf\{n > 0 : X_n^{m,d} = x\} = \inf\{n > 0 : X_{m+nd} = x\}.$$

Using the fact that

$$\mathbb{P}_x(\{X_n = x\}) = p_n(x, x) = 0 \quad \forall n \neq 0, d, 2d, \dots,$$

show that

$$\phi_x = d\phi_x^{0,d}, \quad \mathbb{P}_x\text{-almost surely.}$$

Given that, for all m in $\mathbb{N}$, $X^{m,d}$ is a discrete-time chain with the same one-step matrix P_d (10), use the above to conclude that x is transient (null/positive recurrent) for X if and only if it is transient (respctively, null/positive recurrent) for $X^{m,d}$ and m in $\mathbb{N}$.

The limiting behaviour of the time-averages is indifferent to periodicity. You may be wondering why the notion of periodicity doesn't affect the limiting behaviour of the time averages while it does affect that of the ensemble averages. The answer is best understood through an example: consider a chain that alternates between two states (say x and y) at each step. The state space $(\{x, y\})$ consists of a single periodic class with period two. If the initial distribution is concentrated on one of these two states (say x), then the time-varying law will keep alternating between them ($p_0 = 1_x, p_1 = 1_y, p_2 = 1_x, p_3 = 1_y, \dots$). Consequently, p_n does not converge. On the other hand, the empirical distribution (ϵ_N in (16.1)) averages out these oscillations and settles down:

$$\begin{aligned} \epsilon_1 &= 1_x, & \epsilon_2 &= \frac{1}{2}1_x + \frac{1}{2}1_y, & \epsilon_3 &= \frac{2}{3}1_x + \frac{1}{3}1_y, & \epsilon_4 &= \frac{1}{2}1_x + \frac{1}{2}1_y, & \epsilon_5 &= \frac{3}{5}1_x + \frac{2}{5}1_y, \\ \epsilon_6 &= \frac{1}{2}1_x + \frac{1}{2}1_y, & \epsilon_7 &= \frac{4}{7}1_x + \frac{3}{7}1_y, & \dots & \rightarrow & \frac{1}{2}1_x + \frac{1}{2}1_y. \end{aligned}$$

20. Limits of the time-varying law; tightness; ergodicity; coupling. The aim of this section is to describe the long-term behaviour of the chain's time-varying law $(p_n)_{n \in \mathbb{N}}$ (introduced in Section 4). We do this via two theorems, the proofs of which can be found at the end of the section. The first one shows that the probability that the chain is in any state not positive recurrent tends to zero as time progresses:

(1) THEOREM. *For any state x that is not positive recurrent,*

$$\lim_{n \rightarrow \infty} p_n(x) = 0.$$

The intuition here is that if x is transient or null recurrent, visits to x eventually become so rare that not only does the fraction of time any one path spends in x approaches zero (Section 16: $\epsilon_N(x) \rightarrow 0$ as $N \rightarrow \infty$, $\mathbb{P}_\gamma$ -almost surely) but so does the fraction $p_n(x)$ of the entire ensemble of paths. Be careful here: if x is null recurrent and, say, the chain starts at x , then every path will revisit x infinitely many times (Theorem 14.2), however the frequency of these visits drops quickly enough that the fraction of paths in x at any given moment decays to zero.

The second theorem builds on Theorem 1 to show that, if the chain is aperiodic, then the time-varying law converges to a weighted combination of the ergodic distributions (c.f. Section 17), where the weight awarded to each one is the probability that the chain gets absorbed in the closed communicating class associated with it:

(2) THEOREM (The limits of the time-varying law). *Let $\{\mathcal{C}_i : i \in \mathcal{I}\}$ denote the collection of positive recurrent closed communicating classes and let π_i be the ergodic distribution of $\mathcal{C}_i$ for each i in $\mathcal{I}$.*

(i) *If the chain is aperiodic, then time-varying law $(p_n)_{n \in \mathbb{N}}$ (4.2) converges pointwise with limit*

$$(3) \quad \lim_{n \rightarrow \infty} p_n = \sum_{i \in \mathcal{I}} \mathbb{P}_\gamma(\{\phi_{\mathcal{C}_i} < \infty\}) \pi_i =: \pi_\gamma,$$

where $\phi_{\mathcal{C}_i}$ denotes the first entrance time to $\mathcal{C}_i$ (Definition 13.1).

- (ii) Conversely, if the chain is periodic there exists at least one initial distribution such that the time-varying law does not converge pointwise.

Theorem 2(ii) shows that the time-varying law does not converge if the chain is periodic. However, by sampling every d steps a periodic chain with period d , we obtain an aperiodic chain (Exercise 19.10). Applying Theorem 2(i) to the sampled chain, we obtain the following generalisation:

(4) EXERCISE (The limits of the time-varying law: the periodic case). Suppose that the period d (19.8) of the chain is finite. Let $X^{0,d}$ be the discrete-time chain with one-step matrix P_d and initial distribution γ obtained by sampling X every d steps (19.9). Show that a state y is accessible from another state x for $X^{0,d}$ only if it is accessible for X . Conclude that each positive recurrent class of $X^{0,d}$ is contained in one of X and label the positive recurrent classes $\{\mathcal{C}_{i,j}^d : i \in \mathcal{I}, j \in \mathcal{J}_i\}$ of $X^{0,d}$ such that

$$\mathcal{C}_{i,j}^d \subseteq \mathcal{C}_i \quad \forall j \in \mathcal{J}_i, \quad i \in \mathcal{I},$$

where $\{\mathcal{C}_i : i \in \mathcal{I}\}$ denotes the set of positive recurrent classes of X . Given that $X^{0,d}$ and X share the same positive recurrent states (Exercise 19.11), argue that

$$(5) \quad \bigcup_{j \in \mathcal{J}_i} \mathcal{C}_{i,j}^d = \mathcal{C}_i \quad \forall i \in \mathcal{I}.$$

Next, tweaking the arguments in Exercise 19.11, show that

$$d\mathbb{E}_x \left[\phi_x^{0,d} \right] = \mathbb{E}_x \left[\phi_x^{d/d(x)} \right] \quad \forall x \notin \mathcal{T},$$

where $d(x)$ denotes the period (19.1) of a state x , ϕ_x^k denotes to the k th entrance time (13.1) to x of X , $\phi_x^{0,d}$ the first entrance time (19.12) to x of $X^{0,d}$, and $\mathcal{T}$ the set of states that are not positive recurrent (for both X and $X^{0,d}$). Apply the regenerative property (15.2) of X to the above and obtain that

$$\mathbb{E}_x \left[\phi_x^{0,d} \right] = \frac{\mathbb{E}_x [\phi_x]}{d(x)} \quad \forall x \notin \mathcal{T}.$$

Using the characterisation of ergodic distributions in terms of mean return times (Theorem 17.10(i)) and the above, conclude that

$$(6) \quad \pi_{i,j}^d(x) = d_i 1_{\mathcal{C}_{i,j}^d}(x) \pi_i(x) \quad \forall x \in \mathcal{S}, \quad j \in \mathcal{J}_i, \quad i \in \mathcal{I},$$

where d_i denotes the period of the class $\mathcal{C}_i$ (Definition 19.6), $\pi_{i,j}^d$ the ergodic distribution of $X^{0,d}$ associated with $\mathcal{C}_{i,j}^d$, and π_i the ergodic distribution of X associated with $\mathcal{C}_i$. Summing (6) over x in $\mathcal{C}_{i,j}^d$, show that

$$(7) \quad \pi_i(\mathcal{C}_{i,j}^d) = \frac{1}{d_i} \quad \forall j \in \mathcal{J}_i, \quad i \in \mathcal{I}.$$

Using Exercise 19.10, argue that, for any given m , the process $X^{m,d}$ obtained by sampling X every d steps starting from m is an aperiodic discrete-time chain possessing the same positive recurrent closed communicating classes and ergodic distributions as $X^{0,d}$ does. Using this fact, Theorem (2)(i), and (5–6), derive the following generalisation of (3):

$$(8) \quad \lim_{n \rightarrow \infty} p_{m+nd} = \sum_{i \in \mathcal{I}} \left(\sum_{j \in \mathcal{J}_i} \mathbb{P}_\gamma \left(\left\{ \phi_{\mathcal{C}_{i,j}^d}^{m,d} < \infty \right\} \right) 1_{\mathcal{C}_{i,j}^d} \right) d_i \pi_i =: \pi_\gamma^m,$$

for all $m < d$, where $\phi_A^{m,d}$ denotes the first entrance time (19.12) of $X^{m,d}$ to a set A and the convergence is pointwise.

Use the closedness of $\mathcal{R}_+ = \cup_{i \in \mathcal{I}} \mathcal{C}_i$ for X to argue that, with probability one, $X^{m,d}$ eventually enters $\mathcal{R}_+$ if and only if X does (here, use Proposition 13.9 and the strong Markov property). Consequently, apply Proposition 13.11 and (6)–(7) to argue that

$$(9) \quad \pi_\gamma^m(\mathcal{S}) = \mathbb{P}_\gamma \left(\left\{ \phi_{\mathcal{R}_+}^{m,d} < \infty \right\} \right) = \mathbb{P}_\gamma \left(\left\{ \phi_{\mathcal{R}_+} < \infty \right\} \right).$$

Convergence in total variation and tightness. The question ‘under what circumstances does (3) hold in total variation?’ has a straightforward answer involving the notion of *tightness*³:

(10) **DEFINITION (Tightness).** A sequence $(\rho_n)_{n \in \mathbb{N}}$ of probability distributions on $\mathcal{S}$ is tight if and only if for every $\varepsilon > 0$ there exists a finite set F such that $\rho_n(F) \geq 1 - \varepsilon$ for all n in $\mathbb{N}$.

We then have the following corollary of Theorem 2:

(11) **COROLLARY.** Suppose that the period d in (19.8) of the chain is finite and let $\mathcal{R}_+$ be the set of positive recurrent states. The following are equivalent:

- (i) The chain enters $\mathcal{R}_+$ with probability one: $\mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}) = 1$.
- (ii) The time-varying law $(p_n)_{n \in \mathbb{N}}$ is tight.
- (iii) (Aperiodic case) The limit (3) holds in total variation: with $\|\cdot\|$ as in (18.4),

$$(12) \quad \lim_{n \rightarrow \infty} \|p_n - \pi_\gamma\| = 0.$$

Proof. To not turn this proof into a notational horror show, we only consider the aperiodic case. To argue the equivalence of (i) and (ii) for the periodic case, use (8) and (9) analogously to how we use (3) and (13) below.

(i) $\Leftrightarrow$ (iii) Given that $\mathcal{R}_+$ equals the union $\cup_{i \in \mathcal{I}} \mathcal{C}_i$ of the positive recurrent classes and that, because the classes are closed sets, the probability of both $\{\phi_{\mathcal{C}_i} < \infty\}$ and $\{\phi_{\mathcal{C}_j} < \infty\}$ occurring is zero if $i \neq j$ (Proposition 13.11), the mass of the limit in (3) equals the probability that the chain enters the set of positive recurrent states:

$$(13) \quad \begin{aligned} \pi_\gamma(\mathcal{S}) &= \sum_{i \in \mathcal{I}} \mathbb{P}_\gamma(\{\phi_{\mathcal{C}_i} < \infty\}) = \mathbb{P}_\gamma \left(\bigcup_{i \in \mathcal{I}} \{\phi_{\mathcal{C}_i} < \infty\} \right) = \mathbb{P}_\gamma(\{\phi_{\cup_{i \in \mathcal{I}} \mathcal{C}_i} < \infty\}) \\ &= \mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}). \end{aligned}$$

Scheffé’s lemma (Lemma 18.5) then implies that (i) holds if and only if (iii) does.

(i) $\Leftrightarrow$ (ii) Equations (3) and (13) imply that

$$\lim_{n \rightarrow \infty} p_n(F) = \pi_\gamma(F) \leq \pi_\gamma(\mathcal{S}) = \mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\})$$

for any finite set F and it follows that the time-varying law is tight only if $\mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}) = 1$.

For the converse, fix any $\varepsilon > 0$ and suppose that $\mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}) = 1$. Given that the limit (3) holds in total variation (as we have just shown), we can find an N such that

$$\|p_n - \pi_\gamma\| \leq \frac{\varepsilon}{2} \quad \forall n > N.$$

Using (13), pick a finite set F large enough that

$$p_n(F) \geq 1 - \varepsilon \quad \forall n \leq N, \quad \pi_\gamma(F) \geq 1 - \frac{\varepsilon}{2}.$$

The above inequalities then imply that

$$p_n(F) \geq 1_{\{n \leq N\}} p_n(F) + 1_{\{n > N\}} p_n(F) \geq 1_{\{n \leq N\}} (1 - \varepsilon) + 1_{\{n > N\}} (\pi_\gamma(F) - \|p_n - \pi_\gamma\|) \geq 1 - \varepsilon,$$

for all natural numbers n . Because the ε was arbitrary, we have that $(p_n)_{n \in \mathbb{N}}$ is tight. $\square$

³Here, we are topologising the state space using the discrete metric so that the compact sets are the finite sets.

Positive Tweedie recurrence revisited. In Section 18, we characterised positive Tweedie recurrent chains (Definition 18.8) in terms of the limits of the time-varying law ϵ_N in (16.1). Armed with Corollary 11, it is straightforward to improve this characterisation:

(14) COROLLARY (Characterising positive Tweedie recurrent chains). *Let d in (19.8) denote the period of the chain. The following are equivalent:*

- (i) *The chain is positive Tweedie recurrent.*
- (ii) *The empirical distribution converges in total variation to ϵ_∞ in (18.2) with $\mathbb{P}_\gamma$ -probability one, for all initial distributions γ .*
- (iii) *(Finite d case) The time-varying law $(p_n)_{n \in \mathbb{N}}$ is tight, for all initial distributions γ .*
- (iv) *(Aperiodic case) The time-varying law converges in total variation (i.e. (12)), for all initial distributions γ .*

Proof. This follows directly from Corollaries 18.3 and 11 and the definition of positive Tweedie recurrence in (18.8). $\square$

Corollary 14 shows that a chain with finite period is positive Tweedie recurrent if and only if for each initial distribution γ and ε in $(0, 1]$, we can find a large enough finite set F such that the chain has at least $1 - \varepsilon$ probability of being in F at any given time-step (i.e., $p_n(F) \geq 1 - \varepsilon$ for all $n \geq 0$): further reinforcing the view that a chain is stable if and only if it is positive Tweedie recurrent.

Ergodicity and positive Harris recurrence. In the case of an aperiodic positive Harris recurrent chain (Definition 18.8), Theorem 17.10 and Corollary 14 show that the chain has a unique stationary distribution π and that, regardless of the initial distribution γ , both the time-varying law (4.1) and the empirical distribution (16.1) converge to it:

$$\lim_{N \rightarrow \infty} \epsilon_N = \lim_{n \rightarrow \infty} p_n = \pi \quad \mathbb{P}_\gamma\text{-almost surely,}$$

where the convergence is in total variation. That is, the *time averages* ϵ_N converge to the *space averages* p_n and the chain is said to be *ergodic*.

The coupling inequality. To prove Theorems 1 and 2, we use a technique called *coupling*. It involves two sequences of random variables $X = (X_n)_{n \in \mathbb{N}}$ and $X' = (X'_n)_{n \in \mathbb{N}}$, defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$, for which there exists a random time $\sigma_c : \Omega \rightarrow \mathbb{N}_E$ such that X and X' are identical from σ_c onwards:

$$X_n(\omega) = X'_n(\omega) \quad \text{if } n \geq \sigma_c(\omega).$$

The time σ_c is said to be a *coupling time* of X and X' . The key result here is the celebrated *coupling inequality*:

(15) THEOREM (The coupling inequality). *If σ_c denotes a coupling time of X and X' , p_n denotes the time-varying law of X , and p'_n that of X' , then*

$$\|p_n - p'_n\| \leq \mathbb{P}(\{\sigma_c > n\}) \quad \forall n \in \mathbb{N}.$$

Proof. For any $A \subseteq \mathcal{S}$ and $n \in \mathbb{N}$,

$$\begin{aligned} |\mathbb{P}(\{X_n \in A\}) - \mathbb{P}(\{X'_n \in A\})| &\leq \mathbb{E}[|1_{\{X_n \in A\}} - 1_{\{X'_n \in A\}}|] \leq \mathbb{E}[1_{\{X_n \neq X'_n\}}] \leq \mathbb{E}[1_{\{\sigma_c > n\}}] \\ &= \mathbb{P}(\{\sigma_c > n\}). \end{aligned}$$

Taking the supremum over $A \subseteq \mathcal{S}$ in the above completes the proof. $\square$

Coupling for the proofs of Theorems 1 and 2. In the proofs of Theorems 1 and 2 that follow, we use two independently generated copies of the chain, X^1 and X^2 , with initial distributions γ_1 and γ_2 (respectively). We construct these copies by running Algorithm 1 with the same one-step matrix P but using two independent sets of random variables $X_0^1, U_1^1, U_2^1, \dots$ and $X_0^2, U_1^2, U_2^2, \dots$ such that $X_0^1 \sim \gamma_1$ and $X_0^2 \sim \gamma_2$ and we denote the underlying probability space by $(\Omega, \mathcal{F}, \mathbb{P})$. Additionally, we set σ_c be the first time these two chains coincide,

$$\sigma_c := \inf\{n \geq 0 : X_n^1 = X_n^2\},$$

and set X to be X^1 and X' to be

$$X'_n(\omega) = \begin{cases} X_n^2(\omega) & \text{if } n < \sigma_c(\omega) \\ X_n^1(\omega) & \text{if } n \geq \sigma_c(\omega) \end{cases} \quad \forall \omega \in \Omega, \quad n \in \mathbb{N}.$$

The independence of X^1 and X^2 imply that X' is also Markovian:

(16) EXERCISE. *Following steps analogous to those taken in Exercise 19.10, show that X' is a Markov chain with initial distribution γ_2 and one-step matrix P .*

Because the definition of X' implies that σ_c is a coupling time for X and X' , the coupling inequality (15) shows that

$$(17) \quad \|\gamma_1 P_n - \gamma_2 P_n\| \leq \mathbb{P}(\{\sigma_c > n\}) \quad \forall n \in \mathbb{N}.$$

To use the above to prove Theorems 1–2, we need to argue that the coupling time is almost surely finite. To this end, we use the *product chain* $\tilde{X} := (X^1, X^2)$:

(18) EXERCISE. *Following steps analogous to those taken in Exercise 19.10, show that $\tilde{X} := (X^1, X^2)$ is a Markov chain taking values in $\mathcal{S}^2$ with initial distribution*

$$\tilde{\gamma}((x_1, x_2)) := \gamma_1(x_1)\gamma_2(x_2) \quad \forall x_1, x_2 \in \mathcal{S}.$$

and one-step matrix

$$(19) \quad \tilde{p}((x_1, x_2), (y_1, y_2)) := p(x_1, y_1)p(x_2, y_2) \quad \forall x_1, x_2, y_1, y_2 \in \mathcal{S}.$$

The coupling time σ_c of X and X' is the point in time that the product chain first enters the diagonal $\{(x, x) : x \in \mathcal{S}\}$ of its state space $\mathcal{S}^2$. For this reason, σ_c is bounded above by the product chain's entrance time $\phi_{(x,x)}$ to any given state (x, x) on the diagonal:

$$(20) \quad \sigma_c \leq \phi_{(x,x)} := \inf\{n > 0 : \tilde{X}_n = (x, x)\} \quad \forall x \in \mathcal{S}.$$

In summary, our ability to prove that $\gamma_1 P_n$ and $\gamma_2 P_n$ converge to each other hinges on whether we are able to show that the time it takes the product chain to enter any given state (x, x) on the diagonal is almost surely finite if its starting location is sampled from $\gamma_1 \gamma_2$. This is where aperiodicity comes into play:

(21) LEMMA. *Let $\mathcal{C}$ be an aperiodic closed communicating class $\mathcal{C}$. Then, $\mathcal{C}^2$ is a closed communicating class of the product chain $\tilde{X} := (X^1, X^2)$. If, additionally, $\mathcal{C}^2$ is recurrent (for the product chain) and the supports of γ_1 and γ_2 are contained in $\mathcal{C}$, then σ_c is almost surely finite.*

Proof. Iterating (19), we find that

$$(22) \quad \tilde{p}_n((x_1, x_2), (y_1, y_2)) := p_n(x_1, y_1)p_n(x_2, y_2) \quad \forall x_1, x_2, y_1, y_2 \in \mathcal{S},$$

where $(\tilde{p}_n(x, y))_{x, y \in \mathcal{S}^2}$ denotes the n -step matrix of the product chain (defined analogously to $(p_n(x, y))_{x, y \in \mathcal{S}}$ in (4.5)). Because $p_n(x, y) = 0$ for all $n \geq 0$ if and only if y is not accessible from x (use (4.5) and (13.6) to argue this), the above implies that $\mathcal{C}^2$ is closed (for $\tilde{X}$).

Now, suppose that x_1, x_2, y_1, y_2 belong to $\mathcal{C}$. Because $\mathcal{C}$ is aperiodic, there exists an $n(x_1)$ such that $p_n(x_1, x_1) > 0$ for all n greater than $n(x_1)$ (Proposition 19.3). Because y_2 is accessible from x_2 (for X), we can find an n larger than $n(x_1)$ such that $p_n(x_2, y_2) > 0$ (once again, use (4.5) and (13.6)). Thus, (22) shows that (x_1, y_2) is accessible from (x_1, x_2) . Repeating the same argument shows (y_1, y_2) is accessible from (x_1, y_2) and it follows that (y_1, y_2) is accessible from (x_1, x_2) . Because x_1, x_2, y_1, y_2 were arbitrary, we have that $\mathcal{C}^2$ is a communicating class as desired.

Finiteness of σ_c for the recurrent case then follows by setting x in (20) to be a state in $\mathcal{C}$, applying (1.2), and recalling that $\mathbb{P}_x(\{\phi_y < \infty\}) = 1$ if x and y are recurrent states belonging to the same class of any chain (Theorem 14.6). $\square$

A proof of Theorem 1. We do this proof in four steps, beginning by swiftly dispatching with the case of a transient state:

Step 1: x is transient. In this case, Corollary 14.9 shows that

$$\sum_{n=1}^{\infty} p_n(x) < \infty$$

and it follows that $p_n(x)$ tends to zero as n approaches to infinity. $\square$

The null recurrent states take a bit more work. For the time being we focus on the case that the chain starts at the state in question and that the state is aperiodic:

Step 2: x is null recurrent and aperiodic and the chain starts at x . Let $\mathcal{C}$ be the aperiodic closed communicating class that x belongs to (Theorems 14.6 and 19.5) and recall that $\mathcal{C}^2$ is a closed communicating class of the product chain $\tilde{X}$ (Lemma 21). Setting $x_1 = x_2 = y_1 = y_2 = x$ in (22) we have that

$$p_n(x, x)^2 = \tilde{p}_n((x, x), (x, x)) \quad \forall n \in \mathbb{N}.$$

For this reason, applying Step 1 to $\tilde{X}$, we find that $p_n(x, x) \rightarrow 0$ as $n \rightarrow \infty$ if $\mathcal{C}^2$ is transient.

If $\mathcal{C}^2$ instead is recurrent, then the coupling inequality (17) and Lemma 21 show that

$$(23) \quad \lim_{n \rightarrow \infty} \|\gamma_1 P_n - \gamma_2 P_n\| = 0,$$

where $\gamma_1 := 1_x$ and $\gamma_2 := \gamma_1 P$ (to apply the lemma, note that γ_2 has support in $\mathcal{C}$ because $\mathcal{C}$ is closed for X).

For $p_n(x, x)$ not to converge to zero, there must exist a strictly increasing sequence $(n_k)_{k \in \mathbb{N}}$ and a constant $c > 0$ such that

$$\lim_{k \rightarrow \infty} p_{n_k}(x, x) = c.$$

Using a routine diagonal argument (see the end of the step), we find a subsequence $(n_l)_{l \in \mathbb{N}}$ of $(n_k)_{k \in \mathbb{N}}$ such that, for all y in $\mathcal{S}$, $p_{n_l}(x, y)$ converges as l tends to infinity; we denote the limit $\rho(y)$. The non-negativeness of $p_{n_l}(x, y)$ ensures that $\rho(y)$ is non-negative for each y in $\mathcal{S}$, while Fatou's lemma shows that

$$(24) \quad \sum_{y \in \mathcal{S}} \rho(y) \leq \lim_{l \rightarrow \infty} \sum_{y \in \mathcal{S}} p_{n_l}(x, y) = 1.$$

However, (23) shows that $p_{n_l+1}(x, y)$ also converges to $\rho(y)$ as n tends to infinity, for all y in $\mathcal{S}$. Calling on Fatou's lemma once again, we have that

$$\rho(y) = \lim_{l \rightarrow \infty} p_{n_l+1}(y) = \lim_{l \rightarrow \infty} \left(\sum_{x' \in \mathcal{S}} p_{n_l}(x, x') p(x', y) \right) \geq \sum_{x' \in \mathcal{S}} \rho(x') p(x', y) \quad \forall y \in \mathcal{S}.$$

Because $\rho(x) = c$, iteratively applying the above inequality and Tonelli's theorem, we find that

$$\begin{aligned}
\sum_{y \in \mathcal{S}} \frac{\rho(y)}{c} &= 1 + \sum_{z_1 \neq x} \frac{\rho(z_1)}{c} \geq 1 + \sum_{z_1 \neq x} \sum_{z_2 \in \mathcal{S}} \frac{\rho(z_2)}{c} p(z_2, z_1) \\
&= 1 + \sum_{z_1 \neq x} p(x, z_1) + \sum_{z_1 \neq x} \sum_{z_2 \neq x} \frac{\rho(z_2)}{c} p(z_2, z_1) \\
&\geq \cdots \geq 1 + \sum_{n=2}^m \left(\sum_{z_{n-1} \neq x} \sum_{z_{n-2} \neq x} \cdots \sum_{z_1 \neq x} p(x, z_{n-1}) p(z_{n-1}, z_{n-2}) \cdots p(z_2, z_1) \right. \\
&\quad \left. + \sum_{z_n \neq x} \cdots \sum_{z_1 \neq x} \frac{\rho(z_n)}{c} p(z_n, z_{n-1}) \cdots p(z_2, z_1) \right) \\
&\geq 1 + \sum_{n=2}^m \sum_{z_{n-1} \neq x} \cdots \sum_{z_1 \neq x} p(x, z_{n-1}) p(z_{n-1}, z_{n-2}) \cdots p(z_2, z_1) \\
&= \mathbb{P}_x(\{\phi_x > 0\}) + \sum_{n=1}^{m-1} \mathbb{P}_x(\{\phi_x > n\}) \quad \forall m > 0.
\end{aligned}$$

Taking the limit $m \rightarrow \infty$, we find that

$$\sum_{y \in \mathcal{S}} \rho(y) \geq c \sum_{n=0}^{\infty} \mathbb{P}_x(\{\phi_x > n\}) = c \sum_{n=0}^{\infty} n \mathbb{P}_x(\{\phi_x = n\}) = c \mathbb{E}_x[\phi_x] = \infty.$$

The above contradicts (24) and so it must be the case that $p_n(x, x)$ converges to zero as n tends to infinity.

The diagonal argument: Label the elements of $\mathcal{S}$ as $x, x_0, x_1, \dots$, keeping x as before. Because the sequence $(p_{n_k}(x_0))_{k \in \mathbb{N}}$ is contained in the interval $[0, 1]$, the Bolzano-Weierstrass Theorem tells us that $(p_{n_k}(x_0))_{k \in \mathbb{N}}$ has a converging subsequence $(p_{n_{j_0}}(x_0))_{j_0 \in \mathbb{N}}$. Repeating the same argument for x_1 and $(p_{n_{j_0}})_{j_0 \in \mathbb{N}}$, we find a convergent subsequence $(p_{n_{j_1}}(x_1))_{j_1 \in \mathbb{N}}$ of $(p_{n_{j_0}}(x_1))_{j_0 \in \mathbb{N}}$, and so on. Letting

$$p_{n_l} := p_{n_{j_l}} \quad \forall l \in \mathbb{N},$$

we obtain our pointwise convergent sequence. □

The remainder of the proof is downhill. Next, we get rid of the aperiodicity assumption:

Step 3: x is null recurrent and the chain starts at x . Let d be x 's period and recall x is aperiodic and null recurrent for the chain $X^{m,d}$ (19.9) obtained by sampling X every d steps starting from step $m \in \{0, 1, \dots, d-1\}$ (Exercises 19.10–19.11). For this reason, Step 2 shows that

$$\lim_{n \rightarrow \infty} p_{m+nd}(x, x) = \mathbb{P}_x\left(\left\{X_n^{m,d} = x\right\}\right) = 0 \quad \forall m \in \{0, 1, \dots, d-1\}.$$

In other words, for each $\varepsilon > 0$ and m in $\{0, 1, \dots, d-1\}$, there exists an N_m such that $p_{m+nd}(x) \leq \varepsilon$ for all $n \geq N_m$. For this reason,

$$p_n(x) \leq \varepsilon \quad \forall n \geq \max\{N_0 d, 1 + N_1 d, \dots, d-1 + N_{d-1} d\}.$$

As ε was arbitrary, the result follows. □

We are nearly done: we just need to apply the strong Markov property and port Step 3 from the start-at- x case to the arbitrary-initial-distribution- γ case:

Step 4: x is null recurrent. For any positive n , X_n equals x only if ϕ_x is no greater than n . For this reason, applying Lemma 8.3, Exercise 13.4, and the strong Markov property (Theorem 12.4 with $\varsigma := k$, $Z := 1_{\{\phi_x=k\}}$, and $F((z_m)_{m \in \mathbb{N}}) := 1_x(z_{n-k})$), yields

$$\begin{aligned}
 p_n(x) &= \mathbb{P}_\gamma(\{\phi_x \leq n, X_n = x\}) = \sum_{k=1}^{\infty} 1_{\{k \leq n\}} \mathbb{P}_x(\{\phi_x = k, X_{\phi_x} = x, X_n = x\}) \\
 &= \sum_{k=1}^{\infty} 1_{\{k \leq n\}} \mathbb{P}_\gamma(\{\phi_x = k, X_{\phi_x} = x\}) \mathbb{P}_x(\{X_{n-k} = x\}) \\
 (25) \quad &= \sum_{k=1}^{\infty} 1_{\{k \leq n\}} \mathbb{P}_\gamma(\{\phi_x = k\}) p_{n-k}(x, x) \quad \forall n > 0.
 \end{aligned}$$

Given that $p_n(x, x) \rightarrow 0$ as $n \rightarrow \infty$ (Step 3) and that

$$\begin{aligned}
 1_{\{k \leq n\}} \mathbb{P}_\gamma(\{\phi_x = k\}) p_{n-k}(x, x) &\leq \mathbb{P}_\gamma(\{\phi_x = k\}) \quad \forall k > 0, \\
 \sum_{k=1}^{\infty} \mathbb{P}_\gamma(\{\phi_x = k\}) &= \mathbb{P}_\gamma(\{\phi_x < \infty\}) < \infty,
 \end{aligned}$$

taking the limit $n \rightarrow \infty$ in (25) and applying dominated convergence, we obtain the desired $p_n(x) \rightarrow 0$ as $n \rightarrow \infty$. $\square$

A proof of Theorem 2. We do this proof one part at a time:

(i) Given Theorem 1, we only need to argue here that the limit (3) limit holds for states x belonging to a positive recurrent class $\mathcal{C}_i$. Suppose that we are able to show that

$$(26) \quad \lim_{n \rightarrow \infty} p_n(x, x) = \pi_i(x).$$

Because the event $\{\phi_x < \infty\}$ coincides with $\{\phi_{\mathcal{C}_i} < \infty\}$ with $\mathbb{P}_\gamma$ -probability one (Corollary 14.8), taking the limit $n \rightarrow \infty$ in (25) and applying dominated convergence yields

$$\lim_{n \rightarrow \infty} p_n(x) = \left(\sum_{k=1}^{\infty} \mathbb{P}_\gamma(\{\phi_x = k\}) \right) \pi_i(x) = \mathbb{P}_\gamma(\{\phi_x < \infty\}) \pi_i(x) = \mathbb{P}_\gamma(\{\phi_{\mathcal{C}_i} < \infty\}) \pi_i(x).$$

Given that no other ergodic distribution has support on x (Theorem 17.10(i)), we may rewrite the above as (3).

To argue (26), set $\gamma_1 := 1_x$ and $\gamma_2 := \pi_i$ and recall that $\mathcal{C}_i^2$ is a closed communicating class of the product chain $\tilde{X}$ (Lemma 21). Because π_i is a stationary distribution, $\gamma_2 P_n = \pi_i$ for all natural numbers n (Theorem 17.4). Given that π_i has support contained in $\mathcal{C}_i$ (Theorem 17.10(i)), the coupling inequality (17) and Lemma 21 imply that (26) holds if $\mathcal{C}^2$ is recurrent. This follows easily as $(x_1, x_2) \mapsto \pi(x_1)\pi(x_2)$ is a stationary distribution of the product chain (use (17.4) and (19) to argue this) and Theorem 17.7 shows that a state is positive recurrent if and only if there exists a stationary distribution with support on the state.

(ii) Let x be any state in a periodic positive recurrent class $\mathcal{C}_i$ with period d so that

$$(27) \quad p_n(x, x) = 0 \quad \forall n \neq 0, d, 2d, \dots,$$

and initialise the chain at x (i.e., $\gamma := 1_x$). The state x is aperiodic and positive recurrent for the chain $X^{0,d}$ (19.9) obtained by sampling X every d steps (Exercises 19.10–19.11). Thus, (i) and Theorem 17.10(i) imply that $\mathbb{P}_x(\{X_n^{0,d} = x\})$ converges to a positive constant as n grows unbounded. Given that $X_n^{0,d} = X_{nd}$, it follows from (27) that $p_n(x, x)$ does not converge as n tends to infinity.

21. Kendall's theorem on geometric recurrence and convergence*. A matter that has been extensively studied is under what conditions the time-varying law converges geometrically fast with the numbers of steps. Key in answering this question are the *geometrically recurrent* states, that is, the states x whose return time distribution has light tails: $\mathbb{E}_x [\theta^{\phi_x}] < \infty$ for some real number $\theta > 1$. These states can alternatively be characterised in terms of the convergence of the n -step matrix:

(1) **THEOREM** (Kendall's Theorem, [Kendall, 1959](#)). *Suppose that the chain is aperiodic. Let x denote any state belonging to a positive recurrent class $\mathcal{C}$ and let π be the ergodic distribution associated with $\mathcal{C}$.*

- (i) $p_n(x, x)$ converges geometrically fast to $\pi(x)$: $|p_n(x, x) - \pi(x)| = \mathcal{O}(\kappa^{-n})$ for some $\kappa > 1$.
- (ii) x is geometrically recurrent.

The remainder of this section is dedicated to the proof of Kendall's theorem and to the open problem of its generalisation. We begin with the former.

The renewal equation. The proof of Theorem 1 builds on the celebrated *renewal equation* linking the time-varying distribution with the return and entrance times:

$$\begin{aligned}
 p_n(x) &= \sum_{k=1}^{\infty} \mathbb{P}_{\gamma}(\{\phi_x^k = n\}) = \sum_{k=1}^{\infty} \sum_{m=0}^{n-1} \mathbb{P}_{\gamma}(\{\phi_x^{k-1} = m, \phi_x^k = n\}) \\
 &= \mathbb{P}_{\gamma}(\{\phi_x = n\}) + (1 - 1_1(n)) \sum_{k=2}^{\infty} \sum_{m=1}^{n-1} \mathbb{P}_{\gamma}(\{\phi_x^{k-1} = m, \phi_x^k - \phi_x^{k-1} = n - m\}) \\
 &= \mathbb{P}_{\gamma}(\{\phi_x = n\}) + (1 - 1_1(n)) \sum_{k=2}^{\infty} \sum_{m=1}^{n-1} \mathbb{P}_{\gamma}(\{\phi_x^{k-1} = m\}) \mathbb{P}_x(\{\phi_x = n - m\}) \\
 (2) \quad &= \mathbb{P}_{\gamma}(\{\phi_x = n\}) + (1 - 1_1(n)) \sum_{m=1}^{n-1} p_m(x) \mathbb{P}_x(\{\phi_x = n - m\}) \quad \forall n > 0, \quad x \in \mathcal{S},
 \end{aligned}$$

where the penultimate equality follows from the strong Markov property:

(3) **EXERCISE.** *Setting $\varsigma := \phi^{k-1}$, $Z := 1_{\{\phi^{k-1}=n\}}$, and $F((z_l)_{l \in \mathbb{N}}) := 1_m(\inf\{l > 0 : \sum_{l'=1}^l 1_x(z_{l'}) = 1\})$ and applying Lemma 8.3, Theorem 12.4, Lemma 12.8(ii), and Exercise 13.4, show that*

$$\mathbb{P}_{\gamma}(\{\phi_x^k = n, \phi_x^{k+1} - \phi_x^k = m\}) = \mathbb{P}_{\gamma}(\{\phi_x^k = n\}) \mathbb{P}_x(\{\phi_x = m\}) \quad \forall n, m, k \geq 0.$$

Multiplying (2) by z^n and summing over all natural numbers n yields the z -transform version of the renewal equation:

$$(4) \quad G(z) = H_{\gamma}(z) + H_x(z)G(z) \quad \forall z \in D_1,$$

where $D_1 := \{z \in \mathbb{C} : |z| < 1\}$ denotes the open unit disk in the complex plane $\mathbb{C}$,

$$G(z) := \sum_{n=1}^{\infty} p_n(x) z^n, \quad H_{\gamma}(z) := \sum_{n=1}^{\infty} \mathbb{P}_{\gamma}(\{\phi_x = n\}) z^n, \quad H_x(z) := \sum_{n=1}^{\infty} \mathbb{P}_x(\{\phi_x = n\}) z^n, \quad \forall z \in D_1,$$

and we have used the equations

$$\begin{aligned}
 \sum_{n=2}^{\infty} \sum_{m=1}^{n-1} p_m(x) \mathbb{P}_x(\{\phi_x = n - m\}) z^n &= \sum_{m=1}^{\infty} p_m(x) z^m \sum_{n=m+1}^{\infty} \mathbb{P}_x(\{\phi_x = n - m\}) z^{n-m} \\
 &= \sum_{m=1}^{\infty} p_m(x) z^m \sum_{n=1}^{\infty} \mathbb{P}_x(\{\phi_x = n\}) z^n \quad \forall z \in D_1,
 \end{aligned}$$

A proof of Kendall's theorem. We do this proof in three steps:

(5) LEMMA (Step 1). *Suppose that the chain is aperiodic and let x be any positive recurrent state and π_γ be the limit of the time-varying law in (20.3). For any $\kappa > 1$, we have that $|p_n(x) - \pi_\gamma(x)| = \mathcal{O}(\kappa^{-n})$ if and only if G in (4) has an analytic extension on the disc $D_\kappa := \{z \in \mathbb{C} : |z| < \kappa\}$ except for a simple pole at $z = 1$.*

Proof. It is not difficult to check that $|p_n(x) - \pi_\gamma(x)| = \mathcal{O}(\kappa^{-n})$ if and only if

$$\sum_{n=0}^{\infty} |p_n(x) - \pi_\gamma(x)| r^n < \infty \quad \forall 1 < r < \kappa.$$

In this case, the triangle inequality yields

$$\sum_{n=2}^{\infty} |p_n(x) - p_{n-1}(x)| |z|^n \leq \sum_{n=0}^{\infty} |p_n(x) - \pi(x)| |z|^n + |z| \sum_{n=0}^{\infty} |\pi(x) - p_n(x)| |z|^n < \infty \quad \forall z \in D_\kappa,$$

and it follows that the function

$$(6) \quad F(z) := \sum_{n=2}^{\infty} (p_n(x) - p_{n-1}(x)) z^n + p_1(x) z \quad \forall z \in D_\kappa$$

is analytic. Because, by definition, G is analytic on D_1 and

$$(7) \quad F(z) = (1 - z)G(z),$$

for all z in D_1 , we can extend $z \mapsto (1 - z)G(z)$ analytically in D_κ . Because the above equation then holds for all z in D_κ , it follows that G has no singularities in D_κ except for a simple pole at $z = 1$.

Conversely, if, for some $\kappa > 1$, G is analytic on D_κ except for a simple pole at 1, then (7) allows us to extend F (defined by replacing D_κ in (6) with D_1) analytically in D_κ . It follows that the radius of convergence of the series in (6) is at least κ and, by the the equation holds if we replace “:=” with “=”. Because the chain is aperiodic, $p_n(x)$ tends to $\pi_\gamma(x)$ as n approaches infinity (Theorem 20.2(i)) which implies that

$$\begin{aligned} \sum_{n=0}^{\infty} |p_n(x) - \pi_\gamma(x)| r^n &= \sum_{n=0}^{\infty} \left| \sum_{m=n+1}^{\infty} (p_m(x) - p_{m-1}(x)) \right| r^n \leq \sum_{m=1}^{\infty} |p_m(x) - p_{m-1}(x)| \left(\sum_{n=0}^{m-1} r^n \right) \\ &= \sum_{m=1}^{\infty} |p_m(x) - p_{m-1}(x)| \frac{r^m - 1}{r - 1} \leq \frac{1}{r - 1} \sum_{m=1}^{\infty} |p_m(x) - p_{m-1}(x)| r^m \end{aligned}$$

for all $1 < r < \kappa$. Cauchy's inequality implies that the right-hand side is finite for all $1 < r < \kappa$, and it follows that $|p_n(x) - \pi_\gamma(x)|$ is $\mathcal{O}(\kappa^{-n})$. $\square$

Setting $\gamma := 1_x$ in Lemma 5 shows that $|p_n(x, x) - \pi(x)|$ is $\mathcal{O}(\kappa^{-n})$ if and only if

$$(8) \quad G(z) = \sum_{n=1}^{\infty} p_n(x, x) z^n$$

has an analytic extension on D_κ except for a simple pole at $z = 1$. To relate the existence of this extension to the tails of the return time distribution we proceed as follows:

(9) LEMMA (Step 2). *Suppose that the chain is aperiodic and that x is a geometrically recurrent state whose entrance time distribution, restricted to the event $\{\phi_x < \infty\}$, has light tails:*

$$\mathbb{E}_x [\theta^{\phi_x}] < \infty, \quad \mathbb{E}_\gamma [1_{\{\phi_x < \infty\}} \theta^{\phi_x}] < \infty,$$

for some $\theta > 1$. In this case, the function G in (4) is analytic on D_κ for some $\kappa > 1$ except for a simple pole at $z = 1$.

Proof. Because

$$\begin{aligned} \sum_{n=1}^{\infty} \mathbb{P}_{\gamma}(\{\phi_x = n\}) |z|^n &\leq \sum_{n=1}^{\infty} \mathbb{P}_{\gamma}(\{\phi_x = n\}) \theta^n = \mathbb{E}_{\gamma} \left[1_{\{\phi_x < \infty\}} \theta^{\phi_x} \right], \\ \sum_{n=1}^{\infty} \mathbb{P}_x(\{\phi_x = n\}) |z|^n &\leq \sum_{n=1}^{\infty} \mathbb{P}_x(\{\phi_x = n\}) \theta^n = \mathbb{E}_x \left[\theta^{\phi_x} \right], \quad \forall z \in D_{\theta} \end{aligned}$$

our premise implies that H_{γ} and H_x in (4) are analytic on D_{θ} .

Because $\mathbb{P}_x(\{\phi_x = n\}) \geq p_n(x, x)$ and the chain is aperiodic, Proposition 19.3 shows that $\mathbb{P}_x(\{\phi_x = n\}) > 0$ for all sufficiently large n . For this reason,

$$\operatorname{Re}(H_x(z)) = \sum_{n=1}^{\infty} \mathbb{P}_x(\{\phi_x = n\}) \operatorname{Re}(z^n) < \sum_{n=1}^{\infty} \mathbb{P}_x(\{\phi_x = n\}) = 1$$

for any z in the closed unit disc $\bar{D}_1 := \{z \in \mathbb{C} : |z| \leq 1\}$ except 1. Consequently, 1 is the only zero of $H_x(z) - 1$ in $\bar{D}_1$ and, thus, there exists a $1 < \kappa \leq \theta$ such that the only zero of $H_x(z) - 1$ in D_{κ} is $z = 1$ (here, use the Bolzano-Weierstrass theorem and the principle of permanence). This zero is simple because

$$\lim_{z \rightarrow 1} \frac{H_x(z) - 1}{z - 1} = \frac{dH_x}{dz}(1) = \sum_{n=1}^{\infty} n \mathbb{P}_x(\{\phi_x = n\}) = \mathbb{E}_x[\phi_x] \geq 1.$$

In other words, $1 - H_x(z) = (1 - z)K(z)$ for some function K that is analytic and non-zero on D_{κ} . The renewal equation (4) and (7) then show that

$$(10) \quad F(z) = \frac{G(z)(1 - H_x(z))}{K(z)} = \frac{H_{\gamma}(z)}{K(z)} \quad z \in D_1.$$

Because we have chosen κ to be no greater than θ and because H_{γ} and H_x are analytic on D_{θ} , it follows that K and F have analytic extensions on D_{κ} . It then follows from (7) that G has an analytic extension on D_{κ} . $\square$

Setting $\gamma := 1_x$ in Lemma 9, we find that Theorem 1(ii) holds only if there exists some $\kappa > 1$ such that G in (8) is analytic on D_{κ} , except for a simple pole at $z = 1$. Given Lemma 5, we need only argue the converse to finish off the proof of Kendall's theorem:

(11) LEMMA (Step 3). *Let x be a positive recurrent state. If, for some $\kappa > 1$, G in (8) has an analytic extension on D_{κ} , except for a simple pole at $z = 1$, then Theorem 1(ii) is satisfied.*

Proof. Fixing the initial distribution γ to be 1_x , the renewal equation (4) reduces to

$$G(z) = 1 + G(z)H_x(z) \quad \forall z \in D_1.$$

Multiplying both sides by $1 - z$ and applying (7) yields

$$F(z) = 1 - z + F(z)H_x(z) \quad \forall z \in D_1$$

and it follows that $F(z) \neq 0$ for all z belonging to the closed unit disc $\bar{D}_1$ aside from, perhaps, $z = 1$. However,

$$F(1) = \lim_{n \rightarrow \infty} p_n(x, x) = \pi(x)$$

which is non-zero unless $\pi(x) = 1$ (in which case the claim is trivial as x must be an absorbing state). From the Bolzano-Weierstrass theorem and the principle of permanence it then follows that the zero z_1 of F closest to the origin must have magnitude strictly greater than one (i.e., $|z_1| > 1$). Because F is analytic on D_{κ} and $H_x(z) = (F(z) - 1 + z)/F(z)$ for all z in $D_{|z_1|}$, H_x in (4) has an analytical extension on $D_{\kappa \wedge |z_1|}$. In particular,

$$\mathbb{E}_x \left[\theta^{\phi_x} \right] = \sum_{n=1}^{\infty} \mathbb{P}_x(\{\phi_x = n\}) \theta^n = H_x(\theta) < \infty \quad \forall 1 < \theta < \kappa \wedge |z_1|,$$

as desired. $\square$

An open question. Lemmas 5 and 9 show that for any given initial distribution γ and positive recurrent state x , $p_n(x)$ converges geometrically fast to its limit $\pi_\gamma(x)$ in (20.3),

$$(12) \quad |p_n(x) - \pi_\gamma(x)| \text{ is } \mathcal{O}(\kappa^{-n}) \text{ for some } \kappa > 1,$$

if x is geometrically recurrent and its entrance time distribution (restricted to the event $\{\phi_x < \infty\}$) has light tails:

$$\mathbb{E}_x [\theta^{\phi_x}] < \infty, \quad \mathbb{E}_\gamma [1_{\{\phi_x < \infty\}} \theta^{\phi_x}] < \infty \text{ for some } \theta > 1.$$

Trivially, if the chain never visits x (i.e., $\mathbb{P}_\gamma(\{\phi_x < \infty\}) = 0$), then $p_1(x) = p_2(x) = \dots = \pi_\gamma(x) = 0$ and the convergence of $p_n(x)$ is geometric. Putting these together and using our convention that $0 \cdot \infty = 0$, we have the following:

(13) THEOREM. *If the chain is aperiodic and x is a positive recurrent state satisfying*

$$(14) \quad \mathbb{E}_\gamma [1_{\{\phi_x < \infty\}} \theta^{\phi_x^2}] = \mathbb{E}_\gamma [1_{\{\phi_x < \infty\}} \theta^{\phi_x}] \mathbb{E}_x [\theta^{\phi_x}] < \infty$$

for some $\theta > 1$, then $p_n(x)$ converges geometrically fast to its limit $\pi_\gamma(x)$ (i.e. (12) holds for some $\kappa > 1$).

Proof. Given Lemmas 5 and 9, the theorem follows directly from the strong Markov property (in particular, (22.6) in the following section with $A := \{x\}$, $B := \emptyset$ and $k := l := 1$). $\square$

To the best of my knowledge, the question of whether or not (14) is not only sufficient but necessary for (12) to hold remains open. It seems to me that it should be: I have been unable to cook up a situation in which geometric convergence of $p_n(x)$ to a positive recurrent state x for an aperiodic chain would not imply geometric convergence of $p_n(x, x)$. Unless, of course, $\mathbb{P}_\gamma(\{\phi_x < \infty\}) = 0$, in which case the claim is trivial. Were $p_n(x, x)$ to converge geometrically fast, then Kendall's Theorem (Theorem 1) would show that x is geometrically recurrent and it would follow from (10) that H_γ has an analytic extension on a disk D_κ with $\kappa > 1$. In particular, we would have (14) as it would be the case that

$$\mathbb{E}_\gamma [1_{\{\phi_x < \infty\}} \theta^{\phi_x}] = H_\gamma(\theta) < \infty \quad \forall \theta < \kappa.$$

If you know the answer to this question, I'd love to hear about it.

22. Geometric trials arguments*. As we have seen throughout Sections 16–21, we are often interested in the time elapsed until the chain enters some set B (e.g. a particular state, a certain class, the union of the positive recurrent classes, etc.). The Foster-Lyapunov criteria in the following sections yield information on the chain's visits to some set F . However, more often than not F will not be the set we are actually interested in: $F \neq B$. Fortunately, if B is accessible from F (Definition 13.5) and F is finite, then the chance that a visit to F results in a visit to B may be bounded below by a constant independent of which state in F the chain visits. The existence of this constant implies that the probability that the chain has not yet visited B decays geometrically with the number of visits to F :

(1) LEMMA (Geometric trials property). *If F is finite and $F \rightarrow B$ for a second set B , then there exists constants m and $\varepsilon > 0$ independent of the initial distribution γ such that*

$$\mathbb{P}_\gamma(\{\phi_F^{nm} < \phi_B\}) \leq (1 - \varepsilon)^{n-1} \quad \forall n > 0,$$

where ϕ_B denotes the first entrance time to B and ϕ_B^k the k th entrance time to F (Definition 13.1).

Lemma 1 lays the foundation for the following three geometric-trials-type arguments that allow us to turn information on the chain's visits to F into information on its visits to B :

(2) LEMMA (Geometric trials arguments). *Given a finite subset F of the state space, suppose that $F \rightarrow B$ for some other subset B and let ϕ_F and ϕ_B denote the first entrance times to F and B (Definition 13.1).*

- (i) *If $\mathbb{P}_\gamma(\{\phi_F < \infty\}) = 1$ and $\mathbb{P}_x(\{\phi_F < \infty\}) = 1$ for all x in F , then $\mathbb{P}_\gamma(\{\phi_B < \infty\}) = 1$.*
- (ii) *If $\mathbb{E}_\gamma[\phi_F] < \infty$ and $\mathbb{E}_x[\phi_F] < \infty$ for all x in F , then $\mathbb{E}_\gamma[\phi_B] < \infty$.*
- (iii) *If there exists a constant $\theta > 1$ such that $\mathbb{E}_\gamma[\theta^{\phi_F}] < \infty$ and $\mathbb{E}_x[\theta^{\phi_F}] < \infty$ for all x in F , then there exists second constant $\vartheta > 1$ such that $\mathbb{E}_\gamma[\vartheta^{\phi_B}] < \infty$.*

Proofs of Lemmas 1 and 2. The key to these proofs are applications of the strong Markov property that express expectations involving the $(k + l)$ th entrance time to a set in terms of expectations involving the k th entry time and others involving the l th entrance time:

(3) LEMMA. *For any subsets A and B of the state space, constant $\vartheta > 1$, and natural numbers k and l ,*

$$(4) \quad \mathbb{P}_\gamma\left(\left\{\phi_A^{k+l} < \phi_B\right\}\right) = \sum_{x \in A} \mathbb{P}_\gamma\left(\left\{\phi_A^k < \phi_B, X_{\phi_A^k} = x\right\}\right) \mathbb{P}_x\left(\left\{\phi_A^l < \phi_B\right\}\right),$$

$$(5) \quad \mathbb{E}_\gamma\left[1_{\{\phi_A^k < \phi_B\}}(\phi_A^{k+l} - \phi_A^k)\right] = \sum_{x \in A} \mathbb{P}_\gamma\left(\left\{\phi_A^k < \phi_B, X_{\phi_A^k} = x\right\}\right) \mathbb{E}_x\left[\phi_A^l\right],$$

$$(6) \quad \mathbb{E}_\gamma\left[1_{\{\phi_A^k < \phi_B\}}\vartheta^{\phi_A^{k+l}}\right] = \sum_{x \in A} \mathbb{E}_\gamma\left[1_{\{\phi_A^k < \phi_B, X_{\phi_A^k} = x\}}\vartheta^{\phi_A^k}\right] \mathbb{E}_x\left[\vartheta^{\phi_A^l}\right],$$

where ϕ_B denotes the first entrance time to B and ϕ_A^k the k th entrance time to A (Definition 13.1).

Proof. To argue (4)–(6), we will show that

$$(7) \quad \mathbb{E}_\gamma\left[Z 1_{\{\phi_A^k < \phi_B\}} g(\phi_A^{k+l} - \phi_A^k, \phi_B - \phi_A^k)\right] = \sum_{x \in A} \mathbb{E}_\gamma\left[Z 1_{\{\phi_A^k < \phi_B, X_{\phi_A^k} = x\}}\right] \mathbb{E}_x\left[g(\phi_A^l, \phi_B)\right]$$

for any given non-negative function $g : \mathbb{N}_E^2 \rightarrow \mathbb{R}_E$ and non-negative $\mathcal{F}_{\phi_A^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable random variable Z . Setting $g(n, m) := 1_{\{0, \dots, m-1\}}(n)$ and $Z := 1$ in (7), we obtain (4). For (5), use $g(n, m) := n$ and $Z := 1$, while for (6), use $g(n, m) := \vartheta^n$ and $Z := \vartheta^{\phi_A^k}$.

To prove (7), note that X_n does not belong to B for $n = 1, 2, \dots, \phi_A^k$ if $\phi_A^k < \phi_B$, and so

$$(8) \quad \begin{aligned} \phi_B &= \inf \left\{ n > 0 : \sum_{m=1}^n 1_B(X_m) = 1 \right\} = \inf \left\{ n > \phi_A^k : \sum_{m=\phi_A^k+1}^n 1_B(X_m) = 1 \right\} \\ &= \phi_A^k + G_B^1(X_{\phi_A^k}) \quad \text{on} \quad \{\phi_A^k < \phi_B\}, \end{aligned}$$

where $X_{\phi_A^k}$ denotes the ϕ_A^k -shifted chain (12.3) and, for any subset S of $\mathcal{S}$ and $j > 0$, G_S^j denotes the $\mathcal{E}/2^{\mathbb{N}_E}$ -measurable (Lemma 12.8(ii)) function defined by

$$G_S^j(x) := \inf \left\{ n > 0 : \sum_{m=1}^n 1_S(x_m) = j \right\} \quad \forall x \in \mathcal{P}.$$

Similarly, because the number of entrances to A by time ϕ_A^k is exactly k ($\sum_{m=1}^{\phi_A^k} 1_A(X_m) = k$) and ϕ_A^{k+l} is greater than ϕ_A^k by definition, we have that

$$\begin{aligned} \phi_A^{k+l} &= \inf \left\{ n > 0 : \sum_{m=1}^n 1_A(X_m) = k+l \right\} = \inf \left\{ n > \phi_A^k : \sum_{m=\phi_A^k+1}^n 1_A(X_m) = l \right\} \\ (9) \quad &= \phi_A^k + G_A^l(X^{\phi_A^k}). \end{aligned}$$

The event $\{\phi_A^k < \phi_B\}$ belongs to the pre- ϕ_A^k sigma-algebra $\mathcal{F}_{\phi_A^k}$ (Definition 8.1) because we are able to deduce whether the chain has entered B once by the time of the k th entry to F if we observe the chain up until the latter moment (for a formal argument, use Lemma 8.3 and Exercise 13.4). For these reasons, we can apply the strong Markov property (Theorem 12.4) to obtain (7):

$$\begin{aligned} &\mathbb{E}_\gamma \left[Z1_{\{\phi_A^k < \phi_B\}} g(\phi_A^{k+l} - \phi_A^k, \phi_B - \phi_A^k) \right] \\ &= \sum_{x \in A} \mathbb{E}_\gamma \left[Z1_{\{\phi_A^k < \phi_B, \phi_A^k < \infty, X_{\phi_A^k} = x\}} g(\phi_A^{k+l} - \phi_A^k, \phi_B - \phi_A^k) \right] \\ &= \sum_{x \in A} \mathbb{E}_\gamma \left[Z1_{\{\phi_A^k < \phi_B, \phi_A^k < \infty, X_{\phi_A^k} = x\}} g(G_A^l(X^{\phi_A^k}), G_B^1(X^{\phi_A^k})) \right] \\ &= \sum_{x \in A} \mathbb{E}_\gamma \left[Z1_{\{\phi_A^k < \phi_B, \phi_A^k < \infty, X_{\phi_A^k} = x\}} \right] \mathbb{E}_x \left[g(G_A^l(X), G_B^1(X)) \right] \\ &= \sum_{x \in A} \mathbb{E}_\gamma \left[Z1_{\{\phi_A^k < \phi_B, X_{\phi_A^k} = x\}} \right] \mathbb{E}_x \left[g(\phi_A^l, \phi_B) \right], \end{aligned}$$

where the first equality follows from the fact that $\{\phi_A^k < \phi_B\}$ is contained in $\{\phi_A^k < \infty\}$ (as ϕ_A^k is strictly less than ϕ_B only if the ϕ_A^k is finite), the second from (8)–(9), the third from the strong Markov property, and the fourth from the definitions of ϕ_A^l and ϕ_B and the fact that $\{\phi_A^k < \phi_B\}$ is contained in $\{\phi_A^k < \infty\}$. $\square$

With the above out of the way, we turn our attention to the proofs of Lemmas 1 and 2:

Proof of Lemma 1. Pick any x in F and let y in B be such that $x \rightarrow y$. Because $\mathbb{P}_x(\{\phi_y = \infty\}) < 1$, there exists some $0 < m_x < \infty$ and $\varepsilon_x > 0$ such that

$$\mathbb{P}_x(\{n < \phi_B\}) \leq \mathbb{P}_x(\{n < \phi_y\}) \leq 1 - \varepsilon_x \quad \forall n \geq m_x.$$

Let $m := \max\{m_x : x \in F\}$ and $\varepsilon := \min\{\varepsilon_x : x \in F\}$. Finiteness of F ensures that $m < \infty$ and $\varepsilon > 0$, and it follows from the above that

$$\mathbb{P}_x(\{n < \phi_B\}) \leq 1 - \varepsilon \quad \forall n \geq m, \quad x \in F.$$

By definition, $\phi_F^n \geq n$ and the above implies that

$$\mathbb{P}_x(\{\phi_F^n < \phi_A\}) \leq 1 - \varepsilon \quad \forall n \geq m, \quad x \in F.$$

Setting $A := F$, $k := (n-1)m$, and $l := m$ in (4), the result follows by induction:

$$\begin{aligned} \mathbb{P}_\gamma(\{\phi_F^{nm} < \phi_B\}) &= \sum_{x \in F} \mathbb{P}_\gamma \left(\left\{ \phi_F^{(n-1)m} < \phi_B, X_{\phi_F^{(n-1)m}} = x \right\} \right) \mathbb{P}_x(\{\phi_F^m < \phi_B\}) \\ &\leq (1 - \varepsilon) \sum_{x \in F} \mathbb{P}_\gamma \left(\left\{ \phi_F^{(n-1)m} < \phi_B, X_{\phi_F^{(n-1)m}} = x \right\} \right) \\ &= (1 - \varepsilon) \mathbb{P}_\gamma \left(\left\{ \phi_F^{(n-1)m} < \phi_B \right\} \right) = \dots \leq (1 - \varepsilon)^{n-1} \mathbb{P}_\gamma(\{\phi_F^m < \phi_B\}) \\ &\leq (1 - \varepsilon)^{n-1}. \end{aligned}$$

$\square$

Proof of Lemma 2(i). Setting $\varsigma := \phi_F^{k-1}$ in Lemma 13.12 shows that the chain keeps visiting F :

$$\begin{aligned} \mathbb{P}_\gamma \left(\left\{ \phi_F^k < \infty \right\} \right) &= \sum_{x \in F} \mathbb{P}_\gamma \left(\left\{ \phi_F^{k-1} < \infty, X_{\phi_F^{k-1}} = x \right\} \right) \mathbb{P}_x \left(\left\{ \phi_F < \infty \right\} \right) \\ &= \sum_{x \in F} \mathbb{P}_\gamma \left(\left\{ \phi_F^{k-1} < \infty, X_{\phi_F^{k-1}} = x \right\} \right) = \mathbb{P}_\gamma \left(\left\{ \phi_F^{k-1} < \infty \right\} \right) \\ &= \cdots = \mathbb{P}_\gamma \left(\left\{ \phi_F < \infty \right\} \right) = 1 \quad \forall k > 0. \end{aligned}$$

For this reason, and letting m and $\varepsilon > 0$ be as in Lemma 1, we have that

$$\mathbb{P}_\gamma \left(\left\{ \phi_B = \infty \right\} \right) = \mathbb{P}_\gamma \left(\left\{ \phi_F^{nm} < \phi_B, \phi_B = \infty \right\} \right) \leq \mathbb{P}_\gamma \left(\left\{ \phi_F^{nm} < \phi_B \right\} \right) \leq (1 - \varepsilon)^{n-1} \quad \forall n > 0.$$

The result follows by taking the limit $n \rightarrow \infty$ in the above. $\square$

Proof of Lemma 2(ii). Let m and $\varepsilon > 0$ be as in Lemma 1. Because, by definition, $\phi_F^0 = 0$ and $\phi_F^{nm} \geq nm \rightarrow \infty$ as $n \rightarrow \infty$, we may re-write ϕ_B as

$$\phi_B = \sum_{n=0}^{\infty} 1_{\{\phi_B > \phi_F^{nm}\}} (\phi_B \wedge \phi_F^{(n+1)m} - \phi_F^{nm}) \leq \sum_{n=0}^{\infty} 1_{\{\phi_B > \phi_F^{nm}\}} (\phi_F^{(n+1)m} - \phi_F^{nm}).$$

For this reason,

$$\begin{aligned} \mathbb{E}_\gamma [\phi_B] &\leq \mathbb{E}_\gamma [\phi_F^m] + \sum_{n=1}^{\infty} \mathbb{P}_\gamma \left(\left\{ \phi_B > \phi_F^{nm} \right\} \right) \left(\max_{x \in F} \mathbb{E}_x [\phi_F^m] \right) \\ &\leq \mathbb{E}_\gamma [\phi_F^m] + \left(\max_{x \in F} \mathbb{E}_x [\phi_F^m] \right) \sum_{n=0}^{\infty} (1 - \varepsilon)^n \\ &\leq \mathbb{E}_\gamma [\phi_F] + (m - 1) \left(\max_{x \in F} \mathbb{E}_x [\phi_F] \right) + m \left(\max_{x \in F} \mathbb{E}_x [\phi_F] \right) \frac{1}{\varepsilon} < \infty, \end{aligned}$$

where the first and third inequalities follow from (5) and Tonelli's theorem and the second from the Lemma 1. $\square$

Proof of 2(iii). Let m and ε be as in Lemma 1 and define $M_\vartheta := \max_{x \in F} \mathbb{E}_x [\vartheta^{\phi_F}]$ for all $\vartheta > 1$. Setting $1 < \vartheta \leq \theta$ and $\delta_\vartheta := \log(\vartheta)/\log(\theta)$, Jensen's inequality implies that

$$M_\vartheta = \max_{x \in F} \mathbb{E}_x \left[(\theta^{\phi_F})^{\delta_\vartheta} \right] \leq M_\theta^{\delta_\vartheta} < \infty.$$

For this reason, M_ϑ tends to 1 as ϑ approaches 1 from above.

As shown in the proof of (i), the k th entrance time ϕ_F^k to F is finite $\mathbb{P}_\gamma$ -almost surely, for all $k > 0$. For this reason, setting $B := \emptyset$ (so that $\phi_B = \infty$) in (6), we find that

$$\begin{aligned} \mathbb{E}_\gamma \left[\vartheta^{\phi_F^k} \right] &= \sum_{x \in F} \mathbb{E}_\gamma \left[1_{\{\phi_F^{k-1} < \infty, X_{\phi_F^{k-1}} = x\}} \vartheta^{\phi_F^{k-1}} \right] \mathbb{E}_x \left[\vartheta^{\phi_F} \right] \leq M_\vartheta \mathbb{E}_\gamma \left[\vartheta^{\phi_F^{k-1}} \right] \\ (10) \quad &\leq \cdots \leq M_\vartheta^{k-1} \mathbb{E}_\gamma \left[\vartheta^{\phi_F} \right] \quad \forall k > 0. \end{aligned}$$

Because $\phi_F^n \geq n$ by definition, $\phi_F^n \rightarrow \infty$ as $n \rightarrow \infty$ and it follows that

$$(11) \quad \vartheta^{\phi_B} = \sum_{n=0}^{\infty} 1_{\{\phi_F^{nm} < \phi_B\}} \left(\vartheta^{\phi_B \wedge \phi_F^{(n+1)m}} - \vartheta^{\phi_F^{nm}} \right) \leq \sum_{n=0}^{\infty} 1_{\{\phi_F^{nm} < \phi_B\}} \left(\vartheta^{\phi_F^{(n+1)m}} - \vartheta^{\phi_F^{nm}} \right).$$

However, putting (6) and (10) together, we have that

$$\begin{aligned}
 (12) \quad \mathbb{E}_\gamma \left[1_{\{\phi_F^{nm} < \phi_B\}} \left(\vartheta^{\phi_F^{(n+1)m}} - \vartheta^{\phi_F^{nm}} \right) \right] &= \sum_{x \in F} \mathbb{E}_\gamma \left[1_{\{\phi_F^{nm} < \phi_B, X_{\phi_F^{nm}} = x\}} \vartheta^{\phi_F^{nm}} \right] \mathbb{E}_x \left[\vartheta^{\phi_F^m} - 1 \right] \\
 &\leq \left(\max_{x \in F} \mathbb{E}_x \left[\vartheta^{\phi_F^m} \right] - 1 \right) \mathbb{E}_\gamma \left[1_{\{\phi_F^{nm} < \phi_B\}} \vartheta^{\phi_F^{nm}} \right] \\
 &\leq (M_\vartheta^m - 1) \mathbb{E}_\gamma \left[1_{\{\phi_F^{nm} < \phi_B\}} \vartheta^{\phi_F^{nm}} \right] \quad \forall n \geq 0.
 \end{aligned}$$

It then follows from (11)–(12) and Tonelli's theorem that

$$\mathbb{E}_\gamma \left[\vartheta^{\phi_B} \right] \leq (M_\vartheta^m - 1) \sum_{n=0}^{\infty} \mathbb{E}_\gamma \left[1_{\{\phi_F^{nm} < \phi_B\}} \vartheta^{\phi_F^{nm}} \right].$$

To bound the sum we use the inequality $ab \leq \alpha^n a^2 + \alpha^{-n} b^2$, for any non-negative a, b, n , and $\alpha > 0$. In particular, we have that

$$\begin{aligned}
 \sum_{n=0}^{\infty} \mathbb{E}_\gamma \left[1_{\{\phi_F^{nm} < \phi_B\}} \vartheta^{\phi_F^{nm}} \right] &\leq \sum_{n=0}^{\infty} \alpha^n \mathbb{E}_\gamma \left[\vartheta^{2\phi_F^{nm}} \right] + \sum_{n=0}^{\infty} \alpha^{-n} \mathbb{P}_\gamma (\{\phi_F^{nm} < \phi_B\}) \\
 &\leq \mathbb{E}_\gamma \left[\vartheta^{2\phi_F} \right] \sum_{n=0}^{\infty} (\alpha M_{\vartheta^2}^m)^n + \sum_{n=0}^{\infty} \alpha^{-n} \mathbb{P}_\gamma (\{\phi_F^{nm} < \phi_B\}),
 \end{aligned}$$

where the second inequality follows from (10). Lemma 1 then shows that

$$\sum_{n=0}^{\infty} \mathbb{E}_\gamma \left[1_{\{\phi_F^{nm} < \phi_B\}} \vartheta^{\phi_F^{nm}} \right] \leq \mathbb{E}_\gamma \left[\vartheta^{2\phi_F} \right] \sum_{n=0}^{\infty} (\alpha M_{\vartheta^2}^m)^n + 1 + \frac{1}{\alpha - 1 + \varepsilon},$$

for any $1 - \varepsilon < \alpha < 1$. Choosing an $\vartheta \leq \sqrt{\theta}$ sufficiently close to 1 so that $M_{\vartheta^2}^m < 1/\alpha$ makes the right-hand side finite as desired. $\square$

23. Foster-Lyapunov criteria I: the criterion for recurrence*. In this section, we derive the criterion for recurrence: Theorem 2 below. It involves a *norm-like* function meaning a $\mathbb{R}_E$ -valued function v on $\mathcal{S}$ that tends to infinity as its argument tends to infinity:

(1) DEFINITION (Norm-like function). *A function $v : \mathcal{S} \rightarrow \mathbb{R}_E$ is norm-like if its sub-level sets are finite:*

$$\mathcal{S}_r := \{x \in \mathcal{S} : v(x) < r\} \quad \text{is finite for all } r \in \mathbb{R}.$$

These types of functions are also sometimes called *inf-compact*. A useful fact is that if ρ is a probability distribution on $\mathcal{S}$ and v is norm-like, then the expectation $\rho(v)$ is well-defined as, in this case,

$$\rho(v \wedge 0) = \sum_{x \in \mathcal{S}_0} v(x) \rho(x) \geq \left(\min_{x \in \mathcal{S}_0} v(x) \right) \rho(\mathcal{S}_0) \geq \min_{x \in \mathcal{S}_0} v(x) > -\infty.$$

The criterion goes as follows:

(2) THEOREM (The criterion for recurrence). *If there exists a real-valued norm-like function v on $\mathcal{S}$ and a finite set F such that*

$$(3) \quad Pv(x) \leq v(x) \quad \forall x \notin F,$$

then the chain is Tweedie recurrent. Conversely, if the chain is Tweedie recurrent and has finitely many closed communicating classes, then there exists (v, F) satisfying (3) with v real-valued and norm-like and F finite.

If there are infinitely many closed communicating classes, it may be the case that no (v, F) satisfying the theorem's premise exists even if the chain is Tweedie recurrent. For instance, let $K = (k(x, y))_{x, y \in \mathbb{N}}$ be the one-step matrix of an irreducible and recurrent $\mathbb{N}$ -valued chain W and let $X = (X^1, X^2)$ be a chain on $\mathbb{N}^2$ with one-step matrix

$$p((x_1, x_2), (y_1, y_2)) = \begin{cases} k(x_2, y_2) & \text{if } y_1 = x_1 \\ 0 & \text{if } y_1 \neq x_1 \end{cases} \quad \forall x_1, x_2, y_1, y_2 \in \mathbb{N}.$$

That is, X remains in the slice $\mathbb{N}_i := \{(i, x) : x \in \mathbb{N}\}$ of $\mathbb{N}^2$ indexed by its first coordinate $i := X_0^1$, moving up and down this slice in the same manner that W moves up and down in $\mathbb{N}$. Because W is irreducible and recurrent, it follows that X is Tweedie recurrent with closed communicating classes $\mathbb{N}_1, \mathbb{N}_2, \dots$. Suppose that there exists a (v, F) satisfying the premise of Theorem 2. If v is not non-negative, then replace it by $v - (\min_{x \in \mathcal{S}} v(x))$ and note that the theorem's premise still holds. Because F is finite, there exists infinitely many slices $\mathbb{N}_i$ that do not intersect with F . For any such $\mathbb{N}_i$, (3) implies that $v_i(x) := v(i, x)$ is *superharmonic* for K :

$$Kv_i(x) \leq v_i(x) \quad \forall x \in \mathbb{N}.$$

As is well-known, c.f. (Asmussen, 2003, Prop. I.5.1), any non-negative superharmonic function is necessarily constant. Thus, $v(i, x) = v_i(x) = c$ for all x in $\mathbb{N}$ and some real number c showing that v cannot be norm-like.

A proof of Theorem 2. We do each direction separately. For the forward direction, we need the following lemma showing we lose nothing by assuming that the set F in (3) contains only recurrent states (in which case, $\phi_{F \cup \mathcal{R}} = \phi_{\mathcal{R}}$).

(4) LEMMA. *Let F be a finite set such that $\mathbb{P}_\gamma(\{\phi_F < \infty\}) = 1$ for all initial distributions γ . The same inequality holds if we remove all transient states from F .*

Proof. Pick any transient state $z \in F$ and let $F_z := F \setminus \{z\}$ denote F with z removed from it. It must be the case that F_z is accessible from z (i.e., $\{z\} \rightarrow F$ as in Definition 13.5). Otherwise, $\mathbb{P}_z(\{\phi_z < \infty\}) = \mathbb{P}_z(\{\phi_F < \infty\}) = 1$ contradicting the transience of z .

Clearly, F itself is accessible from any given state x in F . Thus, either F_z is accessible from x , or z is accessible from x . In the latter case, F_z is accessible from x because it is accessible from z . It then follows from the transitivity of $\rightarrow$ that F_z is accessible from every x in F (i.e., $F \rightarrow F_z$) and, so, Lemma 22.2(i) that $\mathbb{P}_\gamma(\{\phi_{F_z} < \infty\}) = 1$ for all initial distributions γ . Because F is finite, it contains at most finitely many transient states. Hence, the result follows by repeatedly replacing F with F_z throughout the above argument. $\square$

The proof of the forward direction then goes as follows:

Proof of Theorem 2 (the forward direction). Given that $\mathbb{P}_\gamma = \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x$, it suffices to show that

$$(5) \quad \mathbb{P}_x(\{\phi_{\mathcal{R}} < \infty\}) = 1 \quad \forall x \in \mathcal{S}.$$

Because of Lemma 4, we may assume that F is contained in $\mathcal{R}$. To argue the above, set $\mathcal{D}$ to be the set of transient states such that the exit time σ in (11.1) equals the hitting time of $\mathcal{R}$. Because $F \subseteq \mathcal{R}$, inequality in (2) implies that

$$\hat{P}v(x) \leq v(x) \quad \forall x \in \mathcal{S},$$

where $\hat{P} = (\hat{p}(x, y))_{x, y \in \mathcal{S}}$ is as in (11.6). Recall that $\hat{P}$ is a one-step matrix and that its associated chain $\hat{X}$ (c.f. discussion after Theorem 11.3) is identical to X except that each state inside $\mathcal{R}$ has been turned into an absorbing state. Iterating the above, we find that

$$\hat{P}_n v(x) \leq v(x) \quad \forall x \in \mathcal{S}, \quad n > 0,$$

where $\hat{P}_n = (\hat{p}_n(x, y))_{x, y \in \mathcal{S}}$ denotes the n -step matrix of $\hat{X}$. We then have that

$$(6) \quad \sum_{z \in \mathcal{S}_r} \hat{p}_n(x, z) = 1 - \sum_{z \notin \mathcal{S}_r} \hat{p}_n(x, z) \geq 1 - \frac{1}{r} \sum_{z \notin \mathcal{S}_r} \hat{p}_n(x, z) v(z) \geq 1 - \frac{1}{r} \hat{P}_n v(x) \geq 1 - \frac{v(x)}{r} \quad \forall x \in \mathcal{S},$$

where $\mathcal{S}_r$ denotes the r th sublevel set of v (Definition 1).

Because the dynamics of X and $\hat{X}$ are identical except that the states in $\mathcal{R}$ are absorbing for $\hat{X}$, any state that is transient for X is also transient for $\hat{X}$ (to formally argue this, use Theorem 11.3). For this reason, Theorem 20.1 implies that

$$\lim_{n \rightarrow \infty} \hat{p}_n(x, z) = 0 \quad \forall x \in \mathcal{S}, \quad z \notin \mathcal{R},$$

and it follows that

$$\lim_{n \rightarrow \infty} \sum_{z \in \mathcal{S}_r} \hat{p}_n(x, z) = \lim_{n \rightarrow \infty} \sum_{z \in \mathcal{S}_r \cap \mathcal{R}} \hat{p}_n(x, z) \quad \forall x \in \mathcal{S}, \quad r > 0.$$

The limit on the right-hand side exists because Theorem 11.7 shows that

$$\hat{p}_n(x, z) = \mathbb{P}_x(\{\sigma \leq n, X_\sigma = z\}) \quad \forall x \in \mathcal{S}, \quad z \in \mathcal{R}, \quad n \geq 0.$$

Moreover, taking the limit $n \rightarrow \infty$ in (6), we find that

$$\begin{aligned} \mathbb{P}_x(\{\sigma < \infty\}) &= \lim_{n \rightarrow \infty} \sum_{z \in \mathcal{R}} \mathbb{P}_x(\{\sigma \leq n, X_n = z\}) = \lim_{n \rightarrow \infty} \sum_{z \in \mathcal{R}} \mathbb{P}_x(\{\sigma \leq n, X_\sigma = z\}) \\ &= \lim_{n \rightarrow \infty} \sum_{z \in \mathcal{R}} \hat{p}_n(x, z) \geq \lim_{n \rightarrow \infty} \sum_{z \in \mathcal{S}_r \cap \mathcal{R}} \hat{p}_n(x, z) \geq 1 - \frac{v(x)}{r} \quad \forall x \in \mathcal{S}, \quad r > 0. \end{aligned}$$

Taking the limit $r \rightarrow \infty$ in the above, we find that $\mathbb{P}_x(\{\sigma < \infty\}) = 1$ for all $x \in \mathcal{S}$.

Consider now the entrance time $\phi_{\mathcal{R}}$ to $\mathcal{R}$ (for X). By its definition,

$$\mathbb{P}_x(\{\phi_{\mathcal{R}} < \infty\}) = \mathbb{P}_x(\{\sigma < \infty\}) = 1 \quad \forall x \notin \mathcal{R}.$$

For the states inside $\mathcal{R}$, we use Proposition 13.10 to obtain

$$\mathbb{P}_x(\{\phi_{\mathcal{R}} < \infty\}) = \sum_{z \in \mathcal{R}} p(x, z) + \sum_{z \notin \mathcal{R}} \mathbb{P}_x(\{X_1 = z\}) \mathbb{P}_z(\{\phi_{\mathcal{R}} < \infty\}) = \sum_{z \in \mathcal{R}} p(x, z) + \sum_{z \notin \mathcal{R}} p(x, z) = 1,$$

for all $x \in \mathcal{S}$, completing the proof of (5). $\square$

We split the proof of the reverse direction into three lemmas:

(7) LEMMA. Let $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ be a sequence of increasing truncations that approach the state space $(\cup_{r=1}^\infty \mathcal{S}_r = \mathcal{S})$ and $(\sigma_r)_{r \in \mathbb{Z}_+}$ be the corresponding sequence of exit times:

$$(8) \quad \sigma_r(\omega) := \inf\{n \geq 0 : X_n(\omega) \notin \mathcal{S}_r\} \quad \forall \omega \in \Omega, \quad r > 0.$$

For any initial distribution γ ,

$$\lim_{r \rightarrow \infty} \sigma_r = \infty \quad \mathbb{P}_\gamma\text{-almost surely.}$$

Proof. The limit $\sigma_\infty := \lim_{r \rightarrow \infty} \sigma_r$ exists as the sequence $(\sigma_r)_{r \in \mathbb{Z}_+}$ is increasing. By definition,

$$\mathbb{P}_\gamma(\{\sigma_r \leq k\}) = \sum_{j=0}^k \mathbb{P}_\gamma(\{\sigma_r = j\}) \leq \sum_{j=0}^k \mathbb{P}_\gamma(\{X_j \notin \mathcal{S}_r\}) = \sum_{j=0}^k p_j(\mathcal{S}_r^c)$$

and, given that $\cup_{r=1}^\infty \mathcal{S}_r = \mathcal{S}$, downwards monotone convergence shows that

$$\mathbb{P}_\gamma(\{\sigma_\infty \leq k\}) = \lim_{r \rightarrow \infty} \mathbb{P}_\gamma(\{\sigma_r \leq k\}) = \sum_{j=0}^k \lim_{r \rightarrow \infty} p_j(\mathcal{S}_r^c) = 0.$$

Taking the limit $k \rightarrow \infty$ and applying monotone convergence completes the proof. $\square$

(9) LEMMA. Suppose that the chain is Tweedie recurrent and has $n \in \mathbb{N}$ closed communicating classes labelled $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_n$. If $F = \{x_1, x_2, \dots, x_n\}$, where x_1 denotes a state in $\mathcal{C}_1$, x_2 one in $\mathcal{C}_2$, etc., then $\mathbb{P}_\gamma(\{\phi_F < \infty\}) = 1$ for all initial distributions γ .

Proof. By definition, ϕ_F is at most ϕ_{x_m} , and so

$$\begin{aligned} \{\phi_F \neq \phi_{x_m} < \infty\} &= \{\phi_F < \phi_{x_m} < \infty\} = \bigcup_{k \neq m} \{\phi_F = \phi_{x_k} < \phi_{x_m} < \infty\} \\ &\subseteq \bigcup_{k \neq m} \{\phi_{x_k} < \infty, \phi_{x_m} < \infty\} \subseteq \bigcup_{k \neq m} \{\phi_{\mathcal{C}_k} < \infty, \phi_{\mathcal{C}_m} < \infty\} \quad \forall m = 1, 2, \dots, n. \end{aligned}$$

Taking expectations and applying Proposition 13.11 then yields $\mathbb{P}_\gamma(\{\phi_F \neq \phi_{x_m} < \infty\}) = 0$ for all m . It follows that

$$\mathbb{P}_\gamma(\{\phi_F < \infty\}) = \sum_{m=1}^n \mathbb{P}_\gamma(\{\phi_F = \phi_{x_m} < \infty\}) = \sum_{m=1}^n \mathbb{P}_\gamma(\{\phi_{x_m} < \infty\})$$

Using Proposition 13.11 and Corollary 14.8 then completes the proof:

$$\mathbb{P}_\gamma(\{\phi_F < \infty\}) = \sum_{m=1}^n \mathbb{P}_\gamma(\{\phi_{\mathcal{C}_m} < \infty\}) = \mathbb{P}_\gamma\left(\bigcup_{m=1}^n \{\phi_{\mathcal{C}_m} < \infty\}\right) = \mathbb{P}_\gamma(\{\phi_{\mathcal{R}} < \infty\}) = 1,$$

for all initial distributions γ . □

The following lemma then completes the proof of Theorem 24.4:

(10) LEMMA. If F is a finite set such that $\mathbb{P}_\gamma(\{\phi_F < \infty\}) = 1$ for all initial distributions γ , then there exists a non-negative norm-like function v on $\mathcal{S}$ such that (3) holds.

Proof. Let $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ be any sequence of increasing finite truncations containing F ($F \subseteq \mathcal{S}_1$) that approach the state space ($\bigcup_{r=1}^\infty \mathcal{S}_r = \mathcal{S}$) and let $(\sigma_r)_{r \in \mathbb{Z}_+}$ be the corresponding sequence of exit times (8). Additionally, let

$$\sigma_F(\omega) := \inf\{n \geq 0 : X_n(\omega) \in F\} \quad \forall \omega \in \Omega$$

be the hitting time of F and, for any given state x ,

$$v_r(x) := \mathbb{P}_x(\{\sigma_r \leq \sigma_F\})$$

be the probability that the chain leaves $\mathcal{S}_r$ without hitting F if it starts at x . By definition, $v_r(x) = 1$ if x lies outside of $\mathcal{S}_r$ and we have that

$$(11) \quad P v_r(x) = \sum_{y \in \mathcal{S}} p(x, y) \mathbb{P}_y(\{\sigma_r \leq \sigma_F\}) \leq \sum_{y \in \mathcal{S}} p(x, y) = 1 = v_r(x) \quad \forall x \notin \mathcal{S}_r.$$

If instead x lies inside of $\mathcal{S}_r$ but outside of F , then

$$\sigma_r = \inf\{n \geq 1 : X_n \notin \mathcal{S}_r\}, \quad \sigma_F = \inf\{n \geq 1 : X_n \in F\}, \quad \mathbb{P}_x\text{-almost surely.}$$

For this reason, an application of the strong Markov property similar to that in the proof of Lemma 13.12 yields

$$\begin{aligned} P v_r(x) &= \sum_{y \in \mathcal{S}} p(x, y) v_r(y) = \sum_{y \in \mathcal{S}} \mathbb{P}_x(\{X_1 = y\}) \mathbb{P}_y(\{\sigma_r \leq \sigma_F\}) \\ &= \sum_{y \in \mathcal{S}} \mathbb{P}_x(\{X_1 = y, \inf\{n \geq 1 : X_n \notin \mathcal{S}_r\} \leq \inf\{n \geq 1 : X_n \in F\}\}) \\ &= \mathbb{P}_x(\{\sigma_r \leq \sigma_F\}) = v_r(x) \end{aligned}$$

for all x in $\mathcal{S}_r$ but outside F . The above and (11) show that v_r satisfies the inequality in (2).

However, v_r is clearly not norm-like given that it is bounded. To obtain a norm-like function let $(r_k)_{k \in \mathbb{Z}_+}$ be any increasing sequence of positive integers and note that

$$\sum_{y \in \mathcal{S}} p(x, y) \sum_{k=1}^{\infty} v_{r_k}(y) \leq \sum_{k=1}^{\infty} \sum_{y \in \mathcal{S}} p(x, y) v_{r_k}(y) \leq \sum_{k=1}^{\infty} v_r(x) \quad \forall x \notin F$$

because of Fatou's lemma. That is, the function v defined by

$$v(x) := \sum_{k=1}^{\infty} v_{r_k}(x) \quad \forall x \in \mathcal{S}$$

also satisfies the inequality in (2). Moreover, because $v_{r_k}(x) = 1$ for all x outside $\mathcal{S}_{r_k}$, $v(x)$ is at least N if x does not belong to $\mathcal{S}_{r_N}$ and it follows that v is norm-like.

For v to meet all the conditions in the theorem's premise, we only have left to show that it is finite (i.e., $v(x) < \infty$ for all $x \in \mathcal{S}$). The trick here is to pick the sequence $(r_k)_{k \in \mathbb{Z}_+}$ carefully. In particular, our premise implies that

$$\mathbb{P}_x(\{\sigma_F < \infty\}) = \mathbb{P}_x(\{\phi_F < \infty\}) = 1 \quad \forall x \notin F.$$

Given that σ_r tends to ∞ with $\mathbb{P}_x$ -probability one (Lemma 7), it follows from the above that

$$\lim_{r \rightarrow \infty} v_r(x) = \lim_{r \rightarrow \infty} \mathbb{P}_x(\{\sigma_r \leq \sigma_F\}) = \mathbb{P}_x(\{\sigma_F = \infty\}) = 0 \quad \forall x \in \mathcal{S}.$$

For this reason, finiteness of $\mathcal{S}_1, \mathcal{S}_2, \dots$ allows us to find an increasing sequence $(r_k)_{k \in \mathbb{Z}_+}$ tending to infinity such that

$$v_{r_k}(x) \leq \frac{1}{2^k} \quad \forall x \in \mathcal{S}_k, \quad k > 0.$$

With this choice, we have that, for any given $x \in \mathcal{S}$,

$$v(x) = \sum_{k=1}^{\infty} v_{r_k}(x) = \sum_{k=1}^{K-1} v_{r_k}(x) + \sum_{k=K}^{\infty} \frac{1}{2^k} < \infty,$$

where K is large enough that x belongs to $\mathcal{S}_K$. □

Notes and references. The sufficiency of Theorem 2 was first shown in (Foster, 1953) for irreducible chains and singleton sets F . This was later generalised to arbitrary finite sets in (Pakes, 1969). The generalisation for non-irreducible chains was proven in (Tweedie, 1975b). The necessity was first shown in (Mertens et al., 1978) for irreducible and aperiodic chains. To the best of my knowledge, it took another fifteen years until the theorem's necessity was proven for the general case in the first edition of (Meyn and Tweedie, 2009) (where it was shown for a more general class of discrete-time Markov processes).

24. Foster-Lyapunov criteria II: Foster's theorem*. A fantastically useful result is Theorem 1 below, commonly known as *Foster's theorem*:

(1) **THEOREM** (Foster's theorem). *If there exists a non-negative real-valued function v on $\mathcal{S}$, a finite subset F of $\mathcal{S}$, and a real number b satisfying*

$$(2) \quad Pv(x) \leq v(x) - 1 + b1_F(x) \quad \forall x \in \mathcal{S},$$

then the chain is positive Tweedie recurrent and has a finite number of closed communicating classes. Conversely, if the chain is positive Tweedie recurrent and if the state space contains no transient states and only a finite number of closed communicating classes, then there exists (v, F, b) satisfying (2) with v real-valued and non-negative, F finite, and b real.

This theorem is one of the few tools we have in practice to establish positive Tweedie recurrence of a chain (and, consequently, the existence of stationary distributions, the convergence of the empirical distribution, etc.). It is a consequence of Theorem 4 below which shows that for a fixed F , the existence of a v and b satisfying (2) is the same as the mean entrance time to F being finite for all deterministic initial positions.

Notice that Theorem 1 shows that the existence of the (v, F, b) satisfying (2) is equivalent to positive Tweedie recurrence in the case of no transient states and only a finite number of closed communicating classes. The importance of a finite number of closed communicating classes is easy to see: the set F must contain at least one state from each closed communicating class. Otherwise, the chain would not be able to reach F whenever it starts inside of a closed communicating class that does not intersect with F and the entrance time to F would be infinite. As an example, the chain with one-step matrix P being the identity matrix is trivially positive recurrent, however $Pv(x) - v(x) = 0 \geq -1$ for all states x and functions v . Consequently, if the state space is infinite, then (2) will never be satisfied for a finite F .

The no transient states requirement is more subtle. Positive Tweedie recurrence asks that the chain reaches a positive recurrent state with probability one. The criterion instead requires that the mean amount of time it takes the chain to reach the positive recurrent states is finite. As the example below shows the latter is a stronger demand. This is one of the unfortunately numerous corners of Markov process theory where two concepts are almost the same, but not quite (so close, and yet...).

(3) EXAMPLE. Consider again the gambler's ruin problem introduced in Section 3 and suppose that the coin is unbiased ($a = 1/2$). As shown at the end of Section 11, for any initial distribution, the chain has probability one of entering the absorbing state 0. Because $\{0\}$ is the only closed communicating class, it follows that the chain is positive Harris recurrent. However, it is not difficult to use the Markov property to verify that, unless the chain starts at 0, the average amount of steps it takes the chain to enter 0 is infinite. The reason why is that the chain makes very long trips away from 0 before eventually entering 0. Furthermore, these long trips ensure that, for any given finite set F , there exists a state x such that the average number of steps it takes the chain to enter F if it starts at x is infinite and it follows from Theorem 4 below that no (v, F, b) satisfying the premise of (2) exist.

More directly, suppose that there exists a non-negative real-valued function v satisfying

$$Pv(x) = \frac{1}{2}v(x-1) + \frac{1}{2}v(x+1) \leq v(x) - 1$$

for all states x outside some finite set F . Letting z denote the largest state in F , we have that

$$v(x+1) - v(x) \leq v(x) - v(x-1) - 2 \quad \forall x > z.$$

Iterating the above, we find that

$$v(z+n) - v(z+n-1) \leq v(z+n-1) - v(z+n-2) - 2 \leq \cdots \leq v(z+1) - v(z) - 2(n-1).$$

For this reason,

$$\begin{aligned} v(z+n) &= v(z) + \sum_{m=1}^n (v(z+m) - v(z+m-1)) \leq v(z) + \sum_{m=1}^n (v(z+1) - v(z) - 2(m-1)) \\ &= v(z) + n(v(z+1) - v(z)) - n(n-1) = v(z) + n(v(z+1) - v(z) + 1) - n^2. \end{aligned}$$

Taking the limit $n \rightarrow \infty$ shows that $v(x) \rightarrow -\infty$ as $x \rightarrow \infty$ contradicting the non-negativeness of v and proving that no function v and set F satisfying the premise of (4) exist even though the chain is positive Harris recurrent.

A proof of Foster's theorem. As mentioned before, Theorem 4 really is a consequence of the following:

(4) THEOREM. *Given any finite set F , there exists $v : \mathcal{S} \rightarrow [0, \infty)$ and $b \in \mathbb{R}$ satisfying (2) if and only if $\mathbb{E}_x[\phi_F] < \infty$ for all $x \in \mathcal{S}$. In this case, $\mathbb{E}_\gamma[\phi_F] < \infty$ whenever the initial distribution γ satisfies $\gamma(v) < \infty$.*

The key to proving the above is the following lemma.

(5) LEMMA. *Let F be a subset of the state space and $\theta > 0$. The minimal non-negative solution $v : \mathcal{S} \rightarrow [0, \infty]$ to the inequality*

$$(6) \quad Pv(x) \leq \theta^{-1}v(x) - 1 \quad \forall x \notin F$$

is given by $u(x) = 0$ for all x in F and

$$(7) \quad u(x) = \theta \sum_{n=0}^{\infty} \mathbb{P}_x(\{\phi_F > n\}) \theta^n \quad \forall x \notin F.$$

Proof. For any natural number n , let

$$f_n(x, y) := \mathbb{P}_x(\{\phi_F > n, X_n = y\}) \quad \forall x, y \notin F,$$

so that

$$\mathbb{P}_x(\{\phi_F > n\}) = \sum_{y \notin F} f_n(x, y) \quad \forall x \notin F.$$

Applying the Markov property (Theorem 12.4), we find that

$$\begin{aligned} \mathbb{P}_x(\{\phi_F > n, X_n = y\}) &= \sum_{z \notin F} \mathbb{P}_x(\{X_1 = z, \phi_F - 1 > n - 1, X_{1+(n-1)} = y\}) \\ &= \sum_{z \notin F} p(x, z) f_{n-1}(z, y) \end{aligned}$$

if $n > 0$. For this reason,

$$f_n(x, y) = 1_x(y)1_0(n) + (1 - 1_0(n)) \sum_{z \notin F} p(x, z) f_{n-1}(z, y) \quad \forall x, y \notin F.$$

Thus, we have that u in (7) satisfies

$$\begin{aligned} \theta^{-1}u(x) &= \sum_{n=0}^{\infty} \sum_{y \notin F} f_n(x, y) \theta^n = 1 + \sum_{n=1}^{\infty} \sum_{y \notin F} \sum_{z \notin F} p(x, z) f_{n-1}(z, y) \theta^n \\ &= 1 + \sum_{z \notin F} p(x, z) \left(\theta \sum_{n=1}^{\infty} \sum_{y \notin F} f_{n-1}(z, y) \theta^{n-1} \right) = 1 + Pu(x) \quad \forall x \notin F \end{aligned}$$

proving that u is a solution to (6). To prove minimality of u , let v be any other non-negative

solution to (6). By definition, $v(x) \geq 0 = u(x)$ for all $x \in F$. For all other states, note that

$$\begin{aligned}
v(x) &\geq \theta + \theta \sum_{x_1 \in \mathcal{S}} p(x, x_1) v(x_1) \geq \theta + \theta \sum_{x_1 \notin F} p(x, x_1) v(x_1) \\
&\geq \theta + \theta^2 \sum_{x_1 \notin F} p(x, x_1) + \theta^2 \sum_{x_1 \notin F} \sum_{x_2 \in \mathcal{S}} p(x, x_1) p(x_1, x_2) v(x_2) \\
&\geq \theta + \theta^2 \sum_{x_1 \notin F} p(x, x_1) + \theta^2 \sum_{x_1 \notin F} \sum_{x_2 \notin F} p(x, x_1) p(x_1, x_2) v(x_2) \\
&\geq \dots \\
&\geq \theta + \theta^2 \sum_{x_1 \notin F} p(x, x_1) + \theta^3 \sum_{x_1 \notin F} \sum_{x_2 \notin F} p(x, x_1) p(x_1, x_2) + \dots \\
&\quad + \theta^l \sum_{x_1 \notin F} \dots \sum_{x_{l-1} \notin F} p(x, x_1) \dots p(x_{l-2}, x_{l-1}) + \theta^l \sum_{x_1 \notin F} \dots \sum_{x_l \notin F} p(x, x_1) \dots p(x_{l-1}, x_l) v(x_l) \\
&= \theta \mathbb{P}_x(\{\phi_F > 0\}) + \theta^2 \mathbb{P}_x(\{\phi_F > 1\}) + \theta^3 \mathbb{P}_x(\{\phi_F > 2\}) + \dots + \theta^l \mathbb{P}_x(\{\phi_F > l\}) \\
&\quad + \theta^l \sum_{x_1 \notin F} \dots \sum_{x_{l-1} \notin F} p(x, x_1) \dots p(x_{l-1}, x_l) v(x_l) \quad \forall x \notin F.
\end{aligned}$$

Because non-negativity of v implies that the right most term is non-negative, taking the limit $l \rightarrow \infty$ then shows that $v(x) \geq u(x)$ as desired. $\square$

Proving Theorem 4 is now straightforward:

Proof of Theorem 4. Suppose that (2) is satisfied. Setting $\theta := 1$ in (5) shows that

$$\mathbb{E}_x[\phi_F] \leq v(x) \quad \forall x \notin F.$$

For states inside F , an application of the strong Markov property similar to that in the proof of Proposition 13.10 yields

$$\mathbb{E}_x[\phi_F] = 1 + \sum_{z \notin F} p(x, z) \mathbb{E}_z[\phi_F] \leq 1 + \sum_{z \notin F} p(x, z) v(z) \leq 1 + P v(x) \leq v(x) + b \quad \forall x \in F.$$

Given that $\mathbb{P}_\gamma = \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x$, multiplying the above two inequalities by $\gamma(x)$ and summing over x in $\mathcal{S}$ then shows that $\mathbb{E}_\gamma[\phi_F]$ is finite whenever $\gamma(v)$ is finite.

Conversely, suppose that $\mathbb{E}_x[\phi_F] < \infty$ for all x in $\mathcal{S}$. Lemma 5 shows that u in (7) (with $\theta = 1$) satisfies (2) for all states x outside of F . For states inside, we apply the Markov property as before:

$$P u(x) = \sum_{z \in \mathcal{S}} p(x, z) u(z) = \sum_{z \notin F} p(x, z) \mathbb{E}_z[\phi_F] = \mathbb{E}_x[\phi_F] - 1 \quad \forall x \in F.$$

In other words, u satisfies (2) with $b := \max\{\mathbb{E}_x[\phi_F] : x \in F\}$. $\square$

To make full use of Theorem 4 and prove Foster's theorem, we require one final result:

(8) LEMMA. *Let F be a finite set such that $\mathbb{E}_x[\phi_F] < \infty$ for all x in $\mathcal{S}$. The same inequality holds if we remove all transient states from F .*

Proof. Replace Lemma 22.2(i) with Lemma 22.2(ii) in the proof of Lemma 23.4. $\square$

We are now in a great position to prove Foster's theorem (Theorem 1):

Proof of Theorem 1. To prove the forward direction, suppose that there exists (v, F, b) as in the premise. Theorem 4 shows that $\mathbb{E}_x[\phi_F]$ is finite for all x in $\mathcal{S}$. Lemma 8 shows that, without any loss of generality, we may assume that F contains only recurrent states. Thus,

$$\mathbb{E}_x[\phi_{\mathcal{R}}] \leq \mathbb{E}_x[\phi_F] < \infty \quad \forall x \in \mathcal{S},$$

where $\mathcal{R}$ is the set of recurrent states, and it follows that the chain is positive Tweedie recurrent. Furthermore, Proposition 13.11 implies that

$$\mathbb{E}_x[\phi_{F \cap \mathcal{C}}] = \mathbb{E}_x[\phi_F] < \infty \quad \forall x \in \mathcal{C},$$

where $\mathcal{C}$ is any closed communicating class. The intersection $F \cap \mathcal{C}$ must then be non-empty (otherwise, $\phi_{F \cap \mathcal{C}} = \infty$ contradicting the above). Given that F is finite and closed communicating classes are disjoint by definition, it follows that there exists only a finite number of closed communicating classes.

Conversely, suppose that only finitely many closed communicating classes $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_l$ exist and that there are no transient states. Positive Tweedie recurrence implies that all of these classes are positive recurrent. Pick any states $x_1 \in \mathcal{C}_1, x_2 \in \mathcal{C}_2, \dots, x_l \in \mathcal{C}_l$ and set $F := \{x_1, x_2, \dots, x_l\}$. Because x_j is only accessible from states in $\mathcal{C}_j$,

$$\mathbb{E}_x[\phi_F] = \mathbb{E}_x[\phi_{x_j}] \quad \forall x \in \mathcal{C}_j.$$

Fix any such state $x \in \mathcal{C}_j$. Because $\mathbb{E}_x[\phi_x] < \infty$ and $x \rightarrow x_j$, Lemma 22.2(ii) shows that $\mathbb{E}_x[\phi_{x_j}]$ is finite. Because $\cup_{j=1}^l \mathcal{C}_j = \mathcal{S}$ (as there are no transient states), the above then shows that $\mathbb{E}_x[\phi_F]$ is finite for all x in $\mathcal{S}$ and the existence of the (v, F, b) satisfying (2) follows from Theorem 4. $\square$

Notes and references. The forward direction of Theorem 1, for irreducible chains and general F , first shows up in F. G. Foster's comments in the discussion of D. G. Kendall's famous queueing paper (Kendall, 1951a). Foster does not give a proof therein but instead promises that one will be included in an upcoming paper. It seems that he was referring to his well-known paper (Foster, 1953) published two years later where he proves both the forward and reverse directions of Theorem 1 for irreducible chains and singleton F s. A version of the generalisation of the forward direction to both non-irreducible chains with multiple closed communicating classes and arbitrary F s was first shown in (Mauldon, 1957)—although the generalisation to arbitrary F s in the irreducible case is often attributed to (Pakes, 1969) (it was also stated, but not shown, in (Kingman, 1961)). In particular, J. G. Mauldon showed that the existence of (v, F, b) satisfying (2) implies that the (pointwise) limit

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} p_n,$$

has unit mass for all initial distributions. Because bounded convergence and Theorem 20.2 imply that this limit is π_γ in (18.2), Corollary 18.3 then shows that the chain is Tweedie recurrent. However, this point seems to have been made explicit much later in (Tweedie, 1975b). I have not encountered the converse given in Theorem 1 elsewhere, but the underlying ideas are the same as those behind the converses in (Foster, 1953) for the irreducible case and (Tweedie, 1975a,b) for the general case.

25. Foster-Lyapunov criteria III: the criterion for geometric convergence*.

Lemma 24.5 begs us to study the more general inequality

$$(1) \quad Pv(x) \leq \theta^{-1}v(x) - 1 + b1_F(x) \quad \forall x \in \mathcal{S}.$$

instead of the special case $\theta = 1$ considered in Section 24. The lemma shows that, for any given $\theta > 0$, the inequality's minimal (possibly infinite-valued) non-negative solution is given by $u(x) = 0$ for all $x \in F$ and

$$(2) \quad u(x) = \theta \sum_{n=0}^{\infty} \mathbb{P}_x(\{\phi_F > n\}) \theta^n = \theta \mathbb{E}_x \left[\sum_{n=0}^{\infty} 1_{\{\phi_F > n\}} \theta^n \right] = \theta \mathbb{E}_x \left[\sum_{n=0}^{\phi_F-1} \theta^n \right] \quad \forall x \notin F.$$

For $\theta \neq 1$, we have that

$$(3) \quad u(x) = \theta \mathbb{E}_x \left[\frac{\theta^{\phi_F} - 1}{\theta - 1} \right] = \frac{\theta}{\theta - 1} (\mathbb{E}_x[\theta^{\phi_F}] - 1) \quad \forall x \notin F.$$

The $\theta > 1$ analogue of then Theorem 24.4 follows immediately:

(4) **THEOREM.** *Given any finite set F and $\theta > 1$, there exists $v : \mathcal{S} \rightarrow [0, \infty)$ and $b \in \mathbb{R}$ satisfying (1) if and only if $\mathbb{E}_x[\theta^{\phi_F}] < \infty$ for all $x \in \mathcal{S}$. In this case, $\mathbb{E}_\gamma[\theta^{\phi_F}] < \infty$ whenever the initial distribution satisfies $\gamma(v) < \infty$.*

Proof. Given (2)–(3), the proof is entirely analogous to that of Theorem 24.4. $\square$

The above theorem shows that (1) is satisfied for some $\theta > 1$, then the return time distribution of F has a geometrically decaying tail (i.e., a *light tail*) for any deterministic starting state. Conversely, if the tails are heavy for any one starting state, then (1) will not be satisfied. Kendall's theorem (Theorem 21.1) then relates the light tails with the geometric convergence of the time-varying law, and we obtain:

(5) **THEOREM** (The geometric criterion). *If there exists a real-valued non-negative function v on $\mathcal{S}$, a finite set F , and a real number b satisfying (1) for some $\theta > 1$, then the chain is positive Tweedie recurrent. Moreover, if the chain is aperiodic and $\gamma(v) < \infty$, then the time varying law converges geometrically fast: there exists some $\kappa > 1$ such that*

$$(6) \quad \|p_n - \pi_\gamma\| = \mathcal{O}(\kappa^{-n}),$$

where π_γ denotes the limiting stationary distribution in (20.3) and $\|\cdot\|$ denotes the total variation norm in (18.4).

Conversely, if the chain is aperiodic and positive Tweedie recurrent, there exists no transient states and only a finite number of closed communicating classes, and (6) holds whenever the chain starts at a fixed state (i.e., whenever $\gamma = 1_x$ for some x in $\mathcal{S}$), then there exists (v, F, b, θ) satisfying (24.2) with v real-valued and non-negative, F finite, b real, and $\theta > 1$.

It is straightforward to see that the finitely-many-closed-communicating-classes requirement in the converse is necessary: consider a chain on an infinite state space whose one-step matrix is the identity matrix.

An open question. To the best of my knowledge, whether or not the no-transient-states requirement in the converse is necessary remains an open question. I believe that the answer here lies in that of the open question discussed in Section 21.

First-entrance last-exit decomposition. For the proof of Theorem 5 we require the so-called *first-entrance last-exit decomposition* of the n -step matrix:

$$(7) \quad p_n(x, y) = p_n^z(x, y) + \sum_{k=1}^{n-1} \sum_{l=1}^k p_l^z(x, z) p_{k-l}(z, z) p_{n-k}^z(z, y) \quad \forall x, y \in \mathcal{S}, \quad n > 0,$$

where z is any given state in $\mathcal{S}$ and $p_n^z(x, y)$ denotes the probability that the chain transitions from x to y without visiting z during time-steps $1, 2, \dots, n-1$:

$$(8) \quad p_n^z(x, y) = \mathbb{P}_x(\{X_n = y, \phi_z \geq n\}) \quad \forall x, y \in \mathcal{S}, \quad n > 0.$$

The above are often referred to as the *taboo probabilities*. The decomposition (7) follows by noting that

$$\begin{aligned} \{X_n = y\} &= \{X_n = y, X \text{ doesn't visit } z \text{ during time-steps } 1, 2, \dots, n-1\} \\ &\cup \bigcup_{k=1}^{n-1} \{X_n = y, X \text{ last visits } z \text{ at time } k\} \\ &= \{X_n = y, X \text{ doesn't visit } z \text{ during time-steps } 1, 2, \dots, n-1\} \\ &\cup \bigcup_{k=1}^{n-1} \left(\bigcup_{l=1}^k \{X_n = y, X \text{ first visits } z \text{ at time } l \text{ and last visits } z \text{ at time } k\} \right) \end{aligned}$$

and that these events are disjoint.

(9) EXERCISE. *Taking expectations of the above and applying (4.4)–(4.5) and Tonelli's Theorem, prove (7).*

As we will in the proof of Theorem 5, this decomposition turns out to be the key to strengthening the convergence of $p_n(x, x)$ for individual states x (as in Kendall's Theorem, Theorem 21.1) to convergence in total variation of the entire time-varying law. In particular, suppose that z belongs to a positive recurrent class $\mathcal{C}$ with ergodic distribution π . Because $\mathbb{E}_z[1_y(X_0)] = 1_y(z) = \mathbb{E}_z[1_y(X_{\phi_z})]$ for any recurrent state z , Theorem 17.10(i) shows that

$$\begin{aligned} \pi(y) &= \frac{1}{\mathbb{E}_z[\phi_z]} \mathbb{E}_z \left[\sum_{n=0}^{\phi_z-1} 1_y(X_n) \right] = \pi(z) \mathbb{E}_z \left[\sum_{n=0}^{\phi_z-1} 1_y(X_n) \right] = \pi(z) \mathbb{E}_z \left[\sum_{k=1}^{\infty} 1_{\{\phi_z \geq k\}} 1_y(X_k) \right] \\ &= \pi(z) \sum_{k=1}^{\infty} p_k^z(x, y) \quad \forall z \in \mathcal{C}, \quad y \in \mathcal{S}. \end{aligned}$$

For this reason, (8) gives us the following bounds on $|p_n(x, A) - \pi(A)|$ that we will need to establish the convergence in total variation of Theorem 5:

$$\begin{aligned} (10) \quad |p_n(x, A) - \pi(A)| &\leq p_n^z(x, A) + \sum_{k=1}^{n-1} \left| \sum_{l=1}^k p_l^z(x, z) p_{k-l}(z, z) - \pi(z) \right| p_{n-k}^z(z, A) \\ &\quad + \pi(z) \sum_{k=n}^{\infty} p_k^z(x, A) \quad \forall x \in \mathcal{S}, \quad z \in \mathcal{C}, \quad A \subseteq \mathcal{S}, \quad n > 0, \end{aligned}$$

where $p_n(x, A) := \sum_{y \in A} p(x, y)$.

A proof of Theorem 5. For this proof we require one final ingredient: the geometric analogue of Lemma 24.8 which tells us that we lose nothing by assuming that the finite set F in (1) does not contain any transient states.

(11) LEMMA. *Let F be a finite set such that $\mathbb{E}_\gamma[\theta^{\phi_F}] < \infty$ and $\mathbb{E}_x[\theta^{\phi_F}] < \infty$ for all $x \in \mathcal{S}$ and some $\theta > 1$. There exists a $\vartheta > 1$ such that, after removing all transient states from F , $\mathbb{E}_\gamma[\vartheta^{\phi_F}] < \infty$ and $\mathbb{E}_x[\vartheta^{\phi_F}] < \infty$ for all x in $\mathcal{S}$.*

Proof. Replace Lemma 22.2(i) with Lemma 22.2(iii) in the proof of Lemma 23.4. $\square$

Armed with the above, Kendall's Theorem, the geometric trials arguments of Section 22, Foster's Theorem, and Theorem 4, the proof of the forward direction in Theorem 5 reduces to careful bookkeeping and (numerous) applications Markov's inequality:

Proof of Theorem 5 (the forward direction). Suppose that there exists (v, F, b) satisfying (1) for some $\theta > 1$ and that $\gamma(v) < \infty$. Theorem 24.1 shows that the chain is positive Tweedie recurrent with finitely many closed communicating classes and that each class intersects with F (for this last bit, see the Theorem 24.1's proof). Theorem 4 and Lemma 11 imply that there exists $\vartheta > 1$ such that, after removing all transient states from F ,

$$(12) \quad \mathbb{E}_\gamma[\vartheta^{\phi_F}] < \infty, \quad \mathbb{E}_x[\vartheta^{\phi_F}] < \infty \quad \forall x \in \mathcal{S}.$$

Let $\{\mathcal{C}_i : i \in \mathcal{I}\}$ the set of closed communicating classes $\mathcal{C}_i$ and $F_i := \mathcal{C}_i \cap F$ be their (non-empty) intersections with F . Pick any $A \subseteq \mathcal{S}$ and note that

$$(13) \quad |p_n(A) - \pi_\gamma(A)| \leq \mathbb{P}_\gamma(\{\phi_F > n, X_n \in A\}) + |\mathbb{P}_\gamma(\{\phi_F \leq n, X_n \in A\}) - \pi_\gamma(A)|.$$

Using Markov's inequality, we find that

$$(14) \quad \mathbb{P}_\gamma(\{\phi_F > n, X_n \in A\}) \leq \mathbb{P}_\gamma(\{\phi_F > n\}) \leq \mathbb{E}_\gamma[\vartheta^{\phi_F}] \vartheta^{-n}.$$

To deal with the second term in (13), note that

$$(15) \quad \mathbb{P}_\gamma(\{\phi_F \leq n, X_n \in A\}) = \sum_{i \in \mathcal{I}} \mathbb{P}_\gamma(\{\phi_{F_i} \leq n, X_n \in A\})$$

given that F contains no transient states (hence, $F = \cup_{i \in \mathcal{I}} F_i$) and that

$$(16) \quad \mathbb{P}_\gamma(\{\phi_{F_i} < \infty, \phi_{F_j} < \infty\}) = 0 \quad \forall i \neq j,$$

as the classes are closed (Proposition 13.11). For this reason,

$$(17) \quad \begin{aligned} |\mathbb{P}_\gamma(\{\phi_F \leq n, X_n \in A\}) - \pi_\gamma(A)| &\leq \sum_{i \in \mathcal{I}} |\mathbb{P}_\gamma(\{\phi_{F_i} \leq n, X_n \in A\}) - \mathbb{P}_\gamma(\{\phi_{\mathcal{C}_i} < \infty\}) \pi_i(A)| \\ &\leq \sum_{i \in \mathcal{I}} |\mathbb{P}_\gamma(\{\phi_{F_i} \leq n, X_n \in A\}) - \mathbb{P}_\gamma(\{\phi_{F_i} \leq n\}) \pi_i(A)| \\ &\quad + \sum_{i \in \mathcal{I}} (\mathbb{P}_\gamma(\{\phi_{\mathcal{C}_i} < \infty\}) - \mathbb{P}_\gamma(\{\phi_{F_i} \leq n\})) \pi_i(A). \end{aligned}$$

Using (16), Proposition 13.11, and Markov's inequality, we easily control the second sum:

$$(18) \quad \begin{aligned} \sum_{i \in \mathcal{I}} (\mathbb{P}_\gamma(\{\phi_{\mathcal{C}_i} < \infty\}) - \mathbb{P}_\gamma(\{\phi_{F_i} \leq n\})) \pi_i(A) &\leq \sum_{i \in \mathcal{I}} (\mathbb{P}_\gamma(\{\phi_{\mathcal{C}_i} < \infty\}) - \mathbb{P}_\gamma(\{\phi_{F_i} \leq n\})) \\ &= \mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}) - \mathbb{P}_\gamma(\{\phi_F \leq n\}) = 1 - \mathbb{P}_\gamma(\{\phi_F \leq n\}) = \mathbb{P}_\gamma(\{\phi_F > n\}) \leq \frac{\mathbb{E}_\gamma[\vartheta^{\phi_F}]}{\vartheta^n}, \end{aligned}$$

where $\mathcal{R}_+$ denotes the set of positive recurrent states and $\mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}) = 1$ because the chain is positive Tweedie recurrent. To control the first sum in (17), apply the strong Markov property similarly as in (20.25) to obtain

$$(19) \quad \mathbb{P}_\gamma(\{\phi_{F_i} \leq n, X_n \in A\}) = \sum_{x \in F_i} \sum_{m=0}^n \mathbb{P}_\gamma(\{\phi_{F_i} = m, X_{\phi_{F_i}} = x\}) p_{n-m}(x, A) \quad \forall i \in \mathcal{I}, \quad A \subseteq \mathcal{S},$$

and, consequently,

$$(20) \quad \sum_{i \in \mathcal{I}} |\mathbb{P}_\gamma(\{\phi_{F_i} \leq n, X_n \in A\}) - \mathbb{P}_\gamma(\{\phi_{F_i} \leq n\}) \pi_i(A)| \\ \leq \sum_{i \in \mathcal{I}} \sum_{x \in F_i} \sum_{m=0}^n \mathbb{P}_\gamma(\{\phi_{F_i} = m, X_{\phi_{F_i}} = x\}) |p_{n-m}(x, A) - \pi_i(A)|, \quad \forall A \subseteq \mathcal{S}.$$

To proceed, note that $\mathbb{E}_x[\vartheta^{\phi_{F_i}}] = \mathbb{E}_x[\vartheta^{\phi_F}] < \infty$ for each x in F_i and i in $\mathcal{I}$ as the classes are closed sets. Because any x in F_i is accessible from F_i , and because $F = \cup_{i \in \mathcal{I}} F_i$ is finite, Lemma 22.2(iii) and Kendall's Theorem (Theorem 21.1) imply that there exists a $\kappa \in (1, \vartheta]$ and $C_0 \in [0, \infty)$ such that

$$\mathbb{E}_x[\kappa^{\phi_x}] < \infty, \quad |p_n(x, x) - \pi_i(x)| \leq C_0 \kappa^{-n}, \quad \forall x \in F_i, \quad i \in \mathcal{I}, \quad n > 0,$$

where π_i denotes the ergodic distribution associated with $\mathcal{C}_i$. Setting $z := x$ in (10), we find that

$$(21) \quad |p_n(x, A) - \pi_i(A)| \leq p_n^x(x, A) + \sum_{k=1}^{n-1} \left| \sum_{l=1}^k p_l^x(x, x) p_{k-l}(x, x) - \pi_i(x) \right| p_{n-k}^x(x, A) \\ + \pi_i(z) \sum_{k=n}^{\infty} p_k^x(x, A) \quad \forall x \in F_i, \quad i \in \mathcal{I}, \quad A \subseteq \mathcal{S}, \quad n > 0.$$

Markov's inequality then yields bounds on the first two terms on the right-hand side of (21):

$$(22) \quad p_n^x(x, A) = \mathbb{P}_x(\{\phi_x \geq n, X_n \in A\}) \leq \mathbb{P}_x(\{\phi_x \geq n\}) \leq \mathbb{E}_x[\kappa^{\phi_x}] \kappa^{-n},$$

$$(23) \quad \sum_{k=n}^{\infty} p_k^x(x, A) \leq \sum_{k=n}^{\infty} \mathbb{P}_x(\{\phi_x \geq k\}) \leq \frac{\mathbb{E}_x[\kappa^{\phi_x}]}{\kappa - 1} \kappa^{-n}, \quad \forall x \in F, \quad A \subseteq \mathcal{S}, \quad n > 0.$$

For the middle term, we have that

$$\left| \sum_{l=1}^k p_l^x(x, x) p_{k-l}(x, x) - \pi_i(x) \right| \leq \sum_{l=1}^k p_l^x(x, x) |p_{k-l}(x, x) - \pi_i(x)| + \pi_i(x) \left| 1 - \sum_{l=1}^k p_l^x(x, x) \right| \\ \leq C_0 \kappa^{-k} \sum_{l=1}^k p_l^x(x, x) \kappa^l + \pi_i(x) \mathbb{P}_x(\{\phi_x \geq k\}) \leq \mathbb{E}_x[\kappa^{\phi_x}] (C_0 + \pi_i(x)) \kappa^{-k} \quad \forall x \in F_i, \quad i \in \mathcal{I},$$

for all x in F_i and i in $\mathcal{I}$. Applying Markov's inequality once again,

$$\sum_{k=1}^{n-1} \left| \sum_{l=1}^k p_l^x(x, x) p_{k-l}(x, x) - \pi_i(x) \right| p_{n-k}^x(x, A) \leq \mathbb{E}_x[\kappa^{\phi_x}] (C_0 + \pi_i(x)) \sum_{k=1}^{n-1} \kappa^{-k} p_{n-k}^x(x, A) \\ \leq \mathbb{E}_x[\kappa^{\phi_x}] (C_0 + \pi_i(x)) \kappa^{-n} \sum_{k=1}^{n-1} \kappa^k \mathbb{P}_x(\{\phi_x \geq k\}) \leq \mathbb{E}_x[\kappa^{\phi_x}]^2 (C_0 + \pi_i(x)) n \kappa^{-n}, \quad \forall x \in F_i, \quad i \in \mathcal{I}.$$

Because κ is no greater than ϑ and F is finite, the above, (12), and (21)–(23) show that

$$(24) \quad |p_n(x, A) - \pi_i(A)| \leq (C_1 + C_2 n) \kappa^{-n} \quad \forall x \in F_i, \quad i \in \mathcal{I}, \quad A \subseteq \mathcal{S}, \quad n \geq 0,$$

for some constants $C_1, C_2 \in [0, \infty)$. Pick any $\iota \in (1, \kappa)$ and note that $n \mapsto n(\kappa/\iota)^{-n}$ is bounded over $\mathbb{N}$. Hence, the above implies that there exists a third constant $C_3 \in [0, \infty)$ such that

$$|p_n(x, A) - \pi_i(A)| \leq C_3 \iota^{-n} \quad \forall x \in F_i, \quad i \in \mathcal{I}, \quad A \subseteq \mathcal{S}, \quad n \geq 0.$$

Plugging the above into (20) and using (16), we obtain

$$\begin{aligned}
 (25) \quad & \sum_{i \in \mathcal{I}} |\mathbb{P}_\gamma(\{\phi_{F_i} \leq n, X_n \in A\}) - \mathbb{P}_\gamma(\{\phi_{F_i} \leq n\}) \pi_i(A)| \\
 & \leq C_3 \iota^{-n} \sum_{i \in \mathcal{I}} \sum_{x \in F_i} \sum_{m=0}^n \mathbb{P}_\gamma(\{\phi_{F_i} = m, X_{\phi_{F_i}} = x\}) \iota^m \\
 & = C_3 \iota^{-n} \sum_{m=0}^n \mathbb{P}_\gamma(\{\phi_F = m\}) \iota^m \leq \mathbb{E}_\gamma \left[\iota^{\phi_F} \right] C_3 \iota^{-n} \quad \forall A \subseteq \mathcal{S}, \quad n \geq 0.
 \end{aligned}$$

Because ι is less than κ and κ is no greater than ϑ , the above, (12)–(14), and (17)–(18) imply that

$$|p_n(A) - \pi_\gamma(A)| \leq C_4 \iota^{-n} \quad \forall A \subseteq \mathcal{S}, \quad n \geq 0.$$

for some constant $C_4 \in [0, \infty)$. Taking the supremum over $A \subseteq \mathcal{S}$, completes the proof. $\square$

The proof of the reverse direction is nothing new:

Proof of Theorem 5 (the reverse direction). Given Kendall's Theorem (Theorem 21.1) and Theorem 4, this proof is analogous to that of the reverse direction of Foster's theorem (Theorem (24.1)), with Lemma 22.2(iii) replacing Lemma 22.2(ii) therein. $\square$

Notes and references. Theorem 5 dates back to (Popov, 1977) where both directions were proven for irreducible chains. The reverse direction in Theorem 5 follows directly from that in the irreducible case by restricting the state space to the closed communicating classes. As for the forward direction, it was shown in (Nummelin and Tuominen, 1982) that, for positive Harris recurrent chains (and potentially-uncountable state space generalisations thereof) with stationary distribution π , the existence of (v, F, b, θ) satisfying Theorem 5's premise implies that

$$\|p_n - \pi\| = \mathcal{O}(\kappa^{-n})$$

for some $\kappa > 1$, whenever the chain starts a state x such that $\pi(x) > 0$. Note that, in the countable case, this also follows easily from the irreducible case by restricting the state space to the support of π . In (Tweedie, 1983), the same result is given for all states x (instead of only those in π 's support), however the proof given therein cites an unpublished draft of (Nummelin and Tuominen, 1982) (while the published manuscript only considers starting states in the support of π). For initial distributions γ satisfying $\gamma(v) < \infty$, the result was proven in (Meyn and Tweedie, 1992).

The first-entrance last-exit decomposition (7) seems to date back to (Nummelin and Tweedie, 1978) and (Nummelin, 1978) where it was given for potentially-uncountable state space generalisations of discrete-time chains.

Continuous-time chains I: the basics

Continuous-time Markov processes on countable state spaces are known as *continuous-time Markov chains* or *continuous-time chains* for short. They are very similar to discrete-time chains except that the amount of time elapsed between any two consecutive transitions, known as the *waiting time*, is an exponentially distributed random variable whose mean is a function of the current state.

The aim of this chapter is to introduce the basic tools and concepts of continuous-time chain theory. Its development parallels that of its discrete-time counterpart (Sections 1–12) and the chapter overview given there largely applies here as well. The main differences are:

- We need to account for the possibility of continuous-time chains *exploding*: the waiting time between consecutive jumps drops fast enough that infinitely many jumps accumulate in a finite amount of time and the chain *explodes*. As we will see in Section 26, the moment of explosion is the instant in time by which the chain has left every single finite subset of the state space, hence justifying the name given to this event.
- We need to take a bit of time close to the beginning of the chapter (Section 27) to prove some fundamental properties specific to continuous-time chains: conditioned on the chain's history, a) the waiting time until the next jump and the state it jumps to are independent, b) the next state visited depends only the current state, and c) the waiting time is exponentially distributed and its mean depends only on the current state.
- Using these properties, we dispatch with the proofs of the Markov and strong Markov properties one after the other and early on (Section 30). In this case, the machinery required for the latter is no heavier than that required for the former. It does, however, mean that we need to introduce the notion of stopping times earlier (Section 28) than we did for discrete-time chains. Moreover, because the possibility of explosions makes conditioning on the chain's location X_t at time t using conditional expectation a bit of a pain (the chain may have exploded by time t !), we directly skip to the analogues of the ugly-but-useful versions of these properties given in Section 12 which seamlessly overcome this issue. Of course, for this we also need to introduce the path space and path law a bit earlier (Section 29) than we did before.
- We skip the continuous-time martingale property for it does not buy us anything in this book that its discrete-time counterpart does not already do.
- The question of whether the particular construction of the chains matters has a more complicated answer than in the discrete-time case and is left to the very end of the chapter (Section 37).

26. The Kendall-Gillespie (KG) algorithm and the chain's definition. For the type of continuous-time chains we will consider throughout the book (*minimal chains with sample paths that are right-continuous in the discrete topology*), it doesn't actually matter how the chain is defined (Section 37) and we might as well choose a definition that we find convenient. The aim of this section is to describe the construction we will use throughout the book involving an algorithm commonly used to simulate the chain in practice .

Rate matrices, jump probabilities, and jump rates. The starting point of our here is a totally stable⁴ and conservative rate matrix $Q := (q(x, y))_{x, y \in \mathcal{S}}$ indexed by a countable state

⁴Not to be confused with the notions of stability previously discussed. These have nothing to do with each other: a chain with a rate matrix that is not totally stable is one that will leave certain states immediately after entering them, see (Rogers and Williams, 2000a). This, however, does not rule out the existence of stationary distributions, convergence of the averages, etc.

space $\mathcal{S}$. That is, a matrix of real numbers indexed by $\mathcal{S}$ satisfying

$$(1) \quad q(x, y) \geq 0 \quad \forall x \neq y, \quad q(x, x) = - \sum_{y \neq x} q(x, y) < \infty \quad \forall x \in \mathcal{S}.$$

Throughout this book, we say “a rate matrix Q ” as an abbreviation for “a totally stable and conservative rate matrix Q ” and employ $q(x)$ as shorthand for $-q(x, x)$.

Using the rate matrix, we define the *jump matrix* $P = (p(x, y))_{x, y \in \mathcal{S}}$,

$$(2) \quad p(x, y) := \begin{cases} (1 - 1_x(y))q(x, y)/q(x) & \text{if } q(x) > 0, \\ 1_x(y) & \text{otherwise,} \end{cases} \quad \forall x, y \in \mathcal{S},$$

and a vector $(\lambda(x))_{x \in \mathcal{S}}$ of *jump rates*,

$$(3) \quad \lambda(x) := \begin{cases} q(x) & \text{if } q(x) > 0 \\ 1 & \text{if } q(x) = 0 \end{cases} \quad \forall x \in \mathcal{S}.$$

(4) EXERCISE. *Convince yourself that P in (2) is a one-step matrix of discrete-time chain (i.e., that it satisfies (1.1)).*

The technical set-up. To define the chain, we additionally require a measurable space $(\Omega, \mathcal{F})$, a collection $(\mathbb{P}_x)_{x \in \mathcal{S}}$ of probability measures on $(\Omega, \mathcal{F})$, an $\mathcal{F}/2^{\mathcal{S}}$ -measurable function X_0 from Ω to $\mathcal{S}$, a sequence of $\mathcal{F}/\mathcal{B}((0, 1))$ -measurable functions $U_1, U_2, \dots$ from Ω to $(0, 1)$, and one of $\mathcal{F}/\mathcal{B}((0, 1))$ -measurable functions $\xi_1, \xi_2, \dots$ from Ω to $(0, 1)$ such that, under $\mathbb{P}_x$,

1. $\{X_0 = x\}$ occurs almost surely: $\mathbb{P}_x(\{X_0 = x\}) = 1$;
2. for each $n > 0$, the random variable U_n is uniformly distributed on $(0, 1)$: $\mathbb{P}_x(\{U_n \leq u\}) = u$ for all $u \in (0, 1)$;
3. for each $n > 0$, the random variable ξ_n is exponentially distributed on $(0, 1)$ with mean one: $\mathbb{P}_x(\{\xi_n > s\}) = e^{-s}$ for all $s \in (0, \infty)$;
4. and the random variables $X_0, U_1, U_2, \dots, \xi_1, \xi_2, \dots$ are independent:

$$\begin{aligned} & \mathbb{P}_x(\{X_0 = y, U_1 \leq u_1, \dots, U_n \leq u_n, \xi_1 > s_1, \dots, \xi_m > s_m\}) \\ &= \mathbb{P}_x(\{X_0 = y\}) \mathbb{P}_\gamma(\{U_1 \leq u_1\}) \dots \mathbb{P}_\gamma(\{U_n \leq u_n\}) \mathbb{P}_\gamma(\{\xi_1 > s_1\}) \dots \mathbb{P}_\gamma(\{\xi_m > s_m\}) \end{aligned}$$

for all $y \in \mathcal{S}$, $u_1, \dots, u_n \in (0, 1)$, $s_1, \dots, s_m \in (0, \infty)$, and $n, m > 0$;

for each x in $\mathcal{S}$. Technically, we can construct this space, measures, and random variables using Theorem 1.3 similarly as we did in Section 1 for the discrete-time case.

Below, we will set the chain's starting location to be X_0 . For this reason, $\mathbb{P}_x$ will describe the statistics of the chain were it to start from x . To instead describe situations in which the chain's starting location was sampled from a given probability distribution $\gamma = (\gamma(x))_{x \in \mathcal{S}}$ on $\mathcal{S}$ (the *initial distribution*), we define the probability measure $\mathbb{P}_\gamma$ on $(\Omega, \mathcal{F})$ via

$$(5) \quad \mathbb{P}_\gamma(A) := \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x(A) \quad \forall A \in \mathcal{F}.$$

Analogous arguments to those given after (1.2) show that X_0 has law γ under $\mathbb{P}_\gamma$ and that 2–4 of the above list holds if we replace $\mathbb{P}_x$ with $\mathbb{P}_\gamma$. As in the discrete-time case, we use $\mathbb{E}_x$ (resp. $\mathbb{E}_\gamma$) to denote expectation with respect to $\mathbb{P}_x$ (resp. $\mathbb{P}_\gamma$).

The Kendall-Gillespie algorithm and the chain's definition. We construct our chain recursively by running the *Kendall-Gillespie algorithm* given below as Algorithm 2. It goes as follows: sample a state x from the initial distribution γ and start the chain at x . Wait an exponentially distributed amount of time with mean $1/\lambda(x)$, where $\lambda(x)$ is as in (3); sample y from the probability distribution $p(x, \cdot)$ in (2); and update the chain's state to y (we say that the chain *jumps* from x to y , we call the time waited the *waiting time* and the instant at which it jumps the *jump time*). Repeat these steps starting from y instead of x . All random variables sampled must be independent of each other.

Algorithm 2 The Kendall-Gillespie Algorithm on $\mathcal{S} = \{x_0, x_1, x_2, \dots\}$

```

1:  $Y_0 := X_0$ 
2: for  $n = 1, 2, \dots$  do
3:   sample  $U_n \sim \text{uni}_{(0,1)}$  independently of  $\{X_0, \xi_1, \dots, \xi_{n-1}, U_1, \dots, U_{n-1}\}$ 
4:   sample  $\xi_n \sim \exp(1)$  independently of  $\{X_0, \xi_1, \dots, \xi_{n-1}, U_1, \dots, U_n\}$ 
5:    $i := 0$ 
6:   while  $U_n > \sum_{j=0}^i p(Y_{n-1}, x_j)$  do
7:      $i := i + 1$ 
8:   end while
9:    $Y_n := x_i$ 
10:   $S_n := \xi_n / \lambda(Y_{n-1})$ 
11: end for

```

Technically, the algorithm constructs the *waiting times* $S_1, S_2, \dots$ and the *jump chain* (or *embedded chain*) $Y := (Y_n)_{n \in \mathbb{N}}$ and the chain is defined to be a piecewise constant interpolation of the jump chain:

$$(6) \quad X_t(\omega) := Y_n(\omega) \quad \forall t \in [T_n(\omega), T_{n+1}(\omega)), \quad \omega \in \Omega, \quad n \geq 0,$$

where T_n denotes the n th *jump time*:

$$(7) \quad T_0(\omega) := 0 \quad \forall \omega \in \Omega, \quad T_n(\omega) := \sum_{m=1}^n S_m(\omega) \quad \forall \omega \in \Omega, \quad n > 0.$$

Explosions and regular rate matrices. The chain is defined only up until the *explosion time* (also known as the *escape time* and the *time of the first infinity*),

$$(8) \quad T_\infty(\omega) := \lim_{n \rightarrow \infty} T_n(\omega) = \sum_{m=1}^{\infty} S_m(\omega) \quad \forall \omega \in \Omega,$$

In other words, X_t is defined only on $\{t < T_\infty\}$ and not on all of Ω .

To understand the meaning behind T_∞ , let $\mathcal{S}_1 \subseteq \mathcal{S}_2 \subseteq \dots$ be any increasing sequence of finite subsets (or *truncations*) of $\mathcal{S}$ that approach $\mathcal{S}$ (i.e., $\cup_{r=1}^{\infty} \mathcal{S}_r = \mathcal{S}$) and, for each $r > 0$, let τ_r be the time that the chain X first leaves $\mathcal{S}_r$:

$$(9) \quad \tau_r(\omega) := \inf\{t \in [0, T_\infty(\omega)) : X_t(\omega) \notin \mathcal{S}_r\} \quad \forall \omega \in \Omega.$$

Because our truncations are an increasing, $(\tau_r)_{r \in \mathbb{Z}_+}$ is an increasing sequence of random variables. Therefore, the limit $\lim_{r \rightarrow \infty} \tau_r$ exists for everywhere. Moreover, this limit is T_∞ :

(10) **THEOREM.** *If $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ is an increasing sequence of finite sets such that $\cup_{r=1}^{\infty} \mathcal{S}_r = \mathcal{S}$, then τ_r tends to T_∞ , $\mathbb{P}_\gamma$ -almost surely, for all initial distributions γ .*

The limiting random variable $\lim_{r \rightarrow \infty} \tau_r$ is the point in time by which the chain has left all of the truncations in the sequence $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$. Because the truncations approach $\mathcal{S}$, every finite subset of the state space is contained in all truncations sufficiently far along in the sequence. Thus, the limit $\lim_{r \rightarrow \infty} \tau_r$ is the instant in time that the chain has left all finite subsets of the state space. The above tells us that this limit is (almost surely) equal to T_∞ regardless of the particular sequence of finite truncations $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ in its definition. For these reasons, we interpret T_∞ as the point in time that the chain leaves the state space, or, in other words, *explodes* (see Figure 1 Part I introduction for an example). The proof of Theorem 10 builds on the following lemma and can be found at the end of the section.

(11) LEMMA. *The random variables T_∞ and $\sum_{n=0}^{\infty} 1/\lambda(Y_n)$ are either both finite or both infinite, with $\mathbb{P}_\gamma$ -probability one, for any initial distribution γ .*

Proof. See the end of Section 27. □

An example of explosive chain. Consider a chain taking values on $\mathbb{N}$ that moves one state up every jump with rate $\lambda(x) := 2^x$ for all x in $\mathbb{N}$. That is, a chain with rate matrix

$$Q = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & \dots \\ 0 & -2 & 2 & 0 & 0 & \dots \\ 0 & 0 & -4 & 4 & 0 & \dots \\ 0 & 0 & 0 & -8 & 8 & \ddots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots \end{bmatrix}.$$

In this case,

$$\sum_{n=0}^{\infty} \frac{1}{\lambda(Y_n)} = \sum_{n=x}^{\infty} \frac{1}{2^n} = \frac{2}{2^x} \quad \mathbb{P}_x\text{-almost surely, } \forall x \in \mathbb{N}.$$

For this reason, (5) and Lemma 11 implies that T_∞ is finite, $\mathbb{P}_\gamma$ almost surely, for all initial distribution γ .

Absorbing states and fictitious jumps. Our definition of the jump matrix in (2) implies that any state x satisfying $q(x) = 0$ is *absorbing* in the sense that once inside such an x , the chain may never leave:

(12) PROPOSITION. *For any x in $\mathcal{S}$,*

$$\mathbb{P}_x(\{T_\infty = \infty, X_t = x, \forall t \in [0, \infty)\}) = \begin{cases} 1 & \text{if } q(x) = 0 \\ 0 & \text{if } q(x) \neq 0 \end{cases}.$$

Proof. See the end of Section 29. □

Note that Algorithm 2 keeps the chain ‘jumping’ in an absorbing state x . This is nothing more than a notational trick that frees us from having to keep track of the number of jumps until the chain gets absorbed: the jumps are fictitious in the sense that they take the chain from x to x and nothing changes.

A proof of Theorem 10. Let $\tau_\infty := \lim_{r \rightarrow \infty} \tau_r$ be the limit of the random times. First, note that τ_r 's definition in (9) implies that

$$(13) \quad \tau_r(\omega) \in [0, T_\infty(\omega)) \cup \{\infty\} \quad \forall \omega \in \Omega$$

and that X lies outside $\mathcal{S}_r$ at the time τ_r if τ_r is finite or, equivalently, less than T_∞ (this follows from the right-continuity of the chain's paths, e.g., see Proposition 34.9 later on). Because the

truncations are increasing, it follows that $\{\tau_l < T_\infty, X_{\tau_l} \in \mathcal{S}_r\}$ is empty if $l \geq r$. For this reason, bounded convergence implies that

$$\mathbb{P}_\gamma(\{\tau_\infty < T_\infty, X_{\tau_\infty} \in \mathcal{S}_r\}) = \lim_{l \rightarrow \infty} \mathbb{P}_\gamma(\{\tau_l < T_\infty, X_{\tau_l} \in \mathcal{S}_r\}) = 0 \quad \forall r > 0.$$

In turn, monotone convergence shows that

$$(14) \quad \mathbb{P}_\gamma(\{\tau_\infty < T_\infty\}) = \mathbb{P}_\gamma(\{\tau_\infty < T_\infty, X_{\tau_\infty} \in \mathcal{S}\}) = \lim_{r \rightarrow \infty} \mathbb{P}_\gamma(\{\tau_\infty < T_\infty, X_{\tau_\infty} \in \mathcal{S}_r\}) = 0.$$

Next, suppose that ω is such that $T_\infty(\omega)$ is finite but $\tau_\infty(\omega)$ is infinite. It follows from (13) that there exists a positive integer R such that $\tau_R = \infty$. In other words, $X_t(\omega)$ belongs to $\mathcal{S}_R$ for all t in $[0, T_\infty(\omega))$. In particular,

$$Y_n(\omega) = X_{T_n(\omega)}(\omega) \in \mathcal{S}_R \quad \forall n \geq 0.$$

Finiteness of $\mathcal{S}_R$ then implies that

$$\sum_{n=0}^{\infty} \frac{1}{\lambda(Y_n(\omega))} \geq \left(\inf_{x \in \mathcal{S}_R(\omega)} \frac{1}{\lambda(x)} \right) \left(\sum_{n=0}^{\infty} 1 \right) = \infty.$$

Because Lemma 11 shows that the event that $\sum_{n=0}^{\infty} \frac{1}{\lambda(Y_n)}$ is infinite and T_∞ is finite has zero probability of occurring, it follows from the above that

$$(15) \quad \mathbb{P}_\gamma(\{T_\infty < \tau_\infty = \infty\}) = 0.$$

Lastly, given that τ_r is bounded above by τ_∞ , if ω is such that $\tau_\infty(\omega)$ is finite, then $\tau_r(\omega)$ is finite. In this case, (13) implies that

$$\tau_r(\omega) < T_\infty(\omega) \quad \forall r > 0.$$

Taking the limit $r \rightarrow \infty$ shows that $\tau_\infty(\omega)$ is at most $T_\infty(\omega)$. For this reason, the event $\{T_\infty < \tau_\infty < \infty\}$ is empty and, thus,

$$(16) \quad \mathbb{P}_\gamma(\{T_\infty < \tau_\infty < \infty\}) = 0.$$

Putting (14)–(16) together completes the proof.

Notes and references. In the physical and life sciences, Algorithm 2 is typically referred to as the *Stochastic Simulation Algorithm* or the *Gillespie Algorithm* after D. T. Gillespie who introduced it to chemical physics literature (Gillespie, 1976) and argued (Gillespie, 1977) in favour of its use in the modelling of reacting chemical species. The ideas underlying the algorithm (Properties 1–3 in the ensuing section) seem to have been first delineated in the ‘40s by W. Feller (Feller, 1940) and J. L. Doob (Doob, 1942, 1945). The first published implementation of the algorithm was carried out in 1950 by D. G. Kendall (Kendall, 1950) (see also his student F. G. Foster in the discussion of (Kendall, 1951a)).

In applications, the possibility of an explosion is often regarded as a pathological technicality with no relevance to real life. You may have encountered before an argument of the type “this real life phenomena clearly cannot explode, hence neither can its model”. Even if something seems reasonable based on physical arguments, models are abstractions of real life and, unfortunately, can be far from faithful representations of it. This is particularly prominent in the context of modelling where the model is not yet fitted. Indeed, many fitting algorithms involve automated sweeps of parameter sets, and the algorithms can easily step through parameter values that lead to poor models (e.g., an exploding model of a phenomenon impossible of exhibiting this type of behaviour).

Theorem 10 as given here seems to have been first shown in (Kuntz, 2017), although I would take this with a grain of salt. Lemma 11 which makes up the bulk of the theorem’s proof goes back to (Chung, 1967), if not earlier.

27. Three important properties of the jump chain and waiting times. The main aim of this section is to prove three important facts regarding the chain defined in previous section. Namely, when conditioned on the chain's history up until (and including) the n th jump time T_n ,

1. the next state Y_{n+1} visited by the chain has distribution $(p(Y_n, y))_{y \in \mathcal{S}}$, where Y_n denotes its current state;
2. the amount of time S_{n+1} elapsed until the next jump is exponentially distributed with mean $1/\lambda(Y_n)$; and
3. Y_{n+1} and S_{n+1} are independent of each other.

The first property implies that Y is a discrete-time chain and, hence, justifies the names of *jump chain* and *embedded chain* afforded to it.

These properties are not an artifice of our definitions in Section 26: all $\mathcal{S}$ -valued continuous-time processes satisfying the Markov property with right-continuous sample paths that explode at most once have them (Section 37). Furthermore, these properties, together with λ and P , fully determine the law induced by the chain on the space of paths (Section 29). In other words, the paths of any continuous-time chain with right-continuous paths that explode at most once, jump matrix P , and jump rates λ are statistically indistinguishable from those of the chain in (26.6)–(26.7). This is the reason why, as mentioned in Section 26, we lose nothing by focusing on the particular algorithmic construction of the chain introduced there.

The filtration generated by the jump chain and jump times. To argue 1–3 above, we must first introduce the filtration $(\mathcal{G}_n)_{n \in \mathbb{N}}$ generated by the jump chain and jump times. It describes the information that can be gleaned by observing the chain's progress from one jump time to another. In particular, for each n , $\mathcal{G}_n$ consists of all events whose occurrence can be determined by observing the chain's path up until (and including) T_n and, so, $\mathcal{G}_n$ formalises the idea of the chain's history up until T_n . In full:

(1) DEFINITION (The filtration generated by the jump chain and jump times). *Let $(\mathcal{G}_n)_{n \in \mathbb{N}}$ be the filtration defined by*

$$\mathcal{G}_n := \text{the sigma-algebra generated by } (Y_0, Y_1, \dots, Y_n, T_0, T_1, \dots, T_n) \quad \forall n \geq 0,$$

meaning that $\mathcal{G}_n$ is the smallest sigma-algebra such that $Y_0, Y_1, \dots, Y_n, T_1, \dots, T_n$ are $\mathcal{G}_n$ -measurable random variables.

Because we know the times $T_1, \dots, T_n$ at which the first n jumps occur if and only if we know the amount of time S_1 elapsed until the first jump and the amounts $S_2, \dots, S_n$ elapsed between the next $n - 1$ jumps,

$$(2) \quad \mathcal{G}_n = \text{the sigma-algebra generated by } (Y_0, Y_1, \dots, Y_n, S_1, \dots, S_n) \quad \forall n > 0.$$

(3) EXERCISE. *Verify that (2) holds.*

A proof of Properties 1–3. Using the filtration, we formalise Properties 1–3 as

1. $\mathbb{P}_\gamma(\{Y_{n+1} = y, S_{n+1} > s\} | \mathcal{G}_n) = \mathbb{P}_\gamma(\{Y_{n+1} = y\} | \mathcal{G}_n) \mathbb{P}_\gamma(\{S_{n+1} > s\} | \mathcal{G}_n)$, $\mathbb{P}_\gamma$ -almost surely, for all states y , non-negative numbers s , and initial distributions γ ;
2. $\mathbb{P}_\gamma(\{Y_{n+1} = y\} | \mathcal{G}_n) = p(Y_n, y)$, $\mathbb{P}_\gamma$ -almost surely, for all y, s, γ ; and
3. $\mathbb{P}_\gamma(\{S_{n+1} > s\} | \mathcal{G}_n) = e^{-\lambda(Y_n)s}$, $\mathbb{P}_\gamma$ -almost surely, for all y, s, γ .

We prove a slight generalisation of the above:

(4) THEOREM. *For any given natural number n , let W be a $\mathcal{G}_n/\mathcal{B}(\mathbb{R})$ -measurable random variable, f a bounded real-valued function on $\mathcal{S}$, and g a bounded $\mathcal{B}(\mathbb{R}^2)/\mathcal{B}(\mathbb{R})$ -measurable function. For any initial distribution γ ,*

$$\begin{aligned} \mathbb{E}_\gamma [f(Y_{n+1})g(S_{n+1}, W)|\mathcal{G}_n] &= \mathbb{E}_\gamma [f(Y_{n+1})|\mathcal{G}_n] \mathbb{E}_\gamma [g(S_{n+1}, W)|\mathcal{G}_n], \quad \mathbb{P}_\gamma\text{-almost surely}; \\ \mathbb{E}_\gamma [f(Y_{n+1})|\mathcal{G}_n] &= Pf(Y_n), \quad \mathbb{P}_\gamma\text{-almost surely}; \\ \mathbb{E}_\gamma [g(S_{n+1}, W)|\mathcal{G}_n] &= \int_0^\infty g(s, W)\lambda(Y_n)e^{-\lambda(Y_n)s}ds, \quad \mathbb{P}_\gamma\text{-almost surely}; \end{aligned}$$

where $P = (p(x, y))_{x, y \in \mathcal{S}}$ denotes the jump matrix (26.2), $\lambda = (\lambda(x))_{x \in \mathcal{S}}$ the jump rates (26.3), and

$$Pf(x) := \sum_{y \in \mathcal{S}} p(x, y)f(y) \quad \forall x \in \mathcal{S}.$$

Proof. The definitions of Y_{n+1} and S_{n+1} in the Kendall-Gillespie Algorithm (Algorithm 2 in Section 26) imply that

$$f(Y_{n+1}) = \tilde{f}(U_{n+1}, Y_n), \quad g(S_{n+1}, W) = g(\xi_{n+1}/\lambda(Y_n), W),$$

where

$$\tilde{f}(u, x) := \sum_{j=0}^{\infty} f(x_j) 1_{\{\sum_{k=0}^{j-1} p(x, x_k) \leq u < \sum_{k=0}^j p(x, x_k)\}} \quad \forall x \in \mathcal{S}, \quad u \in (0, 1),$$

and we are using the enumeration of the state space introduced in the algorithm. By the definitions of U_{n+1} , ξ_{n+1} , and $\mathcal{G}_n$, the first two are independent of the third. For this reason, the same arguments as those given in (Williams, 1991, Section 9.10) show that

$$\mathbb{E}_\gamma [f(Y_{n+1})g(S_{n+1}, W)|\mathcal{G}_n] = \mathbb{E}_\gamma [\tilde{f}(U_{n+1}, Y_n)g(\xi_{n+1}/\lambda(Y_n), W)|\mathcal{G}_n] = h(Y_n, W) \quad \mathbb{P}_\gamma\text{-almost surely},$$

where

$$h(x, w) = \mathbb{E}_\gamma [\tilde{f}(U_{n+1}, x)g(\xi_{n+1}/\lambda(x), w)] \quad \forall x \in \mathcal{S}, \quad w \in \mathbb{R}.$$

Because U_{n+1} and ξ_{n+1} are independent,

$$h(x, w) = h_1(x)h_2(x, w) \quad \text{where} \quad h_1(x) := \mathbb{E}_\gamma [\tilde{f}(U_{n+1}, x)], \quad h_2(x, w) := \mathbb{E}_\gamma [g(\xi_{n+1}/\lambda(x), w)],$$

for all x in $\mathcal{S}$ and w in $\mathbb{R}$. Similarly, the arguments in (Williams, 1991, Section 9.10) imply that

$$h_1(Y_n) = \mathbb{E}_\gamma [\tilde{f}(U_{n+1}, Y_n)|\mathcal{G}_n], \quad h_2(Y_n, W) = \mathbb{E}_\gamma [g(\xi_{n+1}/\lambda(Y_n), W)|\mathcal{G}_n], \quad \mathbb{P}_\gamma\text{-almost surely}.$$

The result then follows from U_{n+1} 's uniform distribution, ξ_{n+1} 's unit-mean exponential distribution, and Fubini's theorem:

$$\begin{aligned} h_1(x) &= \sum_{j=0}^{\infty} f(x_j) \mathbb{P}_\gamma \left(\left\{ \sum_{k=0}^{j-1} p(x, x_k) \leq u < \sum_{k=0}^j p(x, x_k) \right\} \right) = \sum_{j=0}^{\infty} f(x_j) p(x, x_j) = Pf(x), \\ h_2(x, w) &= \int_0^\infty g(t/\lambda(x), w) e^{-t} dt = \int_0^\infty g(s, w) \lambda(x) e^{-\lambda(x)s} ds \quad \forall x \in \mathcal{S}, \quad w \in \mathbb{R}, \end{aligned}$$

where we have used the change of variable $s := t/\lambda(x)$ in the last equation. $\square$

Paying our dues: a proof of Lemma 26.11. This proof relies on the following consequence of the discrete-time martingale convergence theorem:

(5) THEOREM. (Levy's extension of the Borel-Cantelli lemmas, (Williams, 1991, Section 12.15)). Suppose that $W = (W_n)_{n \in \mathbb{N}}$ is a sequence of non-negative random variables bounded above by one (i.e., $W_n \leq 1$ for all $n \geq 0$) and adapted to some filtration $(\mathcal{F}_n)_{n \in \mathbb{N}}$ (i.e., W_n is $\mathcal{F}_n/\mathcal{B}(\mathbb{R})$ -measurable for all $n \geq 0$). With probability one, the sums

$$\sum_{n=1}^{\infty} W_n \quad \text{and} \quad \sum_{n=1}^{\infty} \mathbb{E}[W_n | \mathcal{F}_{n-1}],$$

are both finite or both infinite.

(6) EXERCISE. Prove (5) by generalising the proof given in (Williams, 1991, Section 12.15) for the special case $W_n = 1_{E_n}$ with $E_n \in \mathcal{F}_n$.

We are now in a great position to prove Lemma 26.11:

Proof of Lemma 26.11. It is not difficult to check that

$$T_{\infty} = \lim_{m \rightarrow \infty} T_m = \sum_{n=1}^{\infty} S_n$$

is finite if and only if

$$\sum_{n=1}^{\infty} S_n \wedge 1$$

is finite. Theorem 5 implies that, with $\mathbb{P}_{\gamma}$ -probability one, the above sum is finite if and only if

$$\sum_{n=1}^{\infty} \mathbb{E}_{\gamma}[S_n \wedge 1 | \mathcal{G}_{n-1}]$$

is finite. Because S_n is exponentially distributed with mean $1/\lambda(Y_{n-1})$ (Theorem 4),

$$\begin{aligned} \mathbb{E}_{\gamma}[S_n \wedge 1 | \mathcal{G}_{n-1}] &= \int_0^1 s \lambda(Y_{n-1}) e^{-\lambda(Y_{n-1})s} ds + \int_1^{\infty} \lambda(Y_{n-1}) e^{-\lambda(Y_{n-1})s} ds \\ &= \frac{1}{\lambda(Y_{n-1})} (1 - e^{-\lambda(Y_{n-1})}), \quad \mathbb{P}_{\gamma}\text{-almost surely, } \forall n > 0. \end{aligned}$$

For this reason,

$$(7) \quad \sum_{n=1}^{\infty} \mathbb{E}_{\gamma}[S_n \wedge 1 | \mathcal{G}_{n-1}] = \sum_{n=0}^{\infty} \frac{1}{\lambda(Y_n)} (1 - e^{-\lambda(Y_n)}) \leq \sum_{n=0}^{\infty} \frac{1}{\lambda(Y_n)} \quad \mathbb{P}_{\gamma}\text{-almost surely,}$$

and to complete the proof we need only to show that the rightmost sum is finite whenever the middle one is. The one-sided limit comparison test implies that, for any sequence of positive numbers $(\alpha_n)_{n \in \mathbb{N}}$, the sum $\sum_{n=0}^{\infty} \frac{1}{\alpha_n}$ is finite if $\sum_{n=0}^{\infty} \frac{1}{\alpha_n} (1 - e^{-\alpha_n})$ is finite unless

$$\limsup_{n \rightarrow \infty} \frac{1}{1 - e^{-\alpha_n}} = \infty,$$

or, equivalently, there exists a subsequence $(\alpha_{n_k})_{k \in \mathbb{N}}$ such that

$$\lim_{k \rightarrow \infty} \alpha_{n_k} = 0.$$

However, this is impossible if $\sum_{n=0}^{\infty} \frac{1}{\alpha_n} (1 - e^{-\alpha_n})$ is finite as L'Hôpital's rule would imply that

$$\lim_{k \rightarrow \infty} \frac{1}{\alpha_{n_k}} (1 - e^{-\alpha_{n_k}}) = \lim_{k \rightarrow \infty} e^{-\alpha_{n_k}} = 1$$

and $\sum_{n=0}^{\infty} \frac{1}{\alpha_n} (1 - e^{-\alpha_n})$ would be infinite.

□

Notes and references. The proof of Lemma 26.11 given above is a slightly more annotated version of that given for Theorem 1 in (Chung, 1967, p.237).

28. The filtration generated by the chain and stopping times. The filtration $(\mathcal{F}_t)_{t \geq 0}$ generated by the chain describes the information that may be gleaned from observing the chain's progress. For any point in time $t \geq 0$, $\mathcal{F}_t$ captures the chain's history up until (and including) t : an event A belongs to $\mathcal{F}_t$ if and only if we are able to determine whether or not it has occurred by time t from observing the chain's position up until (and including) t . Formally:

(1) DEFINITION (The filtration generated by the chain). *Let $(\mathcal{F}_t)_{t \geq 0}$ be the filtration defined by*

$$\mathcal{F}_t := \text{the sigma-algebra generated by } \{T_n \leq s, Y_n = x\} \quad \forall s \in [0, t], \quad x \in \mathcal{S}, \quad n \in \mathbb{N},$$

meaning the smallest sigma-algebra containing $\{T_n \leq s, Y_n = x\}$ for all $s \leq t$, $x \in \mathcal{S}$, and $n \in \mathbb{N}$.

In other texts, you may find $(\mathcal{F}_t)_{t \geq 0}$ defined slightly differently:

(2) EXERCISE. *Show that, for any $t \in [0, \infty)$, the sets*

$$\{X_s = x, s < T_\infty\} \quad \forall s \leq t, \quad x \in \mathcal{S}$$

generate $\mathcal{F}_t$ in (1). Hint: (26.6)–(26.7) implies that

$$T_{n+1}(\omega) = \inf\{r \in \mathbb{Q} \cap (T_n(\omega), \infty) : X_r(\omega) \neq X_{T_n(\omega)}(\omega)\} \quad \forall \omega \in \Omega, \quad n \geq 0,$$

Stopping times. A non-negative random variable η is an $(\mathcal{F}_t)_{t \geq 0}$ -stopping time if and only if it is the time that some event occurs and we are able to determine whether the event has occurred by observing the chain up until that instant. For example, η can be the first (or second or k th) time that the chain visits some subset of the state space of interest, but not the last time that it enters the subset. Formally:

(3) DEFINITION (Stopping times). *Given the filtration $(\mathcal{F}_t)_{t \geq 0}$ generated by the chain (c.f. (1)), a random variable $\eta : \Omega \rightarrow [0, \infty]$ is said to be a $(\mathcal{F}_t)_{t \geq 0}$ -stopping time if, for each $t \geq 0$, the event $\{\eta \leq t\}$ belongs to $\mathcal{F}_t$. With the stopping time we associated a sigma-algebra $\mathcal{F}_\eta$ defined by*

$$\forall A \in \mathcal{F}, \quad A \in \mathcal{F}_\eta \Leftrightarrow A \cap \{\eta \leq t\} \in \mathcal{F}_t \quad \forall t \geq 0.$$

(4) EXERCISE. *Convince yourself that $\mathcal{F}_\eta$ is a sigma-algebra.*

The reasons behind the name ‘stopping time’ afforded to these random variables are the same as in the discrete-time case (Section 8). Similarly, the *pre- η sigma-algebra* $\mathcal{F}_\eta$ is the collection of events whose occurrence we are able to glean by tracking the chain's position up until the stopping time η . For instance, if η is the second time that the chain visits a given state x , then the event that the chain visits x once (or at least once, or twice, or at least twice) belongs to $\mathcal{F}_\eta$ but the event that the chain visits x three (or four, or ...) does not belong to $\mathcal{F}_\eta$.

The following lemma is very useful. It formalises three simple facts: (i) from observing the chain up until a stopping time ϑ , we can deduce whether or not the event associated with another stopping η occurs no later than (before than, or at) ϑ ; (ii) if we can deduce whether an event A occurs from observing the chain up until η , we can deduce whether A and the event associated with η both occur no later than ϑ from observing the chain up until ϑ ; and (iii) if η occurs no later than ϑ and we can deduce whether an event A occurs by observing the chain up until η , then we can deduce whether A occurs from observing the chain up until time ϑ .

(5) LEMMA. *If η and ϑ are $(\mathcal{F}_t)_{t \geq 0}$ -stopping times, then*

- (i) The events $\{\eta \leq \vartheta\}$, $\{\eta < \vartheta\}$, and $\{\eta = \vartheta\}$ belong to $\mathcal{F}_\vartheta$.
- (ii) Given any event A in $\mathcal{F}_\eta$, the event $A \cap \{\eta \leq \vartheta\}$ belongs to $\mathcal{F}_\vartheta$.
- (iii) In particular, if $\eta \leq \vartheta$, then $\mathcal{F}_\eta \subseteq \mathcal{F}_\vartheta$.

Proof. Given that

$$\{\eta < \vartheta\} \cap \{\vartheta \leq t\} = \bigcap_{s \in [0, t) \cap \mathbb{Q}} \{\eta \leq s\} \cap \{s < \vartheta\} \cap \{\vartheta \leq t\} \in \mathcal{F}_t \quad \forall t \in [0, \infty),$$

this proof is analogous to that of the lemma's discrete-time counterpart (Lemma 8.3) and we skip it. $\square$

(6) EXERCISE. Show that $(\mathcal{F}_t)_{t \geq 0}$ -stopping times η and states x in $\mathcal{S}$, $\{\eta < T_\infty, X_\eta = x\}$ belongs to $\mathcal{F}_\eta$. Hint: re-write $\{\eta < T_\infty, X_\eta = x\}$ as

$$\{\eta < T_\infty, X_\eta = x\} = \bigcup_{n=0}^{\infty} \{T_n \leq \eta < T_{n+1}, Y_n = x\},$$

and use Lemma 5(i, ii).

Does $\mathcal{F}_{T_n} = \mathcal{G}_n$? An open problem. It follows directly from Definitions 1 and 3 that any jump time T_n is a stopping time. The pre- T_n sigma algebra $\mathcal{F}_{T_n}$ contains all events whose occurrence or non-occurrence can be deduced from observing the chain's path up until (and including) the jump time T_n . Because this path segment is characterised by the first $n+1$ states $Y_0, \dots, Y_n$ visited and first n jump times $T_1, \dots, T_n$ (recall (26.6)–(26.7)), it seems natural for $\mathcal{F}_{T_n}$ to coincide with $\mathcal{G}_n$ in Definition 27.1 (i.e., with the collection of events whose occurrence or non-occurrence can be deduced from observing $Y_0, \dots, Y_n, T_1, \dots, T_n$). Indeed, it is straightforward to show that $\mathcal{G}_n$ is contained in $\mathcal{F}_{T_n}$:

(7) PROPOSITION. $\mathcal{G}_n \subseteq \mathcal{F}_{T_n}$ for all $n \geq 0$.

Proof. $\mathcal{G}_n$ is generated by sets of the form

$$A := \{Y_0 = x_0, Y_1 = x_1, \dots, Y_n = x_n, T_0 \leq t_0, T_1 \leq t_1, \dots, T_n \leq t_n\},$$

where $x_0, \dots, x_n \in \mathcal{S}$ and $t_0, \dots, t_n \in [0, \infty)$. Fix any $t \geq 0$. Because $\mathcal{F}_t$ is closed under finite intersections, it follows from its definition that

$$A \cap \{T_n \leq t\} = \{Y_0 = x_0, Y_1 = x_1, \dots, Y_n = x_n, T_0 \leq t_0 \wedge t, T_1 \leq t_1 \wedge t, \dots, T_n \leq t_n \wedge t\}$$

belongs to $\mathcal{F}_t$. In other words, A belongs to $\mathcal{F}_{T_n}$ and the proposition follows. $\square$

I have so far been unable to find a simple argument for the converse. One possible way forward could involve (Freedman, 1983, Proposition 6.157(c)). However, at least at first glance, it seems like this approach would require tinkering with the definition of the underlying space $(\Omega, \mathcal{F})$ (Section 26) to ensure that it is Borel.

29. The path space and the path law. The *law of the process* (or *path law* for short) is the distribution induced by the chain on the *path space* (i.e., the set of possible paths).

The path space. Recall that in (26.6)–(26.7), we defined the chain X in terms of the jump chain $Y := (Y_n)_{n \in \mathbb{N}}$ and the waiting times $S := (S_n)_{n \in \mathbb{Z}_+}$. To simplify the technicalities, throughout this book we identify X with S and Y in the sense that we view X as a random variable taking values in the set

$$\mathcal{P} := \mathcal{S}^{\mathbb{N}} \times [0, \infty)^{\mathbb{Z}_+}$$

of all possible sequences of states and waiting times:

$$(1) \quad X(\omega) = (Y(\omega), S(\omega)) \in \mathcal{P} \quad \forall \omega \in \Omega.$$

To be able to talk about random variables taking values in the *path space* $\mathcal{P}$, we must assign it a sigma-algebra. We choose the sigma-algebra $\mathcal{E}$ generated by the cylinder sets, i.e., sets of the form

$$(2) \quad \{x_0\} \times \{x_1\} \times \cdots \times \{x_n\} \times \mathcal{S} \times \mathcal{S} \times \cdots \times (s_1, \infty) \times (s_2, \infty) \times \cdots \times (s_n, \infty) \times [0, \infty) \times [0, \infty) \times \cdots,$$

where n is any positive integer, $x_0, x_1, \dots, x_n$ are any states in $\mathcal{S}$, and $s_1, s_2, \dots, s_n$ are any non-negative real numbers. It is not difficult to check that Y_n and S_m are $\mathcal{F}/\mathcal{E}$ -measurable functions for every $n \geq 0$ and $m > 0$. It follows that the map $X : \Omega \rightarrow \mathcal{P}$ defined by (1) is $\mathcal{F}/\mathcal{E}$ -measurable.

(3) EXERCISE. Show that X is an $\mathcal{F}/\mathcal{E}$ -measurable function.

The path law. Given the above exercise,

$$(4) \quad \mathbb{L}_\gamma(A) := \mathbb{P}_\gamma(\{X \in A\}) \quad \forall A \in \mathcal{E},$$

is a well-defined probability measure on $(\mathcal{P}, \mathcal{E})$ known as the *path law* of X , where

$$\{X \in A\} := \{\omega \in \Omega : X(\omega) \in A\}$$

denotes the preimage of A under X . Just as with $\mathbb{P}_x$ and $\mathbb{E}_x$, we write $\mathbb{L}_x$ as a shorthand for $\mathbb{L}_\gamma$ with $\gamma = 1_x$. The path law is characterised as follows:

(5) THEOREM. The path law $\mathbb{L}_\gamma$ defined in (4) is the only measure on $(\mathcal{P}, \mathcal{E})$ such that

$$(6) \quad \mathbb{L}_\gamma((2)) = \gamma(x_0)p(x_0, x_1) \cdots p(x_{n-1}, x_n)e^{-\lambda(x_0)s_1}e^{-\lambda(x_1)s_2} \cdots e^{-\lambda(x_{n-1})s_n}$$

for all positive integers n , states $x_0, x_1, \dots, x_n \in \mathcal{S}$, and real numbers $s_1, s_2, \dots, s_n \geq 0$.

Proof. Because sets of the type (2) form a π -system that generates $\mathcal{E}$, Lemma 5.6 tells us that there is at most one probability measure on $(\mathcal{P}, \mathcal{E})$ satisfying (6) for all $n, x_0, x_1, \dots, x_n, s_1, s_2, \dots, s_n$. Given that $\mathbb{L}_\gamma(\mathcal{P}) = \mathbb{P}_\gamma(\Omega) = 1$, we need only to show that (6) holds for all $n, x_0, x_1, \dots, x_n, s_1, s_2, \dots, s_n$.

To do so, define the events

$$A_m := \{Y_0 = x_0, \dots, Y_m = x_m, S_1 > s_1, \dots, S_m > s_m\}, \quad \forall m = 1, 2, \dots, n.$$

It follows from Exercise 27.3 that A_m belongs to $\mathcal{G}_m$ in (Definition 27.1). For this reason, the definition of $\mathbb{L}_\gamma$ in (4) implies that

$$(7) \quad \begin{aligned} \mathbb{L}_\gamma((2)) &= \mathbb{P}_\gamma(A_{n-1} \cap \{Y_n = x_n, S_n > s_n\}) \\ &= \mathbb{E}_\gamma[\mathbb{P}_\gamma(A_{n-1} \cap \{Y_n = x_n, S_n > s_n\} | \mathcal{G}_{n-1})] \\ &= \mathbb{E}_\gamma[1_{A_{n-1}} \mathbb{P}_\gamma(\{Y_n = x_n, S_n > s_n\} | \mathcal{G}_{n-1})]. \end{aligned}$$

where the last two equalities follow from the tower and take-out-what-is-known properties of conditional expectation (Theorem 0.3(iv, v)). Setting $f(x) := 1_{x_n}(x)$, $g(s, w) := 1_{(s_n, \infty)}(s)$, and $W := 1$ in Theorem 27.4 we find that

$$\begin{aligned}\mathbb{P}_\gamma(\{Y_n = x_n, S_n > s_n\} | \mathcal{G}_{n-1}) &= \mathbb{P}_\gamma(\{Y_n = x_n\} | \mathcal{G}_{n-1}) \mathbb{P}_\gamma(\{S_n > s_n\} | \mathcal{G}_{n-1}) \\ &= p(Y_{n-1}, x_n) e^{-\lambda(Y_{n-1})s_n} \quad \mathbb{P}_\gamma\text{-almost surely,}\end{aligned}$$

Plugging the above into (7) and exploiting that $Y_{n-1} = x_{n-1}$ on A_{n-1} , we obtain

$$\mathbb{L}_\gamma((2)) = \mathbb{E}_\gamma[1_{A_{n-1}} p(Y_{n-1}, x_n) e^{-\lambda(Y_{n-1})s_n}] = \mathbb{P}_\gamma(A_{n-1}) p(x_{n-1}, x_n) e^{-\lambda(x_{n-1})s_n}.$$

Iterating the above backwards, we find that

$$\mathbb{L}_\gamma((2)) = \mathbb{P}_\gamma(A_0) p(x_0, x_1) e^{-\lambda(x_0)s_1} p(x_1, x_2) e^{-\lambda(x_1)s_2} \dots p(x_{n-1}, x_n) e^{-\lambda(x_{n-1})s_n},$$

and (6) follows by re-arranging and noting that

$$\mathbb{P}_\gamma(A_0) = \mathbb{P}_\gamma(\{Y_0 = x_0\}) = \mathbb{P}_\gamma(\{X_0 = x_0\}) = \gamma(x_0).$$

□

More dues: a proof of Proposition 26.12. Our first use of the machinery set up in this section is proving Proposition 26.12 given in Section 26. Because the definition of the jump matrix in (26.2) implies that $p(x, x) = 1$ if and only if $q(x) = 0$, the chain's definition (26.6)–(26.8), downwards monotone convergence, (5) imply that

$$\begin{aligned}(8) \quad \mathbb{P}_x(\{X_t = x, \forall t \in [0, T_\infty)\}) &= \mathbb{P}_x(\{Y_n = x, \forall n \in \mathbb{N}\}) = \lim_{m \rightarrow \infty} \mathbb{P}_x\left(\bigcap_{n=0}^m \{Y_n = x\}\right) \\ &= \lim_{m \rightarrow \infty} p(x, x)^m = \begin{cases} 1 & \text{if } q(x) = 0 \\ 0 & \text{if } q(x) \neq 0 \end{cases},\end{aligned}$$

and it follows that $\mathbb{P}_x(\{T_\infty = \infty, X_t = x, \forall t \in [0, \infty)\}) = 0$ unless $q(x) = 0$. If $q(x) = 0$, then the above shows that $Y_n = x$ for all $n \geq 0$, $\mathbb{P}_\gamma$ -almost surely. Moreover, the jump rate $\lambda(x)$ is one by its definition in (26.3) and, so, Kolmogorov's strong law of large numbers (Theorem 16.4) implies that

$$T_\infty = \lim_{n \rightarrow \infty} T_n = \lim_{n \rightarrow \infty} \sum_{m=1}^n S_m = \sum_{m=1}^{\infty} S_m = \sum_{m=1}^{\infty} \frac{\xi_m}{\lambda(Y_n)} = \sum_{m=1}^{\infty} \frac{\xi_m}{\lambda(x)} = \sum_{m=1}^{\infty} \xi_m = \infty \quad \mathbb{P}_\gamma\text{-a.s.}$$

The above and (8) then imply that $\mathbb{P}_x(\{T_\infty = \infty, X_t = x, \forall t \in [0, \infty)\}) = 1$, as desired.

Measurable functions on $\mathcal{P}$. In the next section, we will give the Markov property in terms of $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable functions on $\mathcal{P}$. Here, we take a moment to get acquainted with some of these functions and a feel for what the others may be. To start with, the definition of $\mathcal{E}$ implies that *coordinate functions*

$$c_n^Y(x) = y_n, \quad c_n^S(x) = s_n, \quad \forall x := (y, s) \in \mathcal{P}$$

are $\mathcal{E}/2^{\mathcal{S}}$ -measurable and $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable, respectively. Because linear combinations and limits of real-valued measurable functions are measurable,

$$c_0^T(x) := 0, \quad c_n^T(x) = \sum_{m=1}^n s_m, \quad c_\infty^T(x) = \lim_{m \rightarrow \infty} c_n^T(x) = \sum_{m=1}^{\infty} s_m, \quad \forall x \in \mathcal{P},$$

are also $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable. We now have measurable functions whose composition with X give us the n th state visited by the chain, the n th waiting time, the n th jump time, and the explosion time:

$$(9) \quad Y_n(\omega) = c_n^Y(X(\omega)), \quad S_n(\omega) = c_n^S(X(\omega)), \quad T_n(\omega) = c_n^T(X(\omega)), \quad T_\infty(\omega) = c_\infty^T(X(\omega))$$

for all ω in Ω . Similarly, for any $f : \mathcal{S} \rightarrow \mathbb{R}$,

$$(10) \quad 1_{\{t < T_\infty(\omega)\}} f(X_t(\omega)) = \sum_{n=0}^{\infty} 1_{\{T_n(\omega) \leq t < T_{n+1}(\omega)\}} f(Y_n(\omega)) = c_t^f(X(\omega)) \quad \forall \omega \in \Omega,$$

where we are using the notation for partially-defined functions introduced in (0.1) and

$$c_t^f(x) := \sum_{n=0}^{\infty} 1_{[0,t]}(c_n^T(x)) 1_{(t,\infty)}(c_{n+1}^T(x)) f(c_n^Y(x)) \quad \forall x \in \mathcal{P},$$

is a $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable function given that it is obtained by composing, adding, multiplying, and taking limits of measurable functions. Setting f to be the indicator function 1_z of any given state z , we have that the indicator function of the event that the chain is in z at time t , is the composition $c_t^z(X)$ of the chain X and the $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable function

$$(11) \quad c_t^z(x) := \sum_{n=0}^{\infty} 1_{[0,t]}(c_n^T(x)) 1_{(t,\infty)}(c_{n+1}^T(x)) 1_z(c_n^Y(x)) \quad \forall x \in \mathcal{P}.$$

(12) EXERCISE. *If you'd like a challenge in measurability-checking-pedantry(!), show that the sets*

$$(13) \quad \{x \in \mathcal{P} : c_{t_1}^{z_1}(x) = 1, c_{t_2}^{z_2}(x) = 1, \dots, c_{t_n}^{z_n}(x) = 1\}$$

for all $t_1 \leq t_2 \leq \dots \leq t_n \in [0, \infty)$, $z_1, z_2, \dots, z_n \in \mathcal{S}$, and $n \in \mathbb{N}$, generate $\mathcal{E}$.

We finish this section with the following rather dull lemma that will cover most of our measurability-checking necessities throughout the ensuing treatment of continuous-time chains.

(14) LEMMA. *The following are $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable functions*

(i) *For any real-valued function f on $\mathcal{S}$,*

$$F_-(x) := \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} c_{m+1}^S(x) f(c_m^Y(x)),$$

$$F_+(x) := \limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} c_{m+1}^S(x) f(c_m^Y(x)), \quad \forall x \in \mathcal{P}.$$

(ii) *For any subset A of $\mathcal{S}$ and positive integer k ,*

$$F(x) := \inf \left\{ n > 0 : \sum_{m=1}^n 1_A(c_m^Y(x)) = k \right\}, \quad \forall x \in \mathcal{P}.$$

(iii) *For any non-negative function f on $\mathcal{S}$,*

$$G(x) = \sum_{m=0}^{F(x)-1} c_{m+1}^S(x) f(c_m^Y(x)) \quad \forall x = (y, s) \in \mathcal{P},$$

where F is as in (ii).

(iv) For any given t in $[0, \infty)$ and real-valued function f on $\mathcal{S}$,

$$F_t(x) := \sum_{n=0}^{\infty} 1_{[0,t]}(c_{n+1}^T(x)) c_{n+1}^S(x) f(c_n^Y(x)) \\ + \sum_{n=0}^{\infty} 1_{[0,t]}(c_n^T(x)) 1_{(t,\infty)}(c_{n+1}^T(x)) f(c_n^Y(x)) (t - c_n^T(x)) \quad \forall x \in \mathcal{P},$$

and

$$F_-(x) := \liminf_{t \rightarrow \infty} \frac{F_t(x)}{t}, \quad F_+(x) := \limsup_{t \rightarrow \infty} \frac{F_t(x)}{t}, \quad \forall x \in \mathcal{P}.$$

Proof. This proof is analogous to that of the result's discrete-time counterpart (Lemma 12.8) and may be skipped.

(i) Because finite sums and products of measurable functions are measurable, we have that $x \mapsto \frac{1}{n} \sum_{m=0}^{n-1} c_{m+1}^S(x) f(c_m^Y(x))$ is $\mathcal{E}/\mathcal{B}(\mathbb{R})$ -measurable, for each $n > 0$. Because the limit infimum and supremum of measurable functions are measurable, the result follows.

(ii) By definition, $F(x)$ counts the number of jumps until the path x enters the set A for the k th time. For this reason,

$$F(x) = \infty \cdot 1_{\{g_\infty(x) < k\}} + \sum_{n=1}^{\infty} n 1_{\{g_n(x) < k\}} 1_{\{g_n(x) = k\}} \\ = \infty \cdot \lim_{N \rightarrow \infty} 1_{\{g_N(x) < k\}} + \lim_{N \rightarrow \infty} \sum_{n=1}^N n 1_{\{g_n(x) < k\}} 1_{\{g_n(x) = k\}} \quad \forall x \in \mathcal{P},$$

where

$$g_n(x) := \sum_{m=1}^n 1_A(c_m^Y(x)) \quad \forall x \in \mathcal{P}$$

denotes the number of times the path x has entered A by time $n > 0$. Because any finite sum of measurable functions is measurable, g_n is $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable. The result then follows from the above by exploiting once again that the sum, product, composition, limit superior, and limit inferior of measurable functions are all measurable.

(iii) Because any finite sum of measurable functions is measurable,

$$g_n(x) := \sum_{m=0}^n c_{m+1}^S(x) f(c_m^Y(x)) \quad \forall x \in \mathcal{P}$$

defines an $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable for any natural number n . Furthermore, as the limit of measurable functions is measurable, $g_\infty := \lim_{n \rightarrow \infty} g_n$ is also $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Pick any A in $\mathcal{B}(\mathbb{R}_E)$ and note that

$$\left\{ (n, x \in \mathbb{N}_E \times \mathcal{P} : \sum_{m=0}^n c_{m+1}^S(x) f(c_m^Y(x)) \in A \right\} \\ = (\{\infty\} \times \{g_\infty \in A\}) \cup \left(\bigcup_{n=0}^{\infty} \{n\} \times \{g_n \in A\} \right).$$

Because the right-hand side belongs to the product sigma-algebra $2^{\mathbb{N}_E} \times \mathcal{E}$, it follows that

$$H(n, x) := \sum_{m=0}^n c_{m+1}^S(x) f(c_m^Y(x)) \quad \forall x \in \mathcal{P}$$

defines an $2^{\mathbb{N}_E} \times \mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable function. Because $G(x) = H(F(x) - 1, x)$ for all x in $\mathcal{P}$, with F as in (ii), the result then follows as the composition of measurable functions is measurable.

(iv) This follows directly from the facts that the sum, product, limit, limit inferior, and limit superior of measurable functions are measurable. $\square$

30. The Markov and strong Markov properties. Stating and proving the Markov property for the continuous-time case is somewhat more involved than it is for the discrete-time case: X_t is only defined on the event $\{t < T_\infty\}$ that no explosion has occurred by time t , and, so, conditioning on X_t requires a bit of care. We have already faced a similar issue: when dealing with the strong Markov property of discrete time chains $(W_n)_{n \in \mathbb{N}}$ (c.f. Section 12), we had to condition on the chain's state W_ς at a stopping time ς but W_ς was only defined on the event $\{\varsigma < \infty\}$ that the stopping time was finite. We follow here the steps we took there to resolve the issue. If necessary, you should brush up on Sections 2 and 12: the concepts discussed therein are equally applicable here and guide the development of this section.

Shifting the chain left by η . To describe the chain's future from a stopping time η onwards, define the *shifted chain* $X^\eta := (Y^\eta, S^\eta)$ as the function mapping from $\{\eta < T_\infty\}$ to the path space $\mathcal{P}$ (c.f. Section 29) given by

$$\begin{aligned} (1) \quad S^\eta(\omega) &:= (S_{n+1}(\omega) - (\eta(\omega) - T_n(\omega)), S_{n+2}(\omega), S_{n+3}(\omega), \dots), \\ (2) \quad Y^\eta(\omega) &:= (Y_n(\omega), Y_{n+1}(\omega), Y_{n+2}(\omega), \dots), \quad \forall \omega \in \{T_n \leq \eta < T_{n+1}\}, \quad n \geq 0, \end{aligned}$$

This $\mathcal{P}$ -valued function X^η describes the chain *shifted* leftwards in time by η amount: setting X_t^η to be the piecewise-constant interpolation of X^η (i.e., replace Y, S in (26.6)–(26.7) with Y^η, S^η), we find that

$$(3) \quad X_t^\eta(\omega) = X_{\eta+t}(\omega) \quad \forall \omega \in \{\eta + t < T_\infty\}, \quad \forall t \in [0, \infty).$$

Similarly, the *jump times* $T_0^\eta, T_1^\eta, \dots$ and explosion time of the shifted chain X^η , defined by

$$T_0^\eta(\omega) := 0, \quad T_n^\eta(\omega) := \sum_{m=1}^n S_m^\eta(\omega) \quad \forall n > 0, \quad T_\infty^\eta(\omega) := \sum_{m=1}^{\infty} S_m^\eta(\omega), \quad \forall \omega \in \{t < T_\infty\},$$

satisfy

$$(4) \quad T_{n'}^\eta(\omega) = T_{n+n'}(\omega) - \eta(\omega) \quad \forall \omega \in \{T_n \leq \eta < T_{n+1}\}, \quad n, n' \geq 0,$$

$$(5) \quad T_\infty^\eta(\omega) = T_\infty(\omega) - \eta(\omega) \quad \forall \omega \in \{\eta < T_\infty\}.$$

(6) EXERCISE. Prove (3) and show that (29.9)–(29.10) hold if we replace $Y, S, T, T_\infty, X_t, t, \Omega$ therein with $Y^\eta, S^\eta, T^\eta, T_\infty^\eta, X^\eta, \eta, \{\eta < T_\infty\}$.

The Markov property. We now have all we need to state and prove the Markov property. Note that whenever we write $1_{\{t < T_\infty\}} F(X^t)$ in what follows, we are using the notation for partially-defined functions introduced in (0.1).

(7) THEOREM (The Markov property). Let $(\mathcal{F}_t)_{t \geq 0}$ be the filtration generated by the chain (Definition 28.1), $\mathcal{E}$ be the cylinder sigma-algebra on the path space $\mathcal{P}$ (Section 5), and t belongs to $[0, \infty)$. Suppose that Z is an $\mathcal{F}_t/\mathcal{B}(\mathbb{R}_E)$ -measurable random variable, F is a $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable function, and that Z and F are both non-negative, or both bounded. If x any state in $\mathcal{S}$, then $Z 1_{\{t < T_\infty, X_t = x\}} F(X^t)$ is $\mathcal{F}/\mathcal{B}(\mathbb{R}_E)$ -measurable, where X^t denotes the t -shifted chain (1)–(2). Moreover,

$$(8) \quad \mathbb{E}_\gamma [Z 1_{\{t < T_\infty, X_t = x\}} F(X^t)] = \mathbb{E}_\gamma [Z 1_{\{t < T_\infty, X_t = x\}}] \mathbb{E}_x [F(X)] \quad \forall x \in \mathcal{S}.$$

For the theorem's proof, we need the following:

(9) LEMMA. Let $(\mathcal{G}_n)_{n \in \mathbb{N}}$ be the filtration generated by the jump chain and jump times (Definition 27.1) and $(\mathcal{F}_t)_{t \geq 0}$ that generated by the chain (Definition 28.1). For any given t in $[0, \infty)$, A in $\mathcal{F}_t$, and n in $\mathbb{N}$, there exists an A_n in $\mathcal{G}_n$ such that

$$A \cap \{t < T_{n+1}\} = A_n \cap \{t < T_{n+1}\}.$$

Proof. Because $\mathcal{G}_n$ is a sigma-algebra,

$$\mathcal{G}_t := \{A \in \mathcal{F}_t : A \cap \{t < T_{n+1}\} = A_n \cap \{t < T_{n+1}\} \text{ for some } A_n \in \mathcal{G}_n\}$$

is also a sigma-algebra. For any $s \leq t$, $m \geq 0$, and x in $\mathcal{S}$,

$$\{T_m \leq s, Y_m = x\} \cap \{t < T_{n+1}\}$$

is either empty (if $m > n$) and we set $A_n := \emptyset$ or $\mathcal{G}_m$ is contained in $\mathcal{G}_n$ (if $m \leq n$) and we set $A_n := \{T_m \leq s, Y_m = x\}$. In either case, A_n belongs to $\mathcal{G}_n$ and

$$\{T_m \leq s, Y_m = x\} \cap \{t < T_{n+1}\} = A_n \cap \{t < T_{n+1}\}$$

showing that $\{T_m \leq s, Y_m = x\}$ belongs to $\mathcal{G}_t$. Because these sets generate $\mathcal{F}_t$ and $\mathcal{G}_t$'s definition implies that $\mathcal{G}_t \subseteq \mathcal{F}_t$, it follows that $\mathcal{G}_t = \mathcal{F}_t$. $\square$

Proof of Theorem 7. Given Exercise 28.6 and Theorem 29.5, this proof is entirely analogous that of Theorem 12.4 as long as we are able to show that

$$(10) \quad \mathbb{P}_\gamma(A \cap \{t < T_\infty, X_t = x, X^t \in B\}) = \mathbb{P}_\gamma(A \cap \{t < T_\infty, X_t = x\}) \mathbb{L}_x(B),$$

for all A in $\mathcal{F}_t$, x in $\mathcal{S}$, and cylinder sets B (that is, sets B of the form in (29.2)), where $\mathbb{L}_x$ denotes the path law (29.4) starting from x . To do so, fix any such A , x , and B and note that

$$(11) \quad \mathbb{P}_\gamma(A \cap \{t < T_\infty, X_t = x, X^t \in B\}) = \sum_{m=0}^{\infty} \mathbb{P}_\gamma(A_m \cap \{T_m \leq t < T_{m+1}, Y_m = x\} \cap B_m),$$

where A_m is the set in Lemma 9 belonging to $\mathcal{G}_m$ and

$$B_m := \{Y_m = x_0, Y_{m+1} = x_1, \dots, Y_{m+n} = x_n, S_{m+1} - (t - T_m) > s_1, S_{m+2} > s_2, \dots, S_{m+n} > s_n\}.$$

Because $B_m \subseteq \{t < T_{m+1}\}$ and $A_m \cap \{T_m \leq t, Y_m = x\}$ belongs to $\mathcal{G}_m$, the tower and take-out-what-is-known properties of conditional expectation (Theorem 0.3(iv, v)) imply that

$$(12) \quad \mathbb{P}_\gamma(A_m \cap \{T_m \leq t < T_{m+1}, Y_m = x\} \cap B_m) = \mathbb{P}_\gamma(A_m \cap \{T_m \leq t, Y_m = x\} \cap B_m)$$

$$(13) \quad \begin{aligned} &= \mathbb{E}_\gamma[\mathbb{P}_\gamma(A_m \cap \{T_m \leq t, Y_m = x\} \cap B_m | \mathcal{G}_m)] \\ &= \mathbb{E}_\gamma[1_{A_m \cap \{T_m \leq t, Y_m = x\}} \mathbb{P}_\gamma(B_m | \mathcal{G}_m)]. \end{aligned}$$

Conditioning on $\mathcal{G}_{m+n-1}, \mathcal{G}_{m+n-2}, \dots, \mathcal{G}_{m+1}$ and making a repeated use of Theorem 27.4 similar to that in proof of Theorem 29.5, we find that

$$\begin{aligned} \mathbb{P}_\gamma(B_m | \mathcal{G}_m) &= 1_{Y_m}(x_0) p(x_0, x_1) \dots p(x_{n-1}, x_n) e^{-\lambda(x_0)(s_1 + t - T_m)} e^{-\lambda(x_1)s_2} \dots e^{-\lambda(x_{n-1})s_n} \\ &= e^{-\lambda(Y_m)(t - T_m)} \mathbb{L}_{Y_m}(B) = \mathbb{P}_\gamma(\{T_{m+1} > t\} | \mathcal{G}_m) \mathbb{L}_{Y_m}(B). \end{aligned}$$

Plugging the above into (12) and applying the take-out-what-is-known and tower properties, we find that

$$\begin{aligned} \mathbb{P}_\gamma(A_m \cap \{T_m \leq t < T_{m+1}, Y_m = x\} \cap B_m) &= \mathbb{E}_\gamma[1_{A_m \cap \{T_m \leq t < T_{m+1}, Y_m = x\}} \mathbb{L}_{Y_m}(B)] \\ &= \mathbb{P}_\gamma(A_m \cap \{T_m \leq t < T_{m+1}, Y_m = x\}) \mathbb{L}_x(B) \\ &= \mathbb{P}_\gamma(A \cap \{T_m \leq t < T_{m+1}, X_t = x\}) \mathbb{L}_x(B). \end{aligned}$$

Combining the above with (11), we obtain (10) and the result follows. $\square$

The strong Markov property. Just as in the discrete-time case (Section 12), the Markov property holds for all stopping times η instead of only for deterministic times t :

(14) **THEOREM** (The strong Markov property). *Let $(\mathcal{F}_t)_{t \geq 0}$ be the filtration generated by the chain (Definition 27.1), $\mathcal{E}$ be the cylinder sigma-algebra on the path space $\mathcal{P}$ (Section 5), η be a $(\mathcal{F}_t)_{t \in [0, \infty)}$ -stopping time (Definition 28.3), and $\mathcal{F}_\eta$ be its associated sigma-algebra. Suppose that Z is an $\mathcal{F}_\eta/\mathcal{B}(\mathbb{R}_E)$ -measurable random variable, F is a $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable function, and that Z and F are both non-negative, or both bounded. If x any state in $\mathcal{S}$, then $Z1_{\{\eta < T_\infty, X_\eta = x\}}F(X^\eta)$ is $\mathcal{F}/\mathcal{B}(\mathbb{R}_E)$ -measurable, where X^η denotes the η -shifted chain (1)–(2). Moreover,*

$$(15) \quad \mathbb{E}_\gamma [Z1_{\{\eta < T_\infty, X_\eta = x\}}F(X^\eta)] = \mathbb{E}_\gamma [Z1_{\{\eta < T_\infty, X_\eta = x\}}] \mathbb{L}_x(F) \quad \forall x \in \mathcal{S}.$$

Proof. Given Exercise 28.6 and Theorem 29.5, the proof is entirely analogous that of the theorem's discrete-time counterpart (Theorem 12.4) as long as we are able to show that

$$(16) \quad \mathbb{P}_\gamma (A \cap \{\eta < T_\infty, X_\eta = x, X^\eta \in B\}) = \mathbb{P}_\gamma (A \cap \{\eta < T_\infty, X_\eta = x\}) \mathbb{L}_x(B),$$

for all A in $\mathcal{F}_\eta$, x in $\mathcal{S}$, and cylinder sets B (that is, sets B of the form in (29.2)), where $\mathbb{L}_x$ denotes the path law (29.4) starting from x . To do so, fix any such A , x , and B and consider the discretisations

$$(17) \quad \eta_k := \sum_{l=1}^{\infty} \frac{l}{k} 1_{\{(l-1)/k < \eta \leq l/k\}} \quad \forall k \in \mathbb{Z}_+$$

of η . It is straightforward to check that $(\eta_k)_{k \in \mathbb{Z}_+}$ is a decreasing sequence of $(\mathcal{F}_t)_{t \in [0, \infty)}$ -stopping times with limit η . Moreover, the right-continuity (w.r.t. the discrete topology on $\mathcal{S}$) of the paths of X , see (26.6)–(26.8), imply that

$$(18) \quad \lim_{k \rightarrow \infty} 1_{\{\eta_k < T_\infty, X_{\eta_k} = x, X^{\eta_k} \in B\}}(\omega) = 1_{\{\eta < T_\infty, X_\eta = x, X^\eta \in B\}}(\omega) \quad \forall \omega \in \Omega,$$

$$(19) \quad \lim_{k \rightarrow \infty} 1_{\{\eta_k < T_\infty, X_{\eta_k} = x\}}(\omega) = 1_{\{\eta < T_\infty, X_\eta = x\}}(\omega) \quad \forall \omega \in \Omega.$$

Because A belongs to $\mathcal{F}_\eta$ and η is bounded above η_k , Lemma 28.5(iii) shows that A belongs to $\mathcal{F}_{\eta_k}$. For this reason, the Markov property (Theorem 7) and Tonelli's theorem imply that

$$\begin{aligned} \mathbb{P}_\gamma (A \cap \{\eta_k < T_\infty, X_{\eta_k} = x, X^{\eta_k} \in B\}) &= \sum_{l=0}^{\infty} \mathbb{P}_\gamma \left(A \cap \left\{ \eta_k = \frac{l}{k}, \frac{l}{k} < T_\infty, X_{\frac{l}{k}} = x, X^{\frac{l}{k}} \in B \right\} \right) \\ &= \sum_{l=0}^{\infty} \mathbb{P}_\gamma \left(A \cap \left\{ \eta_k = \frac{l}{k}, \frac{l}{k} < T_\infty, X_{\frac{l}{k}} = x \right\} \right) \mathbb{L}_x(B) \\ &= \mathbb{P}_\gamma (A \cap \{\eta_k < T_\infty, X_{\eta_k} = x\}) \mathbb{L}_x(B) \quad \forall k > 0. \end{aligned}$$

Bounded convergence and (18)–(19) imply (16) and the result follows. $\square$

(20) **EXERCISE.** *Convince yourself that η_k in (17) is an $(\mathcal{F}_t)_{t \geq 0}$ -stopping time and that (18)–(19) hold.*

31. The transition probabilities and the semigroup property. All finite-dimensional distributions of the chain may be expressed in terms of the initial distribution γ and the collection $(P_t)_{t \geq 0}$ of matrices $P_t = (p_t(x, y))_{x, y \in \mathcal{S}}$ whose (x, y) -entry is the probability that the chain is in state y at time t if it starts in state x :

$$(1) \quad p_t(x, y) := \mathbb{P}_x (\{X_t = y, t < T_\infty\}) \quad \forall x, y \in \mathcal{S}, \quad t \in [0, \infty).$$

In particular:

(2) THEOREM. For all times positive integers $n > 0$, times $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$, and states $x_0, x_1, \dots, x_n$,

$$\begin{aligned} \mathbb{P}_\gamma(\{X_0 = x_0, X_{t_1} = x_1, X_{t_2} = x_2, \dots, X_{t_n} = x_n, t_n < T_\infty\}) \\ = \gamma(x_0)p_{t_1}(x_0, x_1)p_{t_2-t_1}(x_1, x_2) \dots p_{t_n-t_{n-1}}(x_{n-1}, x_n). \end{aligned}$$

Re-arranging the equation in (2), we find that probability that the chain travels from x to y in t amount of time equals $p_t(x, y)$ regardless of when this transition occurs:

$$(3) \quad \mathbb{P}_\gamma(\{X_{s+t} = y, s+t < T_\infty\} | \{X_s = x, s < T_\infty\}) = \frac{\mathbb{P}_\gamma(\{X_{s+t} = y, s+t < T_\infty\})}{\mathbb{P}_\gamma(\{X_s = x, s < T_\infty\})} = p_t(x, y),$$

for all $s \geq 0$ such that the denominator is non-zero. For this reason, $p_t(x, y)$ is referred to as the probability that the chain *transitions* from x to y in t amount of time and P_t as the t -*transition matrix*. The collection $(P_t)_{t \geq 0}$ of these matrices satisfies the *semigroup property*,

$$(4) \quad p_{t+s}(x, y) = \sum_{z \in \mathcal{S}} p_t(x, z)p_s(z, y) \quad \forall x, y \in \mathcal{S} \quad t, s \geq 0,$$

or $P_{t+s} = P_t P_s$ for all $t, s \geq 0$ in matrix notation, and $(P_t)_{t \geq 0}$ is called the *semigroup of transition probabilities*.

(5) EXERCISE. Using Theorem 2, prove (4).

Proof of Theorem 2. Let

$$A := \{X_0 = x_0, X_{t_1} = x_1, X_{t_2} = x_2, \dots, X_{t_{n-2}} = x_{n-2}, t_{n-2} < T_\infty\}$$

and F be the $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable function c_t^z in (29.11) with $z := x_n$ and $t := t_n - t_{n-1}$. Because A belongs to $\mathcal{F}_{t_{n-1}}$ (Exercise 28.2), applying (29.10), Exercise 30.6, and the Markov property (Theorem 30.7), we find that

$$\begin{aligned} \mathbb{P}_\gamma(\{X_0 = x_0, X_{t_1} = x_1, X_{t_2} = x_2, \dots, X_{t_n} = x_n, t_n < T_\infty\}) \\ = \mathbb{E}_\gamma \left[1_{A \cap \{X_{t_{n-1}} = x_{n-1}, t_{n-1} < T_\infty\}} F(X^{t_{n-1}}) \right] \\ = \mathbb{P}_\gamma(A \cap \{X_{t_{n-1}} = x_{n-1}, t_{n-1} < T_\infty\}) \mathbb{E}_{x_{n-1}}[F(X)] \\ = \mathbb{P}_\gamma(\{X_0 = x_0, X_{t_1} = x_1, X_{t_2} = x_2, \dots, X_{t_{n-1}} = x_{n-1}, t_{n-1} < T_\infty\}) p_{t_n-t_{n-1}}(x_{n-1}, x_n). \end{aligned}$$

Iterating the above argument, we have that

$$\begin{aligned} \mathbb{P}_\gamma(\{X_0 = x_0, X_{t_1} = x_1, X_{t_2} = x_2, \dots, X_{t_n} = x_n, t_n < T_\infty\}) \\ = \mathbb{P}_\gamma(\{X_0 = x_0, X_{t_1} = x_1, X_{t_2} = x_2, \dots, X_{t_{n-1}} = x_{n-1}, t_{n-1} < T_\infty\}) p_{t_n-t_{n-1}}(x_{n-1}, x_n) \\ = \dots = \mathbb{P}_\gamma(\{X_0 = x_0, X_{t_1} = x_1, t_1 < T_\infty\}) p_{t_2-t_1}(x_1, x_2) \dots p_{t_n-t_{n-1}}(x_{n-1}, x_n). \end{aligned}$$

Because $T_\infty = \sum_{n=1}^\infty S_n > 0$ and $\mathbb{P}_\gamma(\{X_0 = x_0\}) = \gamma(x_0)$, applying the Markov property (Theorem 30.7) once again with $A := \Omega$ and F as in (29.11) with $z := x_1$ and $t := t_1$ completes the proof. $\square$

32. The forward and backward equations. Perhaps the most celebrated result in Markov chain theory are *Kolmogorov's forward and backward equations*: two sets of ordinary differential equations satisfied by the transition probabilities. In particular:

(1) **THEOREM** (The forward and backward equations). *Suppose that Q is a stable and conservative rate matrix (i.e., satisfies (26.1)). The transition probabilities are continuously differentiable: for each $x, y \in \mathcal{S}$,*

$$t \mapsto p_t(x, y)$$

is a continuously differentiable function on $[0, \infty)$. Moreover, the transition probabilities satisfy the forward equations:

$$(2) \quad \dot{p}_t(x, y) = \sum_{z \in \mathcal{S}} p_t(x, z) q(z, y) \quad \forall t \in [0, \infty), \quad x, y \in \mathcal{S}, \quad p_0(x, y) = 1_x(y) \quad \forall x, y \in \mathcal{S},$$

and the backward equations:

$$(3) \quad \dot{p}_t(x, y) = \sum_{z \in \mathcal{S}} q(x, z) p_t(z, y) \quad \forall t \in [0, \infty), \quad x, y \in \mathcal{S}, \quad p_0(x, y) = 1_x(y) \quad \forall x, y \in \mathcal{S}.$$

Moreover, the transition probabilities are the minimal non-negative solution of these equations: if $(k_t(x, y))_{x, y \in \mathcal{S}, t \geq 0}$ is a non-negative ($k_t(x, y) \geq 0$ for all $x, y \in \mathcal{S}, t \geq 0$) differentiable function satisfying either (2) or (3), then

$$k_t(x, y) \geq p_t(x, y) \quad \forall x, y \in \mathcal{S}, \quad t \in [0, \infty).$$

The proof of the above theorem is delicate and we postpone it until the end of the section. The forward and backward equations are often written in matrix notation as

$$\dot{P}_t = P_t Q, \quad \dot{P}_t = Q P_t, \quad P_0 = I,$$

where $I := (1_x(y))_{x, y \in \mathcal{S}}$ denotes the identity matrix on $\mathcal{S}$.

Regular rate matrices and the uniqueness of the solutions. The rate matrix Q is said to be *regular* if the chain does not explode regardless of the initial distribution:

(4) **DEFINITION** (Regular rate matrices). *The rate matrix Q is regular if, for all initial distributions γ ,*

$$(5) \quad \mathbb{P}_\gamma(\{T_\infty = \infty\}) = 1.$$

We can also express the regularity of Q in terms of the mass of the time-varying law:

(6) **PROPOSITION.** *If the chain starting position is sampled from γ , then the chain does not explode (i.e., (5) holds) if and only if*

$$p_t(\mathcal{S}) = \sum_{x \in \mathcal{S}} p_t(x) = \sum_{x \in \mathcal{S}} \mathbb{P}_\gamma(\{X_t = x, t < T_\infty\}) = \mathbb{P}_\gamma(\{t < T_\infty\}) = 1$$

for a single $t \in [0, \infty)$, in which case the above holds for all $t \in [0, \infty)$. Thus, the rate matrix is regular if and only if the above holds for all initial distributions γ .

Proof. Monotone convergence implies that $\mathbb{P}_\gamma(\{n < T_\infty\}) \rightarrow \mathbb{P}_\gamma(\{T_\infty = \infty\})$ as $n \rightarrow \infty$ and the result follows as $t \mapsto p_t(\mathcal{S}) = \mathbb{P}_\gamma(\{t < T_\infty\})$ is clearly a non-increasing function of t . $\square$

Consequently, if Q is regular, then the transition probabilities are the unique non-negative solution of either the forward or backward equations with mass no greater than 1 (i.e., such that $\sum_{y \in \mathcal{S}} p_t(x, y) \leq 1$ for all x and t). If Q is not regular, then the backward equations have infinitely-many such solutions and the forward equations may also do, see (Anderson, 1991) and references therein.

A proof of Theorem 1. We do this proof in five steps:

- (Lemma 7) We derive the so-called *forward integral recursion (FIR)* and use it to prove that the transition probabilities satisfy a weak version of the forward equations.
- (Lemma 16) We use the FIR to obtain the *backward integral recursion (BIR)* and use it to prove that the transition probabilities satisfy a weak version of the backward equations.
- (Lemma 20) We use the weak version of the backward equations to show that the transition probabilities are continuously differentiable and satisfy the strong form (3) of the backward equations.
- (Lemma 22) We use the differentiability of the transition [probabilities and the weak version of the forward equation to show that the strong form (2) of these equations holds.
- (Lemma 28) We use the FIR and BIR to show that transition probabilities are minimal among both the solutions of the forward equations and those of the backward equations.

Let's begin:

(7) LEMMA (The FIR and the weak version of the forward equations). *Suppose that Q satisfies (26.1). Given any natural number n , time t , and states x, y , let*

$$(8) \quad p_t^n(x, y) := \mathbb{P}_x(\{X_t = y, t < T_{n+1}\})$$

denote the probability that, having started in x , the chain lies in y at time t and that it has not jumped more than n times during the interval $[0, t]$. These probabilities satisfy the forward integral recursion (FIR):

$$(9) \quad p_t^n(x, y) = 1_x(y)e^{-\lambda(y)t} + \int_0^t \sum_{z \in \mathcal{S}} p_s^{n-1}(x, z)\lambda(z)p(z, y)e^{-\lambda(y)(t-s)}ds$$

for all x, y in $\mathcal{S}$, non-negative t , and positive n , where $(p(x, y))_{x, y \in \mathcal{S}}$ denotes the one-step matrix (26.2) of the jump chain and $(\lambda(x))_{x \in \mathcal{S}}$ the jump rates (26.3). Moreover, the transition probabilities satisfy the integral version of the forward equations:

$$(10) \quad p_t(x, y) = 1_x(y)e^{-\lambda(y)t} + \int_0^t \sum_{z \in \mathcal{S}} p_s(x, z)\lambda(z)p(z, y)e^{-\lambda(y)(t-s)}ds \quad \forall x, y \in \mathcal{S}, \quad t \geq 0.$$

Proof. Because taking the limit $n \rightarrow \infty$ in (9) and applying monotone convergence yields (10), we need only to prove (9). To this end, we decompose $p_t^n(x, y)$ as follows:

$$(11) \quad \begin{aligned} p_t^n(x, y) &= \sum_{m=0}^n \mathbb{P}_x(\{X_t = y, T_m \leq t < T_{m+1}\}) \\ &= \mathbb{P}_x(\{Y_0 = y, t < T_1\}) + \sum_{m=1}^n \mathbb{P}_x(\{Y_m = y, T_m \leq t < T_{m+1}\}), \end{aligned}$$

By the definition of the chain in Section 26,

$$(12) \quad p_t^0(x, y) = \mathbb{P}_x(\{Y_0 = y, t < T_1\}) = \mathbb{P}_x(\{X_0 = y\})\mathbb{P}_x(\{S_1 > t\}) = 1_x(y)e^{-\lambda(y)t}.$$

Next, Theorem 27.4 and the tower and take-out-what-is-known properties of conditional expectation (Theorem 0.3(iv, v)) imply that

$$(13) \quad \begin{aligned} \mathbb{P}_x(\{Y_m = y, T_m \leq t < T_{m+1}\}) &= \mathbb{P}_x(\{Y_m = y, T_m \leq t, t - T_m < S_{m+1}\}) \\ &= \mathbb{E}_x[\mathbb{P}_x(\{Y_m = y, T_m \leq t, t - T_m < S_{m+1}\}|\mathcal{G}_m)] \\ &= \mathbb{E}_x[1_{\{Y_m=y, T_m \leq t\}}\mathbb{P}_x(\{t - T_m < S_{m+1}\}|\mathcal{G}_m)] \\ &= \mathbb{E}_x[1_{\{Y_m=y, T_m \leq t\}}e^{-\lambda(y)(t-T_m)}] \\ &= \sum_{z \in \mathcal{S}} \mathbb{E}_x[1_{\{Y_{m-1}=z, Y_m=y, S_m \leq t-T_{m-1}\}}e^{-\lambda(y)(t-T_m)}], \end{aligned}$$

where $(\mathcal{G}_m)_{m \in \mathbb{N}}$ denotes the filtration generated by the jump chain and jump times (Definition 27.1). Similarly, we have that

$$\begin{aligned}
 (14) \quad & \mathbb{E}_x \left[1_{\{Y_{m-1}=z, Y_m=y, S_m \leq t-T_{m-1}\}} e^{-\lambda(y)(t-T_{m-1})} \right] \\
 &= \mathbb{E}_x \left[1_{\{Y_{m-1}=z, Y_m=y, S_m \leq t-T_{m-1}\}} e^{-\lambda(y)(t-S_m)} e^{\lambda(y)T_{m-1}} \right] \\
 &= \mathbb{E}_x \left[\mathbb{E}_x \left[1_{\{Y_{m-1}=z, Y_m=y, S_m \leq t-T_{m-1}\}} e^{-\lambda(y)(t-S_m)} \middle| \mathcal{G}_{m-1} \right] e^{\lambda(y)T_{m-1}} \right] \\
 &= \mathbb{E}_x \left[1_{\{Y_{m-1}=z\}} \mathbb{P}_x(\{Y_m = y\} | \mathcal{G}_{m-1}) \mathbb{E}_x \left[1_{\{S_m \leq t-T_{m-1}\}} e^{-\lambda(y)(t-S_m)} \middle| \mathcal{G}_{m-1} \right] e^{\lambda(y)T_{m-1}} \right] \\
 &= \mathbb{E}_x \left[1_{\{Y_{m-1}=z\}} p(z, y) \left(\int_0^{t-T_{m-1}} e^{-\lambda(y)(t-s)} \lambda(z) e^{-\lambda(z)s} ds \right) e^{\lambda(y)T_{m-1}} \right], \quad \forall z \in \mathcal{S}
 \end{aligned}$$

Applying the change of variables $r := s + T_{m-1}$, we find

$$\begin{aligned}
 (14) &= \mathbb{E}_x \left[1_{\{Y_{m-1}=z\}} \lambda(z) p(z, y) \int_{T_{m-1}}^t e^{-\lambda(y)(t-r)} e^{-\lambda(z)(r-T_{m-1})} dr \right] \\
 &= \int_0^t \mathbb{E}_x \left[1_{\{Y_{m-1}=z, T_{m-1} \leq r\}} e^{-\lambda(z)(r-T_{m-1})} \right] \lambda(z) p(z, y) e^{-\lambda(y)(t-r)} dr \\
 (15) \quad &= \int_0^t \mathbb{P}_x(\{Y_{m-1} = z, T_{m-1} \leq r < T_m\}) \lambda(z) p(z, y) e^{-\lambda(y)(t-r)} dr, \quad \forall z \in \mathcal{S}.
 \end{aligned}$$

Putting (11)–(15) together and applying Tonelli's Theorem then yields the FIR (9). $\square$

(16) LEMMA (The BIR and the weak version of the time-varying equations). *Suppose that Q satisfies (26.1). The backwards integral recursion (BIR),*

$$(17) \quad p_t^n(x, y) = 1_x(y) e^{-\lambda(x)t} + \int_0^t \lambda(x) e^{-\lambda(x)(t-s)} \sum_{z \in \mathcal{S}} p(x, z) p_s^{n-1}(z, y) ds$$

holds for all x, y in $\mathcal{S}$, non-negative t , and positive n , where p_t^n is as in (8). Moreover, the transition probabilities satisfy the integral version of the backward equations:

$$(18) \quad p_t(x, y) = 1_x(y) e^{-\lambda(x)t} + \int_0^t \lambda(x) e^{-\lambda(x)(t-s)} \sum_{z \in \mathcal{S}} p(x, z) p_s(z, y) ds \quad \forall x, y \in \mathcal{S}, \quad t \in [0, \infty).$$

Proof. Because taking the limit $n \rightarrow \infty$ in (17) and applying monotone convergence yields (18), we need only to prove (17). We do this by inductively showing that

$$(19) \quad p_t^n(x, y) = u_t^n(x, y) \quad \forall x, y \in \mathcal{S}, \quad t \in [0, \infty), \quad n \in \mathbb{N},$$

where u_t^n is obtained by running the BIR:

$$u_t^n(x, y) := \begin{cases} 1_x(y) e^{-\lambda(x)t} & \text{if } n = 0 \\ 1_x(y) e^{-\lambda(x)t} + \int_0^t \lambda(x) e^{-\lambda(x)(t-s)} \sum_{z \in \mathcal{S}} p(x, z) u_s^{n-1}(z, y) ds & \text{if } n > 0 \end{cases}$$

for all $x, y \in \mathcal{S}$ and $t \in [0, \infty)$. Clearly,

$$u_t^0(x, y) = 1_x(y) e^{-\lambda(x)t} = p_t^0(x, y) \quad \forall x, y \in \mathcal{S}, \quad t \in [0, \infty).$$

Furthermore,

$$\begin{aligned}
u_t^1(x, y) &= 1_x(y)e^{-\lambda(x)t} + \int_0^t \lambda(x)e^{-\lambda(x)(t-s)} \sum_{z \in \mathcal{S}} p(x, z) 1_z(y) e^{-\lambda(z)s} ds \\
&= 1_x(y)e^{-\lambda(x)t} + \int_0^t \lambda(x)e^{-\lambda(x)(t-s)} p(x, y) e^{-\lambda(y)s} ds \\
&= 1_x(y)e^{-\lambda(x)t} + \int_0^t e^{-\lambda(x)s'} \lambda(x) p(x, y) e^{-\lambda(y)(t-s')} ds' \\
&= 1_x(y)e^{-\lambda(x)t} + \int_0^t \sum_{z \in \mathcal{S}} 1_x(z) e^{-\lambda(x)s'} \lambda(z) p(z, y) e^{-\lambda(y)(t-s')} ds' = p_t^1(x, y),
\end{aligned}$$

where the third equality follows from the change of variables $s' := t - s$ and the fifth from the FIR (9). Now, suppose that (19) holds for all $n = 0, 1, \dots, m-1$ with $m \geq 2$. In this case,

$$\begin{aligned}
u_t^m(x, y) &= 1_x(y)e^{-\lambda(x)t} + \int_0^t \lambda(x) \sum_{z \in \mathcal{S}} p(x, z) p_s^{m-1}(z, y) e^{-\lambda(x)(t-s)} ds \\
&= 1_x(y)e^{-\lambda(x)t} + \int_0^t \lambda(x) \sum_{z \in \mathcal{S}} p(x, z) \left(1_z(y) e^{-\lambda(y)s} \right. \\
&\quad \left. + \int_0^s \sum_{z' \in \mathcal{S}} p_r^{m-2}(z, z') \lambda(z') p(z', y) e^{-\lambda(y)(s-r)} dr \right) e^{-\lambda(x)(t-s)} ds \\
&= 1_x(y)e^{-\lambda(x)t} + \lambda(x) p(x, y) \int_0^t e^{-\lambda(y)(t-s')} e^{-\lambda(x)s'} ds' \\
&\quad + \sum_{z \in \mathcal{S}} \sum_{z' \in \mathcal{S}} \lambda(x) p(x, z) \lambda(z') p(z', y) \int_0^t \int_0^{t-s'} p_{t-s'-r'}^{m-2}(z, z') e^{-\lambda(y)r'} e^{-\lambda(x)s'} dr' ds',
\end{aligned}$$

where we've used the changes of variables $r' := s - r$ and $s' := t - s$. Similarly,

$$\begin{aligned}
p_t^m(x, y) &= 1_x(y)e^{-\lambda(x)t} + \int_0^t \sum_{z \in \mathcal{S}} u_s^{m-1}(x, z) \lambda(z) p(z, y) e^{-\lambda(y)(t-s)} ds \\
&= 1_x(y)e^{-\lambda(x)t} + \int_0^t \sum_{z \in \mathcal{S}} \left(1_x(z) e^{-\lambda(x)s} \right. \\
&\quad \left. + \int_0^s \lambda(x) \sum_{z' \in \mathcal{S}} p(x, z') u_r^{m-2}(z', z) e^{-\lambda(x)(s-r)} dr \right) \lambda(z) p(z, y) e^{-\lambda(y)(t-s)} ds \\
&= 1_x(y)e^{-\lambda(x)t} + \lambda(x) p(x, y) \int_0^t e^{-\lambda(y)(t-s)} e^{-\lambda(x)s} ds \\
&\quad + \sum_{z \in \mathcal{S}} \sum_{z' \in \mathcal{S}} \lambda(x) p(x, z') \lambda(z) p(z, y) \int_0^t \int_0^{t-s'} u_{t-s'-r'}^{m-2}(z, z') e^{-\lambda(y)r'} e^{-\lambda(x)s'} dr' ds'.
\end{aligned}$$

Comparing the above two expressions, we find that (19) holds for $n = m$ and (17) follows. $\square$

(20) LEMMA (The transition probabilities are continuously differentiable and satisfy the strong version of the backward equations). *Suppose that Q satisfies (26.1). For each $x, y \in \mathcal{S}$,*

$$(21) \quad t \mapsto p_t(x, y)$$

is a continuously differentiable function on $[0, \infty)$. Moreover, the backward equations (3) hold.

Proof. Let $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ be any sequence of finite subsets of the state space such that $\cup_{r=1}^{\infty} \mathcal{S}_r = \mathcal{S}$. Because

$$\sum_{z \in \mathcal{S}} p(x, z) p_s(z, y) \leq \sum_{z \in \mathcal{S}} p(x, z) = 1 \quad \forall x, y \in \mathcal{S}, \quad s \in [0, \infty),$$

the weak backward equations (18) imply that, for each $x, y \in \mathcal{S}$, (21) is a continuous function on $[0, \infty)$. Thus, for any given $x, y \in \mathcal{S}$,

$$\left(s \mapsto \sum_{z \in \mathcal{S}_r} p(x, z) p_s(z, y) \right)_{r \in \mathbb{Z}_+}$$

is a sequence of continuous functions on $[0, \infty)$. Because

$$\left| \sum_{z \in \mathcal{S}} p(x, z) p_s(z, y) - \sum_{z \in \mathcal{S}_r} p(x, z) p_s(z, y) \right| = \sum_{z \notin \mathcal{S}_r} p(x, z) p_s(z, y) \leq \sum_{z \notin \mathcal{S}_r} p(x, z) \quad \forall s \in [0, \infty),$$

the sequence converges uniformly over $[0, \infty)$ to

$$s \mapsto \sum_{z \in \mathcal{S}} p(x, z) p_s(z, y)$$

Thus, the limiting function is continuous on $[0, \infty)$. For this reason, the fundamental theorem of calculus and the weak backward equations (18) imply that (21) is continuously differentiable.

Next, multiplying both sides of the weak equations (18) by $e^{\lambda(x)t}$, taking derivatives, and applying the fundamental theorem of calculus yields

$$\dot{p}_t(x, y) e^{\lambda(x)t} + p_t(x, y) \lambda(x) e^{\lambda(x)t} = \lambda(x) e^{\lambda(x)t} \sum_{z \in \mathcal{S}} p(x, z) p_t(z, y) \quad \forall t \in [0, \infty).$$

Multiplying through by $e^{-\lambda(x)t}$, re-arranging, and using (26.2)–(26.3), we obtain (3). $\square$

(22) LEMMA (The transition probabilities satisfy the strong version of the forward equations). *Suppose that Q satisfies (26.1). The transition probabilities satisfy the forward equation (2).*

Proof. Once we show that, for every $x, y \in \mathcal{S}$, the function

$$(23) \quad s \mapsto \sum_{z \in \mathcal{S}} p_s(x, z) \lambda(z) p(z, y)$$

is a continuous on $[0, \infty)$, the remainder of the proof is entirely analogous to that of Lemma 20 and we skip the details.

To show that (23) is continuous on $[0, \infty)$ it suffices to show that it is continuous on $[0, t]$ for each $t \in [0, \infty)$. To do so, let $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ be any sequence of finite subsets of the state space such that $\cup_{r=1}^{\infty} \mathcal{S}_r = \mathcal{S}$ and consider the sequence of functions

$$(24) \quad \left(s \mapsto \sum_{z \in \mathcal{S}_r} p_s(x, z) \lambda(z) p(z, y) \right)_{r \in \mathbb{Z}_+}$$

converging pointwise to (23). Because the subsets $\mathcal{S}_r$ are finite and the transition probabilities are continuous (Lemma 20), each of the functions in the sequence is continuous. For this reason, we need only to show that the convergence is uniform over $s \in [0, t]$. As we show at the end of the proof,

$$(25) \quad \dot{p}_t(x, y) \geq -\lambda(x) \quad \forall t \in [0, \infty), \quad x, y \in \mathcal{S}.$$

Multiplying through by $e^{\lambda(x)t}$, applying the product rule, and integrating shows that $t \mapsto e^{\lambda(x)t}p_t(x, y)$ is a non-decreasing function on $[0, \infty)$. The uniform convergence follows as

$$\begin{aligned} & \left| \sum_{z \in \mathcal{S}} p_s(x, z) \lambda(z) p(z, y) - \sum_{z \in \mathcal{S}_r} p_s(x, z) \lambda(z) p(z, y) \right| = \sum_{z \notin \mathcal{S}_r} p_s(x, z) \lambda(z) p(z, y) \\ & \leq \sum_{z \notin \mathcal{S}_r} e^{\lambda(x)s} p_s(x, z) \lambda(z) p(z, y) \leq \sum_{z \notin \mathcal{S}_r} e^{\lambda(x)t} p_t(x, z) \lambda(z) p(z, y) \quad \forall s \in [0, t] \end{aligned}$$

and the right-hand side converges to zero as $r \rightarrow \infty$, otherwise

$$\begin{aligned} \sum_{z \in \mathcal{S}} p_s(x, z) \lambda(z) p(z, y) &= e^{-\lambda(x)s} \sum_{z \in \mathcal{S}} e^{\lambda(x)s} p_s(x, z) \lambda(z) p(z, y) \\ &\geq e^{-\lambda(x)s} \sum_{z \in \mathcal{S}} e^{\lambda(x)t} p_t(x, z) \lambda(z) p(z, y) = \infty \quad \forall s \in [t, \infty) \end{aligned}$$

contradicting the weak forward equations (10).

We have one loose end to tie up: proving (25) or, more generally, that

$$(26) \quad |\dot{p}_t(x, y)| \leq q(x) \quad \forall t \in [0, \infty), \quad x, y \in \mathcal{S}.$$

If $q(x) = 0$, the claim is trivial as Proposition 26.12 shows that $p_t(x, y) = 1_x(y)$ for all $t \in [0, \infty)$. Otherwise, the semigroup property (31.4) implies that

$$p_{t+h}(x, y) - p_t(x, y) = \sum_{z \in \mathcal{S}} p_h(x, z) p_t(z, y) - p_t(x, y) = (p_h(x, x) - 1) p_t(x, y) + \sum_{z \neq x} p_h(x, z) p_t(z, y).$$

Because $p_t(x, y) \in [0, 1]$ and $\sum_{z \in \mathcal{S}} p_h(x, z) \leq 1$ for all $x, y \in \mathcal{S}$, it is not difficult to see that the two terms on the right-hand side have opposite signs and are bounded by $1 - p_h(x, x)$. Thus,

$$(27) \quad |p_{t+h}(x, y) - p_t(x, y)| \leq 1 - p_h(x, x) \quad \forall x, y \in \mathcal{S}.$$

By definition,

$$\begin{aligned} p_h(x, x) &= \mathbb{P}_x(\{X_h = x, h < T_\infty\}) \geq \mathbb{P}_x(\{X_h = x, h < T_1\}) = \mathbb{P}_x(\{X_0 = x, h < T_1\}) \\ &= \mathbb{P}_x(\{h < T_1\}) = e^{-q(x)h} \geq 1 - hq(x), \end{aligned}$$

and (26) follows by plugging the above into (27), dividing through by h , and taking the limit $h \rightarrow 0$. □

(28) LEMMA (The transition probabilities are the minimal non-negative solution of the forward and backward equations). *Suppose that Q satisfies (26.1), that, for each $x, y \in \mathcal{S}$,*

$$t \mapsto k_t(x, y)$$

is a non-negative ($k_t(x, y) \geq 0$ for all $t \in [0, \infty)$) differentiable function on $[0, \infty)$. If the collection of these functions satisfies the forward equations,

$$\dot{k}_t(x, y) = \sum_{z \in \mathcal{S}} k_t(x, z) q(z, y) \quad \forall t \in [0, \infty), \quad x \in \mathcal{S}, \quad k_0(x, y) = 1_x(y) \quad \forall x, y \in \mathcal{S},$$

or the backward equations,

$$\dot{k}_t(x, y) = \sum_{z \in \mathcal{S}} q(x, z) k_t(z, y) \quad \forall t \in [0, \infty), \quad x, y \in \mathcal{S}, \quad k_0(x, y) = 1_x(y) \quad \forall x, y \in \mathcal{S},$$

then

$$k_t(x, y) \geq p_t(x, y) \quad \forall x, y \in \mathcal{S}, \quad t \in [0, \infty).$$

Proof. Given the BIR (17), the proof for the case of the backward equations is entirely analogous to that for the case of the forward equations and we focus on the latter. Suppose that we are able to show that $(k_t(x, y))_{x, y \in \mathcal{S}}$ satisfies the weak version of the forward equations, i.e.,

$$(29) \quad k_t(x, y) = 1_x(y)e^{-\lambda(y)t} + \int_0^t \sum_{z \in \mathcal{S}} k_s(x, z)\lambda(z)p(z, y)e^{-\lambda(y)(t-s)}ds \quad \forall t \in [0, \infty), \quad x \in \mathcal{S},$$

and let p_t^n be as in (8). Because the above and (12) imply that

$$k_t(x, y) \geq 1_x(y)e^{-\lambda(y)t} = p_t^0(x, y) \quad \forall t \in [0, \infty), \quad x, y \in \mathcal{S},$$

combining the FIR (9) and (29) we find that

$$\begin{aligned} k_t(x, y) &= 1_x(y)e^{-\lambda(y)t} + \int_0^t \sum_{z \in \mathcal{S}} k_s(x, z)\lambda(z)p(z, y)e^{-\lambda(y)(t-s)}ds \\ &\geq 1_x(y)e^{-\lambda(y)t} + \int_0^t \sum_{z \in \mathcal{S}} p_s^0(x, z)\lambda(z)p(z, y)e^{-\lambda(y)(t-s)}ds = p_t^1(x, y) \quad \forall t \in [0, \infty), \quad x \in \mathcal{S}. \end{aligned}$$

Iterating the above argument forward shows that

$$k_t(x, y) \geq p_t^n(x, y) \quad \forall n \in \mathbb{N}, \quad t \in [0, \infty), \quad x, y \in \mathcal{S}.$$

Taking the limit $n \rightarrow \infty$ and applying monotone convergence then proves the minimality of p_t .

We have one loose end remaining: proving (29). If $q(y) = 0$ this is trivial. Otherwise, $\lambda(y) = q(y)$ and

$$\begin{aligned} \frac{d}{ds}(k_s(x, y)e^{\lambda(y)s}) &= \dot{k}_s(x, y)e^{\lambda(y)s} + \lambda(y)k_s(x, y)e^{\lambda(y)s} = \sum_{z \neq y} k_s(x, z)q(z, y)e^{\lambda(y)s} \\ &= \sum_{z \in \mathcal{S}} k_s(x, z)\lambda(z)p(z, y)e^{\lambda(y)s} \quad \forall s \in [0, \infty), \quad x, y \in \mathcal{S}. \end{aligned}$$

Integrating over $s \in [0, t]$, applying the fundamental theorem of calculus, and multiplying through by $e^{-\lambda(y)t}$ then yields (29). $\square$

33. The time-varying law and its differential equation. Let

$$(1) \quad p_t(x) := \mathbb{P}_\gamma(\{X_t = x, t < T_\infty\})$$

denote the probability that the chain is at state x at time t if its starting state was sampled from γ . Theorem 31.2 gives an explicit expression for the *time-varying law* $(p_t)_{t \geq 0}$ in terms of the initial distribution γ and t -transition matrix $P_t = (p_t(x, y))_{x, y \in \mathcal{S}}$,

$$(2) \quad p_t(x) = \sum_{x' \in \mathcal{S}} \gamma(x')p_t(x', x) \quad \forall x \in \mathcal{S},$$

or $p_t = \gamma P_t$ in matrix notation, for all t in $[0, \infty)$. While theoretically useful, (2) is of little practical use for computing the time-varying law of a chain. A far more useful description in this respect—and one very popular among physicists, engineers, and mathematical biologists—is the following a generalisation of the forward equations (32.2).

(3) **THEOREM** (Analytical characterisation of the time-varying law). *Suppose that Q satisfies (26.1) and that*

$$(4) \quad \sum_{x \in \mathcal{S}} \gamma(x) \left(\sup_{s \in [0, t]} \sum_{x' \neq y} p_s(x, x')q(x', y) \right) < \infty, \quad \forall y \in \mathcal{S}, \quad t \in [0, \infty).$$

The time-varying law is continuously differentiable: for each x in $\mathcal{S}$,

$$t \mapsto p_t(x)$$

is a continuously differentiable function on $[0, \infty)$. Moreover, it satisfies the equations

$$(5) \quad \dot{p}_t(x) = \sum_{x' \in \mathcal{S}} p_t(x') q(x', x) \quad \forall t \in [0, \infty), \quad x \in \mathcal{S}, \quad p_0(x) = \gamma(x) \quad \forall x \in \mathcal{S},$$

or

$$\dot{p}_t = p_t Q \quad \forall t \in [0, \infty), \quad p_0 = \gamma,$$

in matrix notation. Moreover, the time-varying law is the minimal non-negative solution of these equations: if $(k_t(x))_{x \in \mathcal{S}, t \in [0, \infty)}$ is a non-negative ($k_t(x) \geq 0$ for all $x \in \mathcal{S}$, $t \in [0, \infty)$) differentiable function satisfying (5), then

$$k_t(x) \geq p_t(x) \quad \forall x \in \mathcal{S}, \quad t \in [0, \infty).$$

For the theorem's proof, see the end of the section. Condition (4) may seem very technical, but it is straightforward to derive conditions that cover the majority of chains encountered in practice:

(6) PROPOSITION (Condition (4) is mild). Suppose that the rate matrix Q satisfies (26.1). If the rate matrix has bounded columns:

$$\sup_{x \in \mathcal{S}} q(x, y) < \infty \quad \forall y \in \mathcal{S},$$

or if it's diagonal is γ -integrable:

$$(7) \quad \mathbb{E}_\gamma [q(X_0)] = \sum_{x \in \mathcal{S}} q(x) \gamma(x) < \infty,$$

then (4) holds.

Proof. Given that $\sum_{x' \in \mathcal{S}} p_s(x, x') \leq 1$ for all s in $[0, \infty)$, the case of bounded columns is trivial. To argue the case of (7), recall that the transition probabilities satisfy the forward equation (32.2). For this reason,

$$\sum_{x' \neq y} p_s(x, x') q(x', y) \leq p_s(x, y) q(y) + |\dot{p}_s(x, y)| \leq q(y) + |\dot{p}_s(x, y)| \quad \forall s \in [0, \infty), \quad x, y \in \mathcal{S}.$$

Because the derivative $\dot{p}_t(x, y)$ is bounded by $q(x)$ (c.f. (32.26)), the result follows from the above. $\square$

Explosions and uniqueness of solutions. For any given initial distribution γ satisfying (4), Proposition 32.6 and Theorem 3 show that the time-varying law is the unique solution $(k_t)_{t \geq 0}$ of (5) with mass no greater than one (i.e., such that $p_t(\mathcal{S}) \leq 1$ for all t in $[0, \infty)$) if sampling the starting state from γ results in chain that does not explode:

$$\mathbb{P}_\gamma (\{T_\infty = \infty\}) = 1.$$

If the above is not satisfied, then (5) may have more than one solution for the same reasons that the forward equation (32.2) may have more than one solution (Anderson, 1991).

Some open problems. To the best of my knowledge, the question of whether equations (5) hold for all initial distributions γ and chains with stable and conservative rate matrices Q remains unresolved. If one is willing to settle for a weak version of these equations the answer is affirmative: integrating the forward equations (32.10), re-arranging, multiplying through by $\gamma(x)$, and summing over x in $\mathcal{S}$, we find that

$$(8) \quad p_t(x) - \gamma(x) + \int_0^t p_s(x)q(x)ds = \int_0^t \sum_{x' \neq x} p_s(x')q(x', x)ds \quad \forall t \in [0, \infty), \quad x \in \mathcal{S}.$$

Because $\int_0^t p_s(x)q(x)ds \leq tq(x) < \infty$, it follows that

$$\sum_{x' \neq x} p_t(x')q(x', x) < \infty \text{ for Lebesgue almost every } t \in [0, \infty) \text{ and all } x \in \mathcal{S}.$$

For this reason, the Lebesgue differentiation theorem (Tao, 2011, Theorem 1.6.11) and (8) show that, for all x in $\mathcal{S}$, $t \mapsto p_t(x)$ is differentiable Lebesgue almost everywhere and

$$(9) \quad \dot{p}_t(x) = p_t Q(x) \text{ for Lebesgue almost every } t \in [0, \infty) \text{ and all } x \in \mathcal{S}.$$

What condition (4) does is ensure that the integrands in (8) are finite and continuous, in which case it follows from the fundamental theorem of calculus that $t \mapsto p_t(x)$ is a continuously differentiable function satisfying $\dot{p}_t(x) = p_t Q(x)$ for all t in $[0, \infty)$. Indeed, were $p_t(x)$ to be a continuously differentiable function on $[0, \infty)$ for all x in $\mathcal{S}$, it would follow that

$$\begin{aligned} \sup_{s \in [0, t]} \sum_{x \in \mathcal{S}} \gamma(x) \sum_{x' \neq y} p_s(x, x')q(x', y) &\leq \sup_{s \in [0, t]} \sum_{x' \neq y} p_s(x')q(x', y) \\ &\leq \sup_{s \in [0, t]} (\dot{p}_s(y) + p_s(y)q(y)) < \infty, \quad \forall y \in \mathcal{S}, \quad t \in [0, \infty) \end{aligned}$$

as continuous functions are bounded over finite intervals. This is close to (4) but not quite, and the question lingers: is (4) necessary for the time-varying law to be continuously differentiable on $[0, \infty)$?

Now, notice that there is nothing stopping us from picking a Q and γ such that $\gamma Q(x) = \infty$ for at least one x . In this case, (4) will not be satisfied and neither will the conclusions of Theorem 3 ($p_t(x)$ cannot both be differentiable on $[0, \infty)$ and satisfy $\dot{p}_0(x) = p_0 Q(x) = \gamma Q(x) = \infty$). However, tweaking the argument in the theorem's proof, it is possible to show that the time-varying law is continuous differentiable on $(0, \infty)$ and (5) holds if we replace (4) with

$$(10) \quad \sum_{x \in \mathcal{S}} \gamma(x) \left(\sup_{s \in [1/t, t]} \sum_{x' \neq y} p_s(x, x')q(x', y) \right) < \infty, \quad \forall y \in \mathcal{S}, \quad t \in [1, \infty).$$

Whether the above is actually necessary for us to be able to strengthen (8) into ‘for all x in $\mathcal{S}$, $t \mapsto p_t(x)$ is continuously differentiable (or even just differentiable) on $(0, \infty)$ and (5) holds’ also remains open. It may well be that this is the case for all initial distributions and chains with stable and conservative rate matrices regardless of whether (10) is satisfied.

Lastly, for the cases not covered by Proposition 6 we are lacking in tools to verify (4) or (10) in practice. If either of these conditions prove to actually be important, the question of whether they can be rephrased entirely in terms of γ and Q is also worth having a look at. If you know anything more on these issues, please get in touch.

A proof of Theorem 3. The argument showing that the time-varying law p_t is minimal among the set of solutions of the master equation (5) is entirely analogous to those in the proofs of Lemmas 32.7 and 32.28. Thus, we skip it and instead focus on proving that p_t is continuously differentiable and satisfies (5).

Given (8) and the fundamental theorem of calculus, it suffices to show that

$$(11) \quad t \mapsto p_t(x) \quad \text{and} \quad t \mapsto \sum_{x' \neq y} p_t(x')q(x', y)$$

are continuous functions $[0, \infty)$ for all y in $\mathcal{S}$. To do so, let $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ be any given sequence of finite subsets of the state space such that $\cup_{r=1}^{\infty} \mathcal{S}_r = \mathcal{S}$. By equation (2), we have that $t \mapsto p_t(x)$ is the pointwise limit of the sequence

$$\left(t \mapsto \sum_{x' \in \mathcal{S}_r} \gamma(x')p_t(x', x) \right)_{r \in \mathbb{Z}_+}.$$

Because the sets $\mathcal{S}_r$ are finite, the continuity of the transition probability (Theorem 32.1) implies that these functions are continuous. Because the convergence is uniform over ts in $[0, \infty)$,

$$\left| p_t(x) - \sum_{x' \in \mathcal{S}_r} \gamma(x')p_t(x', x) \right| = \sum_{x' \notin \mathcal{S}_r} \gamma(x')p_t(x', x) \leq \gamma(\mathcal{S}_r) \quad \forall r > 0,$$

it follows that $t \mapsto p_t(x)$ is continuous on $[0, \infty)$.

To prove the continuity of the other function in (11), notice that it is the pointwise limit of the sequence

$$\left(t \mapsto \sum_{x \in \mathcal{S}_r} \sum_{x' \neq y} \gamma(x)p_t(x, x')q(x', y) \right)_{r \in \mathbb{Z}_+}$$

of continuous functions (continuity of the above also follows from Theorem 32.1). Because

$$\begin{aligned} \left| \sum_{x' \neq y} p_s(x')q(x', y) - \sum_{x \in \mathcal{S}_r} \sum_{x' \neq y} \gamma(x)p_s(x, x')q(x', y) \right| &= \sum_{x \notin \mathcal{S}_r} \sum_{x' \neq y} \gamma(x)p_s(x, x')q(x', y) \\ &\leq \sum_{x \notin \mathcal{S}_r} \gamma(x) \left(\sup_{s \in [0, t]} \sum_{x' \neq y} p_s(x, x')q(x', y) \right), \end{aligned}$$

our assumption (4) implies that the convergence is uniform over s in $[0, t]$, for every interval $[0, t]$, and it follows that the rightmost function in (11) is continuous on $[0, \infty)$.

34. Dynkin's formula. Integrating the generalised forward equation (33.5) over $[0, t]$ we find that

$$p_t(x) = \gamma(x) + \int_0^t p_s Q(x) ds \quad \forall x \in \mathcal{S}.$$

If f is a real-valued function $\mathcal{S}$ satisfying the appropriate integrability conditions, then multiplying both sides by $f(x)$ and summing over $x \in \mathcal{S}$, we obtain the following the integral version of (33.5):

$$(1) \quad \mathbb{E}_\gamma [f(X_t)1_{\{t < T_\infty\}}] = \gamma(f) + \mathbb{E}_\gamma \left[\int_0^{t \wedge T_\infty} Qf(X_s) ds \right],$$

where $Qf(x) := \sum_{x' \in \mathcal{S}} q(x, x')f(x')$. The aim of this section is to prove a version of Dynkin's formula which states that (1) holds not only for deterministic times t for stopping times η .

A special class of stopping times. To not complicate matters unnecessarily, we only prove Dynkin's formula a special type of stopping times that I call *jump-time-valued*:

(2) **DEFINITION** (Jump-time-valued stopping times). Let $(\mathcal{F}_t)_{t \geq 0}$ denote the filtration generated by the chain (Definition 28.1) and $(\mathcal{G}_n)_{n \in \mathbb{N}}$ that generated by the jump chain and jump times (Definition 27.1). We say that a random variable $\eta : \Omega \rightarrow [0, \infty]$ is a *jump-time-valued* $(\mathcal{F}_t)_{t \geq 0}$ -stopping time if there exists a $(\mathcal{G}_n)_{n \in \mathbb{N}}$ -stopping time ς (Definition 8.1 with $(\mathcal{G}_n)_{n \in \mathbb{N}}$ replacing $(\mathcal{F}_n)_{n \in \mathbb{N}}$ therein) such that

$$(3) \quad \eta(\omega) = \begin{cases} T_{\varsigma(\omega)}(\omega) & \text{if } \varsigma(\omega) < \infty \\ \infty & \text{if } \varsigma(\omega) = \infty \end{cases} \quad \forall \omega \in \Omega.$$

For any such η ,

$$(4) \quad \{\eta < \infty\} = \{\eta < T_\infty\} = \{\varsigma < \infty\}$$

because the jump times are all finite by their definition in (26.7). Moreover these random times are $(\mathcal{F}_t)_{t \geq 0}$ -stopping times (in the sense of Definition 28.3):

(5) **THEOREM.** All jump-time-valued $(\mathcal{F}_t)_{t \geq 0}$ -stopping times (Definition 2) are $(\mathcal{F}_t)_{t \geq 0}$ -stopping times (Definition 28.3), where $(\mathcal{F}_t)_{t \geq 0}$ denotes the filtration generated by the chain.

Proof. For any η satisfying (3),

$$\{\eta \leq t\} = \bigcup_{n=0}^{\infty} \{\varsigma = n\} \cap \{T_n \leq t\} \quad \forall t \in [0, \infty).$$

Because $\{\varsigma = n\}$ belongs to $\mathcal{G}_n$ (Lemma 8.3(i)) and $\mathcal{G}_n \subseteq \mathcal{F}_{T_n}$ (Proposition 28.7), the above shows that η is a $(\mathcal{F}_t)_{t \geq 0}$ -stopping time. $\square$

An open problem. The idea behind jump-time-valued stopping times η is that they are stopping times (Definition 28.3) taking values in the set of jump times (and plus infinity):

$$(6) \quad \eta(\omega) \in \{T_0(\omega), T_1(\omega), T_2(\omega), \dots\} \cup \{\infty\} \quad \forall \omega \in \Omega.$$

In other words, the event that occurs at such a stopping time is one that may only occur when the chain jumps. It seems to me that any stopping time satisfying the above should be of the form in Definition 2. However, the question lingers and its answer hinges on that to the open question of Section 28: ‘does $\mathcal{G}_n = \mathcal{F}_{T_n}$?’ Indeed, any $(\mathcal{F}_t)_{t \geq 0}$ -stopping time η satisfying the above satisfies (3) with

$$(7) \quad \varsigma(\omega) := \begin{cases} n & \text{if } \eta(\omega) = T_n(\omega) \quad \forall n \in \mathbb{N} \\ \infty & \text{if } \eta(\omega) = \infty \end{cases} \quad \forall \omega \in \Omega.$$

Were $\mathcal{F}_{T_n}$ to coincide $\mathcal{G}_n$ for each n , ς would be a $(\mathcal{G}_n)_{n \in \mathbb{N}}$ -stopping time as Lemma 28.5(i) implies that $\{\varsigma \leq n\} = \{\eta \leq T_n\}$ belongs to $\mathcal{F}_{T_n}$ for all n .

An important example: hitting times. The *hitting time* (or *first passage time*) τ_A of a set $A \subseteq \mathcal{S}$ is the first time that the chain X visits the subset (or plus infinity if it never does). Formally,

$$(8) \quad \tau_A(\omega) := \inf\{t \in [0, T_\infty(\omega)) : X_t(\omega) \in A\} \quad \forall \omega \in \Omega.$$

We use the shorthand $\tau_x := \tau_{\{x\}}$ for any x in $\mathcal{S}$. Of course, the first time that the chain enters A is the moment that it jumps into A and it follows that τ_A is a jump-time-valued stopping time:

(9) PROPOSITION. *The hitting time τ_A is a jump-time-valued $(\mathcal{F}_t)_{t \geq 0}$ -stopping time. Furthermore, X_{τ_A} belongs to A if τ_A is finite and (3)–(4) hold if we replace η with τ_A and ς with σ_A , where σ_A denotes the hitting time of A of the jump chain Y (defined by replacing X with Y in (8.5)).*

Proof. By the definition of τ_A ,

$$(10) \quad \{\tau_A < \infty\} = \{\tau_A < T_\infty\} = \bigcup_{n=0}^{\infty} \{T_n \leq \tau_A < T_{n+1}\}.$$

Similarly, ω belongs to $\{T_n \leq \tau_A < T_{n+1}\}$ if only if $T_n(\omega) \leq \tau_A(\omega)$ and there exists an $0 < \varepsilon < T_{n+1}(\omega) - \tau_A(\omega)$ such that $X_{\tau_A(\omega)+\varepsilon}(\omega)$ belongs to A . In this case, the definition of the chain's paths (26.6)–(26.8) implies that

$$X_{\tau_A(\omega)+\varepsilon'} = X_{T_n(\omega)}(\omega) = Y_n(\omega) \in A \quad \forall \varepsilon' \in [0, \varepsilon].$$

For this reason, τ_A 's definition further implies that $\tau_A = T_n$ on $\{T_n \leq \tau_A < T_{n+1}\}$. It follows from (10) that X_{τ_A} belongs to A whenever τ_A is finite and that τ_A takes values in the set of jump times (i.e., that (6) holds if we replace η therein with τ_A). Because

$$\begin{aligned} \{\tau_A = T_n\} &= \{X_{T_0} \notin A, \dots, X_{T_{n-1}} \notin A, X_{T_n} \in A\} = \{Y_0 \notin A, \dots, Y_{n-1} \notin A, Y_n \in A\} \\ &= \{\sigma_A = n\} \quad \forall n \in \mathbb{N}, \\ \{\tau_A = \infty\} &= \{\tau_A < \infty\}^c = \left(\bigcup_{n=0}^{\infty} \{\tau_A = T_n\} \right)^c = \left(\bigcup_{n=0}^{\infty} \{\sigma_A = n\} \right)^c = \{\sigma_A < \infty\}^c \\ &= \{\sigma_A = \infty\}, \end{aligned}$$

ς in (7) (with τ_A replacing η) coincides with σ_A and the result follows from Proposition 8.6 and the fact that $(\mathcal{G}_n)_{n \in \mathbb{N}}$ contains the filtration generated by Y . $\square$

Dynkin's formula. We are now ready to tackle Dynkin's Formula:

(11) THEOREM (Dynkin's formula). *Let $t \in [0, \infty)$, η be as in (3), $\eta_n := \eta \wedge T_n$ for each $n \in \mathbb{N}$, f be a γ -integrable function on $\mathcal{S}$ such that $Qf(x)$ is absolutely convergent for all x in $\mathcal{S}$, and g be a continuously differentiable function on $[0, \infty)$. If there exists a finite set $F \subseteq \mathcal{S}$ such that X_s belongs to F for all $s \in [0, \eta \wedge T_\infty)$, then*

$$(12) \quad \mathbb{E}_\gamma [g(t \wedge \eta_n) f(X_{t \wedge \eta_n})] = g(0) \gamma(f) + \mathbb{E}_\gamma \left[\int_0^{t \wedge \eta_n} (g(s) Qf(X_s) + \dot{g}(s) f(X_s)) ds \right] \quad \forall n \in \mathbb{N}.$$

Proof. Because F is finite and g is continuously differentiable,

$$(13) \quad \begin{aligned} |g(s)| &\leq \sup_{r \in [0, t]} g(r) < \infty, \quad |\dot{g}(s)| \leq \sup_{r \in [0, t]} \dot{g}(r) < \infty, \quad \forall s \in [0, t \wedge \eta], \\ |f(X_s)| &\leq \max_{x \in F} |f(x)| < \infty, \quad |Qf(X_s)| \leq \max_{x \in F} \sum_{y \in \mathcal{S}} |q(x, y)| |f(y)| < \infty, \quad \forall s \in [0, t \wedge \eta), \end{aligned}$$

and it follows that all of the integrals and expectations in (12) are well-defined and finite. Suppose that we have argued for the case of a bounded f and let $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ be a sequence of finite truncation approaching $\mathcal{S}$ (i.e., $\cup_{r=1}^{\infty} \mathcal{S}_r = \mathcal{S}$). Then, for any unbounded f , we recover (12) by replacing f therein with $f 1_{\mathcal{S}_r}$, taking the limit $k \rightarrow \infty$, and applying dominated convergence (note that the inequalities in (13) if we replace f by $f 1_{\mathcal{S}_r}$ in their left-hand sides). Thus, without loss of generality, we assume that f is bounded.

Let ς be the $(\mathcal{G}_n)_{n \in \mathbb{N}}$ -stopping time in (3) associated with η , where $(\mathcal{G}_n)_{n \in \mathbb{N}}$ denotes the filtration generated by the jump chain and the jump times (Definition 27.1), and let $\varsigma_n := \varsigma \wedge n$

for each n in $\mathbb{N}$. It is not difficult to check that $\eta_n = T_{\zeta_n}$. For this reason, it follows from the definition of the chain's paths in (26.6)–(26.8) that

$$(14) \quad \begin{aligned} \int_0^{t \wedge \eta_n} (g(s)Qf(X_s) + \dot{g}(s)f(X_s))ds &= \sum_{m=0}^{\zeta_n-1} \left(\int_{t \wedge T_m}^{t \wedge T_{m+1}} g(s)Qf(X_s)ds + \int_{t \wedge T_m}^{t \wedge T_{m+1}} \dot{g}(s)f(X_s)ds \right) \\ &= \sum_{m=0}^{\zeta_n-1} \left(Qf(Y_m) \int_{t \wedge T_m}^{t \wedge T_{m+1}} g(s)ds + (g(t \wedge T_{m+1}) - g(t \wedge T_m))f(Y_m) \right). \end{aligned}$$

Notice that

$$\int_{t \wedge T_m}^{t \wedge T_{m+1}} g(s)ds = 1_{\{T_m \leq t < T_{m+1}\}} \int_0^{t-T_m} g(T_m + s)ds + 1_{\{T_{m+1} \leq t\}} \int_0^{T_{m+1}-T_m} g(T_m + s)ds$$

Because the $(m+1)$ th waiting time S_{m+1} conditioned on $\mathcal{G}_m$ is exponentially distributed with mean $1/\lambda(Y_m)$ (Theorem 27.4), the take-out-what-is-known property of conditional expectation (Theorem 0.3(v)) implies that

$$\begin{aligned} &\mathbb{E}_\gamma \left[1_{\{T_m \leq t < T_{m+1}\}} \int_0^{t-T_m} g(T_m + s)ds \middle| \mathcal{G}_m \right] \\ &= 1_{\{T_m \leq t\}} \left(\int_0^{t-T_m} g(T_m + s)ds \right) \mathbb{P}_\gamma(\{t - T_m < S_{m+1}\} | \mathcal{G}_m) \\ &= 1_{\{T_m \leq t\}} \left(\int_0^{t-T_m} g(T_m + s)ds \right) e^{-\lambda(Y_m)(t-T_m)}, \quad \mathbb{P}_\gamma\text{-almost surely.} \end{aligned}$$

Similarly,

$$\begin{aligned} &\mathbb{E}_\gamma \left[1_{\{T_{m+1} \leq t\}} \int_0^{T_{m+1}-T_m} g(T_m + s)ds \middle| \mathcal{G}_m \right] \\ &= 1_{\{T_m \leq t\}} \mathbb{E}_\gamma \left[1_{\{S_{m+1} \leq t-T_m\}} \int_0^{S_{m+1}} g(T_m + s)ds \middle| \mathcal{G}_m \right] \\ &= 1_{\{T_m \leq t\}} \int_0^{t-T_m} \lambda(Y_m) e^{-\lambda(Y_m)r} \left(\int_0^r g(T_m + s)ds \right) dr \\ &= 1_{\{T_m \leq t\}} \int_0^{t-T_m} g(T_m + s) \left(\int_s^{t-T_m} \lambda(Y_m) e^{-\lambda(Y_m)r} dr \right) ds \\ &= 1_{\{T_m \leq t\}} \int_0^{t-T_m} g(T_m + s) e^{-\lambda(Y_m)s} ds - 1_{\{T_m \leq t\}} \left(\int_0^{t-T_m} g(T_m + s)ds \right) e^{-\lambda(Y_m)(t-T_m)}, \end{aligned}$$

$\mathbb{P}_\gamma$ -almost surely. Putting the above three together, we have that

$$\begin{aligned} \mathbb{E}_\gamma \left[\int_{t \wedge T_m}^{t \wedge T_{m+1}} g(s)ds \middle| \mathcal{G}_m \right] &= 1_{\{T_m \leq t\}} \int_0^{t-T_m} g(T_m + s) e^{-\lambda(Y_m)s} ds \\ &= \frac{1}{\lambda(Y_m)} 1_{\{T_m \leq t\}} \mathbb{E}_\gamma [1_{\{S_{m+1} \leq t-T_m\}} g(T_{m+1}) | \mathcal{G}_m] \\ &= \frac{1}{\lambda(Y_m)} \mathbb{E}_\gamma [\tilde{g}(T_{m+1}) | \mathcal{G}_m], \quad \mathbb{P}_\gamma\text{-almost surely} \end{aligned}$$

where $\tilde{g}(s) := 1_{\{s \leq t\}} g(s)$ for all s in $[0, \infty)$. For this reason,

$$\mathbb{E}_\gamma \left[Qf(Y_m) \int_{t \wedge T_m}^{t \wedge T_{m+1}} g(s)ds \middle| \mathcal{G}_m \right] = (Pf(Y_m) - f(Y_m)) \mathbb{E}_\gamma [\tilde{g}(T_{m+1}) | \mathcal{G}_m] \quad \mathbb{P}_\gamma\text{-almost surely,}$$

where P denotes the one-step matrix of the jump chain (26.2). Taking expectations of (14) and applying the take-out-what-is-known and tower properties of conditional expectation (Theorem 0.3(iv, v)), we find that

$$\begin{aligned} & \mathbb{E}_\gamma \left[\int_0^{t \wedge \eta_n} (g(s)Qf(X_s) + \dot{g}(s)f(X_s))ds \right] \\ &= \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma_n-1} (\tilde{g}(T_{m+1})Pf(Y_m) - \tilde{g}(T_m)f(Y_m) + 1_{\{T_m \leq t < T_{m+1}\}}g(t)f(Y_m)) \right] \\ &= \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma_n-1} (\tilde{g}(T_{m+1})Pf(Y_m) - \tilde{g}(T_m)f(Y_m)) \right] + \mathbb{E}_\gamma \left[g(t)f(X_t) \sum_{m=0}^{\varsigma_n-1} 1_{\{T_m \leq t < T_{m+1}\}} \right] \\ &= \mathbb{E}_\gamma \left[\sum_{m=0}^{\varsigma_n-1} (\tilde{g}(T_{m+1})Pf(Y_m) - \tilde{g}(T_m)f(Y_m)) \right] + \mathbb{E}_\gamma [1_{\{t < \eta_n\}}g(t)f(X_t)]. \end{aligned}$$

It follows from the above that (12) is satisfied if and only if $\mathbb{E}_\gamma [M_{\varsigma_n}] = 0$, where

$$M_n := \tilde{g}(T_n)f(Y_n) - g(0)f(Y_0) - \sum_{m=0}^{n-1} (\tilde{g}(T_{m+1})Pf(Y_m) - \tilde{g}(T_m)f(Y_m)) \quad \forall n \in \mathbb{N}.$$

If we can argue that $M := (M_n)_{n \in \mathbb{N}}$ is a $(\mathcal{G}_n)_{n \in \mathbb{N}}$ -adapted $\mathbb{P}_\gamma$ -martingale, then an application of Doob's optional stopping theorem of the sort in the proof of Theorem 9.1 yields that $\mathbb{E}_\gamma [M_{\varsigma_n}] = 0$, as desired. Because $\tilde{g}$ and f are bounded functions, it easy to show that M_n is bounded and, hence, $\mathbb{P}_\gamma$ -integrable. Thus, to show that M a martingale we need only to argue that $\mathbb{E}_\gamma [M_{n+1}|\mathcal{G}_n]$ equals M_n , $\mathbb{P}_\gamma$ -almost surely, for each n in $\mathbb{N}$. The conditional independence of T_{n+1} and Y_{n+1} (Theorem 27.4) implies that

$$\begin{aligned} \mathbb{E}_\gamma [\tilde{g}(T_{n+1})f(Y_{n+1})|\mathcal{G}_n] &= \mathbb{E}_\gamma [\tilde{g}(T_{n+1})|\mathcal{G}_n] \mathbb{E}_\gamma [f(Y_{n+1})|\mathcal{G}_n] = \mathbb{E}_\gamma [\tilde{g}(T_{n+1})|\mathcal{G}_n] Pf(Y_n) \\ &= \mathbb{E}_\gamma [\tilde{g}(T_{n+1})Pf(Y_n)|\mathcal{G}_n] \quad \mathbb{P}_\gamma\text{-almost surely,} \quad \forall n \in \mathbb{N}. \end{aligned}$$

It follows from the above that

$$\mathbb{E}_\gamma [M_{n+1} - M_n|\mathcal{G}_n] = \mathbb{E}_\gamma [\tilde{g}(T_{n+1})f(Y_{n+1}) - \tilde{g}(T_{n+1})Pf(Y_n)|\mathcal{G}_n] = 0,$$

implying that M is a martingale and completing the proof. $\square$

35. Stopping distributions and occupation measures. Throughout this section, we use η to denote a jump-time-value $(\mathcal{F}_t)_{t \geq 0}$ stopping time (Definition 34.2). For the same reasons as in the discrete-time case (see Section 8), we say that the chain *stops* at time η . With the stopping time, we associate two measures: the *stopping distribution* μ and *occupation measure* ν defined by

$$\begin{aligned} (1) \quad \mu(A, x) &:= \mathbb{P}_\gamma (\{\eta \in A, X_\eta = x\}) & \forall A \in \mathcal{B}([0, \infty)), \quad x \in \mathcal{S}, \\ (2) \quad \nu(A, x) &:= \mathbb{E}_\gamma \left[\int_{A \cap [0, \eta \wedge T_\infty)} 1_x(X_t)dt \right] & \forall A \in \mathcal{B}([0, \infty)), \quad x \in \mathcal{S}. \end{aligned}$$

In other words, $\mu(A, x)$ is the probability that the chain stops at some time point in A and that it lies in state x when it stops. Similarly, $\nu(A, x)$ is the expected amount of time in A that the chain spends in state x before stopping or exploding. If you care for these things, see the end of the section for the full details of μ and ν 's definition. Otherwise, only notice that, in (1), we are exploiting that η is finite if and only if it is less than T_∞ (c.f. (34.4)).

The mass of the stopping distribution is simply the probability that the stopping time is finite

$$(3) \quad \mu([0, \infty), \mathcal{S}) = \mathbb{P}_\gamma(\{\eta < \infty, X_\eta \in \mathcal{S}\}) = \mathbb{P}_\gamma(\{\eta < \infty\}),$$

which follows by applying the monotone convergence theorem and Tonelli's theorem to (1). Similarly, it follows from (2) that the mass of the occupation measure is

$$(4) \quad \nu([0, \infty), \mathcal{S}) = \mathbb{E}_\gamma \left[\int_0^{\eta \wedge T_\infty} \left(\sum_{x \in \mathcal{S}} 1_x(X_t) \right) dt \right] = \mathbb{E}_\gamma[\eta \wedge T_\infty].$$

If no explosion may occur before the chain stops ($\mathbb{P}_\gamma(\{\eta \leq T_\infty\}) = 1$), then ν 's mass is the mean stopping time.

The marginals. We now turn our attention to the marginals of μ and ν . The *time marginal* μ_T of the stopping distribution defined by

$$(5) \quad \mu_T(A) := \mu(A, \mathcal{S}) = \mathbb{P}_\gamma(\{\eta \in A\}), \quad \forall A \in \mathcal{B}([0, \infty))$$

is the distribution of the stopping time itself. Technically, the above is the distribution of η restricted to $[0, \infty)$ because η may take the value ∞ . However, we recover the full distribution of η from the above with $\mathbb{P}_\gamma(\{\eta = \infty\}) = 1 - \mathbb{P}_\gamma(\{\eta < \infty\}) = 1 - \mu_T([0, \infty))$.

The *space marginals* μ_S and ν_S of the stopping distribution and occupation measure

$$(6) \quad \mu_S(x) := \mu([0, \infty), x) = \mathbb{P}_\gamma(\{\eta < \infty, X_\eta = x\}),$$

$$(7) \quad \nu_S(x) := \nu([0, \infty), x) = \mathbb{E}_\gamma \left[\int_0^{\eta \wedge T_\infty} 1_x(X_t) dt \right],$$

tell us where the chain stops and where it spends time before stopping, respectively. Explicitly, $\mu_S(x)$ is the probability that the chain stops in x , while $\nu_S(x)$ denotes the expected amount of time spent in x before stopping. These space marginals are tied together by a set of linear equations:

(8) **THEOREM.** *Suppose that η is a jump-time-valued $(\mathcal{F}_t)_{t \geq 0}$ -stopping time (Definition 34.2) and that μ_S and ν_S are the space marginals of its stopping distribution and occupation in (6)–(7). If the stopping time is almost surely finite,*

$$(9) \quad \mathbb{P}_\gamma(\{\eta < \infty\}) = 1,$$

then the pair (μ_S, ν_S) satisfies

$$(10) \quad \mu_S(x) + \nu_S(x)q(x) = \gamma(x) + \sum_{y \neq x} \nu_S(y)q(y, x), \quad \forall x \in \mathcal{S}.$$

Proof. Pick any state x and increasing sequence $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ of finite truncations with limit $\mathcal{S}$ (i.e., such that $\cup_{r=1}^\infty \mathcal{S}_r = \mathcal{S}$) and let $(\tau_r)_{r \in \mathbb{Z}_+}$ be the corresponding sequence of exit times in (26.9). It follows from Proposition 34.9 that

$$\eta(\omega) \wedge \tau_r(\omega) = \begin{cases} T_{\varsigma(\omega) \wedge \sigma_r(\omega)}(\omega) & \text{if } \varsigma(\omega) \wedge \sigma_r(\omega) < \infty \\ \infty & \text{if } \varsigma(\omega) \wedge \sigma_r(\omega) = \infty \end{cases} \quad \forall \omega \in \Omega,$$

where ς denotes the discrete-time stopping time in (34.3) associated with η and σ_r denotes the time of exit from $\mathcal{S}_r$ for the jump chain:

$$\sigma_r(\omega) := \inf\{n \geq 0 : Y_n(\omega) \notin \mathcal{S}_r\} \quad \forall \omega \in \Omega, \quad r > 0.$$

In other words, $\eta \wedge \tau_r$ is also jump-time-valued $(\mathcal{F}_t)_{t \geq 0}$ -stopping time. Thus, replacing f by 1_x and η by $\eta \wedge \tau_r$ in Dynkin's formula (34.12), re-arranging, taking the limit $t \rightarrow \infty$, and applying bounded convergence, monotone convergence, and Tonelli's theorem yields

$$(11) \quad \mathbb{P}_\gamma(\{X_{\eta \wedge \tau_r \wedge T_n} = x\}) + \mathbb{E}_\gamma \left[\int_0^{\eta \wedge \tau_r \wedge T_n} 1_x(X_s) ds \right] q(x) \\ = \gamma(x) + \sum_{y \neq x} \mathbb{E}_\gamma \left[\int_0^{\eta \wedge \tau_r \wedge T_n} 1_y(X_s) ds \right] q(y, x) \quad \forall n \geq 0, \quad r > 0.$$

Because Theorem 26.10 shows that $(\tau_r)_{r \in \mathbb{Z}_+}$ is an increasing sequence with $\mathbb{P}_\gamma$ -almost sure limit T_∞ , monotone convergence implies that

$$(12) \quad \lim_{n \rightarrow \infty} \lim_{r \rightarrow \infty} \mathbb{E}_\gamma \left[\int_0^{\eta \wedge \tau_r \wedge T_n} 1_x(X_s) ds \right] = \nu_s(x) \quad \forall x \in \mathcal{S}.$$

Similarly, Lemma 23.7 shows that $(\sigma_r)_{r \in \mathbb{Z}_+}$ is an increasing sequence with $\mathbb{P}_\gamma$ -almost sure limit ∞ . Because $\eta \wedge \tau_r \wedge T_n = T_{\zeta \wedge \sigma_r \wedge n}$ (check!), (9), bounded convergence, and (34.4) imply that

$$\begin{aligned} \lim_{n \rightarrow \infty} \lim_{r \rightarrow \infty} \mathbb{P}_\gamma(\{X_{\eta \wedge \tau_r \wedge T_n} = x\}) &= \lim_{n \rightarrow \infty} \lim_{r \rightarrow \infty} \mathbb{P}_\gamma(\{Y_{\zeta \wedge \sigma_r \wedge n} = x\}) \\ &= \lim_{n \rightarrow \infty} \lim_{r \rightarrow \infty} \mathbb{P}_\gamma(\{\zeta < \infty, Y_{\zeta \wedge \sigma_r \wedge n} = x\}) = \mathbb{P}_\gamma(\{\zeta < \infty, Y_\zeta = x\}) \\ &= \mathbb{P}_\gamma(\{\eta < \infty, X_{T_\zeta} = x\}) = \mathbb{P}_\gamma(\{\eta < \infty, X_\eta = x\}) = \mu_S(x) \end{aligned}$$

for all x in $\mathcal{S}$. Putting (11)–(12) together with the above completes the proof. $\square$

You may be wondering whether (9) is actually necessary for (10) to hold. It is easy to find examples of chains violating (9) for which (10) does not hold.

(13) EXAMPLE. Consider the chain on $\mathcal{S} := \{0, 1\}$, with initial law $\gamma = \alpha\delta_0 + (1 - \alpha)\delta_1$ where $\alpha \in [0, 1]$, and rate matrix

$$Q := \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Both states are absorbing, and so the chain remains forever where it starts (Proposition 26.12) $T_\infty = \infty$ and $X_t = X_0$ for all $t \in [0, \infty)$, $\mathbb{P}_\gamma$ almost surely. For this reason, if τ denotes the hitting time of $\{1\}$,

$$\mathbb{P}_\gamma(\{\tau < \infty\}) = \mathbb{P}_\gamma(\{X_0 = 1\}) = 1 - \alpha.$$

Equations (10) read

$$0 = \alpha, \quad \mu_S(1) = 1 - \alpha.$$

The left-most equation is satisfied if and only if $\alpha = 0$, that is if and only if $\mathbb{P}_\gamma(\{\tau < \infty\}) = 1$.

In the case of the example's chain and stopping time, (9) is indeed necessary and sufficient for (10) to hold. However, for other chains and stopping times, the question remains...

An open question. Is (9) necessary for equations (10) to hold?

The details of μ and ν 's definition. Throughout the above, we didn't actually define μ and ν , just specified in (1)–(2) what values these measures should take on sets of the form $A \times \{x\}$ with A in $\mathcal{B}([0, \infty))$ and x in $\mathcal{S}$. It turns out that this is all that's necessary to fully define μ and ν . To do see this, we require Carathéodory's extension theorem:

(14) THEOREM (Carathéodory's extension theorem, (Williams, 1991, Theorem 1.7)). *Suppose that S is a set and Σ_0 is an algebra on S . That is, Σ_0 is a collections of subsets of S satisfying $S \in \Sigma_0$, $A^c \in \Sigma_0$ for all $A \in \Sigma_0$, and $A \cup B \in \Sigma_0$ for all $A, B \in \Sigma_0$. If $\rho^0 : \Sigma_0 \rightarrow [0, \infty]$ is a countably additive and Σ is the sigma-algebra generated by Σ_0 , then there exists an unsigned measure ρ on (S, Σ) such that*

$$\rho(A) = \rho^0(A) \quad \forall A \in \Sigma_0.$$

Now, consider the algebra

$\Sigma_0 :=$ the collection of all finite unions of sets of the form $A \times B$ with $A \in \mathcal{B}([0, \infty))$, $B \subseteq \mathcal{S}$

generating the product sigma-algebra $\mathcal{X} := \mathcal{B}([0, \infty)) \times 2^{\mathcal{S}}$. We can express any set in Σ_0 as

$$\bigcup_{i \in \mathcal{I}} A_i \times B_i$$

for some (finite) indexing set $\mathcal{I}$, A_i s in $\mathcal{B}([0, \infty))$, and B_i s in $2^{\mathcal{S}}$. Next, consider the countably additive functions on Σ_0

$$\mu^0 \left(\bigcup_{i \in \mathcal{I}} A_i \times B_i \right) := \sum_{x \in \mathcal{S}} \mu \left(\bigcup_{i \in \mathcal{I}_x} A_i, x \right), \quad \nu^0 \left(\bigcup_{i \in \mathcal{I}} A_i \times B_i \right) := \sum_{x \in \mathcal{S}} \nu \left(\bigcup_{i \in \mathcal{I}_x} A_i, x \right)$$

where $\mathcal{I}_x := \{i \in \mathcal{I} : x \in B_i\}$ and $\mu(A, x)$ and $\nu(A, x)$ are as in (1)–(2).

(15) EXERCISE. *Check that Σ_0 is indeed an algebra that generates $\mathcal{X}$ and that μ^0 and ν^0 are countably additive on Σ_0 .*

For this reason, Theorem 14 implies that there exists measures μ and ν on $([0, \infty) \times \mathcal{S}, \mathcal{X})$ satisfying

$$(16) \quad \mu(A \times \{x\}) = \mu(A, x), \quad \nu(A \times \{x\}) = \nu(A, x), \quad \forall A \in \mathcal{B}([0, \infty)), \quad x \in \mathcal{S},$$

with $\mu(A, x)$ and $\nu(A, x)$ as in (1)–(2). Because

$$\Sigma_{-1} := \{A \times \{x\} : A \in \mathcal{B}([0, \infty)), \quad x \in \mathcal{S}\}$$

is a π -system and $\mu([0, \infty) \times \mathcal{S}) = \mathbb{P}_\gamma(\{\eta < \infty\}) \leq 1$, Lemma 5.6 implies that μ is the only measure on $([0, \infty) \times \mathcal{S}, \mathcal{X})$ satisfying the leftmost equation in (16). We call it the *stopping distribution*. Furthermore,

$$(17) \quad \nu(A) = \sum_{n=0}^{\infty} \nu_n(A) \quad \forall A \in \mathcal{X},$$

where

$$\nu_n(A) := \nu([n, n+1) \times \mathcal{S} \cap A) \quad \forall A \in \mathcal{X}, \quad n \in \mathbb{N}.$$

Because

$$\nu_n([0, \infty) \times \mathcal{S}) = \mathbb{E}_\gamma[(n+1) \wedge \eta \wedge T_\infty - n \wedge \eta \wedge T_\infty] \leq 1 \quad \forall n \in \mathbb{N},$$

applying Lemma 5.6 to ν_n and using (17), we find that that ν is the only measure on $([0, \infty) \times \mathcal{S}, \mathcal{X})$ satisfying the rightmost equation in (16). We call it the *occupation measure*.

36. Exit times*. We are often interested in how long the chain takes to exit some given subset, or *domain*, of the state space and what part of the domain's boundary the chain crosses to exit. To study this problem, we single out a subset $\mathcal{D}$ of the state space $\mathcal{S}$ and refer to it as the *domain*. The *exit time* τ from the domain is the instant in time that the chain leaves the domain for the first time:

$$(1) \quad \tau(\omega) := \inf\{t \in [0, T_\infty(\omega)) : X_t(\omega) \notin \mathcal{D}\} \quad \forall \omega \in \Omega.$$

For us, starting outside of the domain counts as ‘exiting’ the domain in which case we set τ to zero. We say that the chain *exits via* x if τ is finite and $X_\tau = x$. Clearly, the exit time is just the hitting time of the domain's complement (compare with (34.8) and (1)). For this reason, Proposition 34.9 implies that τ is a jump-time-value $(\mathcal{F}_t)_{t \geq 0}$ -stopping time and that equations (34.3)–(34.4) hold if we replace η with τ and ς with σ , where σ is the exit time of the jump chain:

$$(2) \quad \sigma(\omega) := \inf\{n \geq 0 : Y_n(\omega) \notin \mathcal{D}\}.$$

Let μ be τ 's *exit distribution* and ν its *occupation measure* defined as in (35.6)–(35.6) with τ replacing η . For each state x , the measures $\mu(dt, x)$ and $\nu(dt, x)$ have densities⁵ $\mu(t, x)$ and $\nu(t, x)$ with respect to the Lebesgue measure on $[0, \infty)$ (I distinguish a measure from its density by writing dt or t in its argument). The density $t \mapsto \mu(t, x)$ is a function such that $\mu(t, x)h$ is the probability that the chain first exits the domain via state x during the time interval $[t, t+h]$, for any small $h > 0$. Similarly (and assuming that the chain is non-explosive), $\nu(t, x)$ is the average fraction of the interval $[t, t+h]$ that the chain spends in state x before exiting the domain. Formally, the relationship between the exit distribution and occupation measure and their densities is:

$$(3) \quad \mu(A, x) = 1_{\mathcal{D}^c}(x) \gamma(x) 1_A(0) + \int_A \mu(t, x) dt \quad \forall A \in \mathcal{B}([0, \infty)), \quad x \in \mathcal{S},$$

$$(4) \quad \nu(A, x) = \int_A \nu(t, x) dt \quad \forall A \in \mathcal{B}([0, \infty)), \quad x \in \mathcal{S},$$

where the term $1_{\mathcal{D}^c}(x) \gamma(x) 1_A(0)$ accounts for the possibility that the chain is started outside of the domain. The densities are characterised in terms of the minimal solution to a set of linear differential equations:

(5) **THEOREM** (Analytical characterisation of μ and ν). *Suppose that*

$$(6) \quad \sup_{x \in \mathcal{D}} q(x, y) < \infty \quad \forall y \in \mathcal{S} \quad \text{or} \quad \sum_{x \in \mathcal{D}} q(x) \gamma(x) < \infty.$$

The exit distribution μ and occupation measure ν decompose as in (3)–(4) and their densities $\nu(t, x)$ and $\mu(t, x)$ are non-negative and continuous functions on $[0, \infty)$, for each x in $\mathcal{S}$. Moreover,

$$(7) \quad \mu(t, x) = 1_{\mathcal{D}^c}(x) \dot{p}_t(x), \quad \nu(t, x) = 1_{\mathcal{D}}(x) \hat{p}_t(x), \quad \forall x \in \mathcal{S}, \quad t \in [0, \infty),$$

where $\hat{p}_t$ is the minimal non-negative solution (as in Theorem 33.3) of

$$(8) \quad \dot{p}_t(x) = \sum_{y \in \mathcal{D}} \hat{p}_t(y) q(y, x) \quad t \in [0, \infty), \quad x \in \mathcal{S}, \quad \hat{p}_0(x) = \gamma(x) \quad \forall x \in \mathcal{S}.$$

⁵Use (33.9) and (21) and (23) further down to verify this fact.

The proof takes some doing and we leave it until the end of the section. The ideas guiding it, however, are simple: consider a second chain $\hat{X}$ obtained by replacing Q in Algorithm 2 with $\hat{Q} = (\hat{q}(x, y))_{x, y \in \mathcal{S}}$ and keeping everything else the same, where

$$(9) \quad \hat{q}(x, y) := \begin{cases} q(x, y) & \text{if } x \in \mathcal{D} \\ 0 & \text{if } x \notin \mathcal{D} \end{cases} \quad \forall x, y \in \mathcal{S}.$$

The chains X and $\hat{X}$ are identical up until (and including) the moment they both simultaneously exit the domain via the same state. Therefore the probability $\mu([0, t], x)$ that X has exited the domain by time t via state $x \notin \mathcal{D}$ is also the probability that $\hat{X}$ exited via x by time t . However, all states outside of the domain are absorbing for $\hat{X}$ and $\hat{X}$ gets trapped in the state it enters upon leaving the truncation. For these reasons, $\mu([0, t], x)$ is the probability $\hat{p}_t(x)$ that $\hat{X}$ is in state x by time t and the first equation in (7) follows. Similarly, once $\hat{X}$ leaves the domain it cannot return, hence the amount of time that $\hat{X}$ spends in a state $x \in \mathcal{D}$ until the moment it exits the domain is the *total* time it spends in that state:

$$\int_0^{t \wedge \tau \wedge T_\infty} 1_x(X_s) ds = \int_0^{t \wedge \tau \wedge \hat{T}_\infty} 1_x(\hat{X}_s) ds = \int_0^{t \wedge \hat{T}_\infty} 1_x(\hat{X}_s) ds.$$

Taking expectations of the above we obtain the second equation in (7). Equation (8) on the other hand, is just the differential equation in Theorem 33.3 satisfied by the time-varying law $\hat{p}_t$ of $\hat{X}$.

The marginals. Equations (7)–(8) imply that $\mu(t, x)$ is non-negative. Thus, (3) and Tonelli's theorem show that the time-marginal $\mu_T(dt)$ in (35.5) of the exit distribution also has a density $\mu_T(t)$ with respect to the Lebesgue measure and that this density is given by $\mu_T(t) = \sum_{x \in \mathcal{S}} \mu(t, x)$:

$$(10) \quad \mu_T(A) = \gamma(\mathcal{D}^c) 1_A(0) + \int_A \mu_T(t) dt \quad \forall A \in \mathcal{B}([0, \infty)).$$

Theorem 5 and Tonelli's theorem then give us expressions for the marginals in terms of the solution of (8):

$$\begin{aligned} \mu_T(t) &= \sum_{x \notin \mathcal{D}} \hat{p}_t(x) \quad \forall t \in [0, \infty), \\ \mu_S(x) &= 1_{\mathcal{D}^c}(x) \left(\lim_{t \rightarrow \infty} \hat{p}_t(x) \right), \quad \nu_S(x) = 1_{\mathcal{D}}(x) \int_0^\infty \hat{p}_t(x) dt, \quad \forall x \in \mathcal{S}. \end{aligned}$$

In the case of the space marginals μ_S and ν_S in (35.6)–(35.7) and an almost surely finite exit time, we have an alternative characterisation:

(11) **THEOREM** (Analytical characterisation of μ_S and ν_S). *If $\mathbb{P}_\gamma(\{\tau < \infty\}) = 1$, then ν_S in (35.7) is the minimal non-negative solution of the equations*

$$(12) \quad q(x) \nu_S(x) = \gamma(x) + \sum_{z \neq x} \nu_S(z) q(z, x) \quad \forall x \in \mathcal{D}, \quad \nu_S(x) = 0 \quad \forall x \notin \mathcal{D},$$

while μ_S in (35.6) is given by

$$(13) \quad \mu_S(x) = \gamma(x) + \sum_{z \in \mathcal{D}} \nu_S(z) q(z, x) \quad \forall x \notin \mathcal{D}, \quad \mu_S(x) = 0 \quad \forall x \in \mathcal{D}.$$

Proof. Proposition 34.9 implies that $X_t(\omega)$ belongs to the domain $\mathcal{D}$ if ω belongs to $\{t < \tau \wedge T_\infty\}$ and that $X_{\tau(\omega)}(\omega)$ does not belong to $\mathcal{D}$ if ω belongs to $\{\tau < \infty\}$. For these reasons,

$$\mu_S(x) = 0 \quad \forall x \in \mathcal{D}, \quad \nu_S(x) = 0 \quad \forall x \notin \mathcal{D},$$

and (12)–(13) follow from Theorem 35.8.

All that remains to be shown is that $\rho(x) \geq \nu_S(x)$ for all x in $\mathcal{D}$ and non-negative solution $\rho = (\rho(x))_{x \in \mathcal{S}}$ of (12). Because τ is jump-time-valued (Proposition 34.9), the definition of the chain's paths implies that

$$\begin{aligned} \int_0^{\tau \wedge T_\infty} 1_x(X_t) dt &= \sum_{n=0}^{\infty} 1_{\{\tau > T_n\}} \int_{T_n}^{T_{n+1}} 1_x(X_t) dt = \sum_{n=0}^{\infty} 1_{\{\tau > T_n\}} (T_{n+1} - T_n) 1_x(X_{T_n}) \\ &= \sum_{n=0}^{\infty} 1_{\{\tau > T_n\}} S_{n+1} 1_x(Y_n) = \sum_{n=0}^{\infty} S_{n+1} 1_{\{Y_0 \in \mathcal{D}, \dots, Y_{n-1} \in \mathcal{D}, Y_n = x\}} \quad \forall x \in \mathcal{D}. \end{aligned}$$

Taking expectations of both sides, we have that

$$\begin{aligned} \nu_S(x) &= \sum_{n=0}^{\infty} \mathbb{E}_\gamma [S_{n+1} 1_{\{Y_0 \in \mathcal{D}, \dots, Y_{n-1} \in \mathcal{D}, Y_n = x\}}] = \sum_{n=0}^{\infty} \mathbb{E}_\gamma [\mathbb{E}_\gamma [S_{n+1} | \mathcal{G}_n] 1_{\{Y_0 \in \mathcal{D}, \dots, Y_{n-1} \in \mathcal{D}, Y_n = x\}}] \\ &= \frac{1}{\lambda(x)} \sum_{n=0}^{\infty} \mathbb{P}_\gamma (\{Y_0 \in \mathcal{D}, \dots, Y_{n-1} \in \mathcal{D}, Y_n = x\}) \\ (14) \quad &= \frac{1}{\lambda(x)} \left(\gamma(x) + \sum_{n=1}^{\infty} \sum_{z_0 \in \mathcal{D}} \cdots \sum_{z_{n-1} \in \mathcal{D}} \gamma(z_0) p(z_0, z_1) \dots p(z_{n-1}, x) \right) \quad \forall x \in \mathcal{D}, \end{aligned}$$

where $(p(x, y))_{x, y}$ denotes the jump matrix (26.2), $(\lambda(x))_{x \in \mathcal{S}}$ the jump rates (26.3), and $\mathcal{G}_n$ is the sigma-algebra generated by $Y_0, \dots, Y_n$ and $S_1, \dots, S_n$. The first equation follows from Tonelli's theorem, the second the take-out-what-is-known and tower properties of conditional expectation (Theorem 0.3(iv, v)), the third from S_{n+1} being exponentially distributed with mean $1/\lambda(Y_n)$ when conditioned on $\mathcal{G}_n$ (Theorem 27.4), and the fourth from the expression given in Theorem 29.5 for the path law.

Suppose that x is an absorbing state and fix n in $\mathbb{N}$, $z_0, z_1, \dots, z_{n-1}$ in $\mathcal{D}$. Because the exit time is almost surely finite, downwards monotone convergence, Theorem 29.5, and Proposition 34.9 imply that

$$\begin{aligned} 0 &= \mathbb{P}_\gamma (\{\tau = \infty\}) = \mathbb{P}_\gamma (\cap_{m=0}^{\infty} \{Y_m \in \mathcal{D}\}) = \lim_{m \rightarrow \infty} \mathbb{P}_\gamma (\{Y_0 \in \mathcal{D}, Y_1 \in \mathcal{D}, \dots, Y_{n+m} \in \mathcal{D}\}) \\ &= \lim_{m \rightarrow \infty} \sum_{x_0 \in \mathcal{D}} \sum_{x_1 \in \mathcal{D}} \cdots \sum_{x_{n+m} \in \mathcal{D}} \lambda(x_0) p(x_0, x_1) \dots p(x_{n-1}, x_{n+m}) \\ &\geq \gamma(z_0) p(z_0, z_1) \dots p(z_{n-1}, x) \left(\lim_{m \rightarrow \infty} \sum_{x_1 \in \mathcal{D}} \sum_{x_2 \in \mathcal{D}} \cdots \sum_{x_m \in \mathcal{D}} p(x, x_1) p(x_1, x_2) \dots p(x_{m-1}, x_m) \right) \\ &= \gamma(z_0) p(z_0, z_1) \dots p(z_{n-1}, x) \left(\lim_{m \rightarrow \infty} \sum_{x_2 \in \mathcal{D}} \cdots \sum_{x_m \in \mathcal{D}} p(x, x_2) \dots p(x_{m-1}, x_m) \right) \\ &= \cdots = \gamma(z_0) p(z_0, z_1) \dots p(z_{n-1}, x). \end{aligned}$$

The decomposition (14) then shows that $\nu_S(x) = 0$ (and so $\rho(x) \geq \nu_S(x)$ by non-negativity of ρ). We now only have left to show that $\rho(x) \geq \nu_S(x)$ for every non-absorbing state x inside the domain (that is, x in $\mathcal{D}_{na} := \{x \in \mathcal{D}_{na} : q(x) > 0\}$). For any such state x , we rewrite (14) as

$$(15) \quad q(x) \nu_S(x) = \gamma(x) + \sum_{n=1}^{\infty} \sum_{z_0 \in \mathcal{D}_{na}} \cdots \sum_{z_{n-1} \in \mathcal{D}_{na}} \gamma(z_0) p(z_0, z_1) \dots p(z_{n-1}, x), \quad \forall x \in \mathcal{D}_{na}.$$

Using the definition of the jump matrix in (26.2) and repeatedly applying (12), we have that

$$\begin{aligned}
 q(x)\rho(x) &= \gamma(x) + \sum_{z_0 \in \mathcal{D}_{na}} \rho(z_0)q(z_0, x) \\
 &= \gamma(x) + \sum_{z_0 \in \mathcal{D}_{na}} \gamma(z_0)p(z_0, x) + \sum_{z_0 \in \mathcal{D}_{na}} \sum_{z_1 \in \mathcal{D}_{na}} \rho(z_0)q(z_0, z_1)p(z_1, x) \\
 &= \cdots = \gamma(x) + \sum_{n=1}^m \sum_{z_0 \in \mathcal{D}_{na}} \cdots \sum_{z_{n-1} \in \mathcal{D}_{na}} \gamma(z_0)p(z_0, z_1) \cdots p(z_{n-1}, x) \\
 &\quad + \sum_{z_0 \in \mathcal{D}_{na}} \cdots \sum_{z_m \in \mathcal{D}_{na}} \rho(z_0)q(z_0, z_1)p(z_1, z_2) \cdots p(z_m, x) \\
 &\geq \gamma(x) + \sum_{l=1}^m \sum_{z_0 \in \mathcal{D}_{na}} \cdots \sum_{z_{l-1} \in \mathcal{D}_{na}} \gamma(z_0)p(z_0, z_1) \cdots p(z_{l-1}, x), \quad \forall x \in \mathcal{D}_{na}.
 \end{aligned}$$

Comparing the above with (15) and taking the limit $m \rightarrow \infty$ shows $\rho(x) \geq \nu_S(x)$ for all $x \in \mathcal{D}_{na}$ as desired. $\square$

The continuous-time version of Example 11.11, with $\mathcal{D}$ being $\{1, 2\}$ for a chain with state space $\{0, 1, 2\}$, initial distribution 1_1 , and rate matrix

$$Q := \begin{bmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

shows that (12) can indeed have other non-negative solutions aside from ν_S even if (35.9) is satisfied. It is not difficult to come up with sufficient conditions for ν_S to be the unique non-negative satisfying the equation

$$\gamma(\mathcal{S} \setminus \mathcal{D}) + \sum_{z \in \mathcal{D}} \sum_{z \neq x} \rho(z)q(z, x) = 1$$

obtained by summing over x in (13) (under assumption (35.9)), see (Kuntz, 2017, Corollary 2.39). However, the continuous-time analogue of the question posed in Section 11 persists:

An open question. Assuming that (35.9) is satisfied, when does (12) have multiple non-negative solutions?

A proof of Theorem 5. The particular definition of the chain we chose in Section 26 comes quite in handy here. Consider a second chain $\bar{X}$ constructed using the Kendall-Gillespie algorithm (Algorithm 2) and the same X_0 , $(\xi_n)_{n \in \mathbb{Z}_+}$, and $(U_n)_{n \in \mathbb{Z}_+}$ as for our original chain X but a different rate matrix $\bar{Q}$. If the rows of the rate matrices Q and $\bar{Q}$ coincide on the domain $\mathcal{D}$, then both chains are updated using the same rules while they remain inside the domain. For this reason, the chains are identical up until the instant that they simultaneously leave the domain for the first time. Formally:

(16) LEMMA. Suppose that Q and $\bar{Q}$ coincide on $\mathcal{D}$:

$$q(x, y) = \bar{q}(x, y), \quad \forall x \in \mathcal{D}, \quad y \in \mathcal{S}.$$

There exists a continuous-time chain $\bar{X}$ with rate matrix $\bar{Q}$ defined on the same underlying space $(\Omega, \mathcal{F})$ as X and with jump chain $\bar{Y} := (\bar{Y}_n)_{n \in \mathbb{N}}$, waiting times $(\bar{S}_n)_{n \in \mathbb{Z}_+}$, jump times $(\bar{T}_n)_{n \in \mathbb{N}}$, and explosion time $\bar{T}_\infty$ such that:

(i) X and $\bar{X}$ exit the domain at the same time:

$$\tau(\omega) = \bar{\tau}(\omega), \quad \sigma(\omega) = \bar{\sigma}(\omega) \quad \forall \omega \in \Omega.$$

where τ and σ defined in (1)–(2) denote the respective times of exit for X and its jump chain Y , and $\bar{\tau}$ and $\bar{\sigma}$ those of $\bar{X}$ and $\bar{Y}$:

$$\bar{\tau}(\omega) := \inf\{t \in [0, \bar{T}_\infty(\omega)) : \bar{X}_t(\omega) \notin \mathcal{D}\}, \quad \bar{\sigma}(\omega) := \inf\{n \geq 0 : \bar{Y}_n(\omega) \notin \mathcal{D}\}, \quad \forall \omega \in \Omega.$$

(ii) Up until (and including) this time, both chains have identical jump chains, waiting times, and jump times, and explosion times:

$$Y_n(\omega) = \bar{Y}_n(\omega), \quad S_n(\omega) = \bar{S}_n(\omega), \quad T_n(\omega) = \bar{T}_n(\omega), \quad \forall \omega \in \{n \leq \sigma\}, \quad n \in \mathbb{N}.$$

(iii) Either chain explodes no later than leaving the domain if and only if the other does, in which case the explosion times are the same:

$$\{T_\infty \leq \tau\} = \{\bar{T}_\infty \leq \bar{\tau}\}, \quad T_\infty(\omega) = \bar{T}_\infty(\omega) \quad \forall \omega \in \{T_\infty \leq \tau\}.$$

(iv) In summary, both chains are identical up until they exit the domain for the first time:

$$\{t \leq \tau, t < T_\infty\} = \{t \leq \bar{\tau}, t < \bar{T}_\infty\}, \quad X_t(\omega) = \bar{X}_t(\omega) \quad \forall \omega \in \{t \leq \tau, t < T_\infty\}, \quad \forall t \in [0, \infty).$$

Proof. Let $\bar{P} := (\bar{p}(x, y))_{x, y \in \mathcal{S}}$ and $\bar{\lambda} := (\bar{\lambda}(x))_{x \in \mathcal{S}}$ denote the jump matrix and jump rates obtained by replacing Q with $\bar{Q}$ in (26.2)–(26.3). To construct $\bar{X}$ we (a) run Algorithm 2 employing the same X_0 , $(\xi_n)_{n \in \mathbb{Z}_+}$, and $(U_n)_{n \in \mathbb{Z}_+}$ as for X but with $\bar{P}$ and $\bar{\lambda}$ replacing P and λ to obtain the chain's waiting times $(\bar{S}_n)_{n \in \mathbb{Z}_+}$ and jump chain $\bar{Y} := (\bar{Y}_n)_{n \in \mathbb{N}}$ and (b) define $\bar{X}$, $(\bar{T}_n)_{n \in \mathbb{N}}$, and $\bar{T}_\infty$ by replacing $(S_n)_{n \in \mathbb{Z}_+}$ and Y with $(\bar{S}_n)_{n \in \mathbb{Z}_+}$ and $\bar{Y}$ in (26.6)–(26.8). Because the rate matrices coincide on $\mathcal{D}$, (26.2)–(26.3) imply that the jump matrices and rates also coincide on $\mathcal{D}$:

$$(p(x, y))_{y \in \mathcal{S}} = (\bar{p}(x, y))_{y \in \mathcal{S}}, \quad \lambda(x) = \bar{\lambda}(x), \quad \forall x \in \mathcal{D}.$$

Algorithm 2 and the above imply that

$$(17) \quad \bar{Y}_{n+1}(\omega) = Y_{n+1}(\omega) \quad \forall \omega \in \{\bar{Y}_n = Y_n \in \mathcal{D}\}.$$

Because $Y_0 = X_0 = \bar{Y}_0$, induction and the above imply that

$$(18) \quad \{Y_0 \in \mathcal{D}, \dots, Y_{n-1} \in \mathcal{D}, Y_n = x\} = \{\bar{Y}_0 \in \mathcal{D}, \dots, \bar{Y}_{n-1} \in \mathcal{D}, \bar{Y}_n = x\} \quad \forall x \in \mathcal{S}, \quad n > 0.$$

Due to the definition of the exit times of the jump chains, we have that

$$\begin{aligned} \sigma &= \infty \cdot 1_{\{Y_0 \in \mathcal{D}, Y_1 \in \mathcal{D}, \dots\}} + \sum_{n=1}^{\infty} n 1_{\{Y_0 \in \mathcal{D}, \dots, Y_{n-1} \in \mathcal{D}, Y_n \notin \mathcal{D}\}}, \\ \bar{\sigma} &= \infty \cdot 1_{\{\bar{Y}_0 \in \mathcal{D}, \bar{Y}_1 \in \mathcal{D}, \dots\}} + \sum_{n=1}^{\infty} n 1_{\{\bar{Y}_0 \in \mathcal{D}, \dots, \bar{Y}_{n-1} \in \mathcal{D}, \bar{Y}_n \notin \mathcal{D}\}}. \end{aligned}$$

Plugging (18) into the above, we obtain the rightmost equation in (i). Since $n \leq \sigma(\omega)$ only if

$$Y_0(\omega) \in \mathcal{D}, \quad Y_n(\omega) \in \mathcal{D}, \quad \dots, \quad Y_{n-1}(\omega) \in \mathcal{D},$$

the leftmost equation in (ii) also follows from (17). The definition of the waiting times (Algorithm 2) and that of the jump times (26.7), then yield the other two equations in (ii). The leftmost equation in (i) then follows from the other equation in (i), the rightmost one in (ii), and Proposition 34.9. This proposition and (34.4) imply that

$$\{T_\infty \leq \tau\} = \{\tau = \infty\} = \{\sigma = \infty\}, \quad \{\bar{T}_\infty \leq \bar{\tau}\} = \{\bar{\tau} = \infty\} = \{\bar{\sigma} = \infty\}$$

For this reason, the leftmost equation in (iii) follows from (i). Similarly, given that $\{T_\infty \leq \tau\} = \{\sigma = \infty\}$, picking any ω in this set and taking the limit $n \rightarrow \infty$ of the rightmost equation in (ii) yields the rightmost one in (iii).

To complete the proof, note that $\{T_\infty \leq \tau\} = \{\tau = \infty\}$ further implies that

$$\{t \leq \tau, t < T_\infty\} = \{t < T_\infty, \tau = \infty\} \cup \{t \leq \tau < T_\infty\} = \{t < T_\infty, \tau = \infty\} \cup \{t \leq \tau < \infty\}.$$

Because the same equations hold if we replace τ and T_∞ with $\bar{\tau}$ and $\bar{T}_\infty$, the leftmost equation in (iv) follows from the leftmost one in (i) and the rightmost one in (iii). Because

$$\{t \leq \tau, t < T_\infty\} = \bigcup_{n=0}^{\infty} \{t \leq \tau, T_n \leq t < T_{n+1}\}$$

and $\{T_n \leq \tau\} = \{\sigma \leq n\}$ (Proposition 34.9), the rightmost equation in (iv) then follows from the leftmost, the definition of the paths of X and $\bar{X}$ in (26.6), and (ii). $\square$

We are now ready to tackle the proof of Theorem 5:

Proof of Theorem 5. Let $\hat{X}$ be as in Lemma 16 with $\hat{Q} = (\hat{q}(x, y))_{x, y \in \mathcal{S}}$ in (9) replacing $\bar{Q}$ in the lemma's premise. The key ingredients in this proof are the facts that $\hat{X}$ and X are identical up until the moment that they simultaneously exit the domain (Lemma 16) and that $\hat{X}$ gets stuck in an absorbing state the instant it leaves the domain. We begin by proving the latter.

If $\hat{Y} = (\hat{Y}_n)_{n \in \mathbb{N}}$ denotes the jump of $\hat{X}$ and $\hat{\sigma}$ its exit time from $\mathcal{D}$, then $\hat{Y}_{\hat{\sigma}(\omega)}(\omega)$ lies outside $\mathcal{D}$ for all ω in $\{\hat{\sigma} < \infty\}$ and the definition of $\hat{Q}$ in (9) implies that

$$\hat{q}(\hat{Y}_{\hat{\sigma}(\omega)}(\omega)) = 0 \quad \forall \omega \in \{\hat{\sigma} < \infty\}.$$

The Kendall-Gillespie algorithm (Algorithm 2) then implies that $\hat{Y}_{\hat{\sigma}(\omega)+1}(\omega) = \hat{Y}_{\hat{\sigma}(\omega)}(\omega)$ and, hence, $\hat{q}(\hat{Y}_{\hat{\sigma}(\omega)+1}(\omega)) = 0$, for all ω in $\{\hat{\sigma} < \infty\}$. Iterating this argument forward, we find that

$$(19) \quad \hat{Y}_{\hat{\sigma}(\omega)+n}(\omega) = \hat{Y}_{\hat{\sigma}(\omega)}(\omega) \notin \mathcal{D} \quad \forall n \in \mathbb{N}, \quad \omega \in \{\hat{\sigma} < \infty\}.$$

Next, because Proposition 34.9 implies that

$$\hat{T}_n \vee \hat{\tau} = \hat{T}_n \vee \hat{T}_{\hat{\sigma}} = \hat{T}_{n \vee \hat{\sigma}} \quad \forall n \in \mathbb{N},$$

where $\hat{\tau}$ denotes the exit time of $\hat{X}$, we have that

$$\{\tau \leq t, \hat{T}_n \leq t < \hat{T}_{n+1}\} = \{\hat{T}_n \vee \tau \leq t < \hat{T}_{n+1}\} = \{\hat{T}_{n \vee \hat{\sigma}} \leq t < \hat{T}_{n+1}\} = \{\hat{\sigma} \leq n, \hat{T}_n \leq t < \hat{T}_{n+1}\}$$

for all n in $\mathbb{N}$ and t in $[0, \infty)$. The above, Proposition 34.9, and (19) then show that $\hat{X}$ does indeed get stuck in the first state it enters upon leaving the domain:

$$\begin{aligned} \{\hat{\tau} \leq t < \hat{T}_\infty, \hat{X}_t = x\} &= \bigcup_{n=0}^{\infty} \{\hat{\tau} \leq t, \hat{T}_n \leq t < \hat{T}_{n+1}, \hat{Y}_n = x\} \\ &= \bigcup_{n=0}^{\infty} \{\hat{\sigma} \leq n, \hat{T}_n \leq t < \hat{T}_{n+1}, \hat{Y}_n = x\} \\ &= \bigcup_{n=0}^{\infty} \{\hat{\sigma} \leq n, \hat{T}_n \leq t < \hat{T}_{n+1}, \hat{Y}_{\hat{\sigma}} = x\} \\ &= \{\hat{\sigma} < \infty, \hat{T}_{\hat{\sigma}} \leq t < \hat{T}_\infty, \hat{Y}_{\hat{\sigma}} = x\} \\ (20) \quad &= \{\hat{\tau} \leq t < \hat{T}_\infty, \hat{X}_{\hat{\tau}} = x\} \quad \forall t \in [0, \infty), \quad x \in \mathcal{S}. \end{aligned}$$

Now, on to the characterisations of μ and ν in (7)–(8):

$$\begin{aligned} \mu([0, t], x) &= \mathbb{P}(\{X_\tau = x, \tau \leq t\}) = \mathbb{P}(\{Y_\sigma = x, \sigma < \infty, T_\sigma \leq t\}) = \mathbb{P}(\{\hat{Y}_{\hat{\sigma}} = x, \hat{\sigma} < \infty, \hat{T}_{\hat{\sigma}} \leq t\}) \\ (21) \quad &= \mathbb{P}(\{\hat{X}_{\hat{\tau}} = x, \hat{\tau} \leq t < \hat{T}_\infty\}) = \mathbb{P}(\{\hat{X}_t = x, \hat{\tau} \leq t < \hat{T}_\infty\}) = \hat{p}_t(x) \quad \forall x \notin \mathcal{D}, \quad t \in [0, \infty), \end{aligned}$$

where $\hat{p}_t$ denotes the time-varying law of $\hat{X}$:

$$\hat{p}_t(x) := \mathbb{P}_\gamma(\{\hat{X}_t = x, t < \hat{T}_\infty\}) \quad \forall x \in \mathcal{S}, \quad t \in [0, \infty).$$

The first equation in (21) follows from the definition of μ , the second from Proposition 34.9, the third from Lemma 16(i, ii), the fourth from Proposition 34.9, the fifth from (20). Because (6) and Proposition 33.6 imply that (33.4) holds for $\hat{Q}$, Theorem 33.3 shows that $\hat{p}_t$ is continuously differentiable and the minimal solution of (8). Exploiting the continuity of $t \mapsto \hat{p}_t(x)$ and applying monotone convergence, we find that

$$\mu([0, t], x) = \lim_{n \rightarrow \infty} \mu([0, t(1 - 1/n)], x) = \lim_{n \rightarrow \infty} \hat{p}_{t(1-1/n)}(x) = \hat{p}_t(x), \quad \forall x \notin \mathcal{D},$$

and the first equation in (7) follows.

To argue the second equation in (7), note that Lemma 16(i, iii, iv) implies that

$$\begin{aligned} (22) \quad \int_0^{t \wedge \tau \wedge T_\infty} 1_x(X_s) ds &= \int_0^{t \wedge \hat{\tau} \wedge \hat{T}_\infty} 1_x(\hat{X}_s) ds = 1_{\{\hat{\tau} \leq t\}} \int_0^{\hat{\tau} \wedge \hat{T}_\infty} 1_x(\hat{X}_s) ds \\ &\quad + 1_{\{\hat{\tau} > t\}} \int_0^{t \wedge \hat{T}_\infty} 1_x(\hat{X}_s) ds \quad \forall x \in \mathcal{D}, \quad t \in [0, \infty). \end{aligned}$$

However, because $\hat{X}_{\hat{\tau}}$ lies outside $\mathcal{D}$ whenever $\hat{\tau}$ is finite (Proposition 34.9), (20) shows that

$$\begin{aligned} 1_{\{\hat{\tau} \leq t\}} \int_{\hat{\tau} \wedge \hat{T}_\infty}^{t \wedge \hat{T}_\infty} 1_x(\hat{X}_s) ds &= \int_0^\infty 1_{\{\hat{\tau} \leq s < t \wedge \hat{T}_\infty\}} 1_x(\hat{X}_s) ds = \int_0^\infty 1_{\{\hat{\tau} \leq s < t \wedge \hat{T}_\infty\}} 1_x(\hat{X}_{\hat{\tau}}) ds \\ &= 1_{\{\hat{\tau} \leq t\}} 1_x(\hat{X}_{\hat{\tau}})(t \wedge \hat{T}_\infty - \hat{\tau}) = 0 \quad \forall x \in \mathcal{D}. \end{aligned}$$

Adding the left-hand side of the above to the right-hand side of (22), taking expectations, using Tonelli's theorem, we find that

$$(23) \quad \nu([0, t], x) = \mathbb{E}_\gamma \left[\int_0^{t \wedge \tau \wedge T_\infty} 1_x(X_s) ds \right] = \mathbb{E}_\gamma \left[\int_0^{t \wedge \hat{T}_\infty} 1_x(\hat{X}_s) ds \right] = \int_0^t \hat{p}_s(x) ds \quad \forall x \in \mathcal{D},$$

and the second equation in (7) follows. □

Notes and references. The treatment in this section follows that in (Kuntz, 2017; Kuntz et al., 2019). However, the simple ideas underpinning the above characterisation are classical. For instance, the following is taken from (Feller, 1971, p.494):

“To show how the distribution of recurrence and first passage times may be calculated we number the states $0, 1, 2, \dots$ and use 0 as pivotal state. Consider a new process which coincides with the original process up to the random epoch of the first visit to 0 but with the state fixed at 0 forever after. In other words, the new process is obtained from the old one by making 0 an absorbing state. Denote the transition probabilities of the modified process by ${}^0P_{ik}(t)$. Then ${}^0P_{00}(t) = 1$. In terms of the original process ${}^0P_{i0}(t)$ is the probability of a first passage from $i \neq 0$ to 0 before epoch t , and ${}^0P_{ik}(t)$ gives the probability of a transition from $i \neq 0$ to $k \neq 0$ without intermediate passage through 0. It is probabilistically clear that the matrix ${}^0P(t)$ should satisfy the same backward and forward equations as $P(t)$ except that $q(0)$ [in the notation of (26.1)] is replaced by 0.”

I particularly like (Syski, 1992) on this subject.

There appears to be some confusion in the literature regarding equations (12)–(13) satisfied by the marginals. Sometimes (for example⁶, (Helses et al., 2001; Helmes, 2002; Lasserre and Prieto-Rumeau, 2004; Lasserre et al., 2006)) it is assumed that (12) has no other solutions aside from ν_S . This assumption turns out to be flawed: as the example given immediately after the proof of Theorem 11 shows, (12) can indeed have other solutions if the domain contains absorbing sets that are unreachable from the support of the initial distribution.

37. On the chain’s definition, minimality, and right-continuity*. In this section, we address the same question we did in Section 7 for discrete-time chains: are all chains made equal? Or do the particulars of the chain’s definition matter?

In the discrete-time case the answer was simply ‘no, the details of the chain’s definition do not matter’: the path law of a discrete-time chain depends on its one-step matrix P and its initial distribution γ but not on its particular construction.

In the continuous-time case, the answer is a bit more complicated. In short, if we only consider chains (i.e., continuous-time processes taking values in countable sets $\mathcal{S}$ and satisfying the Markov property) whose paths possess some basic regularity properties (properties that are typically taken for granted in practice), then the particulars of the definition do not matter. However, in general, they do matter for reasons stemming from the uncountability of the time-axis allowing for a lot of bad behaviour (behaviour that is tamed by the regularity assumptions).

The properties are two: *right-continuity of the paths* and *minimality of the chain*. Right-continuity of the paths (w.r.t. the discrete topology on $\mathcal{S}$) means that (a) upon entering a state, the chain lingers in the state for at least a short amount of time and (b) the chain enters states at well-defined points in time (for example, it cannot be the case that $X_{t'}(\omega) = x$ for all t' in some interval $(t, t + \varepsilon)$ but $X_t(\omega) \neq x$). From an applied point of view, chains whose paths do not satisfy these properties can seem very odd indeed. For instance, there are chains that immediately jump out of every state they enter (Rogers and Williams, 2000a, Sections III.23, 35). What is going on in these cases is that the state space is countable but not ‘discrete’ in the sense that its states are not ‘well-separated’ from each other (for instance, $\mathcal{S}$ might be the set of rational numbers $\mathbb{Q}$). For such chains, the theory is substantially more involved as the discrete topology on $\mathcal{S}$ must be replaced with the more complicated *Ray-Knight* topology. For a lovely account of this theory, see (Rogers and Williams, 2000a). On the other hand, the right-continuous case is simple: the time elapsed between two consecutive jumps is exponentially distributed with parameter $\lambda(x)$ depending on the chain’s state x immediately prior the second jump and the probability $p(x, y)$ that the chain next jumps to y if it lies in x depends only on x and y .

As we have seen in Section 26, chains with right-continuous paths can explode: infinitely many jumps accumulate in a finite amount of time and the chain diverges to infinity in the sense captured by Theorem 26.10. In these cases, it is possible to continue the chain in ways that the Markov property is preserved. For example, at the moment of explosion sample a state from the initial distribution, re-initialise the chain at this state, and continue the sample path by running the Kendall-Gillespie algorithm (Section 26) with a new set of independent random variables. Thus, explosions introduce an ambiguity in the chain’s definition (the process “runs out of instructions” at the explosion time, (Asmussen, 2003, p.43)) and depending on whether the chain is re-initialised and how, we obtain chains with different path laws. Of course, a chain that is restarted has probability to be in any given state x at any given time t no smaller than a chain that is not restarted: the probability of the former being in x at t equals the probability that it has not exploded by t and lies in x at t , plus the probability that it has exploded once

⁶We only consider this issue for chains, while the referenced works consider more general Markov processes. However, with some work, I believe that the arguments presented in this section, and in (Kuntz, 2017, Chapter 2), can be ported over to the more general case.

by t , been re-initialised, and lies in x at time t , etc.; while probability of the latter being in x at t is simply the probability that it has not exploded by t and lies in x at t . For this reason, chains that do not continue past an explosion are called *minimal*.

Imposing right-continuity and minimality removes all ambiguity that otherwise may arise in the chain's definition: in this case, the path law is fully determined by the chain's jump rates $\lambda := (\lambda(x))_{x \in \mathcal{S}}$ and jump matrix $P := (p(x, y))_{x, y \in \mathcal{S}}$ (or, equivalently, its rate matrix) and its initial distribution γ . Thus, for a given λ , P , and γ , it does not matter how we build the chain as long as its vector of jump rates is λ , its jump matrix is P , and its initial distribution is γ . The remainder of this section is dedicated to formalising this fact in Theorem 1 below (see the ensuing material for the terminology used in the theorem's statement). To not overly complicate the exposition, we will focus on the case of chains without absorbing states.

(1) **THEOREM** (Uniqueness of the path law of minimal right-continuous chains). *Suppose that there are no absorbing states (i.e., that (7) below holds). The path law*

$$\mathbb{L}_\gamma(A) := \mathbb{P}_\gamma(\{X \in A\}) \quad \forall A \in \mathcal{E}$$

of any minimal right-continuous chain X with state space $\mathcal{S}$ and initial distribution γ is the only probability measure on $(\mathcal{P}, \mathcal{E})$ satisfying (29.4) for all cylinder sets (i.e., sets of the form (29.2)), where the jump rates $(\lambda(x))_{x \in \mathcal{S}}$ and jump matrix $(p(x, y))_{x, y \in \mathcal{S}}$ are given by

$$(2) \quad \lambda(x) = -\ln(\mathbb{P}_x(\{T_1 > 1\})) \quad \forall x \in \mathcal{S},$$

$$(3) \quad p(x, y) = \mathbb{P}_x(\{X_{T_1} = y\}) \quad \forall x, y \in \mathcal{S}.$$

with T_1 denoting the chain's first jump time (6). Moreover, $\lambda(x)$ is positive and finite for all x in $\mathcal{S}$ and $P = (p(x, y))_{x, y \in \mathcal{S}}$ is a one-step matrix (i.e., satisfies (1.1)) with $p(x, x) = 0$ for all x in $\mathcal{S}$. For these reasons,

$$(4) \quad q(x, y) := \begin{cases} -\lambda(x) & \text{if } x = y, \\ \lambda(x)p(x, y) & \text{if } x \neq y \end{cases} \quad \forall x, y \in \mathcal{S},$$

defines a stable and conservative rate matrix $Q := (q(x, y))_{x, y \in \mathcal{S}}$ (i.e., Q satisfies (26.1)) and we can recover λ and P from Q :

$$\lambda(x) = -q(x, x) \quad \forall x \in \mathcal{S}, \quad p(x, y) = (1_x(y) - 1)q(x, y)/q(x, x) \quad \forall x, y \in \mathcal{S}.$$

Thus, the path law is fully determined by the initial distribution γ and rate matrix Q .

Continuous-time chains. In general, a 'continuous-time chain taking values in countable set $\mathcal{S}$ ' is a model composed of

- an *underlying space*: a measurable space $(\Omega, \mathcal{F})$;
- a set of *underlying probability measures*: a collection $(\mathbb{P}_x)_{x \in \mathcal{S}}$ of probability measures $\mathbb{P}_x$ on $(\Omega, \mathcal{F})$ indexed by *states* x in $\mathcal{S}$;
- a *final time* T_f : an $\mathcal{F}/\mathcal{B}(\mathbb{R}_E)$ -measurable function mapping from Ω to $(0, \infty]$;
- a *chain* X : a collection $(X_t)_{t \geq 0}$ of functions X_t indexed by *times* t in $[0, \infty)$ such that
 1. X is defined up until T_f : for each t in $[0, \infty)$, X_t maps from $\{t < T_f\}$ to $\mathcal{S}$,
 2. X satisfies an appropriate measurability requirement: for each x in $\mathcal{S}$ and t in $[0, \infty)$, $\{t < T_f, X_t = x\}$ belongs to $\mathcal{F}$,
 3. under $\mathbb{P}_x$, X starts at x : for each x in $\mathcal{S}$, $\mathbb{P}_x(\{X_0 = x\}) = 1$ (note that $\{0 < T_f\} = \Omega$ by T_f 's definition),

4. immediately prior T_f , X cannot be in one particular state:

$$\liminf_{t \uparrow T_f(\omega)} 1_x(X_t(\omega)) = 0 \quad \forall x \in \mathcal{S}, \quad \omega \in \{T_f < \infty\}$$

where $t \uparrow T_f(\omega)$ means that we are taking the limit infimum from below (that is, we are only considering ts strictly less than $T_f(\omega)$).

5. the collection X satisfies the Markov property:

$$\begin{aligned} & \mathbb{P}_z(A \cap \{t < T_f, X_t = x\} \cap \{t + s < T_f, X_{t+s} = y\}) \\ &= \mathbb{P}_z(A \cap \{t < T_f, X_t = x\}) \mathbb{P}_x(\{s < T_f, X_s = y\}) \quad \forall x, y, z \in \mathcal{S}, \quad t, s \in [0, \infty), \end{aligned}$$

where A denotes any event whose occurrence we can deduce from observing the chain up until t , i.e. A belongs to the sigma-algebra $\mathcal{F}_t$ generated by the events

$$(5) \quad \{X_s = x, s < T_f\} \quad \forall s \in [0, t], \quad x \in \mathcal{S}.$$

Just as for the chains of Section 26, if the chain's starting position is sampled from a probability distribution $\gamma = (\gamma(x))_{x \in \mathcal{S}}$, then we use $\mathbb{P}_\gamma := \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x$ to describe the chain's statistics.

Chains with right-continuous paths. The chain is said to have *right-continuous* paths if for any t in $[0, \infty)$ and ω in $\{t < T_f\}$, there exists an $0 < \varepsilon < T_f(\omega)$ such that

$$X_{t+\varepsilon'}(\omega) = X_t(\omega) \quad \forall \varepsilon' \in [0, \varepsilon].$$

In this case, the *jump times*

$$(6) \quad T_0(\omega) := 0, \quad T_{n+1}(\omega) := \inf\{t \in [T_n(\omega), T_f(\omega)) : X_t(\omega) \neq X_{T_n(\omega)}(\omega)\} \quad \forall n \in \mathbb{N}, \quad \omega \in \Omega,$$

form an increasing sequence $(T_n(\omega))_{n \in \mathbb{N}}$. We use N to denote the number of the chain's last jump:

$$N(\omega) := \sup\{n \in \mathbb{N} : T_n(\omega) < \infty\} \quad \forall \omega \in \Omega.$$

Due to these definitions,

$$T_{N(\omega)} < T_f(\omega) \quad \text{and} \quad X_t(\omega) = X_{T_{N(\omega)}}(\omega) \quad \forall t \in [T_{N(\omega)}(\omega), T_f(\omega))$$

if the chain stops jumping ($N(\omega) < \infty$). In this case, 4 in the definition of X above implies that the final time must be infinite as

$$\begin{aligned} \{N < \infty, T_f < \infty\} &= \{N < \infty, X_t = X_{T_N} \quad \forall t \in [T_N, T_f), T_f < \infty\} \\ &= \bigcup_{x \in \mathcal{S}} \{N < \infty, X_t = X_{T_N} = x \quad \forall t \in [T_N, T_f), T_f < \infty\} \\ &\subseteq \bigcup_{x \in \mathcal{S}} \left\{ \liminf_{t \uparrow T_f} 1_x(X_t) = 1, T_f < \infty \right\} = \emptyset. \end{aligned}$$

Putting the above together we find that the chain stops jumping if only if it gets stuck in some *absorbing state*:

$$\{N < \infty\} = \{N < \infty, T_f = \infty, X_t = X_{T_N} \quad \forall t \in [T_N, \infty)\}.$$

To not overly complicate the exposition of this section, we assume that this never happens:

$$(7) \quad N(\omega) = \infty \quad \forall \omega \in \Omega.$$

In this case, the jump times' definition and the right-continuity of the paths imply that the *waiting times*,

$$(8) \quad S_n(\omega) := T_n(\omega) - T_{n-1}(\omega) \quad \forall \omega \in \Omega, \quad n \in \mathbb{Z}_+,$$

are all well-defined and positive; and that

$$(9) \quad X_t(\omega) = Y_n(\omega) \quad \forall t \in [T_n(\omega), T_{n+1}(\omega)), \quad \omega \in \Omega, \quad n \in \mathbb{N},$$

where $Y = (Y_n)_{n \in \mathbb{N}}$ is the discrete-time process obtained by sampling the chain at the jump times:

$$(10) \quad Y_n(\omega) := X_{T_n(\omega)}(\omega) \quad \forall \omega \in \Omega, \quad n \in \mathbb{N}.$$

The explosion time and minimal chains with right-continuous paths. We refer to the limit of the jump times as the *explosion time* T_∞ :

$$T_\infty(\omega) = \lim_{n \rightarrow \infty} T_n(\omega).$$

Our assumption (7) that the chain never stops jumping implies that $T_n(\omega) < \infty$ for all n and ω . The definition of the jump times (6) shows that this can only be the cases if $T_n(\omega) < T_f(\omega)$ for all n and ω . Taking the limit $n \rightarrow \infty$ then shows that the explosion time cannot be greater than the final time:

$$T_\infty(\omega) \leq T_f(\omega) \quad \forall \omega \in \Omega.$$

The chain is said to be *minimal* if the explosion time is the final time:

$$(11) \quad T_\infty(\omega) = T_f(\omega) \quad \forall \omega \in \Omega.$$

The strong Markov property. For any minimal chain X with right-continuous paths, (9) shows that the chain is fully determined by its waiting times $(S_n)_{n \in \mathbb{Z}_+}$ in (8) and jump chain $Y := (Y_n)_{n \in \mathbb{N}}$ in (10). That is, we can view the chain X as a function mapping from Ω to path space $\mathcal{P}$ introduced in Section 29. It is not too difficult to show that X is $\mathcal{F}/\mathcal{E}$ -measurable (do it!), where $\mathcal{E}$ denotes the cylinder sigma algebra on $\mathcal{P}$ also introduced in Section 29. Moreover, it follows from 5 in the chain's definition that, for any A in $\mathcal{F}_t$ and set B of the type in (29.13),

$$\begin{aligned} & \mathbb{P}_z(A \cap \{t < T_\infty, X_t = x\} \cap \{X^t \in B\}) \\ &= \mathbb{P}_z(A \cap \{t < T_\infty, X_t = x\} \cap \{t + t_n < T_\infty, X_{t+t_1} = z_1, \dots, X_{t+t_n} = z_n\}) \\ &= \mathbb{P}_z(A \cap \{t < T_\infty, X_t = x\}) \mathbb{P}_x(\{t_n < T_\infty, X_{t_1} = z_1, \dots, X_{t_n} = z_n\}) \\ &= \mathbb{P}_z(A \cap \{t < T_\infty, X_t = x\}) \mathbb{P}_x(\{X \in B\}), \end{aligned}$$

where X^t denotes the t -shifted chain (as in Section 30). Because the sets B of this type generate $\mathcal{E}$ (Exercise 29.12), the same arguments as those given in the proofs of Theorems 30.7 and 30.14 then show that X possesses the strong Markov property: that is, Theorem 30.14 also applies to X (here, it is important notice that the filtration $(\mathcal{F}_t)_{t \geq 0}$ defined by (5) coincides with that in Definition 28.1, see Exercise 28.2).

The waiting times, the jump chain, and the path law. It follows from Exercise 28.2 that the jump times are $(\mathcal{F}_t)_{t \geq 0}$ -stopping times (Definition 28.3). The strong Markov property (30.15) then implies that

$$\begin{aligned} \mathbb{P}_z(A \cap \{Y_n = x, Y_{n+1} = y, S_{n+1} > s\}) &= \mathbb{P}_z(A \cap \{X_{T_n} = x, Y_1^{T_n} = y, S_1^{T_n} > s\}) \\ &= \mathbb{P}_z(A \cap \{X_{T_n} = x\}) \mathbb{P}_x(\{Y_1 = y, S_1 > s\}) \\ &= \mathbb{P}_z(A \cap \{Y_n = x\}) \mathbb{P}_x(\{Y_1 = y, S_1 > s\}) \end{aligned}$$

for all natural numbers n , events A in the pre- T_n sigma-algebra $\mathcal{F}_{T_n}$, states x, y, z in $\mathcal{S}$, and $s \geq 0$. Because Proposition 28.7 shows that the pre- T_n sigma-algebra contains the sigma-algebra $\mathcal{G}_n$ generated by $Y_0, \dots, Y_n$ and $S_1, \dots, S_n$ (Definition 27.1), the above implies that

$$(12) \quad \mathbb{P}_x(\{Y_{n+1} = y, S_{n+1} > s\} | \mathcal{G}_n) = \mathbb{P}_{Y_n}(\{Y_1 = y, S_1 > s\}) \quad \mathbb{P}_x\text{-almost surely,}$$

for all n in $\mathbb{N}$, states x, y in $\mathcal{S}$, and s in $[0, \infty)$. Thus,

$$(13) \quad \mathbb{P}_x(\{Y_{n+1} = y\} | \mathcal{G}_n) = \mathbb{P}_{Y_n}(\{Y_1 = y\}) \quad \mathbb{P}_x\text{-almost surely, } \forall x, y \in \mathcal{S}, \quad n \in \mathbb{N},$$

$$(14) \quad \mathbb{P}_x(\{S_{n+1} > s\} | \mathcal{G}_n) = \mathbb{P}_{Y_n}(\{S_1 > s\}) \quad \mathbb{P}_x\text{-almost surely, } \forall x \in \mathcal{S}, \quad s \in [0, \infty), \quad n \in \mathbb{N}.$$

Now, applying the strong Markov property (30.15), we find that

$$\begin{aligned} \mathbb{P}_x(\{Y_1 = y, S_1 > s\}) &= \mathbb{P}_x(\{s < T_1, X_s = x, Y_1^s = y\}) = \mathbb{P}_x(\{s < T_1, X_s = x\}) \mathbb{P}_x(\{Y_1 = y\}) \\ &= \mathbb{P}_x(\{s < S_1\}) \mathbb{P}_x(\{Y_1 = y\}) \quad \forall x, y \in \mathcal{S}, \quad s \in [0, \infty) \end{aligned}$$

and that

$$\begin{aligned} \mathbb{P}_x(\{S_1 > t + s\}) &= \mathbb{P}_x(\{t < T_1, X_t = x, s < S_1^t\}) = \mathbb{P}_x(\{t < T_1, X_t = x\}) \mathbb{P}_x(\{s < S_1\}) \\ &= \mathbb{P}_x(\{t < S_1\}) \mathbb{P}_x(\{s < S_1\}) \quad \forall x \in \mathcal{S}, \quad t, s \in [0, \infty). \end{aligned}$$

That is, the waiting time distribution is *memoryless* and it follows that $\lambda(x)$ in (2) is finite and that S_1 , under $\mathbb{P}_x$, is exponentially distributed with parameter $\lambda(x)$ (Norris, 1997, Theorem 2.3.1). Thus, we can re-write (12)–(14) as

$$(15) \quad \mathbb{P}_x(\{Y_{n+1} = y, S_{n+1} > s\} | \mathcal{G}_n) = \mathbb{P}_x(\{Y_{n+1} = y\} | \mathcal{G}_n) \mathbb{P}_x(\{S_{n+1} > s\} | \mathcal{G}_n) \quad \mathbb{P}_x\text{-a.s.,}$$

$$(16) \quad \mathbb{P}_x(\{Y_{n+1} = y\} | \mathcal{G}_n) = p(Y_n, y) \quad \mathbb{P}_x\text{-a.s.,}$$

$$(17) \quad \mathbb{P}_x(\{S_{n+1} > s\} | \mathcal{G}_n) = e^{-\lambda(Y_n)s} \quad \mathbb{P}_x\text{-a.s.,}$$

for all n in $\mathbb{N}$, states x, y in $\mathcal{S}$, and s in $[0, \infty)$, where $(p(x, y))_{x \in \mathcal{S}}$ is as in (3).

Given (15)–(17), the same argument as that in the proof of Theorem 29.5 yields Theorem 1.

(18) EXERCISE (The case with absorbing states). *By expanding $(\Omega, \mathcal{F})$, introduce a sequence of i.i.d. unit-mean exponentially distributed random variables $(\tilde{S}_n)_{n \in \mathbb{N}}$ independent of (the old) $\mathcal{F}$. Replace $(S_n)_{n \in \mathbb{Z}_+}$ in (8) with*

$$S_n := \begin{cases} T_n(\omega) - T_{n-1}(\omega) & \text{if } n \leq N(\omega) \\ \tilde{S}_n(\omega) & \text{if } n > N(\omega) \end{cases} \quad \forall \omega \in \Omega, \quad n \in \mathbb{Z}_+$$

and redefine the jump times as

$$T_0(\omega) := 0, \quad T_n(\omega) := \sum_{m=1}^n S_m(\omega) \quad \forall n \in \mathbb{Z}_+.$$

Now, define the jump chain $(Y_n)_{n \in \mathbb{N}}$ via (10). Emulating the steps taken above, show that Theorem 1 holds with the following modifications: $p(x, x)$ may be zero and $Q = (q(x, y))_{x, y \in \mathcal{S}}$ in (4) is replaced by

$$q(x, y) := \begin{cases} 0 & \text{if } p(x, x) > 0 \\ -\lambda(x) & \text{if } x = y, \quad p(x, x) = 0 \\ \lambda(x)p(x, y) & \text{if } x \neq y, \quad p(x, x) = 0 \end{cases} \quad \forall x, y \in \mathcal{S}.$$

Continuous-time chains II: the long-term behaviour

With Sections 26–37 out of the way, we now focus on formalising for continuous-time chains the discussion regarding their long-term behaviour given in the introduction to Part I. The treatment here follows its discrete-time counterpart (Sections 13–25) very closely and the chapter overview given for the discrete-time case applies here almost unchanged. The only real differences are:

- The possibility of explosions throws up a couple of complications: the characterisation of the stationary equations in terms of linear equations is a bit more subtle in the continuous-time case than in the discrete-time one (compare Theorems 17.4 and 43.6) and, to rule out explosions in practice, we require a further Foster-Lyapunov criterion (Section 47).
- We need to spend a little bit of time on deriving a series of results connecting continuous-time chains with the discrete-time *skeleton chains* obtained by sampling the former at regular time intervals (Section 42). These will prove key in porting results from the discrete-time case to the continuous-time one.
- Continuous-time chains are always aperiodic (Section 38). Thus, their time-varying law always converges (Theorem 42.8).

38. Entrance times, accesibility, and closed communicating classes. To figure out which states the chain will visit in the long run, we must first look at which states it visits at all. For this, we need *entrance times*:

(1) **DEFINITION** (Entrance times). *Given any set $A \subseteq \mathcal{S}$ and positive integer k , the k th entrance time ϕ_A^k is the point in time that the chain enters the set A for the k th time:*

$$\varphi_A^0(\omega) := 0 \quad \forall \omega \in \Omega, \quad \varphi_A^k(\omega) := \inf\{t \in (\varphi_A^{k-1}(\omega), T_\infty(\omega)) : X_t \in A\} \quad \forall \omega \in \Omega, \quad k > 0,$$

For the first entrance time, we write φ_A instead of φ_A^1 . Additionally, we use the shorthand $\varphi_x := \phi_{\{x\}}$ (resp. $\varphi_x^k := \phi_{\{x\}}^k$) to denote the first (resp. k th) entrance time of a state x in $\mathcal{S}$.

A bit inconsistently with our terminology for exit times (Section 36), we do not view starting inside A as ‘entering’ A (hence, $\varphi_A > 0$ by definition): a small but important detail. If the chain starts in x , then φ_x is the first time that the chain returns to x and φ_x^k the k th time it does. Hence, φ_x is often referred to as the *first return time* (or, simply, the *return time*) and φ_x^k as the k th *return time*. Because we are to deduce whether the chain has entered a given set yet by continuously monitoring it and because it can only enter a set by jumping into it, entrance times are jump-time-valued stopping time.

(2) **EXERCISE.** *Using similar arguments to those given in the proof of Proposition 34.9 and induction, show that the k th entrance time φ_A^k to A is a jump-time-valued $(\mathcal{F}_t)_{t \geq 0}$ -stopping time, where $(\mathcal{F}_t)_{t \geq 0}$ denotes the filtration generated by the chain (Definition 27.1). Furthermore, prove that (34.3)–(34.4) hold if we replace η with φ_A^k and ς with ϕ_A^k , where ϕ_A^k denotes the k th entrance time to A of the jump chain Y (obtained by replacing X with Y in Definition 13.1).*

Accessibility. Entrance times allow us to formalise the idea the chain can reach, or *access*, one region of the state space from another:

(3) **DEFINITION** (Accessibility). *A state y is accessible from a state x (or $x \rightarrow y$ for short) if and only if $\mathbb{P}_x(\{\varphi_y < \infty\}) > 0$. Similarly, we say that a subset B of the state space is accessible from another subset A (or $A \rightarrow B$) if for every x in A there exists a y in B such that $x \rightarrow y$ (equivalently, if $\mathbb{P}_\gamma(\{\varphi_B < \infty\}) > 0$ for all initial distributions γ with support contained in A).*

We can characterise this relationship in multiple ways:

(4) **THEOREM** (Equivalent definitions of accessibility). *Suppose that $x \neq y$ and let $(p(x, y))_{x, y \in \mathcal{S}}$ and $(p_t(x, y))_{x, y \in \mathcal{S}}$ denote the jump matrix (26.2) and t -transition matrix (31.1), respectively. The following statements are equivalent:*

- (i) $x \rightarrow y$ for X .
- (ii) $x \rightarrow y$ for the jump chain Y .
- (iii) There exists some $x_1, \dots, x_l$ in $\mathcal{S}$ such that $q(x, x_1)q(x_1, x_2) \dots q(x_l, y) > 0$.
- (iv) There exists some $x_1, \dots, x_l$ in $\mathcal{S}$ such that $p(x, x_1)p(x_1, x_2) \dots p(x_l, y) > 0$.
- (v) $p_t(x, y) > 0$ for some t in $(0, \infty)$.
- (vi) $p_t(x, y) > 0$ for all t in $(0, \infty)$.

Proof. Because Exercise 2 shows that φ_y is finite if and only if the first entrance time ϕ_y of the jump chain is finite, (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iv) follows directly from Lemma 13.6. The definition of the jump matrix in (26.2) implies that (iii) $\Leftrightarrow$ (iv). The chain's definition in (26.6)–(26.8) and Tonelli's theorem imply that

$$p_t(x, y) = \mathbb{P}_x(\{X_t = y, t < T_\infty\}) = \sum_{n=1}^{\infty} \mathbb{P}_x(\{Y_n = y, T_n \leq t < T_{n+1}\}) \leq \sum_{n=1}^{\infty} \mathbb{P}_x(\{Y_n = y\}).$$

Because the definition of ϕ_y (Definition 13.1) implies that $\{Y_n = y\} \subseteq \{\phi_y < \infty\}$ for all $n > 0$, it follows from the above that (v) $\Rightarrow$ (ii). Trivially (vi) $\Rightarrow$ (v) and it suffices to show that (iv) $\Rightarrow$ (vi). Theorem 29.5 implies that

$$\begin{aligned} p_s(z, w) &= \mathbb{P}_z(\{X_t = w, t < T_\infty\}) \geq \mathbb{P}_z(\{Y_1 = w, T_1 \leq s < T_2\}) \geq \mathbb{P}_z(\{Y_1 = w, S_1 \leq s < S_2\}) \\ &= p(z, w)(1 - e^{-\lambda(z)s})e^{-\lambda(w)s} > 0 \quad \forall s \in (0, \infty) \end{aligned}$$

for all z and w such that $p(z, w) > 0$. The semigroup property (Theorem 31.2) then shows that

$$p_t(x, y) \geq p_{t/(l+1)}(x, x_1)p_{t/(l+1)}(x_1, x_2) \dots p_{t/(l+1)}(x_l, y) > 0 \quad \forall t \in (0, \infty).$$

□

Theorem 4(iii) leaves clear that $\rightarrow$ is a transitive relation on both $\mathcal{S}$ and its power set. The theorem has another more unexpected consequence:

Continuous-time chains are always aperiodic. If $x \neq y$, Theorem 4(vi) ensures that $p_t(x, y)$ is either zero for all $t \geq 0$ or positive for all $t > 0$. This precludes any type of periodicity (compare the discrete-time case in Section 19): a chain is either unable to transition from x to y or it is able to transition in any amount of time.

Communicating classes and closed sets. Two sets are said to *communicate* if each is accessible from the other. A set to be a *communicating class* if all its subsets communicate:

(5) **DEFINITION** (Communicating class). *A subset $\mathcal{C}$ of $\mathcal{S}$ is a communicating class if and only if $x \rightarrow y$ for all x, y in $\mathcal{C}$ (equivalently $A \rightarrow B$ for all A, B contained in $\mathcal{C}$).*

A subset of the state space is said to be *closed* if no state outside the set is accessible from a state inside the set.

(6) **DEFINITION** (Closed set). *A subset $\mathcal{C}$ of $\mathcal{S}$ is closed if and only if for each x in $\mathcal{C}$ and y in $\mathcal{S}$, $x \rightarrow y$ implies that y belongs to $\mathcal{C}$.*

Closed sets are those from which the chain may never escape:

(7) PROPOSITION. *If the chain starts in a closed set $\mathcal{C}$ (i.e., the initial distribution γ satisfies $\gamma(\mathcal{C}) = 1$), then the chain remains in $\mathcal{C}$:*

$$\mathbb{P}_\gamma(\{X_t \in \mathcal{C} \ \forall t \in [0, T_\infty)\}) = 1.$$

Proof. The chain's definition in (26.6)–(26.8) implies that

$$\{X_t \in \mathcal{C} \ \forall t \in [0, T_\infty)\} = \{Y_n \in \mathcal{C} \ \forall n \in \mathbb{N}\},$$

where $Y = (Y_n)_{n \in \mathbb{N}}$ denotes the jump chain. Because Theorem 4(ii) shows that $\mathcal{C}$ is closed for Y , the proposition then follows directly from the above and its discrete-time counterpart (Proposition 13.9). $\square$

Given the above, the following should come as no surprise:

(8) PROPOSITION (The chain visits at most one of two disjoint closed sets). *If $\mathcal{C}_1$ and $\mathcal{C}_2$ are two disjoint closed sets, then*

$$\mathbb{P}_\gamma(\{\varphi_{\mathcal{C}_1} < \infty, \varphi_{\mathcal{C}_2} < \infty\}) = 0.$$

Proof. Because Exercise 2 shows that the first entrance time of the chain is finite if and only if the corresponding entrance time of the jump chain is finite, this proposition follows directly from its discrete-time counterpart (Proposition 13.11) and Theorem 4. $\square$

Communicating classes that are closed play a very important role in the theory of continuous-time chains: they are the irreducible sets mentioned in the introduction to Part I. For this reason, the chain and its rate matrix are said to be *irreducible* if the entire state space is a single closed communicating class.

Non-explosivity is a class property. As we will see throughout this chapter, states in the same communicating class share many properties (e.g., transience, recurrence, etc.) and we refer to these as *class properties*. One example is non-explosivity: the inability of the chain to explode when it starts in a *non-explosive* state.

(9) THEOREM. *The explosion probabilities $(\mathbb{P}_x(\{T_\infty < \infty\}))_{x \in \mathcal{S}}$ satisfy the equations*

$$(10) \quad \sum_{z \in \mathcal{S}} q(x, z) \mathbb{P}_z(\{T_\infty < \infty\}) = 0, \quad \forall x \in \mathcal{S}.$$

In particular, if $\mathbb{P}_x(\{T_\infty < \infty\}) = 0$ and $x \rightarrow y$, then $\mathbb{P}_y(\{T_\infty < \infty\}) = 0$.

Proof. Applying the Strong Markov Property (Theorem 30.14) to $\eta := T_1$ (with $A := \Omega$ and $F(x) := 1_{[0, t]}(c_n^T(x))$ where c_n^T is as in (29.9)), we find that

$$\mathbb{P}_x(\{X_{T_1} = z, T_{n+1} \leq t\}) = \mathbb{P}_x(\{X_{T_1} = z\}) \mathbb{P}_z(\{T_n \leq t\}) = p(x, z) \mathbb{P}_z(\{T_n \leq t\}),$$

for all z in $\mathcal{S}$, n in $\mathbb{N}$, and t in $[0, \infty)$. Given the definition of the jump matrix in (26.2), taking the limits $n \rightarrow \infty$ and $t \rightarrow \infty$ (in that order), applying monotone convergence, summing over z in $\mathcal{S}$, and re-arranging yields (10). Suppose that $x \rightarrow y$ and let $x_1, \dots, x_l$ be as in Theorem 4(ii). Repeatedly applying (10) yields

$$\begin{aligned} \mathbb{P}_x(\{T_\infty < \infty\}) &= \frac{1}{q(x)} \sum_{z \in \mathcal{S}} q(x, z) \mathbb{P}_z(\{T_\infty < \infty\}) \geq \frac{q(x, x_1)}{q(x)} \mathbb{P}_{x_1}(\{T_\infty < \infty\}) \\ &= \dots \geq \frac{q(x, x_1) \dots q(x_l, y)}{q(x) \dots q(x_l)} \mathbb{P}_y(\{T_\infty < \infty\}), \end{aligned}$$

and it follows that $\mathbb{P}_x(\{T_\infty < \infty\}) = 0$ only if $\mathbb{P}_y(\{T_\infty < \infty\}) = 0$. $\square$

39. Recurrence and transience. The starting point in understanding the long-term behaviour of chains are the notions of *recurrence* and *transience*:

(1) DEFINITION (Recurrent and transient states). *A state x is said to be recurrent if the chain has probability one of returning to it (i.e., $\mathbb{P}_x(\{\phi_x < \infty\}) = 1$). Otherwise, the state is said to be transient.*

Recurrent states are those the chain will keep revisiting (as long as it visit them at least once) while transient ones are those the chain will stop visiting at some point:

(2) THEOREM. *For any initial distribution γ and state x ,*

$$(3) \quad \mathbb{P}_\gamma(\{\varphi_x^k < \infty\}) = \mathbb{P}_\gamma(\{\varphi_x < \infty\})\mathbb{P}_x(\{\varphi_x < \infty\})^k \quad \forall k > 0.$$

Moreover, having started at x , the chain will return to x infinitely many times if and only if x is recurrent:

$$\mathbb{P}_x(\{\varphi_x^k < \infty \quad \forall k \in \mathbb{Z}_+\}) = \begin{cases} 1 & \text{if } x \text{ is recurrent} \\ 0 & \text{if } x \text{ is transient} \end{cases}, \quad \forall x \in \mathcal{S}.$$

Proof. Because, for any set A , the entrance time to A of X is finite if and only if the entrance time to A of its jump chain Y is finite (Exercise 38.2), the theorem follows directly its discrete-time counterpart (Theorem 14.2, note also (14.5)). $\square$

Recurrence and transience are class properties.

(4) THEOREM. *If x is recurrent and $x \rightarrow y$, then y is also recurrent and*

$$\mathbb{P}_x(\{\varphi_y < \infty\}) = \mathbb{P}_y(\{\varphi_x < \infty\}) = 1.$$

For this reason, a) if y is transient and $x \rightarrow y$, then x is also transient and b) a state in a communicating class is recurrent (resp. transient) if and only if all states in the class are recurrent (resp. transient).

Proof. Because, for any set A , the entrance time to A of X is finite if and only if the entrance time to A of its jump chain Y is finite (Exercise 38.2), the theorem follows directly from its discrete-time counterpart (Theorem 14.6). $\square$

Due to the above theorem, we say that a communicating class is *recurrent* (resp. *transient*) if any one (and therefore all) of its states is recurrent (resp. transient). For chains, we use the following slightly more involved classification:

(5) DEFINITION (Recurrent and transient chains). *If the set $\mathcal{R}$ of all recurrent states is empty, then the chain is said to be transient. Otherwise, the chain is said to be Tweedie recurrent if, regardless of the initial distribution, the chain has probability one of entering $\mathcal{R}$:*

$$\mathbb{P}_\gamma(\{\varphi_{\mathcal{R}} < \infty\}) = 1 \quad \text{for all initial distributions } \gamma,$$

where $\varphi_{\mathcal{R}}$ denotes the first entrance time to $\mathcal{R}$ (Definition 38.1). If, additionally, there exists only one recurrent communicating class, then the chain is said to be Harris recurrent. If the class is the entire state space (i.e., if $\mathcal{R} = \mathcal{S}$), then the chain is simply called recurrent.

We conclude the section with the following useful proposition:

(6) PROPOSITION. *If the chain ever enters a recurrent closed communicating class $\mathcal{C}$, it will eventually visit every state in $\mathcal{C}$:*

$$1_{\{\varphi_{\mathcal{C}} < \infty\}} = 1_{\{\varphi_x < \infty\}} \quad \forall x \in \mathcal{C}, \quad \mathbb{P}_\gamma\text{-almost surely.}$$

Proof. Because, for any set A , the entrance time to A of X is finite if and only if the entrance time to A of its jump chain Y is finite (Exercise 38.2), the proposition follows directly from its discrete-time counterpart (Corollary 14.8). $\square$

Notes and references. The reasons behind our choice of nomenclature in this section are the same as those given in its discrete-time counterpart (Section 14, notes and references).

40. The regenerative property. For the same reasons as those given in Section 15 for the discrete-time case, the Markov property implies that the chain *regenerates* every time it visits a recurrent state x in the sense that upon each visit it forgets its entire past and starts afresh. More formally, the segments of paths between consecutive visits to x form an i.i.d. sequence:

(1) **THEOREM** (The regenerative property). *For any non-negative function $f : \mathcal{S} \rightarrow \mathbb{R}_E$, state x , and positive integer k ,*

$$(2) \quad I_k := 1_{\{\varphi_x^{k-1} < \infty\}} \int_{\varphi_x^{k-1}}^{\varphi_x^k \wedge T_\infty} f(X_t) dt$$

defines an $\mathcal{F}_{\varphi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable function, where $\mathcal{F}_{\varphi_x^k}$ denotes the pre- φ_x^k sigma algebra (Definition 28.3). In the above, we are using our convention for partially defined random variables (0.1). If the state x is recurrent, the sequence $(I_k)_{k \in \mathbb{Z}_+}$ is i.i.d. under $\mathbb{P}_x$.

Before proving the theorem, we quickly apply it to that the chain cannot explode if it starts in a recurrent state:

(3) **PROPOSITION** (The chain cannot explode if it starts in a recurrent state). *If x is a recurrent state, $\mathbb{P}_x(\{T_\infty = \infty\}) = 1$.*

Proof. Because Theorem 39.2 shows that all entrance times to x are finite with $\mathbb{P}_x$ -probability one, the definition of the entrance times (Definition 38.1) implies that

$$(4) \quad \varphi_x^k < T_\infty \quad \forall k > 0, \quad \mathbb{P}_x\text{-almost surely.}$$

The law of large numbers (Theorem 16.4) and the regenerative property (Theorem 1 with $f := 1$) imply that

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{\varphi_x^N}{N} &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N (\varphi_x^k - \varphi_x^{k-1}) = \mathbb{E}_x[\varphi_x] > 0 \quad \mathbb{P}_x\text{-almost surely,} \\ \Rightarrow \lim_{k \rightarrow \infty} \varphi_x^k &= \infty \quad \mathbb{P}_x\text{-almost surely.} \end{aligned}$$

For this reason, taking the limit $k \rightarrow \infty$ in (4) completes the proof. $\square$

A proof of Theorem 1. This proof is entirely analogous to that of the theorem's discrete-time counterpart (Theorem 15.2). We also do it in two steps, beginning with the measurability of I_k and moving on to showing that the sequence is i.i.d.

Step 1: I_k^f is $\mathcal{F}_{\varphi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Note that

$$I_k = \lim_{N \rightarrow \infty} \sum_{n=1}^N \sum_{m=0}^{n-1} \sum_{l=m}^{n-1} 1_{\{\varphi_x^k = T_n\}} 1_{\{\varphi_x^{k-1} = T_m\}} S_{l+1} f(Y_l).$$

Because limits, finite products, and finite sums of measurable functions are measurable functions, it suffices to show that each term in the sum is $\mathcal{F}_{\varphi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Because $\varphi_{k-1} \leq \varphi_k$ by definition, Lemma 28.5(i, iii) shows that $1_{\{\varphi_x^{k-1} = T_m\}}$ is $\mathcal{F}_{\varphi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable. Similarly, Lemma 28.5(i, ii) and Proposition 28.7 yields that $1_{\{\varphi_x^k = T_n\}} S_{l+1} f(Y_l)$ is $\mathcal{F}_{\varphi_x^k}/\mathcal{B}(\mathbb{R}_E)$ -measurable for all $l < n$ (as $T_{l+1} \leq \varphi_x^k$ for any such l) and the result follows. $\square$

Step 2: $(I_k^f)_{k \in \mathbb{Z}_+}$ is i.i.d. To simplify the notation, we write I_k for I_k^f throughout this proof. Suppose that x is recurrent, choose any $a_1, \dots, a_k \geq 0$, and let

$$A_k := \{I_1 \geq a_1, \dots, I_{k-1} \geq a_{k-1}, I_k \geq a_k\}.$$

The return times to x are all finite with $\mathbb{P}_x$ -probability one (Theorem 39.2). Furthermore, Exercise 38.2 and the chain's definition (26.6)–(26.8) imply that $X_{\varphi_x^{k-1}} = x$ on $\{\varphi_x^{k-1} < \infty\}$ and that $I_k = G(X_{\varphi_x^{k-1}})$, where $X_{\varphi_x^{k-1}}$ denotes the φ_x^{k-1} -shifted chain (Section 30) and G is the $\mathcal{E}/\mathcal{B}(\mathbb{R}_E)$ -measurable function in Lemma 29.14(iii) (with A and k therein set to $\{x\}$ and 1). For these reasons,

$$\begin{aligned} \mathbb{P}_x(A_k) &= \mathbb{P}_x(\{I_1 \geq a_1, \dots, I_{k-1} \geq a_{k-1}, \varphi_x^{k-1} < \infty, X_{\varphi_x^{k-1}} = x, I_k \geq a_k\}) \\ &= \mathbb{P}_x(\{I_1 \geq a_1, \dots, I_{k-1} \geq a_{k-1}, \varphi_x^{k-1} < \infty, X_{\varphi_x^{k-1}} = x, G(X_{\varphi_x^{k-1}}) \geq a_k\}). \end{aligned}$$

Because $\varphi_x^1 \leq \varphi_x^2 \leq \dots \leq \varphi_x^{k-1}$ by definition, Lemma 28.5 and Step 1 imply that A_{k-1} belongs to $\mathcal{F}_{\varphi_x^{k-1}}$. Thus, the strong Markov property (Theorem 30.14) implies that

$$\mathbb{P}_x(A_k) = \mathbb{P}_x(A_{k-1} \cap \{\phi_x^{k-1} < \infty, X_{\phi_x^{k-1}} = x\}) \mathbb{P}_x(\{G(X) \geq a_k\}) = \mathbb{P}_x(A_{k-1}) \mathbb{P}_x(\{I_1 \geq a_k\}).$$

Iterating the above backwards, we obtain

$$(5) \quad \mathbb{P}_x(A_k) = \mathbb{P}_x(\{I_1 \geq a_1\}) \mathbb{P}_x(\{I_1 \geq a_2\}) \dots \mathbb{P}_x(\{I_1 \geq a_k\}).$$

Setting $a_1 = \dots = a_{k-1} = 0$ we find that

$$\mathbb{P}_x(\{I_k \geq a_k\}) = \mathbb{P}_x(\{I_1 \geq a_k\}).$$

Because k and $a_1, \dots, a_k$ were arbitrary, and $\{[a, \infty] : a \in \mathbb{R}_E\}$ is a π -system that generates $\mathcal{B}(\mathbb{R}_E)$, the above and Lemma 5.6 show that the sequence $(I_k)_{k \in \mathbb{N}}$ is identically distributed. For this reason, we can rewrite (5) as

$$\mathbb{P}_x(A_k) = \mathbb{P}_x(\{I_1 \geq a_1\}) \mathbb{P}_x(\{I_2 \geq a_2\}) \dots \mathbb{P}_x(\{I_k \geq a_k\}).$$

The desired independence then also follows from Lemma 5.6 as $\{[a_1, \infty], \dots, [a_k, \infty] : a_1, \dots, a_k \in \mathbb{R}_E\}$ is a π -system that generates $\mathcal{B}(\mathbb{R}_E^k)$. $\square$

41. The empirical distribution and positive recurrent states. Consider the *empirical distribution* $\epsilon_T := (\epsilon_T(x))_{x \in \mathcal{S}}$ tracking the fraction of time the chain spends in each state, where

$$(1) \quad \epsilon_T(x) := \frac{1}{T} \int_0^{T \wedge T_\infty} 1_x(X_t) dt \quad \forall x \in \mathcal{S},$$

denotes is the fraction of the interval $[0, T]$ that the chain spends in state x . From Theorem 39.2, we already know that the chain eventually stops visiting any given transient state x , and it follows that

$$\lim_{T \rightarrow \infty} \epsilon_T(x) = 0 \quad \mathbb{P}_\gamma\text{-almost surely.}$$

In the case of recurrent states x , the visits do not cease. However, the regenerative property implies that the sequences $(\eta_k)_{k \in \mathbb{Z}_+}$ of time- η_k -spend-in- x -between-two-consecutive-visits and

$(\varphi^{k+1} - \varphi^k)_{k \in \mathbb{Z}_+}$ of time- $(\varphi^{k+1} - \varphi^k)$ -elapsed-between-consecutive-visits are i.i.d. if the chain starts from x . In this case, the law of large numbers (Theorem 16.4) then implies that,

$$\begin{aligned} \epsilon_T(x) &\approx \frac{1}{\varphi_x^K} \int_{n=0}^{\varphi_x^K} 1_x(X_t) dt = \left(\sum_{k=0}^{K-1} (\varphi^{k+1} - \varphi^k) \right)^{-1} \left(\sum_{k=0}^{K-1} \eta_k \right) \\ &= \left(\frac{1}{K} \sum_{k=0}^{K-1} (\varphi^{k+1} - \varphi^k) \right)^{-1} \left(\frac{1}{K} \sum_{k=0}^{K-1} \eta_k \right) \approx \frac{\mathbb{E}_x[\eta_1]}{\mathbb{E}_x[\varphi_x]}, \end{aligned}$$

for all T large enough that the number K of the return time φ_x^K closest to T is large and $|\varphi_x^K - T|/T$ is small, where $\lambda(x)$ denotes the jump rate in (26.3). Because η_1 equals the first waiting time S_1 and S_1 's mean is $1/\lambda(x)$ under $\mathbb{P}_x$, it follows that $\epsilon_T(x) \approx (\lambda(x)\mathbb{E}_x[\varphi_x])^{-1}$ for all large enough T . If the chain does not start at x , then noting that $\epsilon_T(x)$ is non-zero only if the chain ever visits x . The Markov property then implies following:

(2) THEOREM. For all states x in $\mathcal{S}$,

$$\epsilon_\infty(x) := \lim_{T \rightarrow \infty} \epsilon_T(x) = 1_{\{\varphi_x < \infty\}} \frac{1}{\mathbb{E}_x[\varphi_x]} \quad \mathbb{P}_\gamma\text{-almost surely.}$$

(3) EXERCISE. Prove Theorem 2 by following the steps taken in the proof of its discrete-time counterpart (Theorem 16.2). To do so, note that F_-, F_+ in Lemma 29.14(iv) (with $f := 1_x$) are such that

$$F_-(X) = \liminf_{T \rightarrow \infty} \epsilon_T(x), \quad F_+(X) = \limsup_{T \rightarrow \infty} \epsilon_T(x),$$

and replace R_N with

$$R_T := \begin{cases} 0 & \text{if } \varphi_x^0 \leq T < \varphi_x^1 \\ 1 & \text{if } \varphi_x^1 \leq T < \varphi_x^2 \\ \vdots & \vdots \end{cases} \quad \forall T \in [0, \infty)$$

and Theorems 12.4, 14.2, and 15.2 with their continuous-time counterparts (Theorems 30.14, 39.2, and 40.1, respectively).

The theorem shows that the fraction $\epsilon_\infty(x)$ of all time that the chain spends in a state x is negligible unless returns to x occur quickly enough that the mean return time is finite. Or, in other words, unless x is *positive recurrent*:

(4) DEFINITION (Positive and null recurrent states). A state x is said to be *positive recurrent* if its mean return time is finite ($\mathbb{E}_x[\varphi_x] < \infty$). If x is recurrent but its mean return time is infinite ($\mathbb{E}_x[\varphi_x] = \infty$), we say that it is *null recurrent*.

42. Skeleton chains and the Croft-Kingman lemma. As the previous sections illustrate, we are able to prove results describing the long-term behaviour of the empirical distribution in the continuous-time case by emulating the proofs the results' discrete-time counterparts. To prove the results that characterise the long-term behaviour of the time-varying law of continuous-time chains, we can directly use their discrete-time counterparts. Key to unlocking this approach are the so-called δ -skeleton chains $X^\delta = (X_n^\delta)_{n \in \mathbb{N}}$: discrete-time processes obtained by sampling our continuous-time chain X at regular time intervals of length $\delta > 0$, i.e.,

$$(1) \quad X_n^\delta(\omega) := \begin{cases} X_{\delta n}(\omega) & \text{if } \delta n < T_\infty(\omega) \\ \Delta & \text{if } \delta n \geq T_\infty(\omega) \end{cases} \quad \forall \omega \in \Omega, \quad n \in \mathbb{N},$$

where $\Delta \notin \mathcal{S}$ denotes an auxiliary dummy (or *dead* or *infinity*) state in which we leave X^δ for all sampling times past the explosion time of X . It is not difficult to use the Markov property of X to show that

$$(2) \quad \mathbb{P}_\gamma(\{X_{n+1}^\delta = x | X_n^\delta\}) = p^\delta(X_n^\delta, x) \quad \forall x \in \mathcal{S}_E, \quad n \in \mathbb{N},$$

where $\mathcal{S}_E$ denotes the *extended state space* $\mathcal{S} \cup \{\Delta\}$ and $P^\delta = (p^\delta(x, y))_{x, y \in \mathcal{S}_E}$ the one-step matrix given by

$$(3) \quad p^\delta(x, y) := \begin{cases} \mathbb{P}_x(\{X_\delta = y, \delta < T_\infty\}) & \text{if } x, y \in \mathcal{S} \\ \mathbb{P}_x(\{T_\infty \leq \delta\}) & \text{if } x \in \mathcal{S}, y = \Delta \\ 1_\Delta(y) & \text{if } x = \Delta \end{cases} \quad \forall x, y \in \mathcal{S}_E.$$

Theorem 7.1 implies that X^δ is a discrete-time chain with state space $\mathcal{S}_E$, one-step matrix P^δ , and initial distribution $\gamma_E := (1_{\mathcal{S}}(x)\gamma(x))_{x \in \mathcal{S}_E}$ (here, we are using our convention in (0.1) for partially-defined functions). Moreover, because Δ is an absorbing state for X^δ , it is straightforward to check that

$$(4) \quad p_n^\delta(x, y) = p_{\delta n}(x, y) \quad \forall x, y \in \mathcal{S}, \quad n \in \mathbb{N},$$

where $P_n^\delta = (p_n^\delta(x, y))_{x, y \in \mathcal{S}_E}$ denotes the n -step matrix of the skeleton chain (defined by replacing P and $\mathcal{S}$ in (4.5) with P^δ and $\mathcal{S}_E$) and $(p_t(x, y))_{x, y \in \mathcal{S}}$ denotes t -transition matrix in (31.1) of our continuous-time chain X .

(5) EXERCISE. Prove (2) and (4), and show that the P^δ defined in (3) is indeed a one-step matrix (i.e., that $\sum_{y \in \mathcal{S}_E} p^\delta(x, y) = 1$ for all x in $\mathcal{S}_E$).

It follows easily from (4) that the skeleton chain and the original chain share the same communicating classes and closed sets and that the former is aperiodic:

(6) EXERCISE. Pick any $\delta \in (0, \infty)$. Use (4) and Theorem 38.4(vi) to prove that X^δ is aperiodic and that y is accessible from x for X^δ if and only if it is for X , for any states $x, y \in \mathcal{S}$.

The Croft-Kingman lemma and the existence of limits for the time-varying law. Equation (4) spells out the relationship between the n -step matrix of X^δ and the transition probabilities of X evaluated at the sampling times $t = 0, \delta, 2\delta, \dots$. To relate these two for other times, the following lemma proves very useful.

(7) LEMMA (The Croft-Kingman lemma, (Kingman, 1963)). Let f be a continuous real-valued function on $[0, \infty)$. Suppose that for each δ in $(0, \infty)$, the limit

$$L_\delta := \lim_{n \rightarrow \infty} f(\delta n)$$

exists and is finite. In this case, the limit L_δ is independent of δ . Moreover, $\lim_{t \rightarrow \infty} f(t)$ exists and equals L_δ .

Before we prove the lemma we use it to prove that the time-varying p_t converges pointwise as t approaches infinity:

(8) THEOREM (The time-varying law has pointwise limits). For all initial distributions γ and states x , the limit $\lim_{t \rightarrow \infty} p_t(x)$ exists and is finite, where p_t denotes the time-varying law in (33.1) of X .

Proof. As we showed in the proof of the differential equation satisfied by the time-varying law (Theorem 33.3), $t \mapsto p_t(x)$ is a continuous function on $[0, \infty)$ for every x in $\mathcal{S}$. Fix any δ in $(0, \infty)$. Because all skeleton chains are aperiodic (Exercise 6), the pointwise convergence of the time-varying law of aperiodic discrete-time chains (Theorem 20.2), bounded convergence, and (4) imply that

$$(9) \quad \lim_{n \rightarrow \infty} p_{n\delta}(x) = \lim_{n \rightarrow \infty} \sum_{x' \in \mathcal{S}} \gamma(x') p_{n\delta}(x', x) = \lim_{n \rightarrow \infty} \sum_{x' \in \mathcal{S}_E} \gamma_E(x') p_n^\delta(x', x) = \lim_{n \rightarrow \infty} p_n^\delta(x) = \pi_\gamma(x)$$

for all x in $\mathcal{S}$, where $(p_n^\delta)_{n \in \mathbb{N}}$ denotes the time-varying law (4.1) of the skeleton chain and π_γ is its limit in (20.3) (with X^δ replacing X therein). Because δ was arbitrary, the Croft-Kingman lemma tells us that the limit π_γ is the same for all δ and that

$$(10) \quad \lim_{t \rightarrow \infty} p_t(x) = \pi_\gamma(x) \quad \forall x \in \mathcal{S}.$$

□

Without realising it, we have almost characterised the limiting behaviour of the time-varying law. As we will show in the next section, continuous-time chains and their skeletons share the same stationary distributions. For this reason, we are able to rewrite the limits in (10) in terms of the continuous-time chain's stationary distributions and its entrance times to positive recurrent classes just as we did in Theorem 20.2 for the aperiodic discrete-time case.

A proof of the Croft-Kingman lemma. Here, we follow the steps taken in (Kingman, 1963).

(11) THEOREM. *Let A be any unbounded open subset of $(0, \infty)$. There exists a $\delta > 0$ such that δn belongs to A for infinitely many integers n .*

Proof. For any $\alpha \in (0, \infty)$ and set $B \subseteq [0, \infty)$, let αB denote the set $\{\alpha x : x \in B\}$. Consider the open set

$$A_m := \bigcup_{n=m}^{\infty} \frac{1}{n} A \quad \forall m \in \mathbb{Z}_+.$$

Suppose that, for some m , there exists a non-empty open interval $I \subseteq (0, \infty)$ disjoint from A_m . In this case, $nI \cap A = \emptyset$ for all $n \geq m$. Thus, A is disjoint from $B := \bigcup_{n=m}^{\infty} nI$. If you think about it a little bit, you'll see that B contains all large enough numbers (hint: compare the distance between the mid-points of $(n-1)I$ and nI with the length of nI), contradicting our assumption that A is unbounded. Thus, A_m intersects with all open intervals contained in $(0, \infty)$ or, in other words, A_m is dense in $(0, \infty)$. For this reason,

$$\bigcap_{m=1}^{\infty} A_m$$

is the intersection of countably many open dense subsets of $(0, \infty)$ and Baire's theorem tells us this intersection is dense in $(0, \infty)$. In particular, it is non-empty and picking any δ belonging to the intersection we have that δn belongs to A for infinitely many n in $\mathbb{Z}_+$. □

The key to the proof of Lemma 7 is the following corollary of Theorem 11:

(12) COROLLARY (Croft's theorem, (Croft, 1957)). *Let f be a continuous real-valued function on $[0, \infty)$. If*

$$\lim_{n \rightarrow \infty} f(\delta n) = 0$$

for all δ in $(0, \infty)$, then

$$\lim_{t \rightarrow \infty} f(t) = 0.$$

Proof. Suppose that $f(t)$ does not tend to zero as t approaches infinity and pick any c in $(0, \infty)$ such that $\limsup_{t \rightarrow \infty} |f(t)| > c$. For any n in $\mathbb{N}$, there exists a $t_n > n$ such that $|f(t_n)| > c$ and so by continuity there exists an open interval I_n containing t_n such that $|f(t)| > c$ for all t in I_n . Thus, $A := \bigcup_{n=0}^{\infty} I_n$ is an unbounded open set such that $|f(t)| > c$ for all t in A . Theorem 11 then implies that there exists a δ in $(0, \infty)$ such that $|f(\delta n)| > c$ for all n in $\mathbb{N}$ contradicting our premise. Hence, it must be the case that $f(t) \rightarrow 0$ as $t \rightarrow \infty$. □

The proof of the Croft-Kingman lemma is now downhill:

Proof of Lemma 7. For any δ in $(0, \infty)$ and positive integer m , the sequence $(nm\delta)_{n \in \mathbb{Z}_+}$ is a subsequence of $(n\delta)_{n \in \mathbb{Z}_+}$ and so

$$L_{m\delta} = \lim_{n \rightarrow \infty} f(nm\delta) = \lim_{n \rightarrow \infty} f(n\delta) = L_\delta.$$

Thus, for any rational number $r = m/l$, we have that $L_{r\delta} = L_{lr\delta} = L_{m\delta} = L_\delta$. Because $\delta \mapsto L_\delta$ is the pointwise limit of the sequence $(\delta \mapsto f(\delta n))_{n \in \mathbb{Z}_+}$ of continuous functions, there exists at least one δ_* such that $\delta \mapsto L_\delta$ is continuous at δ_* . Fix any δ in $(0, \infty)$ and let $(r_n)_{n \in \mathbb{Z}_+}$ be a sequence of rational numbers converging to δ_*/δ . Then,

$$L := L_{\delta_*} = \lim_{n \rightarrow \infty} L_{r_n \delta} = \lim_{n \rightarrow \infty} L_\delta = L_\delta$$

proving that L is independent of δ and the result follows by applying Corollary 12 to the function $t \mapsto f(t) - L$. $\square$

43. Stationary distributions, ergodic distributions, and a Doeblin-like decomposition. A probability distribution π on $\mathcal{S}$ is said to be a stationary distribution of the chain X if sampling its starting state from π makes X a stationary process. By a stationary process, I mean that the chain's path law is invariant to time shifts:

$$(1) \quad \mathbb{P}_\pi(\{X^t \in A\}) = \mathbb{L}_\pi(A) \quad \forall A \in \mathcal{E},$$

where $\mathcal{E}$ denotes the sigma-algebra generated by the cylinder sets of the path space $\mathcal{P}$, $\mathbb{L}_\pi$ the path law when the initial distribution is π , and X^t the t -shifted chain (see Sections 29 and 30). Setting

$$A := \{(y, s) \in \mathcal{P} : y_0 = x\}$$

in (1) for each state x , we find that initialising the chain by sampling a stationary distribution π ensures that the chain remains with law π for all time:

$$(2) \quad \mathbb{P}_\pi(\{X_t = x, t < T_\infty\}) = \pi(x) \quad \forall x \in \mathcal{S}, \quad t \in [0, \infty).$$

It then follows from (33.2) that π is a stationary distribution only if it is a fixed point of the transition probabilities in (31.1):

$$(3) \quad \pi(x) = \mathbb{P}_\pi(\{X_t = x, t < T_\infty\}) = \sum_{x' \in \mathcal{S}} \pi(x') \mathbb{P}_{x'}(\{X_t = x, t < T_\infty\}) = \pi P_t(x) \quad \forall x \in \mathcal{S},$$

or $\pi = \pi P_t$ in matrix notation, for any given t in $[0, \infty)$. The converse is also true:

(4) **THEOREM** (The fixed points of the transition probabilities are the stationary distributions). *A probability distribution π on $\mathcal{S}$ is a stationary distribution of X if and only if it satisfies (2) for at least one t in $(0, \infty)$, in which case it holds for all t in $[0, \infty)$.*

Using Theorem 17.4, we can rephrase Theorem 4 using the jargon of the previous section: $\pi = (\pi(x))_{x \in \mathcal{S}}$ is a stationary distribution of X if and only if $\pi_E := (1_{\mathcal{S}}(x)\pi(x))_{x \in \mathcal{S}_E}$ is a stationary distribution of at least one skeleton chain X^δ , in which case π_E is a stationary distribution of every skeleton chain X^δ .

Proof of Theorem 4. Given the discussion preceding the theorem's statement, we need only to prove that if π is a probability distribution satisfying $\pi = \pi P_t$ for any given t in $(0, \infty)$, then $\pi = \pi P_s$ holds for all s in $[0, \infty)$ and π is a stationary distribution. We begin with the former: repeatedly applying Tonelli's theorem and the semigroup property (31.4), we find that

$$\begin{aligned} \sum_{x' \in \mathcal{S}} \pi(x') p_{nt}(x', x) &= \sum_{x'' \in \mathcal{S}} \left(\sum_{x' \in \mathcal{S}} \pi(x') p_t(x', x'') \right) p_{(n-1)t}(x'', x) \\ &= \sum_{x'' \in \mathcal{S}} \pi(x') p_{(n-1)t}(x', x) = \cdots = \pi(x) \quad \forall x \in \mathcal{S}, \quad n > 0. \end{aligned}$$

Theorem 42.8 shows that the limits

$$k(x', x) := \lim_{s \rightarrow \infty} p_s(x', x) \quad \forall x, x' \in \mathcal{S},$$

exist and are finite. Bounded convergence then shows that

$$\sum_{x' \in \mathcal{S}} \pi(x') k(x', x) = \lim_{n \rightarrow \infty} \sum_{x' \in \mathcal{S}} \pi(x') p_{nt}(x', x) = \lim_{n \rightarrow \infty} \pi(x) = \pi(x) \quad \forall x \in \mathcal{S}.$$

Another application of bounded convergence and the semigroup property yields

$$\sum_{x'' \in \mathcal{S}} k(x', x'') p_s(x'', x) = \lim_{r \rightarrow \infty} \sum_{x'' \in \mathcal{S}} p_r(x', x'') p_s(x'', x) = \lim_{r \rightarrow \infty} p_{r+s}(x', x) = k(x', x) \quad \forall x, x' \in \mathcal{S}.$$

Combining the above two equations then proves the desired $\pi = \pi P_s$ for any s in $[0, \infty)$:

$$\sum_{x' \in \mathcal{S}} \pi(x') p_s(x', x) = \sum_{x'' \in \mathcal{S}} \pi(x'') \sum_{x' \in \mathcal{S}} k(x'', x') p_s(x', x) = \sum_{x'' \in \mathcal{S}} \pi(x'') k(x'', x) = \pi(x) \quad \forall x \in \mathcal{S}.$$

Marginalising over the equation in (31.2) and exploiting that $\pi = \pi P_t$ for all t yields (1) for sets of the form (29.13). Because these sets generate $\mathcal{E}$ (Exercise 29.12), (1) holds for all A in $\mathcal{E}$. $\square$

The rate matrix's left nullspace and the role of explosions. Summing the (2) over all states x , taking the limit $t \rightarrow \infty$, and applying monotone convergence, we find that the chain is non-explosive when initialised with law π :

$$(5) \quad \mathbb{P}_\pi(\{T_\infty = \infty\}) = 1,$$

The above equation plays a starring role in the relationship between the rate matrix Q and the stationary distributions:

(6) **THEOREM** (The stationary distributions and the rate matrix). *A probability distribution π is a stationary distribution if and only if the chain is non-explosive when initialised with law π (that is, (5) holds) and*

$$(7) \quad \pi Q(x) = 0 \quad \forall x \in \mathcal{S},$$

or $\pi Q = 0$ in matrix notation.

Proof. The definitions in (26.2)–(26.3) of the jump matrix $(p(x, y))_{x, y \in \mathcal{S}}$ and jump rates $(\lambda(x))_{x \in \mathcal{S}}$ imply that π satisfies (7) if and only if it satisfies

$$(8) \quad \pi(x) \lambda(x) = \sum_{z \in \mathcal{S}} \pi(z) \lambda(z) p(z, x) \quad \forall x \in \mathcal{S}.$$

Suppose that π is a stationary distribution. We have already argued that π satisfies (5). Showing it satisfies (7) (equivalently, (8)) is also straightforward: multiplying the integral version (32.10) of the forward equations by $\pi(x')$, summing over x' in $\mathcal{S}$, applying Tonelli's theorem and (3), we find that

$$\begin{aligned} \pi(x) &= \sum_{x' \in \mathcal{S}} \pi(x') p_t(x', x) = \pi(x) e^{-\lambda(x)t} + \int_0^t \sum_{z \in \mathcal{S}} \left(\sum_{x' \in \mathcal{S}} \pi(x') p_s(x', z) \right) \lambda(z) p(z, x) e^{-\lambda(x)(t-s)} ds \\ &= \pi(x) e^{-\lambda(x)t} + \int_0^t \sum_{z \in \mathcal{S}} \pi(z) \lambda(z) p(z, x) e^{-\lambda(x)(t-s)} ds \\ &= \pi(x) e^{-\lambda(x)t} + \sum_{z \in \mathcal{S}} \pi(z) \lambda(z) p(z, x) \frac{1 - e^{-\lambda(x)t}}{\lambda(x)} \quad \forall x \in \mathcal{S}, \quad t \in [0, \infty). \end{aligned}$$

Fixing any $t > 0$, multiplying through by $\lambda(x)(1 - e^{-\lambda(x)t})^{-1}$, and re-arranging we obtain (8).

Conversely, suppose that π is a probability distribution satisfying (5) and (7) and let $p_t^n(x, y)$ be as in (32.8). By definition,

$$\begin{aligned} \sum_{x' \in \mathcal{S}} \pi(x') p_t^0(x', x) &= \sum_{x' \in \mathcal{S}} \pi(x') \mathbb{P}_{x'}(\{X_t = x, t < T_1\}) = \sum_{x' \in \mathcal{S}} \pi(x') \mathbb{P}_{x'}(\{X_0 = x, t < T_1\}) \\ &= \pi(x) \mathbb{P}_x(\{t < T_1\}) \leq \pi(x) \quad \forall x \in \mathcal{S}, \quad t \geq 0. \end{aligned}$$

Induction, the forward integral recursion in (32.9), and (8) then imply that

$$\begin{aligned} \sum_{x' \in \mathcal{S}} \pi(x') p_t^n(x', x) &= \pi(x) e^{-\lambda(x)t} + \int_0^t \sum_{z \in \mathcal{S}} \left(\sum_{x' \in \mathcal{S}} \pi(x') p_s^{n-1}(x', z) \right) \lambda(z) p(z, x) e^{-\lambda(x)(t-s)} ds \\ &\leq \pi(x) e^{-\lambda(x)t} + \int_0^t \sum_{z \in \mathcal{S}} \pi(z) \lambda(z) p(z, x) e^{-\lambda(x)(t-s)} ds \\ &= \pi(x) e^{-\lambda(x)t} + \int_0^t \pi(x) \lambda(x) e^{-\lambda(x)(t-s)} ds = \pi(x) \quad \forall x \in \mathcal{S}, \quad t \geq 0, \quad n \geq 0. \end{aligned}$$

Taking the limit $n \rightarrow \infty$ and applying monotone convergence we find that

$$(9) \quad \sum_{x' \in \mathcal{S}} \pi(x') p_t(x', x) \leq \pi(x) \quad \forall x \in \mathcal{S}.$$

Suppose that the inequality is strict for at least one x . Summing both sides we would have that

$$\mathbb{P}_\pi(\{t < T_\infty\}) = \sum_{x \in \mathcal{S}} \mathbb{P}_\pi(\{X_t = x, t < T_\infty\}) = \sum_{x \in \mathcal{S}} \sum_{x' \in \mathcal{S}} \pi(x') p_t(x', x) < \sum_{x \in \mathcal{S}} \pi(x) = 1$$

which violates (5). Hence, it must be the case that (9) holds with equality. $\square$

It is not difficult to find examples of probability distributions π s satisfying (7) that are not stationary distributions:

(10) EXAMPLE (Miller's example). Consider the chain X with state space $\mathbb{N}$ and rate matrix $(q(x, y))_{x, y \in \mathbb{N}}$ defined by

$$\begin{aligned} q(x, x-1) &= 4^x/2 \quad \forall x > 0, \quad q(x, x) = -(4^x + 4^x/2), \quad q(x, x+1) = 4^x \quad \forall x \geq 0, \\ q(x, y) &= 0 \quad \text{for all other } x, y \geq 0. \end{aligned}$$

It is easy to verify that $\pi := (2^{-(x+1)})_{x \in \mathbb{N}}$ is a probability distribution satisfying (7). However, π cannot be a stationary distribution for otherwise every state would be recurrent (see Theorem 15 in the next section), something impossible given that the chain is twice as likely to jump one state up than one state down (if you're not following this, plug Q into (26.2) and look at the definition of the jump chain in Algorithm 2).

For a more formal argument, note that a state is recurrent for X if and only if it is recurrent for its jump chain Y (Exercise 38.2). Let's examine the return probability $\mathbb{P}_0(\{\phi_0 < \infty\})$ to zero of Y , with ϕ_k denote the first entrance time of Y to a set k (Definition 13.1). Because the chain necessarily jumps to 1 if it starts at zero,

$$\mathbb{P}_0(\{\phi_0 < \infty\}) = \mathbb{P}_1(\{\sigma_0 < \infty\}),$$

where σ_k denotes the hitting time of Y to a state k (8.5). Because $\sigma_k \rightarrow \infty$ as $k \rightarrow \infty$ with $\mathbb{P}_1$ -probability one (Lemma 23.7), monotone convergence implies that

$$\mathbb{P}_1(\{\sigma_0 < \infty\}) = \lim_{k \rightarrow \infty} \mathbb{P}_1(\{\sigma_0 < \sigma_k\}).$$

Because Y behaves identically to the gambler's ruin chain in Section 3 with $a := 2/3$ up until the moment it hits 0, (11.18) implies that

$$\mathbb{P}_1(\{\sigma_0 < \sigma_k\}) = 1 - \frac{1}{2 - 2^{-(k-1)}} \quad \forall k > 0.$$

Thus, $\mathbb{P}_0(\{\phi_0 < \infty\}) = 1/2$ showing that 0 is transient for Y (and, consequently, for X) and that π cannot be a stationary distribution. Moreover, Theorem 6 then implies that $\mathbb{P}_\pi(\{T_\infty < \infty\}) > 0$. Because the chain is irreducible, it follows from Theorem 38.9 that the chain is explosive regardless of the initial distribution:

$$\mathbb{P}_\gamma(\{T_\infty < \infty\}) > 0 \text{ for all probability distributions } \gamma.$$

A stationary distribution has support on a state if and only if the state is positive recurrent. If the chain is a stationary process, the average fraction of time it spends in any given state remains constant throughout time. In particular, sampling the starting location from a stationary distribution π , taking expectations of the empirical distribution ϵ_T (Section 41), and applying (3) we find that

$$(11) \quad \mathbb{E}_\pi[\epsilon_T(x)] = \mathbb{E}_\pi \left[\frac{1}{T} \int_0^{T \wedge T_\infty} 1_x(X_t) dt \right] = \frac{1}{T} \int_0^T \mathbb{P}_\pi(\{t < T_\infty, X_t = x\}) dt = \pi(x)$$

for all x in $\mathcal{S}$ and $T \geq 0$. Given Theorem 41.2, taking the limit $T \rightarrow \infty$ and applying bounded convergence we find that

$$(12) \quad \pi(x) = \lim_{T \rightarrow \infty} \mathbb{E}_\pi[\epsilon_T(x)] = \frac{\mathbb{P}_\pi(\{\varphi_x < \infty\})}{\mathbb{E}_x[\varphi_x]} \quad \forall x \in \mathcal{S}.$$

It follows that a stationary distribution π with support on x (i.e., $\pi(x) > 0$) exists only if x is positive recurrent (Definition 16.3). The intuition here is that states x satisfying $\pi(x) > 0$ are those the chain will keep visiting frequently enough that the fraction of time $\epsilon_T(x)$ it spends in x does not decay to zero whenever the initial position is sampled from π . Because, regardless of the starting location, visits to all states x not positive recurrent eventually become so rare that $\epsilon_T(x)$ collapses to zero (Theorem 41.2) it follows that for $\pi(x)$ to be non-zero, x must be positive recurrent.

The converse is also true and the trick in arguing it involves the stopping distributions and occupation measures of Section 10. In particular, let μ_S and ν_S be the space marginals (35.6)–(35.7) of the stopping distribution and occupation measure associated with the return time φ_x of a recurrent state x and suppose that we started the chain in x . Exercise 38.2 implies that the chain is in x at the moment of return. Because x is recurrent, it follows that μ_S is the point mass 1_x at x . Furthermore, as we fixed the initial distribution γ to also be 1_x , the same proposition and Lemma 35.8 imply that

$$(13) \quad q(x)\nu_S(x) = \sum_{x' \neq x} \nu_S(x')q(x', x) \quad \forall x \in \mathcal{S}.$$

Because the state is recurrent, φ 's definition (Definition 38.1) implies that it is less than T_∞ , $\mathbb{P}_x$ almost surely. For this reason, (35.4) shows that the mass of ν_S is the mean return time $\mathbb{E}_x[\varphi_x]$. If x is positive recurrent, then the mass and, consequently, all entries of ν_S are finite. Thus, we can rearrange (13) into $\nu_S Q = 0$ and we obtain a probability distribution π satisfying $\pi Q = 0$ by normalising ν_S (i.e., $\pi := \nu_S / \nu_S(\mathcal{S})$). Moreover,

$$\begin{aligned} \pi(x) &= \frac{\nu_S(x)}{\nu_S(\mathcal{S})} = \frac{\mathbb{E}_x \left[\int_0^{\varphi_x \wedge T_\infty} 1_x(X_s) ds \right]}{\mathbb{E}_x[\varphi_x]} \geq \frac{\mathbb{E}_x \left[\int_0^{T_1} 1_x(X_s) ds \right]}{\mathbb{E}_x[\varphi_x]} = \frac{\mathbb{E}_x[S_1 1_x(X_0)]}{\mathbb{E}_x[\varphi_x]} \\ &= \frac{1}{\lambda(x) \mathbb{E}_x[\varphi_x]} > 0, \end{aligned}$$

where S_1 denotes the first waiting time (that is exponentially distributed with mean $\lambda(x)$ under $\mathbb{P}_x$, see Section 26). If we can only show that (5) is satisfied, Theorem 3 and the above then show that π a stationary distribution with support on x . To argue (5) note that Proposition 40.3 shows that $\mathbb{P}_x(\{T_\infty = \infty\}) = 1$. Because

$$(14) \quad \mathbb{E}_x[\varphi_x] \pi(y) = \nu_S(y) = \mathbb{E}_x \left[\int_0^{\varphi_x \wedge T_\infty} 1_x(X_s) ds \right] \leq \mathbb{E}_x \left[\int_0^{T_\infty} 1_x(X_s) ds \right] = \int_0^\infty p_t(x, y) ds$$

for all y in $\mathcal{S}$, Theorem 38.4(v) implies that $\pi(y) > 0$ only if y is accessible from x and it follows from Theorem 38.9 that

$$\mathbb{P}_y(\{T_\infty = \infty\}) = 1 \quad \forall y \in \mathcal{S} : \pi(y) > 0.$$

Multiplying the above by $\pi(y)$, summing over all states y , and comparing with (26.5) then yields the missing (5). To make a long story short:

(15) THEOREM. *A state x is positive recurrent if and only if there exists a stationary distribution π such that $\pi(x) > 0$. In this case, one such stationary distribution is given by*

$$(16) \quad \pi(y) = \frac{1}{\mathbb{E}_x[\varphi_x]} \mathbb{E}_x \left[\int_0^{\varphi_x \wedge T_\infty} 1_y(X_s) ds \right] \quad \forall y \in \mathcal{S}.$$

As a freebie, the above also gives us that positive recurrence is a class property:

(17) COROLLARY. *If x is positive recurrent and $x \rightarrow y$, then y is also positive recurrent. Moreover, a state in a communicating class is positive (resp. null) recurrent if and only if all states in the class are positive (resp. null) recurrent.*

Proof. Given Theorems 15, 38.4, and 17.7, we can argue this by following the states taken in the proof of the corollary's discrete-time counterpart (Corollary 17.8). Alternatively, given that π is a stationary distribution for X if and only if $(1_S(x)\pi(x))_{x \in S_E}$ is for any and all skeleton chains X^δ (Theorem 4) and that y is accessible from x for X if and only if it is accessible for any and all skeleton chains X^δ (Theorem 38.4(v)), the corollary follows by applying Corollary 17.8 to any skeleton chain X^δ . $\square$

For this reason, we say that a communicating class is *positive recurrent* (resp. *null recurrent*) if each one (or, equivalently, all) of its states is positive recurrent (resp. null recurrent).

Ergodic distributions; Doeblin-like decomposition; set of stationary distributions.

To wrap up and our treatment of stationary distributions and characterise the set of these, consider the following *Doeblin-like* decomposition of the state space:

$$(18) \quad \mathcal{S} = \left(\bigcup_{i \in \mathcal{I}} \mathcal{C}_i \right) \cup \mathcal{T} = \mathcal{R}_+ \cup \mathcal{T},$$

where $\{\mathcal{C}_i : i \in \mathcal{I}\}$ denotes the (necessarily countable) set of positive recurrent closed communicating classes, which we index with some set $\mathcal{I}$, $\mathcal{R}_+ := \bigcup_{i \in \mathcal{I}} \mathcal{C}_i$ the set of all positive recurrent states, and $\mathcal{T}$ that of all other states. As we have already seen in Theorem 17.7, no stationary distribution has support in $\mathcal{T}$ and there is at least one stationary distribution per $\mathcal{C}_i$. Much more can be said:

(19) THEOREM (Characterising the set of stationary distributions). *Let $\{\mathcal{C}_i : i \in \mathcal{I}\}$ be the collection of positive recurrent closed communicating classes.*

- (i) For each $\mathcal{C}_i$, there exists a single stationary distribution π_i with support contained in $\mathcal{C}_i$ (i.e., with $\pi_i(\mathcal{C}_i) = 1$); π_i is known as the ergodic distribution associated with $\mathcal{C}_i$. It has support on all of $\mathcal{C}_i$ ($\pi_i(x) > 0$ for all states x in $\mathcal{C}_i$) and can be expressed as

$$(20) \quad \pi_i(y) = \frac{1_{\mathcal{C}_i}(y)}{\mathbb{E}_y[\varphi_y]} = \frac{1}{\mathbb{E}_x[\varphi_x]} \mathbb{E}_x \left[\int_0^{\varphi_x \wedge T_\infty} 1_y(X_t) dt \right] \quad \forall y \in \mathcal{S},$$

where x is any state in $\mathcal{C}_i$.

- (ii) A probability distribution π is a stationary distribution of the chain if and only if it is a convex combination of the ergodic distributions:

$$\pi = \sum_{i \in \mathcal{I}} \theta_i \pi_i$$

for some collection $(\theta_i)_{i \in \mathcal{I}}$ of non-negative constants satisfying $\sum_{i \in \mathcal{I}} \theta_i = 1$. For any stationary distribution π , the weight θ_i featuring in the above is the mass that π awards to $\mathcal{C}_i$ or, equivalent, the probability that the chain ever enters $\mathcal{C}_i$ if its starting location was sampled from π :

$$\theta_i = \pi(\mathcal{C}_i) = \mathbb{P}_\pi(\{\varphi_{\mathcal{C}_i} < \infty\}) \quad \forall i \in \mathcal{I}.$$

Proof. (i) For any state x in a recurrent closed communicating class $\mathcal{C}_i$, Proposition 39.6 implies that

$$\mathbb{P}_x(\{\varphi_y < \infty\}) = 1_{\mathcal{C}_i}(y) \quad \forall y \in \mathcal{S}.$$

Plugging the above into (12) we find that there can only exist one stationary distribution π_i with support contained in $\mathcal{C}_i$ (i.e., $\pi_i(x) = 0$ for all $x \notin \mathcal{C}_i$), as for any such π_i

$$\pi_i(x) = \frac{\mathbb{P}_{\pi_i}(\{\varphi_x < \infty\})}{\mathbb{E}_x[\varphi_x]} = \frac{\sum_{x' \in \mathcal{C}_i} \pi_i(x') \mathbb{P}_{x'}(\{\varphi_x < \infty\})}{\mathbb{E}_x[\varphi_x]} = \frac{1_{\mathcal{C}_i}(x)}{\mathbb{E}_x[\varphi_x]} \quad \forall x \in \mathcal{S}.$$

On the other hand, Theorem 15 shows that the right-hand side of (20) defines such a stationary distribution π_i because Theorem 38.4(v), (14), and the closedness of $\mathcal{C}_i$ imply that π_i has support contained $\mathcal{C}_i$.

(ii) Given Proposition 39.6, Theorem 4, and (12), this proof is entirely analogous to that of its discrete-time counterpart (Theorem 17.10(ii)). $\square$

Notes and references. Theorem 6 traces back to (Kendall and Reuter, 1957) and (Miller, 1963) where it is shown that, for regular Q , (3) and (6) are equivalent. The minor extension in Theorem 6 seems to be due to (Kuntz, 2017)—although I still find this suspicious—and the proof given here closely follows (Miller, 1963).

44. Limits of the empirical distribution and positive recurrent chains. Combining the results of the previous sections we obtain a complete description of the empirical distribution ϵ_T (Section 41) that tracks the fraction of time that the chain spends in each state:

(1) **THEOREM (The pointwise limits).** Let ϵ_T denote the empirical distribution in (41.1), $\{\mathcal{C}_i : i \in \mathcal{I}\}$ denote the collection of positive recurrent closed communicating classes $\mathcal{C}_i$ (Section 43), and, for each $i \in \mathcal{I}$, π_i denote the ergodic distribution of $\mathcal{C}_i$ (Theorem 43.19). For any initial distribution γ , we have that

$$(2) \quad \epsilon_\infty := \lim_{T \rightarrow \infty} \epsilon_T = \sum_{i \in \mathcal{I}} 1_{\{\varphi_{\mathcal{C}_i} < \infty\}} \pi_i, \quad \mathbb{P}_\gamma\text{-almost surely,}$$

where the convergence is pointwise and $\varphi_{\mathcal{C}_i}$ denotes the time of first entrance to $\mathcal{C}_i$ (Definition 38.1).

Proof. Because, with $\mathbb{P}_\gamma$ -probability one, the chain will enter a state x (i.e., $\varphi_x < \infty$) in a recurrent closed communicating class $\mathcal{C}_i$ if and only if enters the class (i.e., $\varphi_{\mathcal{C}_i} < \infty$) at all (Proposition 39.6), this follows directly from Theorems 41.2 and 43.19(i). $\square$

It is simple to describes the circumstances under which the limit (2) holds in total variation:

(3) COROLLARY (The convergence in total variation). *The chain enters the set $\mathcal{R}_+$ of positive recurrent states with probability one (i.e., $\mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\}) = 1$) if and only if the limit (2) holds in total variation $\mathbb{P}_\gamma$ -almost surely: with $\|\cdot\|$ as in (18.4),*

$$\lim_{T \rightarrow \infty} \|\epsilon_T - \epsilon_\infty\| = 0, \quad \mathbb{P}_\gamma\text{-almost surely,}$$

Proof. If the chain is non-explosive, then $\epsilon_T(\mathcal{S}) = 1$ with $\mathbb{P}_\gamma$ -probability one, for all T in $[0, \infty)$. Thus, if the limit (2) holds in total variation, then

$$1 = \lim_{n \rightarrow \infty} \epsilon_n(\mathcal{S}) = \epsilon_\infty(\mathcal{S}) = \sum_{i \in \mathcal{I}} 1_{\{\varphi_{\mathcal{C}_i} < \infty\}} \pi_i(\mathcal{S}) = \sum_{i \in \mathcal{I}} 1_{\{\varphi_{\mathcal{C}_i} < \infty\}} \quad \mathbb{P}_\gamma\text{-almost surely.}$$

Because the positive recurrent classes are closed and disjoint, it follows from Proposition 38.8 that

$$(4) \quad \sum_{i \in \mathcal{I}} 1_{\{\varphi_{\mathcal{C}_i} < \infty\}} = 1_{\{\varphi_{\cup_{i \in \mathcal{I}} \mathcal{C}_i} < \infty\}} = 1_{\{\varphi_{\mathcal{R}_+} < \infty\}} \quad \mathbb{P}_\gamma\text{-almost surely.}$$

Putting the above two together, we have that $\mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\}) = 1$.

Conversely, if $\mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\}) = 1$, then Proposition 40.3 and the strong Markov property (Theorem 30.7) imply that the chain is non-explosive:

$$(5) \quad \begin{aligned} \mathbb{P}_\gamma(\{T_\infty = \infty\}) &= \mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < T_\infty, T_\infty = \infty\}) \\ &= \sum_{x \in \mathcal{R}_+} \mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < T_\infty, X_{\varphi_{\mathcal{R}_+}} = x, T_\infty^{\varphi_{\mathcal{R}_+}} - \varphi_{\mathcal{R}_+} = \infty\}) \\ &= \sum_{x \in \mathcal{R}_+} \mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < T_\infty, X_{\varphi_{\mathcal{R}_+}} = x\}) \mathbb{P}_x(\{T_\infty = \infty\}) \\ &= \sum_{x \in \mathcal{R}_+} \mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < T_\infty, X_{\varphi_{\mathcal{R}_+}} = x\}) = \mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\}) = 1, \end{aligned}$$

where $T_\infty^{\varphi_{\mathcal{R}_+}}$ is as in (30.4) and the second and penultimate equations follow from Exercise 38.2. In other words, the empirical distribution ϵ_T has mass one (i.e., $\epsilon_T(\mathcal{S}) = 1$) for all T in $[0, \infty)$, with $\mathbb{P}_\gamma$ -probability one. Furthermore, (4) implies that $\epsilon_\infty(\mathcal{S}) = 1$ $\mathbb{P}_\gamma$ -almost surely. Thus, the pointwise convergence (Theorem 1) and Scheffe's lemma (Lemma 18.5) imply that

$$(6) \quad \lim_{N \rightarrow \infty} \left\| \epsilon_{\delta N} - \sum_{i \in \mathcal{I}} 1_{\{\varphi_{\mathcal{C}_i} < \infty\}} \pi_i \right\| = 0 \quad \mathbb{P}_\gamma\text{-almost surely,}$$

for every δ in $(0, \infty)$. That the limit (2) holds in total variation then follows from Croft's lemma (Corollary 42.12) assuming that we are able to show that

$$T \mapsto \left\| \epsilon_T - \sum_{i \in \mathcal{I}} 1_{\{\varphi_{\mathcal{C}_i} < \infty\}} \pi_i \right\|$$

is a continuous function on $[0, \infty)$, with $\mathbb{P}_\gamma$ -probability one. As the total variation distance between two probability distributions is half the ℓ^1 -distance (see (18.6)), we need only show that

$$(7) \quad T \mapsto \sum_{x \in \mathcal{S}} \left| \epsilon_T(x) - \sum_{i \in \mathcal{I}} 1_{\{\varphi_{\mathcal{C}_i} < \infty\}} \pi_i(x) \right|$$

is a continuous function on $[0, \infty)$, with $\mathbb{P}_\gamma$ -probability one.

Let $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ be an increasing sequence of finite truncations approaching the state space (i.e., such that $\cup_{r=1}^\infty \mathcal{S}_r = \mathcal{S}$). The function in (7) is the pointwise limit of the sequence

$$\left\{ T \mapsto \sum_{x \in \mathcal{S}_r} \left| \epsilon_T(x) - \sum_{i \in \mathcal{I}} 1_{\{\varphi_{C_i} < \infty\}} \pi_i(x) \right| \right\}_{r \in \mathbb{Z}_+}$$

of functions. Because, for all x , $T \mapsto \epsilon_T(x)$ is continuous by its definition in (41.1) and the truncations are finite, all of the functions in the sequence are continuous. For this reason, to show continuity of (7) we need only to argue that the sequence converges uniformly on $[0, T^*]$, for every $T^* \in (0, \infty)$. Fix such a T^* and note that

$$\begin{aligned} \sum_{x \notin \mathcal{S}_r} \left| \epsilon_T(x) - \sum_{i \in \mathcal{I}} 1_{\{\varphi_{C_i} < \infty\}} \pi_i(x) \right| &\leq \sum_{x \notin \mathcal{S}_r} \left(\epsilon_T(x) + \sum_{i \in \mathcal{I}} 1_{\{\varphi_{C_i} < \infty\}} \pi_i(x) \right) \\ (8) \qquad \qquad \qquad &= \frac{1}{T} \int_0^{T \wedge T_\infty} 1_{\mathcal{S}_r^c}(X_t) dt + \sum_{i \in \mathcal{I}} 1_{\{\varphi_{C_i} < \infty\}} \pi_i(\mathcal{S}_r^c). \end{aligned}$$

The rightmost term does not depend on T . Given that the chain enters only one positive recurrent class (Proposition 38.8), this term can be made arbitrarily small by choosing large enough r . Furthermore, (5) implies that the set $S(\omega)$ of states that the chain's path $t \mapsto X_t(\omega)$ visits over the interval $[0, T^* \wedge T_\infty(\omega)]$ is finite (i.e., $T_n(\omega) \geq T^*$ for some sufficiently large n), for $\mathbb{P}_\gamma$ -almost every ω in Ω . Because the truncations are increasing and approach the entire state space ($\cup_{r=1}^\infty \mathcal{S}_r = \mathcal{S}$), it follows that $S(\omega)$ is contained in $\mathcal{S}_r$ for all sufficiently large r and the first term in the right-hand side of (8) (evaluated at ω) is zero for all such r s and T in $[0, T^*]$. $\square$

Positive recurrent chains. Corollary 3 motivates the following definitions:

(9) **DEFINITION** (Positive recurrent chains). *A chain is positive Tweedie recurrent if*

$$\mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\}) = 1 \text{ for all initial distributions } \gamma,$$

where $\mathcal{R}_+$ denotes the set of positive recurrent states. If, additionally, there is only one closed communicating class, then the chain is said to be Harris recurrent. If this class is the entire state space, then the chain is simply said to be positive recurrent.

The corollary shows that the empirical distribution converges in total variation, $\mathbb{P}_\gamma$ -almost surely, for every initial distribution γ if and only if the chain is positive Tweedie recurrent. Using analogous arguments to those given in Section 18 for the discrete-time case, it then follows that:

1. If the chain is positive Tweedie recurrent, then, for any given initial distribution γ and ε in $(0, 1]$, we are always able to find a finite set F such that the chain spends at least $(1 - \varepsilon) \times 100\%$ of all time inside F :

$$\epsilon_\infty(F) := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{T \wedge T_\infty} 1_F(X_t) dt \geq 1 - \varepsilon \quad \mathbb{P}_\gamma\text{-almost surely.}$$

2. Otherwise, there exists at least one initial distribution γ for which a *non-negligible fraction* of X 's paths will spend infinitely more time outside any given finite set F than inside the set:

$$\mathbb{P}_\gamma(\{\epsilon_\infty(F) = 0 \text{ for all finite } F \subseteq \mathcal{S}\}) \geq \mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} = \infty\}) > 0.$$

For these reasons, I believe that positive Tweedie recurrence precisely captures what most practitioners think of when they hear the words ‘a stable chain’.

45. Limits of the time varying-law, tightness, and ergodicity. The main aim of this section is to prove the theorem below spelling out the long-term behaviour of the time-varying law $(p_t)_{t \geq 0}$ (introduced in Section 33).

(1) **THEOREM** (The pointwise limits). *Let $\{\mathcal{C}_i : i \in \mathcal{I}\}$ denote the collection of positive recurrent closed communicating classes and let π_i be the ergodic distribution of $\mathcal{C}_i$ for each $i \in \mathcal{I}$ (c.f. Theorem 43.19). For any initial distribution γ ,*

$$(2) \quad \lim_{t \rightarrow \infty} p_t = \sum_{i \in \mathcal{I}} \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}_i} < \infty\}) \pi_i =: \pi_\gamma,$$

where the convergence is pointwise and $\varphi_{\mathcal{C}_i}$ denotes the time of first entrance time to $\mathcal{C}_i$.

Proof. See the end of the section. □

Convergence in total variation and tightness. The question of when does the limit in (2) hold in total variation has a straightforward answer involving the notion of *tightness*⁷:

(3) **DEFINITION** (Tightness). *A set $\{\rho_t : t \in [0, \infty)\}$ of probability distributions on $\mathcal{S}$ indexed by $t \in [0, \infty)$ is tight if and only if for every $\varepsilon > 0$ there exists a finite set F such that $\rho_t(F) \geq 1 - \varepsilon$ for all t in $[0, \infty)$.*

We then have the following corollary of Theorem 1:

(4) **COROLLARY.** *The following are equivalent:*

- (i) *The chain enters the set of positive recurrent states with probability one: $\mathbb{P}_\gamma(\{\phi_{\mathcal{R}_+} < \infty\}) = 1$.*
- (ii) *The chain is non-explosive ($\mathbb{P}_\gamma(\{T_\infty = \infty\}) = 1$) and the time-varying law is tight.*
- (iii) *The chain is non-explosive ($\mathbb{P}_\gamma(\{T_\infty = \infty\}) = 1$) and the limit (2) holds in total variation: with $\|\cdot\|$ as in (18.4),*

$$(5) \quad \lim_{t \rightarrow \infty} \|p_t - \pi_\gamma\| = 0.$$

Proof. See the end of the section. □

Positive Tweedie recurrence revisited. In Section 44, we characterised positive Tweedie recurrent chains (Definition 44.9) in terms of the limits of the time-varying law ϵ_T in (41.1). Armed with Corollary 4, it is straightforward to improve this characterisation:

(6) **COROLLARY** (Characterising positive Tweedie recurrent chains). *The following conditions are equivalent:*

- (i) *The chain is positive Tweedie recurrent.*
- (ii) *The chain is non-explosive ($\mathbb{P}_\gamma(\{T_\infty = \infty\}) = 1$) and the empirical distribution converges in total variation to ϵ_∞ in (44.2) with $\mathbb{P}_\gamma$ -probability one, for all initial distributions γ .*
- (iii) *The chain is non-explosive ($\mathbb{P}_\gamma(\{T_\infty = \infty\}) = 1$) and the time-varying law is tight, for all initial distributions γ .*
- (iv) *The chain is non-explosive ($\mathbb{P}_\gamma(\{T_\infty = \infty\}) = 1$) and the time-varying law converges in total variation (i.e., (5)), for all initial distributions γ .*

Proof. This follows directly from Corollaries 44.3 and 4 and the definition of positive Tweedie recurrence (Definition 44.9). □

In other words, a chain is positive Tweedie recurrent if and only if for each initial distribution γ and ε in $(0, 1]$, we can find a large enough finite set F such that the chain has at least $1 - \varepsilon$ probability of being at F at any given time (i.e., $p_t(F) \geq 1 - \varepsilon$ for all t): further reinforcing the idea discussed in Section 44 that a chain is stable if and only if it is positive Tweedie recurrent.

⁷Here, we are topologising the state space using the discrete metric so that the compact sets are the finite sets.

Ergodicity and positive Harris recurrence. In the case of a positive Harris recurrent chain with a single closed communicating class, Theorem 43.19 and Corollary 6 show that the chain has a unique stationary distribution π and that, regardless of the initial distribution, both the time-varying law and the empirical distribution converge to it:

$$\lim_{T \rightarrow \infty} \epsilon_T = \lim_{t \rightarrow \infty} p_t = \pi \quad \mathbb{P}_\gamma\text{-almost surely,}$$

for all initial distributions γ , where the convergence is in total variation. That is, the *time averages* ϵ_T converge to the *space averages* p_t and the chain is said to be *ergodic*.

A proof of Theorem 1. In (42.10), we showed that

$$(7) \quad \lim_{t \rightarrow \infty} p_t(x) = \sum_{i \in \mathcal{I}^\delta} \mathbb{P}_\gamma(\{\phi_{\mathcal{C}_i^\delta}^\delta < \infty\}) \pi_i^\delta(x) \quad \forall x \in \mathcal{S},$$

for any δ in $(0, \infty)$, where $\{\mathcal{C}_i^\delta : i \in \mathcal{I}^\delta\}$ denote the collection of positive recurrent closed communicating classes of the skeleton chain X^δ (Section 42), $\{\pi_i^\delta : i \in \mathcal{I}^\delta\}$ the corresponding collection of X^δ 's ergodic distributions, and $\phi_{\mathcal{C}_i^\delta}^\delta$ the first entrance time to $\mathcal{C}_i^\delta$ of X^δ :

$$\phi_A^\delta(\omega) := \inf\{n > 0 : X_n^\delta(\omega) \in A\} \quad \forall \omega \in \Omega, \quad A \subseteq \mathcal{S}.$$

Lemma 13.6, Theorem 38.4(vi), and (42.3) imply that X and X^δ have the same closed communicating classes in $\mathcal{S}$. Similarly, Theorems 17.7, 17.10(i), 43.4, 43.15, and 43.19(i) imply that X and X^δ have the same positive recurrent states in $\mathcal{S}$ and that $\pi = (\pi(x))_{x \in \mathcal{S}}$ is an ergodic distribution for X if and only if $\pi_E = (1_{\mathcal{S}}(x)\pi(x))_{x \in \mathcal{S}_E}$ is an ergodic distribution for X^δ . Thus, we can rewrite the (7) as

$$\lim_{t \rightarrow \infty} p_t(x) = \sum_{i \in \mathcal{I}} \mathbb{P}_\gamma(\{\phi_{\mathcal{C}_i}^\delta < \infty\}) \pi_i(x) \quad \forall x \in \mathcal{S}.$$

Consequently, all we have left to show is that

$$(8) \quad \mathbb{P}_\gamma(\{\phi_{\mathcal{C}}^\delta < \infty\}) = \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}} < \infty\})$$

for any given positive recurrent closed communicating class $\mathcal{C} \subseteq \mathcal{S}$ of X (or, equivalently, of X^δ). Because X^δ is obtained by sampling X every δ units of time (c.f. (42.1)), the above follows from X being unable to leave $\mathcal{C}$ or explode once inside $\mathcal{C}$. In particular, letting $X^{\varphi_{\mathcal{C}}}$ denote the $\varphi_{\mathcal{C}}$ -shifted chain (Section 30), we have that

$$\begin{aligned} & \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}} < \infty, T_\infty = \infty, X_t \in \mathcal{C} \quad \forall t \in [\varphi_{\mathcal{C}}, \infty)\}) \\ &= \sum_{x \in \mathcal{C}} \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}} < T_\infty, X_{\varphi_{\mathcal{C}}} = x, T_\infty^{\varphi_{\mathcal{C}}} = \infty, X_t^{\varphi_{\mathcal{C}}} \in \mathcal{C} \quad \forall t \in [0, \infty)\}) \\ &= \sum_{x \in \mathcal{C}} \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}} < T_\infty, X_{\varphi_{\mathcal{C}}} = x\}) \mathbb{P}_x(\{T_\infty = \infty, X_t \in \mathcal{C} \quad \forall t \in [0, \infty)\}) \\ &= \sum_{x \in \mathcal{C}} \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}} < T_\infty, X_{\varphi_{\mathcal{C}}} = x\}) = \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}} < \infty\}), \end{aligned}$$

where the first equation follows from Exercise 38.2, the second from the strong Markov property (Theorem 30.7), the third from Propositions 38.7 and 40.3, and the fourth from Exercise 38.2. It then follows from the definition of the skeleton chain in (42.1) that

$$(9) \quad \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}} < \infty\}) = \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}} < \infty, T_\infty = \infty, X_{\delta n} \in \mathcal{C} \quad \forall n \geq \varphi_{\mathcal{C}}/\delta\}) \leq \mathbb{P}_\gamma(\{\phi_{\mathcal{C}}^\delta < \infty\}).$$

Similarly, Proposition 38.7's discrete-time counterpart (Proposition 13.9) and the discrete-time strong Markov property (Theorem 12.4) imply that

$$\begin{aligned} \mathbb{P}_\gamma(\{\phi_{\mathcal{C}}^\delta < \infty, X_n^\delta \in \mathcal{C} \ \forall n \geq \phi_{\mathcal{C}}^\delta\}) &= \sum_{x \in \mathcal{C}} \mathbb{P}_\gamma(\{\phi_{\mathcal{C}}^\delta < \infty, X_{\phi_{\mathcal{C}}^\delta}^\delta = x, X_n^\delta \in \mathcal{C} \ \forall n \geq \phi_{\mathcal{C}}^\delta\}) \\ &= \sum_{x \in \mathcal{C}} \mathbb{P}_\gamma(\{\phi_{\mathcal{C}}^\delta < \infty, X_{\phi_{\mathcal{C}}^\delta}^\delta = x\}) \mathbb{P}_\gamma(\{X_n^\delta \in \mathcal{C} \ \forall n \geq 0\}) \\ &= \sum_{x \in \mathcal{C}} \mathbb{P}_\gamma(\{\phi_{\mathcal{C}}^\delta < \infty, X_{\phi_{\mathcal{C}}^\delta}^\delta = x\}) = \mathbb{P}_\gamma(\{\phi_{\mathcal{C}}^\delta < \infty\}). \end{aligned}$$

Thus, the definition of the skeleton chain in (42.1) implies that

$$\mathbb{P}_\gamma(\{\phi_{\mathcal{C}}^\delta < \infty\}) = \mathbb{P}_\gamma(\{\phi_{\mathcal{C}}^\delta < \infty\} \cap \{X_{\delta n} \in \mathcal{C}, \ \delta n < T_\infty, \ \forall n \geq \phi_{\mathcal{C}}^\delta\}) \leq \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}} < \infty\}).$$

Putting the above together with (9) then yields (8).

A proof of Corollary 4. We do this proof in parts:

(i) $\Rightarrow$ (iii) Taking expectations of (44.4), we find that

$$(10) \quad \pi_\gamma(\mathcal{S}) = \sum_{i \in \mathcal{I}} \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}_i} < \infty\}) \pi_i(\mathcal{S}) = \sum_{i \in \mathcal{I}} \mathbb{P}_\gamma(\{\varphi_{\mathcal{C}_i} < \infty\}) = \mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\}).$$

Suppose that (i) holds. Downwards monotone convergence, Fatou's lemma, and (10) imply that the chain is non-explosive:

$$(11) \quad \mathbb{P}_\gamma(\{T_\infty = \infty\}) = \lim_{n \rightarrow \infty} \mathbb{P}_\gamma(\{n < T_\infty\}) = \lim_{n \rightarrow \infty} p_n(\mathcal{S}) \geq \pi_\gamma(\mathcal{S}) = \mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\}) = 1.$$

Thus, $(p_{\delta n})_{n \in \mathbb{N}}$ is a sequence of probability distributions, for any given δ in $(0, \infty)$. For this reason Theorem 1 and Scheffe's lemma (Lemma 18.5) imply that the sequence converges to π_γ in total variation. Thus, if we are able to show that

$$(12) \quad t \mapsto \|p_t - \pi_\gamma\|$$

is a continuous function on $[0, \infty)$, the desired (5) then follows from Croft's theorem (Corollary 42.12). For this, it suffices to show that (12) is a continuous function on $[0, T]$ for any given $T > 0$. Because the total variation distance between two probability distributions is half the ℓ^1 -distance (see (18.6)), we need to show that

$$(13) \quad t \mapsto \sum_{x \in \mathcal{S}} |p_t(x) - \pi_\gamma(x)|$$

is a continuous function on $[0, T]$. Let $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ be a sequence of increasing finite sets approaching the state space (i.e., $\cup_{r=1}^\infty \mathcal{S}_r = \mathcal{S}$) and note that (13) is the pointwise limit of the sequence of functions

$$\left\{ t \mapsto \sum_{x \in \mathcal{S}_r} |p_t(x) - \pi_\gamma(x)| \right\}_{r \in \mathbb{Z}_+}.$$

Because the sets are finite and because $t \mapsto p_t(x)$ is a continuous function on $[0, \infty)$ (see the proof of Theorem 33.3), arguing that the convergence is uniform over $t \in [0, T]$ completes the proof. However,

$$\begin{aligned} \sum_{x \notin \mathcal{S}_r} |p_t(x) - \pi_\gamma(x)| &\leq \sum_{x \notin \mathcal{S}_r} p_t(x) + \sum_{x \notin \mathcal{S}_r} \pi_\gamma(x) = \mathbb{P}_\gamma(\{X_t \notin \mathcal{S}_r\}) + \pi_\gamma(\mathcal{S}_r^c) \\ &\leq \mathbb{P}_\gamma(\{\tau_r \leq t\}) + \pi_\gamma(\mathcal{S}_r^c) \leq \mathbb{P}_\gamma(\{\tau_r \leq T\}) + \pi_\gamma(\mathcal{S}_r^c) \quad \forall t \in [0, T], \end{aligned}$$

and, the uniform convergence follows from Theorem 26.10 and (11).

(iii) $\Rightarrow$ (i) Suppose that (iii) holds. Non-explosiveness implies that $p_n(\mathcal{S}) = \mathbb{P}_\gamma(\{T_\infty > n\}) = 1$ for all positive integers n . For this reason, the total variation convergence implies that the limit π_γ is a probability distribution and it follows from (10) that $\mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\}) = 1$.

(i) $\Rightarrow$ (ii) Suppose that (i) holds and fix any $\varepsilon > 0$. Because (iii) also holds (see above), we can find a T such that

$$(14) \quad \|p_t - \pi_\gamma\| \leq \frac{\varepsilon}{2} \quad \forall t \in (T, \infty).$$

Suppose for now that $t \mapsto p_t$ is a continuous function from $[0, T]$ to the space ℓ^1 of absolutely summable sequences indexed by states x in $\mathcal{S}$,

$$\ell^1 := \left\{ \rho : \mathcal{S} \rightarrow \mathbb{R} : \sum_{x \in \mathcal{S}} |\rho(x)| < \infty \right\},$$

topologised by the total variation norm (here, technically, I'm tacitly using $\rho(A) = \sum_{x \in A} \rho(x)$ to identify ℓ^1 with the space of signed measures on $\mathcal{S}$ with finite total variation). Because the Heine-Borel theorem shows that continuous functions between metric spaces (like $[0, \infty)$ and ℓ^1) are uniformly continuous on compact sets (like $[0, T]$), there exists a δ such that

$$(15) \quad \begin{aligned} \|p_t - p_{\delta n}\| &\leq \frac{\varepsilon}{2} \quad \forall t \in [\delta n, \delta(n+1)] \quad n = 0, 1, \dots, \lfloor T/\delta \rfloor - 1, \\ \|p_t - p_{\delta \lfloor T/\delta \rfloor}\| &\leq \frac{\varepsilon}{2} \quad \forall t \in [\delta \lfloor T/\delta \rfloor, T]. \end{aligned}$$

Because (10)–(11) imply that π_γ and p_t are probability distributions for all t in $[0, \infty)$, we can find a finite set F large enough that

$$p_{\delta n}(F) \geq 1 - \frac{\varepsilon}{2} \quad \forall n = 0, 1, \dots, \lfloor T/\delta \rfloor, \quad \pi_\gamma(F) \geq 1 - \frac{\varepsilon}{2},$$

and it follows from (15) that

$$p_t(F) \geq 1 - \varepsilon \quad \forall t \in [0, T].$$

Combining the above with (14), we obtain

$$\begin{aligned} p_t(F) &\geq 1_{[0, T]}(t)p_t(F) + 1_{(T, \infty)}(t)p_t(F) \\ &\geq 1_{[0, T]}(t)(1 - \varepsilon) + 1_{(T, \infty)}(t)(\pi_\gamma(F) + \|p_t - \pi_\gamma\|) \geq 1 - \varepsilon \quad \forall t \in [0, \infty). \end{aligned}$$

Because the ε was arbitrary, we have that $\{p_t : t \in [0, \infty)\}$ is tight.

To complete the proof we need to show that $t \mapsto p_t$ is a continuous function from $[0, \infty)$ to ℓ^1 . To do so, we need only show that $t \mapsto p_t$ is a continuous function on $[0, T]$ for any given $T > 0$. Let $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ be any sequence of increasing finite sets approaching the state space (i.e., $\bigcup_{r=1}^\infty \mathcal{S}_r = \mathcal{S}$) and note that $t \mapsto p_t$ is the pointwise limit of $(t \mapsto 1_{\mathcal{S}_r} p_t)_{r \in \mathbb{Z}_+}$:

$$p_t(x) = \lim_{r \rightarrow \infty} 1_{\mathcal{S}_r}(x)p_t(x) \quad \forall x \in \mathcal{S}, \quad t \in [0, \infty).$$

Because $t \mapsto p_t(x)$ is a continuous function from $[0, T]$ to $[0, 1]$ for each x in $\mathcal{S}$ (see the proof of Theorem 33.3), $t \mapsto 1_x p_t(x)$ is a continuous function from $[0, T]$ to ℓ^1 for each x in $\mathcal{S}$. Because the truncations are finite, it follows that each of the functions in the sequence $(t \mapsto 1_{\mathcal{S}_r} p_t)_{r \in \mathbb{Z}_+}$ is a continuous function from $[0, T]$ to ℓ^1 and we only need to argue that the convergence is uniform over t in $[0, T]$. However,

$$\begin{aligned} \|p_t - 1_{\mathcal{S}_r} p_t\| &= \sup_{A \subseteq \mathcal{S}} \left| \sum_{x \in A} p_t(x) - \sum_{x \in A} 1_{\mathcal{S}_r}(x)p_t(x) \right| = \sup_{A \subseteq \mathcal{S}} \sum_{x \in A \cap \mathcal{S}_r^c} p_t(x) = \sum_{x \in \mathcal{S}_r^c} p_t(x) \\ &= \mathbb{P}_\gamma(\{t < T_\infty, X_t \notin \mathcal{S}_r\}) \leq \mathbb{P}_\gamma(\{\tau_r \leq t\}) \leq \mathbb{P}_\gamma(\{\tau_r \leq T\}) \quad \forall t \in [0, T], \end{aligned}$$

where τ_r denotes the exit time from $\mathcal{S}_r$ (c.f. (26.9)). Because these exit times approach T_∞ as $r \rightarrow \infty$ (Theorem 26.10), the uniform convergence follows from the above and (11).

(ii) $\Rightarrow$ (i) Suppose that (ii) holds. Theorem 1 and (10) imply that

$$\lim_{t \rightarrow \infty} p_t(F) \leq \mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\})$$

for any finite set F . Because the chain is non-explosive, p_t has mass one (i.e., $p_t(\mathcal{S}) = 1$) for all t in $[0, \infty)$ and the above implies that the time-varying law is tight only if $\mathbb{P}_\gamma(\{\varphi_{\mathcal{R}_+} < \infty\}) = 1$.

46. Exponential recurrence and convergence*. The aim of this section is to prove the continuous-time analogue of Kendall's Theorem 21.1. It shows that, in the non-explosive case, $p_t(x, x)$ converges exponentially fast if and only if the state x is *exponentially recurrent*. That is, if and only if the tails of the return time distribution to x are light:

(1) THEOREM. *Let x denote any state belonging to a positive recurrent class $\mathcal{C}$ and let π be the ergodic distribution associated with $\mathcal{C}$.*

- (i) $p_t(x, x)$ converges geometrically fast to $\pi(x)$: $|p_t(x, x) - \pi(x)| = \mathcal{O}(e^{-\alpha t})$ for some $\alpha > 0$.
- (ii) x is geometrically recurrent: $\mathbb{E}_x[e^{\beta \varphi_x}] < \infty$ for some $\beta > 0$, where φ_x denotes the first entrance time to x .

An open question. Are there necessary and sufficient conditions in terms of the return time φ_x to x and the initial distribution γ for the x -entry of the time-varying law to converge exponentially fast? That is, conditions for

$$|p_t(x) - \pi_\gamma(x)| = \mathcal{O}(e^{-\alpha t}) \text{ to hold for some } \alpha > 0,$$

where π_γ denotes p_t 's limit in (45.2).

A proof of Theorem 1. We use skeleton chains make the jump from Kendall's Theorem to its continuous-time counterpart (Theorem 1). The bulk of the work we need to do here consists of showing that a state is exponentially recurrent if and only if it is geometrically recurrent (Section 21) for at least one skeleton chain:

(2) LEMMA. *For any given δ in $(0, \infty)$ and x in $\mathcal{S}$, let*

$$\phi_x^\delta(\omega) := \inf\{n \in \mathbb{Z}_+ : X_n^\delta(\omega) = x\} \quad \forall \omega \in \Omega$$

denote the first entrance time to x of the skeleton chain X^δ in (42.1). The following statements are equivalent:

- (i) *The return time distribution of X to x has light tails: there exists an $\beta > 0$ such that $\mathbb{E}_x[e^{\beta \varphi_x}] < \infty$.*
- (ii) *There exists a δ in $(0, \infty)$ such that the return time distribution of X^δ to x has light tails: there exists an $\theta > 1$ such that $\mathbb{E}_x[\theta^{\phi_x^\delta}] < \infty$.*

For the lemma's proof, we require the following fact:

(3) EXERCISE. *Recall Exercise 38.2 showing that, for any natural number k and state x , the k th entrance time φ_x^k to x of X is finite if and only if the k th entrance time ϕ_x^k to x of the jump chain is finite in which case $\varphi_x^k = T_{\phi_x^k}$. For any positive integer k , and assuming that the chain starts at x , let*

$$U_k := \begin{cases} S_{\phi_x^{k-1}+1} & \text{on } \{\phi_x^{k-1} < \infty\} \\ 0 & \text{on } \{\phi_x^{k-1} = \infty\} \end{cases} \quad \text{and} \quad W_{k+1} := \begin{cases} \varphi_x^k - \varphi_x^{k-1} - S_{\phi_x^{k-1}+1} & \text{on } \{\phi_x^{k-1} < \infty\} \\ 0 & \text{on } \{\phi_x^{k-1} = \infty\} \end{cases}$$

denote the time spent in state x on the $(k-1)$ th visit and the time elapsed between the $(k-1)$ th and k th visits, respectively (with the convention that these times are zero if the k th visit does not occur). By adapting the proof of the chain's regenerative property (Theorem 40.1), show that, under $\mathbb{P}_x$, the sequence

$$(U_1, W_1, U_2, W_2, \dots)$$

is independent and that the sojourn times U_k are identically distributed with

$$\mathbb{P}_x(\{U_k < t\}) = \mathbb{P}_x(\{U_1 < t\}) = \mathbb{P}_x(\{S_1 < t\}) = 1 - e^{-\lambda(x)t} \quad \forall t \in [0, \infty), \quad k \in \mathbb{Z}_+.$$

Hint: note that U_k may be re-written as I_k^f in (40.2) with $f := 1_x$ and W_k may be re-written as I_k^f with $f := 1_{S \setminus \{x\}}$.

Proof of Lemma 2. The lemma follows trivially if x is an absorbing state. Assume otherwise.

(i) $\Rightarrow$ (ii): Set $\theta := e^\beta$ (so that $\mathbb{E}_x[\theta_x^\varphi] = \mathbb{E}_x[e^{\beta\varphi_x}] < \infty$) and let $(U_k)_{k \in \mathbb{Z}_+}$ and $(W_k)_{k \in \mathbb{Z}_+}$ be the sojourn times and inter-visit times defined in Exercise 3. Throughout this proof fix a δ in $(0, \infty)$ small enough that

$$\mathbb{P}_x(\{S_1 < \delta\}) \theta^\delta \mathbb{E}_x[\theta^{\varphi_x}] < 1.$$

The skeleton chain X^δ can skip visits that the continuous-time chain X to x and, so, it might be the case that $\delta\phi_x^\delta > \varphi_x$. To deal with this possibility, let

$$\sigma(\omega) := \inf\{k \in \mathbb{N} : \{n \in \mathbb{Z}_+ : \varphi_x^k(\omega) \leq \delta n < \varphi_x^k(\omega) + U_{k+1}(\omega)\} \neq \emptyset\} \quad \forall \omega \in \Omega$$

denote the visit of X to x during which X^δ first returns to x . The definition of the skeleton chain X^δ in (42.1) implies that σ is finite if and only if X^δ returns to x at some point, i.e., if and only if ϕ_x^δ is finite. Because, by our premise, x is a positive recurrent state for X and because X and X^δ share the same positive recurrent states (c.f. Theorems 17.7, 43.4, and 43.15), it follows that σ is finite $\mathbb{P}_x$ -almost surely. Thus,

$$\delta\phi_x^\delta \leq 1_{\{\sigma < \infty\}} \varphi_x^\sigma + \delta, \quad \mathbb{P}_x\text{-almost surely.}$$

Given that we may rewrite φ_x^k as $\sum_{k'=1}^k U_{k'} + W_{k'}$ for all $k > 0$, it follows that

$$\mathbb{E}_x[\theta^{\delta\phi_x^\delta}] \leq \theta^\delta + \theta^\delta \mathbb{E}_x[1_{\{0 < \sigma < \infty\}} \theta^{\delta + \sum_{k=1}^\sigma U_k + W_k}] = \theta^\delta + \theta^\delta \sum_{n=1}^\infty \mathbb{E}_x[1_{\{\sigma=n\}} \theta^{\sum_{k=1}^n U_k + W_k}].$$

If the n th sojourn time U_{n+1} is at least δ , then the skeleton chain must necessarily return to x during the continuous-time chain's n th visit (recall that X^δ is obtained by sampling X every δ units of time (42.1)) and it must be the case that $\sigma \leq n$. In short, $\{U_{n+1} \geq \delta\} \subseteq \{\sigma \leq n\}$. Because $\{\sigma = n\} \subseteq \{\sigma > n-1\}$, the above implies that

$$\begin{aligned} \mathbb{E}_x[\theta^{\delta\phi_x^\delta}] &\leq \theta^\delta + \theta^\delta \mathbb{E}_x[\theta^{\varphi_x}] + \theta^\delta \sum_{n=2}^\infty \mathbb{E}_x[1_{\bigcap_{k=1}^{n-1} \{U_{k+1} < \delta\}} \theta^{\sum_{k=1}^n U_k + W_k}] \\ &\leq \theta^\delta + \theta^\delta \mathbb{E}_x[\theta^{\varphi_x}] + \sum_{n=2}^\infty \theta^{\delta n} \mathbb{E}_x[1_{\bigcap_{k=1}^{n-1} \{U_{k+1} < \delta\}} \theta^{U_1 + \sum_{k=1}^n W_k}] \end{aligned}$$

Because the U_k s and W_k s are independent and identically distributed (Exercise 3), we find that

$$\begin{aligned} \mathbb{E}_x[\theta^{\delta\phi_x^\delta}] &\leq \theta^\delta + \theta^\delta \mathbb{E}_x[\theta^{\varphi_x}] + \sum_{n=2}^\infty \theta^{\delta n} \mathbb{P}_x(\{S_1 < \delta\})^{n-1} \mathbb{E}_x[\theta^{S_1}] \mathbb{E}_x[\theta^{W_1}]^n \\ &\leq \theta^\delta + \theta^\delta \mathbb{E}_x[\theta^{\varphi_x}] + \frac{\mathbb{E}_x[\theta^{S_1}]}{\mathbb{P}_x(\{S_1 < \delta\})} \sum_{n=2}^\infty (\theta^\delta \mathbb{P}_x(\{S_1 < \delta\}) \mathbb{E}_x[\theta^{\varphi_x}])^n \\ &= \theta^\delta + \theta^\delta \mathbb{E}_x[\theta^{\varphi_x}] + \frac{\mathbb{E}_x[\theta^{S_1}] \theta^{2\delta} \mathbb{P}_x(\{S_1 < \delta\}) \mathbb{E}_x[\theta^{\varphi_x}]^2}{1 - \theta^\delta \mathbb{P}_x(\{S_1 < \delta\}) \mathbb{E}_x[\theta^{\varphi_x}]} < \infty \end{aligned}$$

and (ii) follows.

(ii) $\Rightarrow$ (i): The skeleton chain X^δ having returned to x does not necessarily imply that X has as well: X may not have left x in the first place. To rule out this possibility, let

$$\phi_x^\delta(\omega) := \inf\{n > 0 : X_n^\delta(\omega) \neq x\}, \quad \tilde{\phi}_x^\delta(\omega) := \inf\{n \geq \phi_x^\delta(\omega) : X_n^\delta(\omega) = x\}, \quad \forall \omega \in \Omega,$$

denote the first that X^δ leaves x and the first time that X^δ returns to x after having first left it, respectively. Because the X must have left and returned to x if the skeleton chain has left and return, $\delta\tilde{\phi}_x^\delta$ is greater than the return time φ_x of X and, so, it suffices to show that there exists a $\beta > 0$ such that

$$(4) \quad \mathbb{E}_x \left[e^{\beta\delta\tilde{\phi}_x^\delta} \right] < \infty.$$

To do so, note that Theorem 38.4 implies that $p_\delta(x, x) < 1$ (because we are assuming that x is not absorbing) and pick a $0 < \beta < \ln(\theta)$ close enough to zero that $p_\delta(x, x)e^{\beta\delta} < 1$. It is then straightforward to check that ϕ_x^δ is $\mathbb{P}_x$ -almost surely finite and it follows by applying the discrete-time strong Markov property (Theorem 12.4, use Lemma 12.8(ii) to check the theorem's measurability requirement), that

$$(5) \quad \begin{aligned} \mathbb{E}_x \left[e^{\beta\delta\tilde{\phi}_x^\delta} \right] &= \sum_{z \neq x} \mathbb{E}_x \left[1_{\{\phi_x^\delta < \infty, X_{\phi_x^\delta}^\delta = z\}} e^{\beta\delta\phi_x^\delta} e^{\beta\delta(\tilde{\phi}_x^\delta - \phi_x^\delta)} \right] \\ &= \sum_{z \neq x} \mathbb{E}_x \left[1_{\{\phi_x^\delta < \infty, X_{\phi_x^\delta}^\delta = z\}} e^{\beta\delta\phi_x^\delta} \right] \mathbb{E}_z \left[e^{\beta\delta\phi_x^\delta} \right], \end{aligned}$$

where the above sums are taken over all z in the extended state space $\mathcal{S}_E$ (Section 42) except for x . But,

$$\begin{aligned} \mathbb{E}_x \left[1_{\{\phi_x^\delta < \infty, X_{\phi_x^\delta}^\delta = z\}} e^{\beta\delta\phi_x^\delta} \right] &= \sum_{n=1}^{\infty} e^{\beta n\delta} \mathbb{P}_x(\{\phi_x^\delta = n, X_n^\delta = z\}) \\ &= \sum_{n=1}^{\infty} e^{\beta n\delta} \mathbb{P}_x(\{X_0^\delta = x, \dots, X_{n-1}^\delta = x, X_n^\delta = z\}) \\ &= \sum_{n=1}^{\infty} e^{\beta n\delta} p_\delta(x, x)^{n-1} p_\delta(x, z) = \frac{e^{\beta\delta} p_\delta(x, z)}{1 - p_\delta(x, x)e^{\beta\delta}} \forall z \neq x. \end{aligned}$$

Plugging the above into (5) and applying the strong Markov property one more time, we have that

$$\begin{aligned} \mathbb{E}_x \left[e^{\beta\delta\tilde{\phi}_x^\delta} \right] &= \frac{e^{\beta\delta}}{1 - p_\delta(x, x)e^{\beta\delta}} \sum_{z \neq x} p_\delta(x, z) \mathbb{E}_z \left[e^{\beta\delta\phi_x^\delta} \right] \\ &= \frac{e^{\beta\delta}}{1 - p_\delta(x, x)e^{\beta\delta}} \sum_{z \neq x} \mathbb{P}_x(\{X_1^\delta = z\}) \mathbb{E}_z \left[e^{\beta\delta\phi_x^\delta} \right] \\ &\leq \frac{e^{\beta\delta}}{1 - p_\delta(x, x)e^{\beta\delta}} \mathbb{E}_x \left[e^{\beta\delta(\phi_x^\delta - 1)} \right] \leq \frac{e^{\beta\delta}}{1 - p_\delta(x, x)e^{\beta\delta}} \mathbb{E}_x \left[e^{\beta\delta\phi_x^\delta} \right]. \end{aligned}$$

Because we have chosen β to be smaller than $\ln(\theta)$, $\mathbb{E}_x \left[e^{\beta\delta\phi_x^\delta} \right] \leq \mathbb{E}_x \left[\theta^{\phi_x^\delta} \right] < \infty$ and (4) follows from the above. $\square$

The remainder of Theorem 1's proof is downhill:

Theorem 1. Lemma 2 shows that x is exponentially recurrent for X if and only if there exists a δ such x is geometrically recurrent for the skeleton chain X^δ . Kendall's Theorem (Theorem 21.1), (42.4), and Theorem 43.4 imply that this is the case if and only if

$$(6) \quad |p_{\delta n}(x, x) - \pi(x)| \leq C_\delta \kappa^{-n} \quad \forall n \in \mathbb{N},$$

for some $C_\delta \geq 0$ and $\kappa > 1$. Because this is clearly the case if there exists some $C \geq 0$ and $\alpha > 0$ such that

$$(7) \quad |p_t(x, x) - \pi(x)| \leq C e^{-\alpha t} \quad \forall t \in [0, \infty),$$

we only have left to show that the above holds for some $C \geq 0$ and $\alpha > 0$ if (6) holds for some $C_\delta \geq 0$ and $\kappa > 1$. In this case, examining the proof of Theorem 25.5 (in particular, (25.21)–(25.23)) and taking supremums over $A \subseteq \mathcal{S}$ in (25.24) we find that, after tweaking C_δ and κ , the inequality (6) can be strengthened to

$$\|p_{\delta n}(x, \cdot) - \pi\| \leq C_\delta \kappa^{-n} \quad \forall n \in \mathbb{N},$$

where $\|\cdot\|$ denotes the total variation norm in (18.4). Now, fix any t and re-write t as $\delta n + s$, where n is an integer and $0 \leq s < \delta$. Proposition 40.3 implies that $p_t(x, \cdot)$, $p_{\delta n}(x, \cdot)$, $p_s(x, \cdot)$ are probability distribution as x is recurrent for X . Because the total variation distance between two probability distributions is half the ℓ^1 -distance (see (18.6)), the semigroup property (31.4) and the equations $\pi P_s = \pi$ (Theorem 43.4) imply that

$$\begin{aligned} \|p_t(x, \cdot) - \pi\| &= \frac{1}{2} \sum_{x' \in \mathcal{S}} |p_t(x, x') - \pi(x')| = \frac{1}{2} \sum_{x' \in \mathcal{S}} \left| \sum_{x'' \in \mathcal{S}} (p_{\delta n}(x, x'') - \pi(x'')) p_s(x'', x') \right| \\ &\leq \frac{1}{2} \sum_{x'' \in \mathcal{S}} |p_{\delta n}(x, x'') - \pi(x'')| \left(\sum_{x' \in \mathcal{S}} p_s(x'', x') \right) = \|p_{\delta n}(x, \cdot) - \pi\| \leq C_\delta \kappa^{-n} \\ &= C_\delta \kappa^{-(t-s)/\delta} = C_\delta \kappa^{s/\delta} \kappa^{-t/\delta} \leq C_\delta \kappa e^{-(\ln(\kappa)/\delta)t}. \end{aligned}$$

Given that $|p_t(x, x) - \pi(x)| \leq \|p_t(x, \cdot) - \pi\|$, it follows that (7) holds with $C := C_\delta \kappa$ and $\alpha := \ln(\kappa)/\delta$. $\square$

47. Foster-Lyapunov criteria I: the criterion for regularity*. The regularity of the rate matrix Q (Definition 32.4) plays an important role in many aspects of the theory. For instance, it implies that the only solutions to the forward and backward equations are the transition probabilities (Section 32), guarantees that all Markov chains with rate matrix Q and initial distribution γ have the same path law (Section 37), ensures that the stationary distributions are characterised by the equations $\pi Q = 0$ (Section 43), and is a basic requirement for the chain to be stable (Sections 44–45). For these reasons, it is important to have a practical test for it:

(1) **THEOREM.** *The rate matrix Q in (26.1) is regular if and only if there exists a non-negative real-valued norm-like (Definition 23.1) function v on $\mathcal{S}$ and a constant c in $\mathbb{R}$ such that*

$$(2) \quad Qv(x) \leq cv(x) \quad \forall x \in \mathcal{S}.$$

In this case, the time-varying law p_t in (33.1) satisfies

$$(3) \quad p_t(v) \leq \gamma(v) e^{ct} \quad \forall t \in [0, \infty)$$

where γ denotes the initial distribution.

A practically useful fact is that the existence of a (v, c) satisfying Theorem 1's premise is equivalent to a seemingly weaker condition: the existence of a real-valued norm-like function u on $\mathcal{S}$ and real numbers c_u and d such that

$$Qu(x) \leq c_u u(x) + d \quad \forall x \in \mathcal{S}.$$

To see this, note that (26.1) implies that $Qe = 0$ for any constant function $e = (e)_{x \in \mathcal{S}}$. Thus, the above inequality holds if we replace u with $u - u_m$ and d with $d + c_u u_m$ where $u_m := \inf_{x \in \mathcal{S}} u(x)$ (this number is finite because u is norm-like):

$$Q(u - u_m)(x) = Qu(x) - Qu_m = Qu(x) \leq c_u u(x) + d = c_u(u(x) - u_m) + d + c_u u_m.$$

In other words, we can assume without loss of generality that u is non-negative. If $d \leq 0$, then (2) holds with $v := u$ and $c := c_u$. Otherwise, if $c_u \leq 0$, (2) holds with $v := u + d$ and $c := 1$ and, if $c_u > 0$, it instead holds with $v := u + d/c_u$ and $c := c_u$.

A proof of Theorem 1. This is a bit delicate and we do it in steps. We start with (3):

Proof of (3). Pick any increasing sequence $(\mathcal{S}_r)_{r \in \mathbb{Z}_+}$ of finite subsets of the state space satisfying $\bigcup_{r=1}^{\infty} \mathcal{S}_r = \mathcal{S}$ and let $(\tau_r)_{r \in \mathbb{Z}_+}$ denote the corresponding sequence of exit times (26.9). Because these are jump-time-valued $(\mathcal{F}_t)_{t \geq 0}$ -stopping times (Proposition 34.9), Dynkin's formula (Theorem 34.11, with $\eta := \tau_r$, $g(t) := e^{-ct}$, and $f := v$) shows that

$$\mathbb{E}_\gamma \left[e^{-ct \wedge \tau_r \wedge T_n} v(X_{t \wedge \tau_r \wedge T_n}) \right] = \gamma(v) + \mathbb{E}_\gamma \left[\int_0^{t \wedge \tau_r \wedge T_n} e^{-cs} Qv(X_s) - ce^{-cs} v(X_s) ds \right],$$

for all t in $[0, \infty)$ and n, r in $\mathbb{Z}_+$. Inequality (2) then shows that

$$\mathbb{E}_\gamma \left[e^{-ct \wedge \tau_r \wedge T_n} v(X_{t \wedge \tau_r \wedge T_n}) \right] \leq \gamma(v) \quad \forall t \in [0, \infty), \quad n, r \in \mathbb{Z}_+.$$

As we will show later on, the chain is regular (in particular, $\mathbb{P}_\gamma(\{T_\infty = \infty\}) = 1$). For this reason, Theorem 26.10 and Fatou's lemma imply that

$$\begin{aligned} e^{-ct} p_t(v) &= \mathbb{E}_\gamma \left[e^{-ct} V(X_t) 1_{\{t < T_\infty\}} \right] \leq \liminf_{r \rightarrow \infty} \liminf_{n \rightarrow \infty} \mathbb{E}_\gamma \left[e^{-ct \wedge \tau_r \wedge T_n} v(X_{t \wedge \tau_r \wedge T_n}) 1_{\{t < T_\infty\}} \right] \\ &= \liminf_{r \rightarrow \infty} \liminf_{n \rightarrow \infty} \mathbb{E}_\gamma \left[e^{-ct \wedge \tau_r \wedge T_n} v(X_{t \wedge \tau_r \wedge T_n}) \right] \leq \gamma(v) \end{aligned}$$

for all t in $[0, \infty)$. Multiplying through by e^{ct} then yields (3). $\square$

For the remainder of this proof, we follow the steps taken in (Spieksma, 2015). To do so, we require a discrete-time chain $Y^\alpha = (Y_n^\alpha)_{n \in \mathbb{N}}$ known as the α -jump chain. It takes values in the *extended state space* $\mathcal{S}_E := \mathcal{S} \cup \{\Delta\}$ obtained by appending an extra state $\Delta \notin \mathcal{S}$ to the state space $\mathcal{S}$. The chain is defined Y^α by running Algorithm 1 in Section 1 using the one-step matrix $P^\alpha = (p^\alpha(x, y))_{x, y \in \mathcal{S}_E}$ on $\mathcal{S}_E$ given by

$$(4) \quad p^\alpha(x, y) := \begin{cases} \frac{q(x, y)}{q(x) + \alpha} & \text{if } x, y \in \mathcal{S}, \quad x \neq y \\ \frac{\alpha}{q(x) + \alpha} & \text{if } x \in \mathcal{S}, y = \Delta \\ p(y) & \text{if } x = \Delta, y \in \mathcal{S} \end{cases} \quad \forall x, y \in \mathcal{S}_E,$$

where $\alpha > 0$ and the *restart distribution* $(p(x))_{x \in \mathcal{S}}$ is any probability distribution on $\mathcal{S}$ satisfying

$$(5) \quad p(x) > 0 \quad \forall x \in \mathcal{S}.$$

This α -jump chain Y^α behaves as follows: if Y^α lies at a state x within the original state space $\mathcal{S}$, it jumps to Δ with probability $\alpha/(q(x) + \alpha)$. Otherwise, Y^α updates its states as the jump chain Y does: it samples $p(x, \cdot)$ in (26.2). Whenever Y^α leaves $\mathcal{S}$, it spends a single step in Δ and then returns to $\mathcal{S}$ by sampling a state from the restart distribution $(p(x))_{x \in \mathcal{S}}$. Because Δ is accessible from all states in $\mathcal{S}$ and (5) implies that every state in $\mathcal{S}$ is accessible from Δ , Y^α is irreducible. For this reason, applying the necessary and sufficient criterion for recurrence of irreducible discrete-time chains yields the following:

(6) LEMMA. *For any $\alpha > 0$, the α -jump chain Y^α is recurrent if and only if there exists a norm-like function $v : \mathcal{S} \rightarrow [0, \infty)$ satisfying (2) with $c := \alpha$.*

Proof. Suppose that Y^α is recurrent. Because Y^α is irreducible, Corollary 14.8 and Lemma 23.10 (with $F := \Delta$) tell us that there exists a non-negative norm-like function $u : \mathcal{S}_E \rightarrow [0, \infty)$ satisfying

$$(7) \quad P^\alpha u(x) \leq u(x) \quad \forall x \in \mathcal{S}.$$

Given that $u(\Delta) \geq 0$, the above implies that

$$\sum_{y \neq x} \frac{q(x, y)u(y)}{q(x) + \alpha} \leq \sum_{y \neq x} \frac{q(x, y)u(y)}{q(x) + \alpha} + \frac{\alpha u(\Delta)}{q(x) + \alpha} = P^\alpha u(x) \leq u(x) \quad \forall x \in \mathcal{S},$$

where, in the above, we are summing over all y in $\mathcal{S}$ except x . Multiplying through by $q(x) + \alpha$ and re-arranging we find that $v := (u(x))_{x \in \mathcal{S}}$ is a non-negative real-valued norm-like function on $\mathcal{S}$ satisfying (2) with $c := \alpha$.

Similarly, if $v : \mathcal{S} \rightarrow [0, \infty)$ is norm-like and satisfies (3), then

$$u(x) := v(x) \quad \forall x \in \mathcal{S}, \quad u(\Delta) := 0,$$

is a non-negative real-valued norm-like function on $\mathcal{S}_E$ satisfying (7) and Theorem 23.2 implies that Y^α is recurrent. $\square$

We now need to show that Y^α is recurrent for some $\alpha > 0$ if and only if Q is regular. Because Y^α is irreducible, Corollary 14.8 implies that Y^α is recurrent if and only if the state Δ is recurrent: $\mathbb{P}_\Delta(\{\phi_\Delta < \infty\}) = 1$, where ϕ_Δ denotes the first entrance time to Δ of Y^α (defined by replacing X with Y^α in Definition 13.1). Because Proposition 13.10 shows that

$$\mathbb{P}_\Delta(\{\phi_\Delta < \infty\}) = p^\alpha(\Delta, \Delta) + \sum_{x \in \mathcal{S}} p^\alpha(\Delta, x) \mathbb{P}_x(\{\phi_\Delta < \infty\}) = \sum_{x \in \mathcal{S}} p(x) \mathbb{P}_x(\{\phi_\Delta < \infty\}),$$

our assumption (5) ensures that Δ is recurrent if and only if

$$(8) \quad \mathbb{P}_x(\{\phi_\Delta < \infty\}) = 1 \quad \forall x \in \mathcal{S}.$$

Applying (4.4), we have that

$$\begin{aligned} \mathbb{P}_x(\{\phi_\Delta < \infty\}) &= \sum_{n=0}^{\infty} \mathbb{P}_x(\{\phi_\Delta = n+1\}) = \sum_{n=0}^{\infty} \mathbb{P}_x(\{Y_1^\alpha \in \mathcal{S}, \dots, Y_n^\alpha \in \mathcal{S}, Y_{n+1}^\alpha = \Delta\}) \\ &= \sum_{n=0}^{\infty} \sum_{x_1 \in \mathcal{S}} \cdots \sum_{x_n \in \mathcal{S}} p^\alpha(x, x_1) \cdots p^\alpha(x_{n-1}, x_n) p^\alpha(x_n, \Delta) \\ (9) \quad &= \sum_{n=0}^{\infty} \sum_{y \in \mathcal{S}} \alpha p_n^\alpha(x, y) p^\alpha(y, \Delta) = \sum_{n=0}^{\infty} \sum_{y \in \mathcal{S}} \frac{\alpha p_n^\alpha(x, y)}{q(y) + \alpha}, \end{aligned}$$

where $\Delta P_n^\alpha = (\Delta p_n^\alpha(x, y))_{x, y \in \mathcal{S}}$ denotes the n th power of the *taboo matrix* $\Delta P^\alpha = (p^\alpha(x, y))_{x, y \in \mathcal{S}}$ obtained by removing the Δ -row and column from P^α .

To finish the proof, we are only missing one final ingredient: the α -resolvent matrix $R^\alpha = (r^\alpha(x, y))_{x, y \in \mathcal{S}}$ defined by

$$r^\alpha(x, y) := \int_0^\infty e^{-\alpha t} p_t(x, y) dt \quad \forall x, y \in \mathcal{S}.$$

Tonelli's theorem implies that

$$\sum_{y \in \mathcal{S}} r^\alpha(x, y) = \int_0^\infty e^{-\alpha t} \sum_{y \in \mathcal{S}} p_t(x, y) dt \quad \forall x \in \mathcal{S},$$

and it follows from Proposition 32.6 that Q is regular if and only if αR^α is a stochastic matrix,

$$\alpha \sum_{y \in \mathcal{S}} r^\alpha(x, y) = \alpha \frac{1}{\alpha} = 1 \quad \forall x \in \mathcal{S},$$

for any and all $\alpha > 0$. As we will show below,

$$(10) \quad r^\alpha(x, y) = \sum_{n=0}^\infty \frac{\Delta p_n^\alpha(x, y)}{q(y) + \alpha} \quad \forall x, y \in \mathcal{S}, \quad \alpha > 0.$$

Because (8)–(10) imply that αR^α is stochastic (equivalently, Q is regular) if and only if Y^α is recurrent, Theorem 1 follows from Lemma 6.

(11) LEMMA. Equation (10) holds.

Proof. The equation holds trivially if x is an absorbing state. Suppose otherwise, fix $\alpha > 0$, and let

$$s_n^\alpha(x, y) := \sum_{m=0}^n \frac{\Delta p_m^\alpha(x, y)}{q(y) + \alpha} \quad \forall x, y \in \mathcal{S}, \quad n \in \mathbb{N},$$

and

$$r_n^\alpha(x, y) := \int_0^\infty e^{-\alpha t} p_t^n(x, y) dt \quad \forall x, y \in \mathcal{S}, \quad n \in \mathbb{N},$$

where $p_t^n(x, y)$ as in (32.8). Because monotone convergence implies that

$$\lim_{n \rightarrow \infty} s_n^\alpha(x, y) = \sum_{m=0}^\infty \frac{\Delta p_m^\alpha(x, y)}{q(y) + \alpha}, \quad \lim_{n \rightarrow \infty} r_n^\alpha(x, y) = r^\alpha(x, y), \quad \forall x, y \in \mathcal{S},$$

it suffices to show that

$$(12) \quad r_n^\alpha(x, y) = s_n^\alpha(x, y) \quad \forall x, y \in \mathcal{S}, \quad n \in \mathbb{N}.$$

Multiplying the backward iterated recursion (32.17) by $e^{-\alpha t}$, integrating over t , and applying Tonelli's theorem yields the Laplace transform version of this recursion (note that x is not absorbing and so $\lambda(x) = q(x)$ and $\lambda(x)p(x, y) = q(x, y)$ for all y in $\mathcal{S}$):

$$r_{n+1}^\alpha(x, y) = \frac{1}{q(x) + \alpha} \left(1_x(y) + \sum_{z \neq x} q(x, z) r_n^\alpha(z, y) \right) \quad \forall y \in \mathcal{S}, \quad n \in \mathbb{N}.$$

Thus, if (12) holds for some n in $\mathbb{N}$,

$$\begin{aligned} r_{n+1}^\alpha(x, y) &= \frac{1_x(y)}{q(y) + \alpha} + \sum_{m=0}^n \sum_{z \neq x} \frac{q(x, z)}{q(x) + \alpha} \frac{\Delta p_m^\alpha(z, y)}{q(y) + \alpha} = \frac{\Delta p_0^\alpha(x, y)}{q(y) + \alpha} + \sum_{m=0}^n \frac{\Delta p_{m+1}^\alpha(z, y)}{q(y) + \alpha} \\ &= s_{n+1}^\alpha(x, y) \quad \forall x, y \in \mathcal{S}. \end{aligned}$$

Because (32.11)–(32.12) imply that

$$r_0^\alpha(x, y) = \int_0^\infty e^{-\alpha t} p_t^0(x, y) dt = \int_0^\infty e^{-\alpha t} 1_x(y) e^{-q(x)t} dt = \frac{1_x(y)}{q(y) + \alpha} = s_0^\alpha(x, y) \quad \forall x, y \in \mathcal{S},$$

the result follows by induction. $\square$

Notes and references. The theorem's sufficiency was first shown in (Chen, 1986). Other early proofs can be found in (Anderson, 1991) and (Meyn and Tweedie, 1993b). The necessity was only shown recently in (Spieksma, 2015).

48. Geometric trial arguments*. In the proofs of the Foster-Lyapunov criteria presented in the ensuing sections, we require the continuous-time versions of the geometric trials arguments in Section 22. Just as in the discrete-time case, these arguments allow us to morph information regarding the chain's visits to a finite set F into information on its visits to any set B that is accessible from F (Definition 38.3). The intuition is the same: because B is accessible from F and F is finite, the chance that a visit to F results in a visit to B may be bounded below by a constant independent of which state in F the chain visits. The existence of this constant implies that the probability that the chain has not yet visited B decays geometrically with the number of visits to F :

(1) LEMMA (Geometric trials property). *If F is finite and $F \rightarrow B$ for a second set B , then there exists constants m and $\varepsilon > 0$ independent of the initial distribution γ such that*

$$\mathbb{P}_\gamma(\{\varphi_F^{nm} < \varphi_B\}) \leq (1 - \varepsilon)^{n-1} \quad \forall n \in \mathbb{Z}_+,$$

where φ_B denotes the first entrance time to B and φ_F^k the k th entrance time to F (Definition 38.1).

Proof. The event $\{\varphi_F^k < \varphi_B\}$ that the continuous-time chain enters any F for at least k times before entering B equals the event $\{\phi_F^k < \phi_B\}$ that the jump-chain does (this a consequence of Exercise 38.2). For this reason, the lemma follows immediately from its discrete-time counterpart (Lemma 22.1). $\square$

Lemma 1 lays the foundation for the following three geometric-trials-type arguments that we use in the following sections to turn information on the chain's visits to F into information on its visits to B :

(2) LEMMA (Geometric trials arguments). *Given a finite subset F of $\mathcal{S}$, suppose that $F \rightarrow B$ for some subset B and let φ_F and φ_B denote the first entrance time to F and B (Definition 13.1).*

- (i) *If $\mathbb{P}_\gamma(\{\varphi_F < \infty\}) = 1$ and $\mathbb{P}_x(\{\varphi_F < \infty\}) = 1$ for all x in F , then $\mathbb{P}_\gamma(\{\varphi_B < \infty\}) = 1$.*
- (ii) *If $\mathbb{E}_\gamma[\varphi_F] < \infty$ and $\mathbb{E}_x[\varphi_F] < \infty$ for all x in F , then $\mathbb{E}_\gamma[\varphi_B] < \infty$.*
- (iii) *If there exists a constant $\alpha > 1$ such that $\mathbb{E}_\gamma[e^{\alpha\varphi_F}] < \infty$ and $\mathbb{E}_x[e^{\alpha\varphi_F}] < \infty$ for all x in F , then there exists second constant $\beta > 1$ such that $\mathbb{E}_\gamma[e^{\beta\varphi_B}] < \infty$.*

Because Exercise 38.2 implies that

$$\{\varphi_F < \infty\} = \{\phi_F < \infty\}, \quad \{\varphi_B < \infty\} = \{\phi_B < \infty\},$$

where ϕ_F and ϕ_B denote the first entrance times to F and B of the jump chain, Lemma 2(i) follows directly its discrete-time counterpart (Lemma 22.2(i)). For Lemma 2(ii)–(iii), we need to emulate the proofs of their discrete-time counterparts...

(3) EXERCISE. Following the steps we took in the proof of Lemma 22.3, show that, for any subsets A and B of the state space, constant $\alpha > 0$, and natural numbers k and l ,

$$\begin{aligned}\mathbb{P}_\gamma \left(\left\{ \varphi_A^{k+l} < \varphi_B \right\} \right) &= \sum_{x \in A} \mathbb{P}_\gamma \left(\left\{ \varphi_A^k < \varphi_B, X_{\varphi_A^k} = x \right\} \right) \mathbb{P}_x \left(\left\{ \varphi_A^l < \varphi_B \right\} \right), \\ \mathbb{E}_\gamma \left[1_{\{\varphi_A^k < \varphi_B\}} (\phi_A^{k+l} - \phi_A^k) \right] &= \sum_{x \in A} \mathbb{P}_\gamma \left(\left\{ \varphi_A^k < \varphi_B, X_{\varphi_A^k} = x \right\} \right) \mathbb{E}_x \left[\varphi_A^l \right], \\ \mathbb{E}_\gamma \left[1_{\{\varphi_A^k < \varphi_B\}} e^{\alpha \varphi_A^{k+l}} \right] &= \sum_{x \in A} \mathbb{E}_\gamma \left[1_{\{\varphi_A^k < \varphi_B, X_{\varphi_A^k} = x\}} e^{\alpha \varphi_A^k} \right] \mathbb{E}_x \left[e^{\alpha \varphi_A^l} \right],\end{aligned}$$

where φ_B denotes the first entrance time to B and φ_A^k the k th entrance time to A (Definition 38.1). To do so, replace Lemma 8.3, Theorem 12.4, Exercise 13.4, and G_S^j therein with Lemma 28.5, Theorem 30.7, Exercise 38.2, and G in Lemma 29.14(iii) (with $A := S$, $k := j$, and $f := 1$). Next, prove Lemma 2(ii)–(iii) by using the above equations and tweaking the proof of Lemma 22.2(ii)–(iii).

49. Foster-Lyapunov criteria II: the criterion for positive recurrence*. The continuous-time counterpart of Foster's theorem is often proves indispensable for establishing positive Tweedie recurrence in practice:

(1) THEOREM (The criterion for positive recurrence). *If the rate matrix is regular and there exists a non-negative real-valued function v on S , a finite subset F of S , and a constant b in $\mathbb{R}$ such that*

$$(2) \quad Qv(x) \leq -1 + b1_F(x) \quad \forall x \in S.$$

then the chain is positive Tweedie recurrent and has a finite number of closed communicating classes. Conversely, if the chain is positive Tweedie recurrent and there are no transient states and only a finite number of closed communicating classes, then there exists (v, F, b) satisfying (2) with v real-valued and non-negative, F finite, and b real.

The criterion is a consequence of Theorem 4 further down which shows that, for a fixed F , the existence of such a v and b satisfying (2) is equivalent to F having a finite mean return time for all deterministic starting locations.

The importance of a finite number of closed communicating classes is easy to see: the set F must contain at least one state from each closed communicating class. Otherwise, the chain would not be able to reach F whenever it starts inside of a closed communicating class that does not intersect with F and the return time to F would be infinite. As an example, the chain with one-step matrix Q being the matrix of zeros is trivially positive Tweedie recurrent, however $Qv(x) = 0 \geq -1$ for all states x and functions v . Consequently, if the state space is infinite, then (2) will never be satisfied for for a finite F .

To see the importance of the no transient states requirement, consider the continuous-time analogue of Example 24.3:

(3) EXAMPLE. Consider a chain with state space $\mathbb{N}$ and rate matrix $Q = (q(x, y))_{x, y \in \mathbb{N}}$ defined by

$$q(0, y) = 0 \quad \forall y \geq 0, \quad q(x, y) = \frac{1_{x-1}(y)}{2} - 1_x(y) + \frac{1_{x+1}(y)}{2} \quad \forall x > 0, y \geq 0.$$

In other words, a chain X that waits a unit mean exponential amount of time at its starting state x , jumps to $y = x - 1$ with 50% probability and to $y = x + 1$ with the remaining 50% probability, waits a unit mean exponential amount of time at y , jumps to $z = y - 1$ with 50% probability and to $z = y + 1$ with the remaining 50% probability, ..., up until the moment it hits 0 where it remains for all time. In other words, X is just like the gambler's ruin chain Y

(Section 3) except that the time waited between jumps is exponentially distributed. Indeed, X 's jump chain is Y which is positive Harris recurrent (Example 24.3). Because the waiting times of X all have unit means, Exercise 38.2 then implies that X and Y have the same mean return times. Thus, X is also positive Harris recurrent. However, inequality (2) implies that

$$\frac{1}{2}v(x-1) + \frac{1}{2}v(x+1) \leq v(x) - 1 \quad \forall x \notin F.$$

In Example 24.3, we showed that any function v satisfying the above must tend to $-\infty$ as its argument approaches ∞ . Thus, even though the chain is positive Harris recurrent there do not exist any v , F , and b satisfying the premise of Theorem 1.

A proof of Theorem 1. Given any F , the existence of such a v and b satisfying (2) is equivalent to F having a finite mean return time for every deterministic initial condition:

(4) THEOREM. Suppose that the rate matrix is regular. Given any finite set F , there exists $v : \mathcal{S} \rightarrow [0, \infty)$ and b in $\mathbb{R}$ satisfying (2) if and only if $\mathbb{E}_x[\varphi_F] < \infty$ for all x in $\mathcal{S}$. In this case, $\mathbb{E}_\gamma[\varphi_F] < \infty$ for all initial distributions γ satisfying $\gamma(v) < \infty$.

The key to proving the above is the following lemma:

(5) LEMMA (Tweedie (1981)). Let F be a subset of the state space and α be a real number satisfying $\alpha < q(x)$ for all x outside F . The minimal (possibly infinite-valued) non-negative solution $v := (v(x))_{x \in \mathcal{S}}$ to the inequality

$$(6) \quad Qv(x) \leq -\alpha v(x) - 1 \quad \forall x \notin F$$

is given by $u(x) = 0$ for all x in F and

$$(7) \quad u(x) = \int_0^\infty \mathbb{P}_x(\{\varphi_F > t, t < T_\infty\}) e^{\alpha t} dt \quad \forall x \notin F.$$

While, u in (7) may come across as rather esoteric at first glance, it is not difficult to re-write it in terms that make the connection with Theorem 4 obvious. In particular, if $\alpha = 0$, then Tonelli's theorem implies that

$$\begin{aligned} u(x) &= \int_0^\infty \mathbb{P}_x(\{\varphi_F > t, t < T_\infty\}) dt = \int_0^\infty \mathbb{P}_x(\{t < \varphi_F \wedge T_\infty\}) dt = \mathbb{E}_x \left[\int_0^\infty 1_{[0, \varphi_F \wedge T_\infty)}(t) dt \right] \\ (8) \quad &= \mathbb{E}_x \left[\int_0^{\varphi_F \wedge T_\infty} 1 dt \right] = \mathbb{E}_x[\varphi_F \wedge T_\infty] \quad \forall x \notin F. \end{aligned}$$

Thus, in the regular case, $u(x)$ is the mean entrance time to F for any x outside F : a fact that will be key for the proof of Lemma 5. We will also require the following generalisations of the FIR (32.9) and BIR (32.17): for any subset F of the state space and positive integer n ,

$$(9) \quad f_t^n(x, y) = 1_x(y) e^{-\lambda(x)t} + \int_0^t \sum_{z \notin F} f_s^{n-1}(x, z) \lambda(z) p(z, y) e^{-\lambda(y)(t-s)} ds \quad \forall x, y \notin F, \quad t \geq 0,$$

$$(10) \quad f_t^n(x, y) = 1_x(y) e^{-\lambda(y)t} + \int_0^t \lambda(x) e^{-\lambda(x)(t-s)} \sum_{z \notin F} p(x, z) f_s^{n-1}(z, y) ds \quad \forall x, y \notin F, \quad t \geq 0,$$

where

$$(11) \quad f_t^n(x, y) := \mathbb{P}_x(\{\varphi_F > t, X_t = y, t < T_{n+1}\}) \quad \forall x, y \notin F, \quad t \geq 0, \quad n \in \mathbb{N}.$$

(12) EXERCISE. Prove (9) by tweaking the proof of Lemma 32.7 and use (9) to adapt the proof of Lemma 32.16 and obtain (10). Hint: for any y outside F , we can re-write the event

$$\{Y_m = y, T_m \leq t < T_{m+1}, \varphi_F > t\}$$

as

$$\{Y_m = y, T_m \leq t < T_{m+1}, Y_1 \notin F, \dots, Y_{m-1} \notin F\}.$$

Proof of Lemma 5. Notice that (6) does not have any non-negative solutions if there exists an absorbing state outside of F . Suppose otherwise and note that the generalised BIR (10) reduces to

$$(13) \quad f_t^n(x, y) = 1_x(y)e^{-q(y)t} + \int_0^t q(x)e^{-q(x)(t-s)} \sum_{z \notin F} p(x, z) f_s^{n-1}(z, y) ds \quad \forall x, y \notin F, \quad t \in [0, \infty),$$

for any positive integer n . Let

$$u_n(x) := \begin{cases} \int_0^\infty \sum_{y \notin F} f_t^n(x, y) e^{\alpha t} dt & \text{if } x \notin F \\ 0 & \text{if } x \in F \end{cases} \quad \forall x \in \mathcal{S}, \quad n \in \mathbb{N},$$

and note that

$$\lim_{n \rightarrow \infty} u_n(x) = u(x) \quad \forall x \in \mathcal{S}$$

by monotone convergence. However, applying (13) and Tonelli's Theorem, we have that

$$\begin{aligned} u_n(x) &= \int_0^\infty \sum_{y \notin F} f_t^n(x, y) e^{\alpha t} dt \\ &= \int_0^\infty e^{(\alpha - q(x))t} dt + \int_0^\infty \left(\int_s^\infty e^{(\alpha - q(x))t} dt \right) e^{q(x)s} \sum_{z \notin F} q(x)p(x, z) \left(\sum_{y \notin F} f_s^{n-1}(z, y) \right) ds \\ &= \frac{1}{q(x) - \alpha} + \frac{1}{q(x) - \alpha} \sum_{z \notin F} (q(x)p(x, z)) \left(\int_0^\infty f_s^{n-1}(z, y) e^{\alpha s} ds \right) \\ (14) \quad &= \frac{1 + \sum_{z \neq x} q(x, z) u_{n-1}(z)}{q(x) - \alpha} \quad \forall x \notin F, \quad n \in \mathbb{Z}_+. \end{aligned}$$

Taking the limit $n \rightarrow \infty$ and re-arranging shows that u satisfies (6).

To prove the minimality of u , we use induction. In particular, let $v = (v(x))_{x \in \mathcal{S}}$ be any other non-negative solution of (6). By definition,

$$v(x) \geq 0 = u(x) \quad \forall x \in F.$$

Re-arranging (6), we find that $(q(x) - \alpha)v(x) \geq 1 + \sum_{z \neq x} q(x, z)v(z)$ for all $x \notin F$. Consequently,

$$v(x) \geq \frac{1}{q(x) - \alpha} = \int_0^\infty \sum_{y \notin F} 1_x(y) e^{-(q(x) - \alpha)t} dt = u_0(x) \quad \forall x \notin F.$$

Moreover, if $v(x) \geq u_{n-1}(x)$ for all x in $\mathcal{S}$, it follows from (6) and (14) that

$$u_n(x) \leq \frac{1 + \sum_{z \neq x} q(x, z)v(z)}{q(x) - \alpha} \leq v(x) \quad \forall x \notin F.$$

By induction, we have that $u_n(x) \geq v(x)$ for all x in $\mathcal{S}$ and n in $\mathbb{N}$. Taking the limit $n \rightarrow \infty$ completes the proof. $\square$

Proof of Theorem 4. Suppose that (2) is satisfied. Because we are assuming that the rate matrix is regular, setting $\alpha := 0$ in Lemma 5 and making use of (8), we find that

$$\mathbb{E}_x [\varphi_F] \leq v(x) < \infty \quad \forall x \notin F.$$

For states x inside F , $\mathbb{E}_x [\varphi_F]$ is trivially finite if x is absorbing. Otherwise, an application of the strong Markov property similar to that in the proof of Theorem 38.9 shows that:

$$\begin{aligned} \mathbb{E}_x [\varphi_F] &= \mathbb{E}_x [T_1] + \sum_{z \notin F} \frac{q(x, z)}{q(x)} \mathbb{E}_z [\varphi_F] = \frac{1}{q(x)} + \sum_{z \notin F} \frac{q(x, z)}{q(x)} \mathbb{E}_z [\varphi_F] \leq \frac{1}{q(x)} \left(1 + \sum_{z \notin F} q(x, z) v(z) \right) \\ &\leq \frac{1}{q(x)} \left(1 + \sum_{z \neq x} q(x, z) v(z) \right) \leq v(x) + \frac{b}{q(x)} < \infty \quad \forall x \in F. \end{aligned}$$

Given that $\mathbb{P}_\gamma = \sum_{x \in \mathcal{S}} \gamma(x) \mathbb{P}_x$, multiplying the above two inequalities by γ and summing over x in $\mathcal{S}$ then yields $\mathbb{E}_\gamma [\varphi_F] \leq \gamma(v) + b\gamma(q^{-1}1_F)$ which is finite as long as $\gamma(v)$ is finite.

Conversely, suppose that $\mathbb{E}_x [\phi_F] < \infty$ for all x in $\mathcal{S}$. Lemma 5 and (8) show that u in (7) (with $\alpha = 0$) satisfies the inequality (2) for all states x not in F . For states inside F that are absorbing the inequality holds trivially. For the non-absorbing ones, we apply the strong Markov property as before:

$$\frac{1}{q(x)} Qu(x) = \sum_{z \notin F} \frac{q(x, z)}{q(x)} u(z) = \sum_{z \notin F} \frac{q(x, z)}{q(x)} \mathbb{E}_z [\varphi_F] = \mathbb{E}_x [\varphi_F] - \frac{1}{q(x)} \quad \forall x \in F.$$

In other words, u satisfies (2) with $b := \max\{q(x)\mathbb{E}_x [\varphi_F] : x \in F\}$. $\square$

To make the jump from Theorem 4 to the criterion, we need the following lemma. It tells us that we lose nothing by assuming that the set F in Theorem 4 contains no transient states:

(15) LEMMA. *Let F be a finite set such that $\mathbb{E}_x [\phi_F] < \infty$ for all x in $\mathcal{S}$. The same inequality holds if we remove all transient states from F .*

Proof. Replace Lemma 22.2(i), Definition 13.5, and the ϕ s in the proof of Lemma 23.4 with Lemma 48.2(ii), Definition 38.3, and φ s. $\square$

Given the above, the proof of Theorem 1 is completely analogous to that of its discrete-time counterpart:

Proof of Theorem 1. Replace Proposition 13.11, Lemma 22.2(ii), Theorem 24.4, Lemma 24.8, (24.2), and the ϕ s in the proof Theorem 24.1 with Proposition 38.8, Lemma 48.2(ii), Theorem 4, Lemma 15, (2), and φ s, respectively. $\square$

50. Foster-Lyapunov criteria III: the criterion for exponential convergence*.

Lemma 49.5 instructs us to study the inequality

$$(1) \quad Qv(x) \leq -\alpha v(x) - 1 + b1_F(x) \quad \forall x \in \mathcal{S}.$$

instead of focusing only on the special case $\alpha = 0$ considered in Section 49. For any $\alpha \neq 0$, the inequality's minimal (possibly infinite-valued) non-negative solution is given by $u(x) = 0$ for all states x in F and

$$(2) \quad u(x) = \int_0^\infty \mathbb{P}_x (\{\varphi_F > t, t < T_\infty\}) e^{\alpha t} dt = \mathbb{E}_x \left[\int_0^\infty 1_{[0, \varphi_F \wedge T_\infty)}(t) e^{\alpha t} \right] = \mathbb{E}_x \left[\int_0^{\varphi_F \wedge T_\infty} e^{\alpha t} \right] \\ \frac{1}{\alpha} (\mathbb{E}_x [e^{\alpha \varphi_F \wedge T_\infty}] - 1) \quad \forall x \notin F.$$

The $\alpha > 0$ version of Theorem 49.4 follows easily:

(3) THEOREM. *Suppose the chain is regular. Given any finite set F and $\alpha > 0$, there exists $v : \mathcal{S} \rightarrow [0, \infty)$ and $b \in \mathbb{R}$ satisfying (1) if and only if $\mathbb{E}_x[e^{\alpha\varphi_F}] < \infty$ for all x in $\mathcal{S}$. In this case, $\mathbb{E}_\gamma[e^{\alpha\varphi_F}] < \infty$ for all initial distributions γ satisfying $\gamma(v) < \infty$.*

Proof. Given (2), the proof is entirely analogous to that of Theorem 49.4. $\square$

The theorem shows that (1) is satisfied if and only if the entrance time distribution of F has an exponentially decaying tail (i.e., a *light tail*) for any deterministic starting state. For this reason, we can relate (1) to the exponential convergence of the time-varying law using the continuous-time analogue of Kendall's theorem (Theorem 46.1):

(4) THEOREM (The geometric criterion). *Suppose the chain is regular. If there exists a real-valued non-negative function v on $\mathcal{S}$, a finite set F , and a real number b satisfying (1) for some $\alpha > 0$, then the chain is positive Tweedie recurrent. Moreover, if the initial distribution γ satisfies $\gamma(v) < \infty$, then the time varying law converges geometrically fast: there exists some $\beta > 0$ such that*

$$(5) \quad \|p_t - \pi_\gamma\| = \mathcal{O}(e^{-\beta t}),$$

where π_γ denotes the limiting stationary distribution in (45.2) and $\|\cdot\|$ the total variation norm in (18.4).

Conversely, if the chain is positive Tweedie recurrent, there exist no transient states and only a finite number of closed communicating classes, and (5) holds whenever the chain starts at a fixed state (i.e., whenever $\gamma = 1_x$ for some x in $\mathcal{S}$), then there exists (v, F, b, α) satisfying (1) with v real-valued and non-negative, F finite, b real, and $\alpha > 0$.

To see why the finitely-many-closed-communicating-classes requirement in the converse is necessary, consider a chain on an infinite state space whose rate matrix is the matrix of zeros.

An open question. To the best of my knowledge, whether or not the no-transient-states requirement in the converse is necessary remains an open question. Similarly as in the discrete-time case, I believe that the answer here lies in that of the open question discussed in Section 46.

Proving Theorem 4. To prove Theorem 4, we require the exponential analogue of Lemma 49.15 which tells us that we lose nothing by assuming that the finite set F does not contain any transient states:

(6) LEMMA. *Let F be a finite set such that $\mathbb{E}_\gamma[e^{\alpha\varphi_F}] < \infty$ and $\mathbb{E}_x[e^{\alpha\varphi_F}] < \infty$ for all $x \in \mathcal{S}$ and some $\alpha > 0$. There exists a $\beta > 0$ such that, after removing all transient states from F , $\mathbb{E}_\gamma[e^{\beta\varphi_F}] < \infty$ and $\mathbb{E}_x[e^{\beta\varphi_F}] < \infty$ for all x in $\mathcal{S}$.*

Proof. Replace Lemma 22.2(i) with Lemma 48.2(iii) in the proof of Lemma 23.4. $\square$

Given the above, the proof of Theorem 4 is entirely analogous to that of its discrete-time counterpart:

(7) EXERCISE. *Prove Theorem 4 by adapting the proof of its discrete-time counterpart (Theorem 25.5). To do so, replace Theorems 24.1, 25.4, and 21.1, Lemmas 22.2 and 25.11, and Proposition 13.11 with Theorems 49.1, 3, and 46.1, Lemmas 48.2 and 6, and Proposition 38.8, respectively. Hint: to prove the continuous-time equivalent of (25.25) proceeds as follows:*

$$\begin{aligned}
(8) \quad \{\varphi_{F_i} \leq t, X_t \in A, t < T_\infty\} &= \bigcup_{n=1}^{\infty} \{\varphi_{F_i} \leq T_n, Y_n \in A, T_n \leq t < T_{n+1}\} \\
&= \bigcup_{x \in F_i} \bigcup_{n=1}^{\infty} \bigcup_{m=1}^n \{\varphi_{F_i} = T_m, Y_m = x, Y_n \in A, T_n \leq t < T_{n+1}\} \\
&= \bigcup_{x \in F_i} \bigcup_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{\varphi_{F_i} = T_m, Y_m = x, Y_n \in A, T_n \leq t < T_{n+1}\}.
\end{aligned}$$

If $n > m$,

$$\begin{aligned}
&\{\varphi_{F_i} = T_m, Y_m = x, Y_n \in A, T_n \leq t < T_{n+1}\} \\
&= \left\{ \varphi_{F_i} = T_m, T_m \leq t, Y_m = x, Y_n \in A, \sum_{k=m+1}^n S_k \leq t - T_m < \sum_{k=m+1}^{n+1} S_k \right\} \\
(9) \quad &= \bigcup_{x_1 \in \mathcal{S}} \cdots \bigcup_{x_{n-m-1} \in \mathcal{S}} \bigcup_{x_{n-m} \in A} \left\{ \varphi_{F_i} = T_m, T_m \leq t, Y_m = x_0, Y_{m+1} = x_1, \dots, \right. \\
&\quad \left. Y_{n-1} = x_{n-m-1}, Y_{n-m} = x_{n-m}, \sum_{k=1}^{n-m} S_{m+k} \leq t - T_m < \sum_{k=1}^{n-m+1} S_{m+k} \right\}
\end{aligned}$$

Using Theorem 27.4, Lemma 28.5, Theorem 34.5, Exercise 38.2, and the definitions of the jump chain Y and of the waiting times $(S_n)_{n \in \mathbb{Z}_+}$ in the Kendall-Gillespie algorithm (Algorithm 2), it is not too difficult to show that

$$\begin{aligned}
(10) \quad \mathbb{P}_\gamma \left(\left\{ \varphi_{F_i} = T_m, T_m \leq t, Y_m = x, Y_{m+1} = x_1, \dots, Y_{n-1} = x_{n-m-1}, Y_{n-m} = x_{n-m}, \right. \right. \\
\left. \left. \sum_{k=1}^{n-m} S_{m+k} \leq t - T_m < \sum_{k=1}^{n-m+1} S_{m+k} \middle| \mathcal{G}_m \right\} \right) = 1_{\{\varphi_{F_i} = T_m, T_m \leq t, Y_m = x\}} g_{x_1, \dots, x_{n-m}}^{n, m, x, t}(T_m)
\end{aligned}$$

$\mathbb{P}_\gamma$ -almost surely, where $(\mathcal{G}_n)_{n \in \mathbb{N}}$ denotes the filtration generated by the jump chain and jump times (Definition 27.1),

$$g_{x_1, \dots, x_{n-m}}^{n, m, x, t}(u) := p(x, x_1) \cdots p(x_{n-m-1}, x_{n-m}) \int_0^{t-u} f_x * f_{x_1} * \cdots * f_{x_{n-m-1}}(s) \int_{t-u-s}^\infty f_{x_{n-m}}(r) dr ds$$

for all u in $[0, t]$, f_z denotes the pdf of an exponential random variable with mean $1/\lambda(z)$ and $*$ denotes the convolution operator. Using Theorem 27.4 (or Theorem 29.5), we have that

$$\begin{aligned}
g_{x_1, \dots, x_{n-m}}^{n, m, x, t}(u) &= \mathbb{P}_x \left(\left\{ Y_1 = x_1, \dots, Y_{n-m-1} = x_{n-m-1}, Y_{n-m} = x_{n-m}, \right. \right. \\
&\quad \left. \left. \sum_{k=1}^{n-m} S_m \leq t - u < \sum_{k=1}^{n-m+1} S_m \right\} \right) \quad \forall u \in [0, t].
\end{aligned}$$

Summing over $x_1, \dots, x_{n-m}$ in A , we find that

$$g_A^{n, m, x, t}(u) := \sum_{x_1 \in \mathcal{S}} \cdots \sum_{x_{n-m-1} \in \mathcal{S}} \sum_{x_{n-m} \in A} g_{x_1, \dots, x_{n-m}}^{n, m, x, t}(u) = \mathbb{P}_x(\{X_{t-u} \in A, T_{n-m} \leq t - u < T_{n-m+1}\}),$$

for all $m > n$. Moreover it follows from (9)–(10) that

$$(11) \quad \mathbb{P}_\gamma(\{\varphi_{F_i} = T_m, Y_m = x, Y_n \in A, T_n \leq t < T_{n+1}\} | \mathcal{G}_m) = 1_{\{\varphi_{F_i} = T_m, T_m \leq t, Y_m = x\}} g_A^{n, m, x, t}(T_m),$$

$\mathbb{P}_\gamma$ -almost surely, for all $m > n$. Using the same kind of argument, we also find that the above two also hold for $n = m$. Because

$$g_A^{x,t}(u) := \sum_{n=m}^{\infty} g_A^{n,m,x,t}(u) = \mathbb{P}_x(\{X_{t-u} \in A, t-u < T_\infty\}) \quad \forall u \in [0, t], \quad x \in F_i, \quad i \in \mathcal{I},$$

it follows from (8) and (11) that

$$\begin{aligned} \mathbb{P}_\gamma(\{\varphi_{F_i} \leq t, X_t \in A, t < T_\infty\}) &= \sum_{x \in F_i} \sum_{m=1}^{\infty} \mathbb{E}_\gamma \left[1_{\{\varphi_{F_i}=T_m, T_m \leq t, Y_m=x\}} g_A^{x,t}(T_m) \right] \\ &= \sum_{x \in F_i} \sum_{m=1}^{\infty} \mathbb{E}_\gamma \left[1_{\{\varphi_{F_i}=T_m, \varphi_{F_i} \leq t, X_{\varphi_{F_i}}=x\}} g_A^{x,t}(\varphi_{F_i}) \right] \\ &= \sum_{x \in F_i} \mathbb{E}_\gamma \left[1_{\{\varphi_{F_i} \leq t, X_{\varphi_{F_i}}=x\}} g_A^{x,t}(\varphi_{F_i}) \right] \quad \forall i \in \mathcal{I}. \end{aligned}$$

Applying Lemma 48.2(iii) and Theorem 3 it is not difficult to show that all states in F_i are exponentially recurrent. The argument given at the end of the proof of Theorem 46.1 then shows that for any x in F_i ,

$$\left| g_A^{x,t}(u) - \pi_i(A) \right| \leq \|p_{t-u}(x, \cdot) - \pi_i\| \leq C_x e^{-\beta_x(t-u)} \quad \forall A \subseteq \mathcal{S}, \quad u \in [0, t], \quad t \in [0, \infty),$$

for some constants $C_x < \infty$ and $\beta_x > 0$ depending on x . Because F is finite and $F = \cup_{i \in \mathcal{I}} F_i$, the above implies that there exists some $C < \infty$ and $\beta > 0$ such that

$$\left| g_A^{x,t}(u) - \pi_i(A) \right| \leq C e^{-\beta(t-u)}, \quad \forall A \subseteq \mathcal{S}, \quad x \in F_i, \quad i \in \mathcal{I}, \quad u \in [0, t], \quad t \in [0, \infty).$$

Because Proposition 38.8 implies that φ_{F_i} is finite if and only if φ_F is and $\varphi_{F_i} = \varphi_F$ and Theorem 4 shows that φ_F is $\mathbb{P}_\gamma$ -almost surely finite, it follows from the above that

$$\begin{aligned} \sum_{i \in \mathcal{I}} |\mathbb{P}_\gamma(\{\varphi_{F_i} \leq t, X_t \in A, t < T_\infty\}) - \mathbb{P}_\gamma(\{\varphi_{F_i} \leq t\}) \pi_i(A)| \\ \leq \sum_{i \in \mathcal{I}} \sum_{x \in F_i} \mathbb{E}_\gamma \left[1_{\{\varphi_{F_i} \leq t, X_{\varphi_{F_i}}=x\}} \left| g_A^{x,t}(\varphi_{F_i}) - \pi_i(A) \right| \right] \\ \leq \sum_{i \in \mathcal{I}} \sum_{x \in F_i} \mathbb{E}_\gamma \left[1_{\{\varphi_{F_i} \leq t, X_{\varphi_{F_i}}=x\}} C e^{-\beta(t-\varphi_{F_i})} \right] \\ = C e^{-\beta t} \sum_{i \in \mathcal{I}} \mathbb{E}_\gamma \left[1_{\{\varphi_{F_i} \leq t\}} e^{\beta \varphi_{F_i}} \right] = C e^{-\beta t} \mathbb{E}_\gamma \left[e^{\beta \varphi_F} \right] \quad \forall t \in [0, \infty). \end{aligned}$$

51. Farewell. I originally wrote the blurb below for my thesis. It feels right here too.

... there is yet much to be done to enable the quantitative analysis of chains with large state spaces ... The idiom ‘there is no rest for the wicked’ comes to mind. However, I believe that quite the opposite is true given how much *fun* these things can be. In the case that this is something you might like to get involved in (or already are), I wish you the best of luck: I am rooting for your success, *especially* in the areas where I did not meet mine (or, at best, met it partially). I hope that these questions and issues bring as much delight to your life as they have to mine.

“The things with which we concern ourselves in science appear in myriad forms, and with a multitude of attributes. For example, if we stand on the shore and look at the sea, we see the water, the waves breaking, the foam, the sloshing motion of the

water, the sound, the air, the winds and the clouds, the sun and the blue sky, and light; there is sand and there are rocks of various hardness and permanence, color and texture. There are animals and seaweed, hunger and disease, and the observer on the beach; there may be even happiness and thought.”

Richard Feynman in the first volume of his lectures on physics.

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SYMBOL INDEX

$=$	the left-hand side equals the right-hand side 2,
$:=$	the left-hand side is defined to be the right-hand side 1,
$\equiv$	the right-hand side is defined to be the left-hand side 1,
$\mathbb{N}$	set of natural numbers 1,
$\mathbb{Q}$	set of rational numbers 1,
$(0, \infty)$	set of positive real numbers 1,
$[0, \infty)$	set of non-negative real numbers 1,
$\mathbb{R}$	set of real numbers 1,
$\mathbb{Z}_+$	set of positive integers 1,
$\mathbb{Z}$	set of integers 1,
$\mathbb{N}_E$	set of extended natural numbers 1,
$\mathbb{R}_E$	set of extended real numbers 1,
$[0, \infty]$	set of extended non-negative real numbers 1,
$(0, \infty]$	set of extended positive real numbers 1,
$\mathbb{Z}_E$	set of extended integers 1,
$\lceil a \rceil$	smallest integer no lesser than a 1,
$\lfloor a \rfloor$	largest integer no greater than a 1,
$a \vee b$	maximum of two real numbers a and b 1,
$a \wedge b$	minimum of two real numbers a and b 1,
2^A	power set of A 2,
$\mathcal{B}(A)$	Borel sigma-algebra on A 2,
1_A	indicator function of A 2,
$\{f \in B\}$	pre-image of B through f 2,
$\ \cdot\ $	total variation norm of $\cdot$ 46,
$\text{supp}(\cdot)$	support of $\cdot$

Notation for discrete-time and continuous-time chains

$\mathbb{E}_\gamma$	expectation with respect to $\mathbb{P}_\lambda$ 11, 81,
$\mathbb{E}_x$	expectation with respect to $\mathbb{P}_x$ 11, 81,
$\mathbb{P}_\gamma$	probability measure describing the chain's statistics if its starting state is sampled from γ 11, 81,
$\mathbb{P}_x$	probability measure on describing the chain's statistics if its starting state is x 10, 81,
γ	initial distribution $(\gamma(x))_{x \in \mathcal{S}}$ 11, 81,
$(\Omega, \mathcal{F})$	underlying measurable space 10, 81,
$\mathcal{S}$	countable state space 10, 81,
$\mathcal{E}$	cylinder sigma-algebra on $\mathcal{P}$ 17, 90,
$\mathbb{L}_\gamma$	path law if the chain's starting state is sampled from γ 17, 90,
$\mathbb{L}_x$	path law if the chain's starting state is x 17, 90,
$\mathcal{P}$	path space 16, 90,

$A \rightarrow B$	set B is accessible from set A 33, 127,
$x \rightarrow y$	state y is accessible from state x 33, 127,
$\mathcal{D}$	domain 23, 115,
μ_S	space marginal of stopping/exit distribution 23, 24, 112, 116,
μ	stopping/exit distribution 22, 23, 111, 115,
ν_S	space marginal of occupation measure 23, 24, 112, 116,
ν	occupation measure 22, 23, 111, 115,
$\mathcal{C}_i$	i th positive recurrent closed communicating class 45, 141,
$\mathcal{T}$	set of states not positive recurrent 45, 141,
$\mathcal{R}_+$	set of positive recurrent states 45, 141,
$\mathcal{R}$	set of recurrent states 36, 130,
ϵ_∞	limit of the empirical distribution 46, 142,
π_γ	limit of the time-varying law 51, 144,
π_i	ergodic distribution associated with $\mathcal{C}_i$ 45, 141,
π	stationary distribution 43, 136,

Notation for discrete-time chains

P	one-step matrix $(p(x, y))_{x, y \in \mathcal{S}}$ 10,
$(U_n)_{n \in \mathbb{Z}_+}$	i.i.d. sequence of $(0, 1)$ -valued uniform random variables used in the chain's definition 10,
X	discrete-time chain $(X_n)_{n \in \mathbb{N}}$ 11,
$\mathcal{F}_\varsigma$	pre- ς sigma-algebra 20,
X^ς	ς -shifted chain 28,
$(\mathcal{F}_n)_{n \in \mathbb{N}}$	filtration generated by the chain 13,
ς	$(\mathcal{F}_n)_{n \in \mathbb{N}}$ -stopping time 19,
P_n	n -step matrix $(p_n(x, y))_{x, y \in \mathcal{S}}$ 16,
ϵ_N	empirical distribution $(\epsilon_N(x))_{x \in \mathcal{S}}$ 40,
p_n	time-varying law $(p_n(x))_{x \in \mathcal{S}}$ 15,
$\phi_A^k (\phi_x^k)$	k th entrance time to a set A (resp. state x) 32,
$\phi_A (\phi_x)$	first entrance time to a set A (resp. state x) 32,
$\sigma_A (\sigma_x)$	hitting time of a set A (resp. state x) 20,
σ_r	exit time from the truncation $\mathcal{S}_r$ 68,
σ	exit time from the domain $\mathcal{D}$ 23,

Notation for continuous-time chains

P	jump matrix $(p(x, y))_{x, y \in \mathcal{S}}$ 81,
Q	rate matrix $(q(x, y))_{x, y \in \mathcal{S}}$ of a continuous-time chain 81,
λ	jump rates $(\lambda(x))_{x \in \mathcal{S}}$ 81,
S_n	n th waiting time 82,
T_∞	explosion time 82,
T_n	n th jump time 82,

$(U_n)_{n \in \mathbb{Z}_+}$	i.i.d. sequence of $(0, 1)$ -valued uniform random variables used in the chain's definition 81,
X	continuous-time chain $(X_t)_{t \geq 0}$ 82,
Y	jump chain $(Y_n)_{n \in \mathbb{N}}$ 82,
$(\xi_n)_{n \in \mathbb{Z}_+}$	i.i.d. sequence of unit-mean exponential random variables used in the chain's definition 81,
$\mathcal{F}_\eta$	pre- η sigma-algebra 88,
X^η	η -shifted chain 94,
η	$(\mathcal{F}_t)_{t \geq 0}$ -stopping time a continuous-time chain 88,
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